

REFCON 6

Reefer Container Monitoring System

User's Guide

User's Guide



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REFCON 6

Reefer Container Monitoring System

Introduction to REFCON 6



1 Introduction to REFCON 6

Key subjects in this topic:

[Welcome to REFCON 6](#)^[8]
[Welcome to the User's Guide](#)^[8]

Related topics:

[Contact information](#)^[8]
[Purposes and User Roles](#)^[9]
[REFCON 6 configurations](#)^[11]
[Using the Online Help](#)^[14]

Welcome to REFCON 6

The REFCON system enables a local operator to automatically monitor and control thousands of reefer containers through a single computer system and provides following functions:

- Monitors alarm and temperature conditions, and notifies the user of any alarms generated by the container controller.
- Registers the history of each container, recording all important parameters and storing the data in accurate, detailed log files.
- Interfaces with the vessel's load calculator or the terminal's administrative system for automatically transferring reefer container data and locations.
- Optionally send commands, such as "defrost" and "change of set point", to the containers directly from REFCON.

This makes REFCON the ideal solution for any terminal and any container vessel - large or small.

REFCON 6 introduces built-in configuration options in many ways, which should make the application able to meet the requirements of a wide range of users. You can read more about this in [REFCON 6 configurations](#)^[11].

Welcome to the User's Guide

Welcome to the REFCON 6 User's Guide.

The Guide is available in three versions:

- An online help, which can be opened by selecting "Help" from REFCON's Help menu, or by selecting the F1 keyboard button. The latter will in many cases open the online help directly on the page most relevant in the context. Note that the online help - to some degree - is adapted to the actual purchase time configuration.
- A print-friendly PDF-file comprising the entire online help, also available online from REFCON's menu. Note that the PDF version is **not** adapted to the actual configuration and thus may describe functions that are not enabled in your specific REFCON system.
- A printed manual, which can be purchased from your REFCON supplier.

1.1 Contact information

Key subjects in this topic:

[About us](#)^[8]
[Getting assistance](#)^[9]

Related topics:

[Introduction](#)^[8]

About us

The REFCON application was developed and delivered by:

Copeland LP
Transportation Solutions
Boeletvej 1,



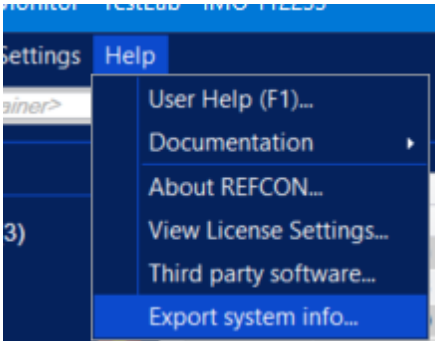
DK-8680 Ry, Denmark.
E-mail: ColdChain.TechnicalServices@copeland.com
www.copeland.com

Getting assistance

Should a problem occur that you are not able to solve, you may send an email to the above address.

In the email, please include everything you have observed, including screen shots and error logs.

Also, please attach the System Info file, if possible. You produce this file from the REFCON Monitor:



In the "Save As" dialog box, choose a drive and folder where you want the file to be stored. We recommend you leave the file name unchanged.

Having selected [Save], the file is created and is now ready to be attached to your email.

1.2 Purposes and User Roles

Key subjects in this topic: Overview^[9] Purpose 1 of REFCON: local cargo supervision^[10] Purpose 2 of REFCON: local claim treatment^[10] Purpose 3 of REFCON: data provider^[10] Summary of user roles^[10] Cargo supervisor role^[10] Claim handler role^[11] Container repairer role^[11] REFCON maintainer role^[11] The IT maintainer role^[11] The Data responsible role^[11]	Related topics: Introduction^[8]
---	---

Overview

Let us discuss the purposes of REFCON and your possible roles when using it.

Of course we, as manufacturers, can not know the specific purposes that a specific customer has in mind when deciding to install a REFCON system. Neither can we know your specific role when using it. But REFCON was designed to comply with the set of known common purposes and the set of common roles described below. The different functions in REFCON often aim at a certain purpose or a certain role, so understanding the purposes and roles would explain why



the functions are included - and when you understand your role in this context you would easier find the right function to use.

Purpose 1 of REFCON: the local cargo supervision system

The original main purpose of REFCON is to make you, the local user, able to treat the cargo in the containers as required while the containers are your responsibility. Complying with this purpose means making sure that:

- all reefer containers are connected to power
- all reefer containers are continuously supervised
- all reefer containers have correct setpoints
- any alarm from the containers is dealt with
- any problem is dealt with as soon as possible.

Purpose 2 of REFCON: the local claim treatment system

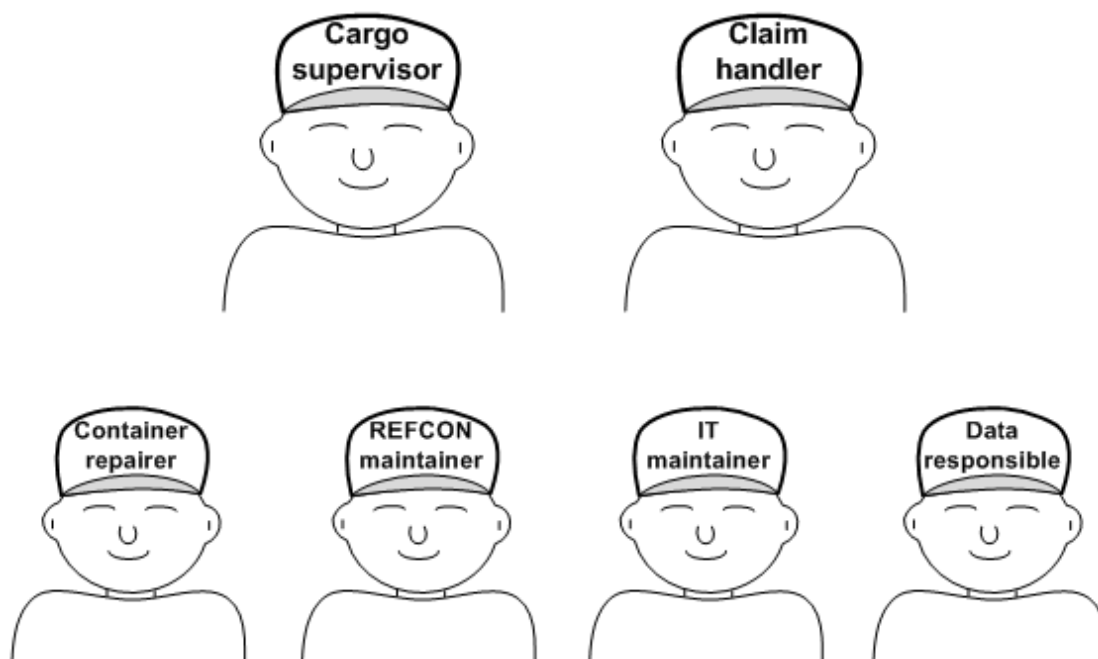
Should a claim occur relating to a container that, while loaded with the specific cargo in question, for some time was connected to your REFCON, the archived log files are supposed to help you proving or rejecting the claim.

Purpose 3 of REFCON: data provider

The last significant purpose of REFCON is to be a data link between all the connected reefer containers and external systems. Examples of external systems are the Global Monitoring Server and the Planning System.

Summary of user roles

Whether one user takes more roles, or there are more users with separate roles, it is relevant to think about the following principle roles for users of REFCON:



As can be seen there are two main roles, corresponding to the purposes 1 and 2 of REFCON. The roles in the second row are service roles necessary to fulfill the purposes and the main roles.

Cargo supervisor role

A user taking the Cargo supervisor role is responsible for the points listed under [Purpose 1](#)^[10] above. This user would take advantage of a wide range of functionality in REFCON, including [the REFCON Monitor](#)^[22] and [the Alarm System](#)^[110].



Claim handler role

A user taking the Claim handler role has the responsibilities mentioned under [Purpose 2](#)^[10] above. This user would specifically take advantage of the [Historic data logging](#)^[160] functionality.

Container repairer role

A user taking the Container repairer role is supposed to take action when a container has mechanical or electrical defect.

- Typically the action is initiated by an alarm (see [the Alarm System](#)^[110]). The alarm text should give an idea of what is wrong. In most cases you will have to check the container manufacturer's manual for instructions.
- If REFCON has no connection to a container, the Container repairer should, in co-operation with the REFCON maintainer, get the communication to work as described in [the Modem Troubleshooting Guide](#)^[278]. You can maximize the value of REFCON by taking this job very seriously.

REFCON maintainer role

A user taking the REFCON maintainer role is responsible for keeping the REFCON system running and working optimally.

This user would need a deep knowledge and understanding of the REFCON system hardware being described in [Understanding and Maintaining REFCON](#)^[302], where you can also find descriptions of the available tools in the REFCON Monitor.

The REFCON maintainer works closely together with the IT maintainer:

The IT maintainer role

In the very most cases, REFCON is connected to a LAN network used for interfacing internally and/or externally.

A user taking the IT maintainer role is responsible for keeping the LAN and the required external systems running and working optimally. For example, in many cases an external mail system is required to transfer data from REFCON to external receivers.

This user would need a thorough understanding of the interfaces relevant to the specific REFCON system. The possible interfaces are sketched in [Understanding and Maintaining REFCON](#)^[302], while detailed descriptions are found in [GMS log interface](#)^[176], [Planning data import](#)^[182], [Planning system communication interface](#)^[202] and [Email notification](#)^[268].

The Data responsible role

A user taking the role as Data responsible must ensure that log data are backed up and secured for possible future claim handling purposes. We recommend using the [Automatic backup and restore function](#)^[328] for this, or alternatively the [Backup function](#)^[43].

1.3 REFCON 6 configurations

Key subjects in this topic:

[Overview](#)^[12]
[Purchase time configuration](#)^[12]
[Runtime configuration](#)^[13]
[Viewing the Purchase time configuration](#)^[13]

Related topics:

[Introduction](#)^[8]



Overview

REFCON 6 is a highly flexible application providing a wide range of functionality. Most users do not need all this functionality. Therefore, REFCON 6 can be configured. Two major levels of configuration exist:

Purchase time configuration

Your REFCON system was delivered with a *license file* that decides the *purchase time configuration*.

The file is named License.xml. You can open it and read it, but you are in no way able to change it, as it is digitally signed. Contact us (see [Contact information](#)^[8]) if you need a change.

To see exactly what is in your license, see [Viewing the Purchase time configuration](#)^[13] below.

The following list shows some of the decisions made in the license file - note that the list should not be considered complete:

Option / information	Type	Description
Terminal or Ship	Flag	Some details in REFCON work different depending on whether your system is in a terminal or on a ship. In the most cases it is smaller differences.
Installation Name	String	The installation name is and must be unique, that is, no other REFCON system in the world is allowed to have the same name. The name is shown in the caption line of the REFCON Monitor. Further it is used to identify the system when transmitting or exporting data to external receivers.
IMO number	String	Ships only: The ship's unique IMO number.
Container Plan enabled or not	Option	This option decides whether the graphical container overviews are included in your system or not.
RDC or not	Option	This option decides whether you can use the handheld REFCON Data Collector with your REFCON system or not. The RDC is powerful to collect data from containers that can for some reason not communicate over the power lines.
Remote container operations enabled or not	Option	This option decides whether you can send commands to container controllers from REFCON, such as changing temperature setpoint or starting defrost.
Email notification enabled or not	Option	This option decides if you can have emails transmitted in case of alarms.
Digital input/output alarm extension enabled or not	Option	If a supported digital input/output card is found in the Main PC AND this option is enabled, alarms are extended through the digital outputs.
Automatic 2-way Planning System Interface enabled or not	Option	This option decides whether REFCON can communicate bi-directionally online via LAN with the Planning System through an integrated protocol. Note that an alternative option is that REFCON communicates similarly through a plugin ^[309] that interfaces to the Planning System.



Option / information	Type	Description
Maximum number of REFCON Monitors	Number	This value decides the maximum number of concurrently running REFCON Monitors.
Maximum number of simultaneously communicating reefer containers	Number	This value decides the maximum number of concurrently communicating reefer containers that REFCON supports. Should more containers be connected, you will be significantly warned, but the containers still are alarm supervised.
Expected number of RMM Masters	Number	Your REFCON system is purchased with a certain number of RMM Masters. REFCON automatically detects the actually connected RMM Masters, and should an extra be connected, this is detected and worked with, too. This number in the license is used to make sure that possible defective RMM Masters, being no longer detected by REFCON, are handled and repaired, rather than ignored.
Expected number of RS2LAN units	Number	Your REFCON system is purchased with a certain number of RS2LAN units, maybe zero. REFCON automatically detects the actually connected RS2LAN units, and should an extra be connected, this is detected and worked with, too. This number in the license is used to make sure that possible defective RS2LAN units, being no longer detected by REFCON, are handled and repaired, rather than ignored.

The license also gives permissions to selected [plugins](#)^[309], if any. Note that if the REFCON Mobile Plugin and/or the REFCON Wireless Hub Plugin is enabled, this enables the REFCON Portable Modem Solution, see [REFCON 6 extension](#)^[218].

Runtime configuration

From the REFCON Monitor it is possible to make quite a number of decisions regarding the way REFCON works. This is primarily done through the [REFCON Setup](#)^[330] dialog box described elsewhere in this online helpmanual.

Viewing the Purchase time configuration

Select "View License Settings" from the "Help" menu to view the contents of your license file:



License settings

The checked options are enabled in your REFCON system:

☒	Ship version of REFCON
☐	Terminal version of REFCON
☒	Planning data import
☐	Automatic 2-way Planning System Interface
☒	Container Plan graphical view
☐	Email notification of alarms
☒	RDC - REFCON Data Collector
☒	Digital Input/Output - alarm extension
☒	Remote container operations enabled
☐	Danbuss on Com1

The following settings are applied:

	Value
Maximum number of REFCON Monitors	5
Maximum number of simultaneously communicating reefer containers	1100
Update expiration date	2020-12-22
Update URL: http://update.myrefcon.com/refcon	
Expected number of RMM Masters	4
Expected number of RS2LAN units	2

Note: Any kind of change to the above settings requires that you obtain a new license file. Please use the buttons below for contact information.

Note that the Installation Name (and corresponding textual information) is not included in the viewer, as this is shown in the caption of the REFCON Monitor.

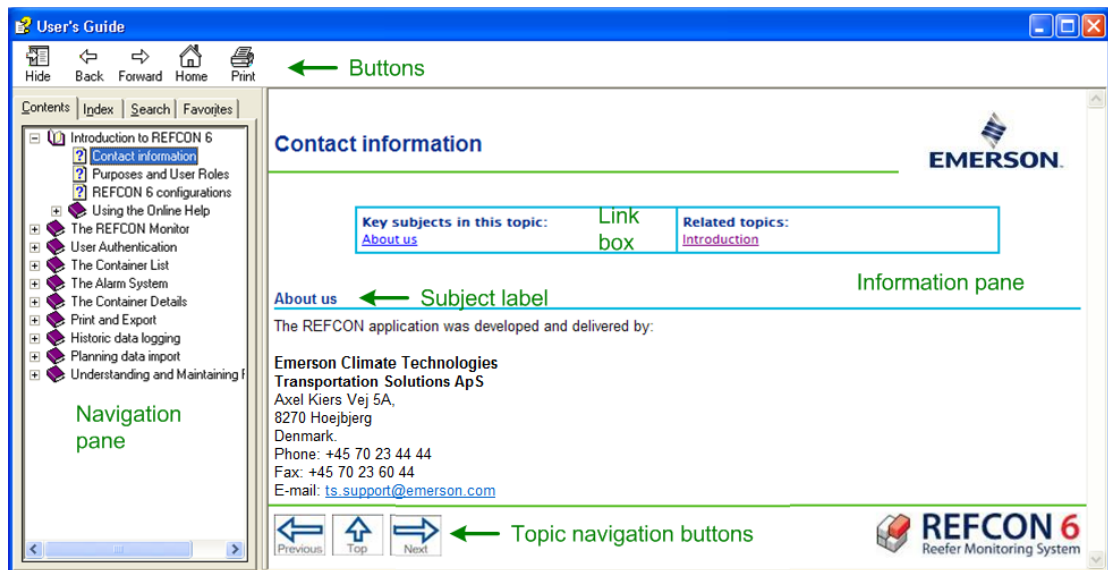
Also note that this window does not show the possible [plugin](#)^[309] or 3rd party Data Service clients being licensed. You can inspect these in the [REFCON Manager](#)^[350], the Plugins tab.

1.4 Using the Online Help

Key subjects in this topic: Overview^[14] The Navigation Pane^[13] Using the Help Contents Tab^[15] Using the Help Index Tab^[13] Using the Help Favorites Tab^[16] The Information Pane^[16] The Buttons^[16]	Related topics: Introduction^[8] Searching the Online Help^[17]
---	---

Overview

In the Help menu, select the "Help" option to have the online help displayed. The Help window appears:



The Help window consists of three main parts:

- The [command buttons](#) ¹⁶¹ at the top.
- The [Navigation Pane](#) ¹⁵¹ to the left.
- The [Information Pane](#) ¹⁶¹ to the right.

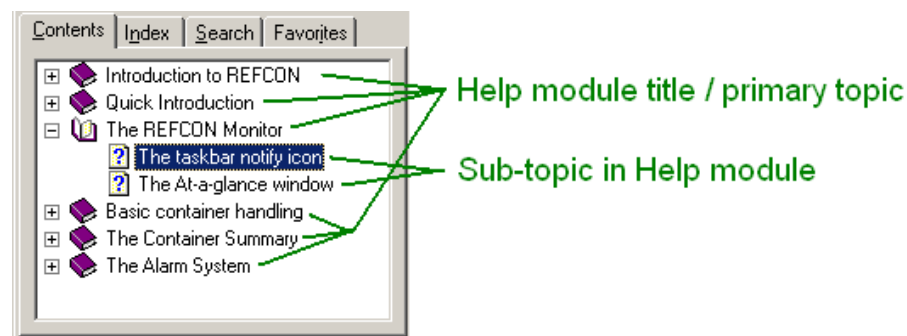
The Navigation Pane

You have four possible entrances to the help information:

- Click the [Contents tab](#) ¹⁵¹ to scroll through a table of contents for Help.
- Click the [Index tab](#) ¹⁵¹ to have all keywords listed in alphabetical order.
- Click the [Search tab](#) ¹⁷¹ to be allowed to enter any free text search string.
- Click the [Favorites tab](#) ¹⁶¹ to be allowed to use and build your own list of important topics.

Using the Help Contents Tab

In the Help window, selecting the **Contents** tab displays the table of contents for the Help system.



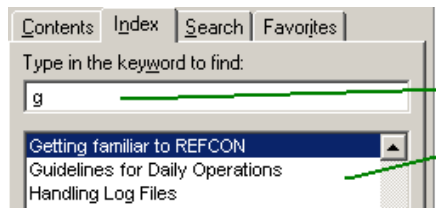
As you see, the Help is organised in *modules*. In the example above, there are 6 modules, of which "The REFCON Monitor" is open. Within each module, you will find a number of sub-topics, in some cases organised in more levels.

Clicking a + sign makes sub-topics shown. Clicking a – sign makes sub-topics hidden.

To display any topic, just click it. This also is true for the module titles.

Using the Help Index Tab

In the Help window, selecting the **Index** tab lists all keywords in alphabetical order.



Enter first characters you're looking for

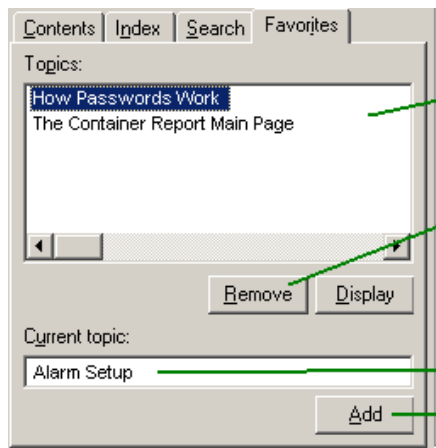
Alphabetically sorted list of topics,
double-click to display information

In the textbox, type in the first characters in the keyword you're looking for. The list scrolls automatically as you type, if the string is found.

Double-click the desired topic, or single-click and select the Display button.

Using the Help Favorites Tab

In the Help window, selecting the **Favorites** tab will allow you to use and build a list of important topics. Note that you share the list with the other users of the PC.



List of favorites,
double-click to display topic

Click to remove highlighted topic from favorites list

The topic being shown right now

Click to add current topic to favorites

To add a topic, you must first open it from one of the other tabs.

The Information Pane

In the larger right side of the Help window, the selected topic is shown. You will find the following ingredients:

- The Title of the topic. You will recognize it from the Navigation Pane.
- The Link box, with the list of subject labels to the left, and a list of related topics to the right.
- The body of the topic, with text and graphics.
- The Topic navigation buttons at the bottom, see [The Buttons](#) ¹⁶.

The Buttons

At the top of the Help window, and at the bottom of the Information Pane, the following buttons are available:

Buttons	Explanation
 Hide /  Show	The Hide button removes the Navigation Pane, making the Help window smaller while the topic information is still visible. Click Show to bring it back.
 Back /  Forward	The Back and Forward buttons operate on your actual chain of visited topic pages. Each time you open a new topic page, it is inserted in the chain after the actually displayed page. The Back button jumps to the page earlier in the chain. Having used the Back button, you can use Forward to return to a page later in the chain.
 Home	Jumps to the very first topic in the very first Help module ("Introduction to REFCON 6").



Buttons	Explanation
	Prints the current topic or a number of topics on the printer you select. To have a collection of topics printed, proceed as follows: 1. Click the Contents tab in the Navigation pane. 2. Select the heading of a topic with sub-topics. 3. Click the Print button and select "Print the selected heading and all subtopics".
 / 	Jumps to the previous / next topic in the current Help module according to the Contents table. If you are already on the first / last topic, the respective button is not shown. Please do not confuse with the Back and Forward buttons upperleft in the Help window.
	Jumps to the first topic in the current Help module according to the Contents table. Please do not confuse with the Home button at the top of the Help window.

1.4.1 Searching the Online Help

Key subjects in this topic:

[Overview](#)^[17]
[Using the Help Search Tab](#)^[17]
[Search Criteria Syntax](#)^[18]
[The List of Topics](#)^[19]

Related topics:

[Using the Online Help](#)^[14]

Overview

You can find the information you are looking for in a number of ways:

- Within REFCON, press the F1 key or select the Help button where available. This typically jumps to the help page most relevant to the context in REFCON.
- Use the Contents or the Index tab, see [Using the Online Help](#)^[14].
- Use the Search function described below.

Using the Help Search Tab

In the Help window, selecting the **Search** tab will allow you to enter any free text search string.



Search criterion textbox

List of previous search criteria

List of operators

Click to do the search

Double-click to inspect topic

Search result highlights

Maybe make additional choices

Proceed as follows:

Step	Action
1	Enter a search criterion (see details below).
2	Maybe change the settings of the three checkboxes at the bottom.
3	Select "List Topics" to do the search – it might take a while.
4	If no hits are found, a dialog box informs about this, and you may try again.
5	If hits are found, a prioritised list appears in the listbox.
6	Double-click a topic to have it shown in the Information pane.
7	Note that your search criterion word(s) are highlighted.

Search Criteria Syntax

If you, as in the above example, type in more than one word, topics will be listed that contain all the words, but they don't have to be in succession. Using the AND operator will have the same function. For example, the criterion **iso container** and the criterion **iso and container** will both find all topics where both the word iso and the word container are found.

Use quotation-marks (") to search for all the words in sequence. For example, the criterion **"iso container"** will only find topics where the two words are found in succession.

Use the OR operator to search for any of the words, that is, topics are listed that contain either the one or the other search word (or more). For example, the criterion **iso or container** will find all topics where either the word iso or the word container are found.

Also you can use wildcard characters:

- Use the asterisk (*) to replace any sequence of characters. For example, **wh*** will find why, who and when.
- Use the question-mark (?) to replace one character. For example, **wh?** will find why and who, but not when.

Note that you can't combine the wildcard characters and the quotation-marks.



The List of Topics

In the list of found topics, the first column shows the topic headings. The second column shows the titles of the Help modules to which the information topics relate.

REFCON 6

Reefer Container Monitoring System

The REFCON Monitor

Transportation Solutions





2 The REFCON Monitor

Key subjects in this topic:

[Overview](#) ^[22]

[Starting the REFCON Monitor](#) ^[22]

[Logging on to REFCON Monitor](#) ^[23]

Related topics:

[The REFCON Monitor base picture](#) ^[24]

[The Mini-status window](#) ^[31]

[Shortcut menus](#) ^[33]

[The Performance Bar](#) ^[27]

[Container Quick Search](#) ^[35]

[The REFCON Clock System](#) ^[36]

[Backup function](#) ^[43]

[Configuration export\(/import\)](#) ^[46]

[Selecting drive and folder on Main PC](#) ^[49]

[The taskbar notify icon](#) ^[53]

Overview

As a REFCON user, you mainly work with the containers through the REFCON Monitor. You work through various pictures and dialog boxes that are described in specific parts of the manual online help.

Certain common principles lie behind all the pictures and the way you operate the system. First of all you will recognize principles origin from Microsoft Windows, that is, things basically work like in most Windows applications.

But on top of this we have defined our own user interface basics, regarding how the screen pictures are organised, and how you operate.

Starting the REFCON Monitor

As standard you start the REFCON Monitor by selecting it in Windows' Start menu:



or by selecting the desktop icon, if installed:



If neither of the above are found, you will have to run through the following first-start procedure. If certain 3rd-party software (Microsoft, Adobe) needs to be installed, you must have administrator rights on the PC.

Step	Action	Comment
0	If this is not the REFCON Main PC, make sure your PC is connected by LAN (ethernet) to the Main PC.	Read more about Main PC and Secondary PC in The REFCON system hardware ^[303] .
1	Obtain the REFCON Monitor link from the REFCON Manager on the Main PC.	See Remote Monitor Configuration ^[356] .
2	Run the REFCON Monitor link in Internet Explorer.	You will see a "web" page as shown after this table.



Step	Action	Comment
3	Click "Install".	An anonymous file named "setup.exe" is to be downloaded.
4	Select "Run" to run "setup.exe".	You may get a warning for "Unknown Publisher". Click "Run" to proceed.
5	Be patient while your PC is checked for software.	You may be asked whether to install REFCON Monitor from "Unknown publisher". Click "Install" to proceed.
6	After a little time you are asked for a PC description. Please enter a significant, unique description of the specific PC.	See dialog box example after this table. The description you enter is in particular important if you have several REFCON Monitors running. See also the System Devices list ³³³ .

The REFCON Monitor installation "web" page looks like this:

Emerson Climate Technologies - Transportation Solutions REFCON 6 Monitor

Name: REFCON 6 Monitor

Publisher: Emerson Climate Technologies - Transportation Solutions ApS

The following prerequisites are required:

- Windows Installer 3.1
- .NET Framework 4.0 (full)

If these components are already installed, you can [launch](#) the application now.
Otherwise, click the button below to install the prerequisites and run the application.

[Install](#)

[Transportation Solutions Customer Support](#) :: [ClickOnce and .NET Framework Resources](#)

The PC Description entry dialog box looks like this:

Enter PC description

Please enter a unique, significant description of this PC, for example, where it is located

Office REFCON PC

OK Cancel

Logging on to REFCON Monitor

Having started REFCON Monitor as described above, as standard you need to log on, please refer to [Logging on to REFCON Monitor](#)⁵⁷.

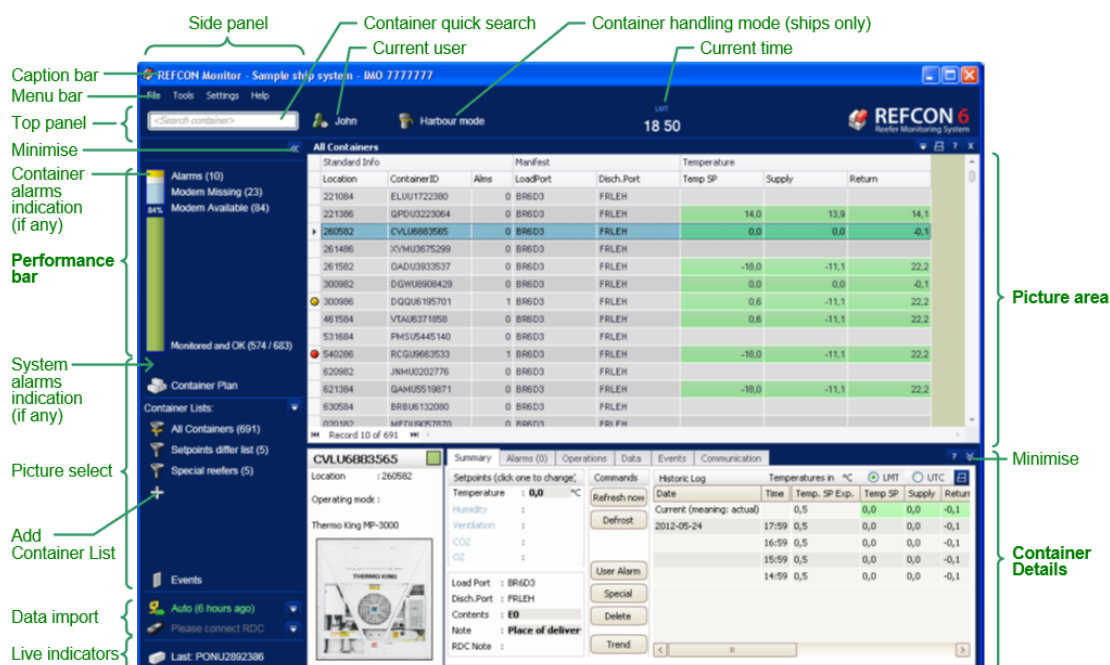


2.1 The REFCON Monitor base picture

Key subjects in this topic: Overview ^[24] The Caption bar ^[24] The Menu bar ^[25] The Top panel ^[25] The Side panel ^[25] Statistics in the Side panel ^[25] The Picture area ^[26] The Container Details ^[26] Increasing the Picture area ^[26] "List changed" indication ^[26]	Related topics: Container Quick Search ^[35] The Performance Bar ^[27] The REFCON Monitor ^[22]
---	---

Overview

When the REFCON Monitor is opened, the base picture is shown:



The various parts of the base picture are described in the following.

The Caption bar

The Caption bar identifies the application and includes the three standard buttons:

Button	Explanation
	Minimises (= hides) the REFCON Monitor. To re-open it, click the REFCON Monitor item in Windows' taskbar (normally bottommost on the screen).
	The first button maximises the REFCON Monitor, making it occupy the entire screen. When so, you can use the second button to restore the REFCON Monitor to only occupy a part of the screen, thus making it possible to work in more applications at the same time.
	Closes the REFCON Monitor. You are asked for confirmation.



Note that unless it is maximised you can:

- resize the REFCON Monitor window by dragging any of its edges or corners;
- move the REFCON Monitor by dragging the Caption bar.

The Menu bar

The actual menu will depend on the context. Feel comfortable to investigate the menu by clicking around!

The Top panel

At the top of the REFCON Monitor, below the Menu, the *Top panel* is utilising the [Container Quick Search](#)^[35] function, the [current user](#)^[56] (if [User Authentication](#)^[56] is enabled), the [handling mode](#)^[117] (ships only), the current time, and the product logo. In rare, special cases, more information may occur.

Click the current time field to

- decide whether to display the LMT time or the UTC time (when the REFCON Monitor starts, it always shows LMT);
- [set the clock](#)^[41].

See [The REFCON Clock System](#)^[36] for further information.

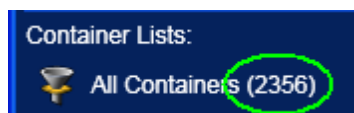
The Side panel

The *Side panel*, being the wide dark blue ribbon to the left, includes overall information and commands for your daily use. From top to bottom the included items are:

- The [Performance Bar](#)^[27].
- The System alarms indicator, which is visible only if there is any system alarm.
- The optional [Container Plan](#)^[96] entry.
- The Container Lists, being described in [Container List](#)^[66].
- The [Events](#)^[171] entry.
- The Planning System data import element, being related to [Planning data import](#)^[182] and, if licensed, to the optional [Planning System Communication Interface](#)^[202] or the optional [Automatic Planning System interface](#)^[214].
- The optional [RDC](#) element.
- The Live indicators. As standard (like in the picture above) there is one indicator that normally shows the ID of the container that right now communicates on the PCT (Power Cable Transmission) system. As this changes frequently, the indicator will normally change rapidly though irregularly, and therefore, if it does not change for some time, it might indicate that something is wrong. However, while a [container remote operation](#)^[137] is taking place, the indicator shows the status of this in stead.

Statistics in the Side panel

Most list items do normally have a counter shown in parentheses:



In general the counter is the number of alarms or containers in that specific list.

The "All Containers" counter illustrated above shows the total number of containers currently registered in REFCON's database:

- All that was included in the latest data transfer from the Planning System, of which some might not yet have started communicating.
- All that are currently communicating.
- All that have stopped communicating but have not yet been deleted from the database.

By [adding your own container lists](#)^[80] you can create statistics as you need.



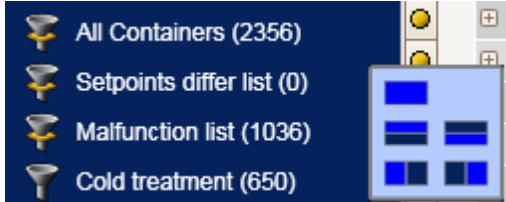
The Picture area

The larger centre-right area of the REFCON Monitor is where the pictures appear that you select in the Side panel or the menu.

It is possible to have one or two pictures open at a time. If two are open, selecting the small Close button (X) upper right in a picture will close this, making the other picture resize to the whole picture area. In contrary, if only one picture is open, the Close button does not work, as you can not close the last picture. If you want to do so, open another instead from the Side panel or the menu.

As the starting point, only one picture is open. When simply clicking another picture item in the Side panel or the menu, this replaces the existing picture.

When you want two pictures to be open, proceed as follows:

Step	Action	Result
1	Move the mouse over the relevant picture item in the Side panel.	You will see that a symbol appears next to the item: 
2	Click the symbol.	The split selection popup appears: 
3	Select one of the split screen symbols.	The new picture opens in the position indicated by the lighter blue.
4	Maybe move the splitter to change the sizes of the pictures. You recognize the splitter by the three spots, appearing horizontally or vertically: 	One picture gets smaller, the other larger.

The Container Details

The larger lower-right area of the REFCON Monitor is occupied by the [Container Details](#)¹³⁰.

Increasing the Picture area

It is possible to make the Picture area larger by sliding aside the Side panel and/or sliding down the Container Details. Use the Minimise buttons upper right in the respective element for this purpose. To restore the Side panel and the Container Details, respectively, re-select the button.

"List changed" indication

It is possible for any user, without logging on, to make **temporary** changes to any of the lists in REFCON.

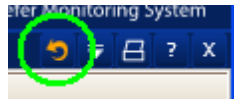
Whereas such changes can be very useful to the current user, they can cause misunderstandings to others. Therefore, a temporarily changed list is distinctly indicated in the REFCON Monitor:



Location	ContainerID	Alms	Disch.Port	Note	Temp. SP Exp.	Temp SP	Supply
020684	MWCUS662704	0	FRLEH	Place of delivery: ...	0,5	0,5	
020484	MHU5613184	0	FRLEH	Place of delivery: ...	0,5	0,5	

As seen, the title of the list gets orange with an asterisk appended, both in the Side panel and in the caption of the list.

You will notice that the "Reset sorting/filtering" button has shown up upper right:



If you click this button, you are asked if you are sure. If so, the list returns to it's standard / last saved version, and your changes are discarded.

2.2 The Performance Bar

Key subjects in this topic:

- [Purpose](#) ²⁷¹
- [Overview](#) ²⁷¹
- [How the Performance Bar works](#) ²⁹¹
- [The Performance Bar colours](#) ²⁹¹
- [Minimum height of coloured area](#) ³⁰¹
- [Dynamic list entry position](#) ³⁰¹
- [Opening a related list](#) ³¹¹

Related topics:

- [Modem Troubleshooting](#) ²⁷⁸¹
- [The REFCON Monitor](#) ²²¹

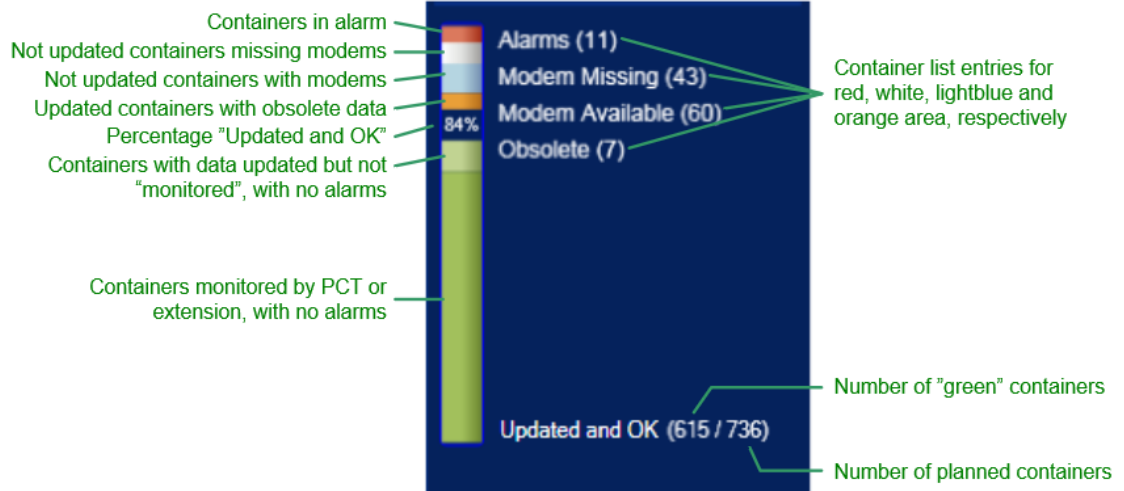
Purpose

The purpose of the Performance Bar is to help you perform your job even smarter. It guides you in your daily work, as it emphasises where your effort is most needed, and gives easy access to the lists of containers that mostly need your attention.

Also it could help the work planner to estimate the work load and similar.

Overview

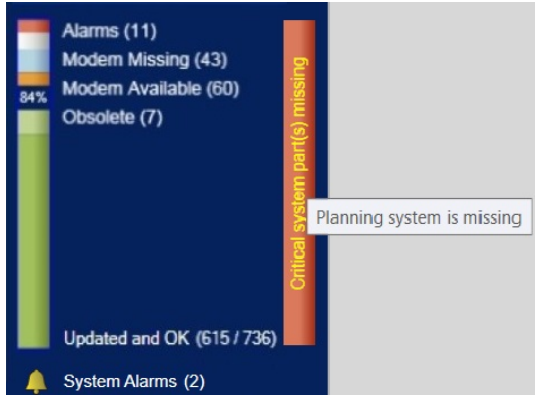
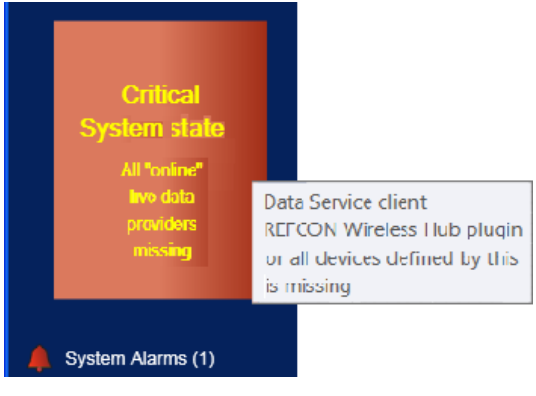
The Performance Bar consists of a vertical bar with up to six coloured areas, together with list entries and textual indications:



You may additionally experience the Performance Bar in any of the following special cases:

No containers at all	Monitoring is perfect !!
Normal starting point. Seen if you have no reefer containers at all for the present.	Ideal situation: All containers are updated satisfactory, and none of them are in alarm. There might be a lighter green area.



Medium severity system alarm(s)	High severity system alarm(s)
	
Essential part(s) of your system are in alarm, but still container monitoring does fully or partly run. The tool tip shows the specific cause - as does the System Alarm Summary ^[112] and the System Devices list ^[333] . You must remedy the problem as soon as possible.	"Online" live data providers, such as the PCT system and the REFCON Wireless Hub system, are licensed and/or detected, but none of these are available. REFCON does therefore not receive any live data as intended, and can therefore not show a reliable Performance Bar. The tool tip shows the specific cause - as does the System Alarm Summary ^[112] and the System Devices list ^[333] . You must remedy the problem as soon as possible.

How the Performance Bar works

REFCON counts the containers periodically with a certain time interval, which means that the updating of the Performance Bar can be slightly delayed.

When counting, REFCON checks the monitoring status for each individual container.

The results are as indicated in the following table.

The Performance Bar colours

The Performance Bar shows up to 6 colours as indicated in the above "real world" example. Interpret the colours as follows:

Colour	Category	Explanation	Recommended actions
 Green	"Monitored" OK	The containers in this group are monitored ^[131] perfectly well by power line communication or extension with data up-to-date, no alarms.	No action needed.
 Lighter green	Data updated OK, not "Monitored"	The containers in this group are updated ^[131] well by the optional RDC or by a similar extension with data up-to-date, no alarms.	You have to repeat the manual RDC inspection or similar extension periodically. You might reduce the need for manual inspection by repairing or mounting modems.
	In alarm	The containers in this group are in alarm. They may or may not be	The alarms must be handled by user.



Colour	Category	Explanation	Recommended actions
Red / flashing		updated well.	
 White	Modem missing	The containers in this group are not updated. They have been reported by the Planning System, and either have they not yet started communicating, or they never will because modems are missing or defect.	Manual inspection is required unless you choose to install modems. You may use the optional RDC or similar extension.
 Light blue	Modem available	The containers in this group are not updated because their modems have no connection to the controllers and communicate by factory ID.	Manual inspection is required unless you choose to repair/replace the modems or their connections. You may use the optional RDC or similar extension.
 Orange	Obsolete	This group covers containers that have been green, but for which data are now obsolete. This can be due to a number of possible reasons, including: • Containers having been plugged off, intendedly or not. • Containers having defective modems or the like. • Planning data being incomplete or not up-to-date. • You are late in doing a manual inspection using RDC or similar extension.	Fix the modems and/or the Planning data, or do a manual inspection.

Minimum height of coloured area

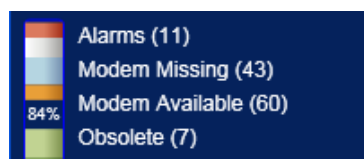
A coloured area has a minimum height of about 4% of the total bar height, because otherwise it could become almost invisible. In particular for the Alarms area it is vital that it is clearly visible. So, if for example just 1% of the containers are in alarm, it still occupies 4% of the total height.

Dynamic list entry position

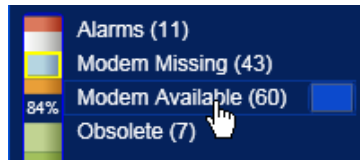
As the starting point, the container list entry text is centred vertically in relation to the corresponding coloured area:



The minimum text height is, however, greater than the minimum area height. This means that in case of small amounts, the two are displaced:



If you are in doubt what text belongs to what coloured area, move the mouse cursor over the text. This makes a border show up at the related area:



Opening a related list

Each coloured area have dedicated list pictures attached:

- For the alarm area, the [Container Alarm Summary](#)^[112].
- For other areas, specific [Container Lists](#)^[66].

To open a list,

- click the text or the [split screen symbol](#)^[26], or
- click the coloured area.

2.3 The Mini-status window

Key subjects in this topic: Overview ^[31] Opening and Closing the Mini-status window ^[31] What the Mini-status window shows ^[32]	Related topics: The REFCON Monitor ^[22]
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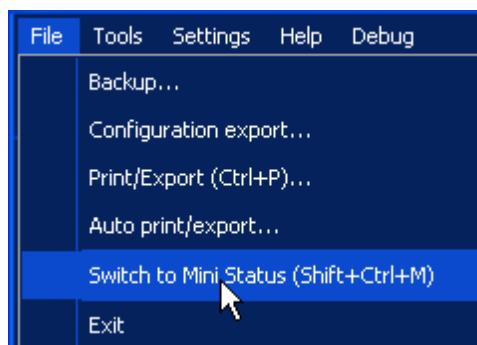
Overview

When designing the Mini-status window, we thought of the situation that you are not working at the REFCON PC, but you are from time to another present in the room where it is situated. By having the Mini-status window open, you can take just a short look from the distance and see if everything is all right or not.

Also we think that the Mini-status window could be useful when you work at the REFCON PC, using another application than REFCON. You might then place the Mini-status window in a corner of the screen and watch it out of the tail of your eye.

Opening and Closing the Mini-status window

You open the Mini-status window by selecting it from the File menu:



You can also use the key combination Ctrl + Shift + M.

You will see that this hides the REFCON Monitor and displays the Mini-status window in stead.

Click anywhere in the Mini-status window except the title bar to change back to the REFCON Monitor.



What the Mini-status window shows

The Mini-status window is divided into four parts:



The upper left part is the title bar, reading REFCON. You can move the Mini-status window by dragging this area.

The centre left part indicates the total number of items in alarm in REFCON right now, that is, the sum of containers in alarm and system devices in alarm. If any of the alarms are unacknowledged, the field is flashing. If there is no alarms at all in REFCON, the field is green:



The lower left part currently shows the ID of the latest inspected container. When running well, with some number of reefer containers connected, the shown ID will change irregularly. See also the PCT indicator in [the Side panel](#)^[25].

The right part is a smaller version of the [Performance Bar](#)^[27].

In case of a system alarm, the Performance Bar gets smaller, making room for the bell indicator:



If the system alarm is even a critical alarm, the Performance Bar is replaced by a distinct warning:



which should make you take action as soon as possible.

2.4 Shortcut menus

Key subjects in this topic:

[Overview](#)^[33]

[Illustrative example](#)^[34]

[The Container Command](#)

[section](#)^[34]

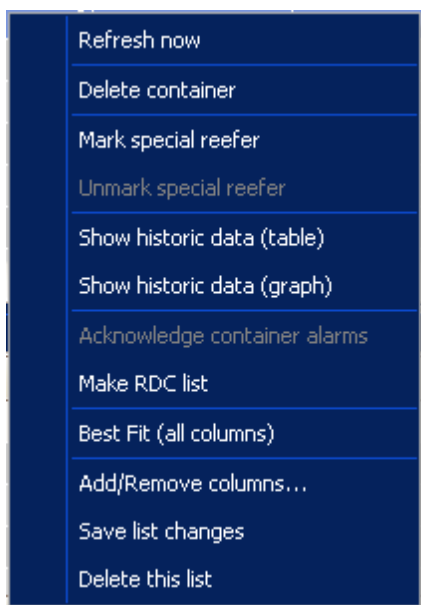
[Other sections](#)^[33]

Related topics:

[The REFCON Monitor](#)^[22]

Overview

In all relevant places, such as list pictures and the Container Plan, Shortcut menus (also known as right-click menus) are available.



In general, the Shortcut menu offers a selection of commands relevant to the context.

To ensure that the Shortcut menus are easy, fast and safe to use, they are designed systematically and uniformly, meaning that the menu is always immediately recognized by the user.

Any Shortcut menu is build up of one or more sections. For example, if a container was right-clicked, the container command section is included.

The order of the sections is always the same. For example, the container command section always comes first if included because it is expected to be most frequently used.



Within each section, commands have been selected according to the following guidelines:

- commands that are (expectedly) used frequently are included
- commands that are considered important in the context are included
- rarely used commands of secondary importance are excluded.

The order of the commands within a section is always the same.

Commands that are irrelevant at the moment are disabled, that is, they are greyed but still visible, thus keeping the uniformity of the menu.

The exception is commands that relate to an option - such commands are included only if the option is activated in the specific REFCON system.

Illustrative example

If you right-click a container, the Shortcut menu will include a section with container relevant commands (see below). This section is exactly the same, no matter where you select the container.

The Container Command section

This Shortcut menu section includes a number of commands relevant to the appointed container or containers. The always included commands are:

Shortcut menu text	Function	Comments
Refresh now	Update all live data for this container via PCT as soon as possible.	See Container Commands ¹⁴³ .
Delete container(s)	Manual container deletion: Removes the appointed container(s) from REFCON's database if allowed (the container is not communicating and has no unacknowledged alarms). The user is asked for accept.	You should normally not use this command. In stead, you should enable automatic deletion, see Automatic Cleaning Up parameters ³³¹ .
Mark special reefer	Selects the "special" alarm priority for the appointed container(s).	See Special reefers ¹¹⁷ .
Unmark special reefer	Selects the "normal" alarm priority for the appointed container(s).	See Special reefers ¹¹⁷ .
Show historic data (tabular view)	Opens the window that displays the container's periodical historic data as a table.	See Historic Log ¹⁶⁹ .
Show historic data (graph)	Opens the window that displays the container's periodical historic data graphically.	See Container Trend ¹⁶⁶ .
Acknowledge container's alarms	Displays the list of alarms on this container, allowing the user to have them acknowledged.	See Acknowledging a container's alarms ¹¹¹ .



Make RDC List	Option: Saves the list of selected containers for use with the optional RDC.	See RDC.
---------------	--	----------

Other sections

Depending on where you display the Shortcut menu, it might include one or more sections other than the [Container Command section](#)^[34]. These sections are described in relation to the pictures where they occur. An example is the [Container List section](#)^[68].

2.5 Container Quick Search

Key subjects in this topic: Overview ^[35] How to search ^[35]	Related topics: The REFCON Monitor ^[22]
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Overview

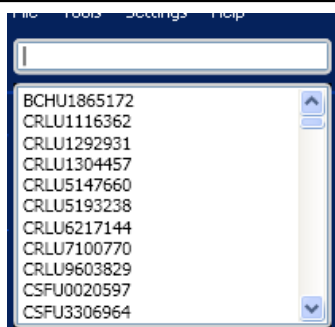
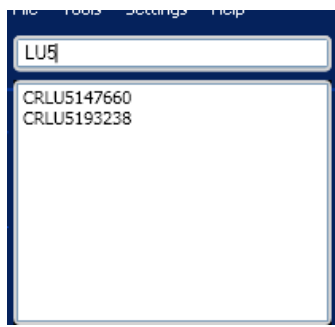
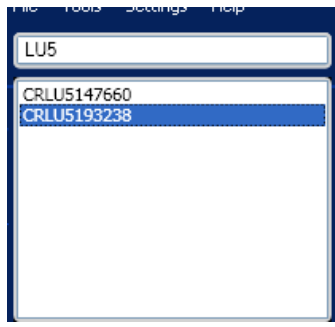
The *Container Quick Search* function allows you to have a specific container selected/displayed quickly.

Note that this search deals with active containers only, that is, the containers listed in "All containers" list. It does **not** search the possible archived containers (History),

How to search

To search for a container, proceed as follows:



Step	Action	Result	Description
1	Click the "<Search container>" element in the Top panel.		The list displays all containers in REFCON's database.
2	Type a search string.		The list changes as you type, still showing all container IDs including the typed string. As shown in the example, this might very well be a string in the middle of the ID.
3	Select the desired container by - Double-clicking it in the "Containers" list, or - Using the Down and Up keys and then the Enter key.		The list closes. The Container Details, all relevant open list pictures and the optional Container Plan, if open, shows/highlights the selected container.

2.6 The REFCON Clock System

Key subjects in this topic:

[Overview](#)^[36]
[UTC is very important](#)^[37]
[UTC time synchronisation is very important](#)^[37]
[Clock Configuration and Clock Setting](#)^[37]
[REFCON Clock System Overview](#)^[37]
[Symbols used in the illustration](#)^[38]
[Use of the Windows date and time](#)^[39]

Related topics:

[Clock Configuration](#)^[39]
[Clock Setting](#)^[41]
[The REFCON Monitor](#)^[22]

Overview

The *REFCON Clock System* provides two dates and times:

- **UTC** (Universal Time), previously known as GMT.
- **LMT** (Local Mean Time).



In fact, the LMT is identical to the UTC with the addition of an offset, called the LMT Offset. The LMT Offset is an integer number of minutes, up to +/- 12 hours (note: it can be up to +14 hours in certain time zones).

If your system includes more than one REFCON PC, you will discover that the current date and time shown in the Top panel on each REFCON Monitor are synchronised.

UTC is very important

The UTC is, along with the LMT, used when time stamping alarms, events and data logs, and when deciding when to make logs and other time controlled actions. Also the UTC is used when time stamping data sent to the Global Monitoring Server. It is vital to the reliability of these data that the UTC is correct and that it runs continuously without adjustments. And as you know, the reliability of data is vital in case of claims and similar.

Therefore, **do never change the UTC unless it is strongly required!** If you are changing time zone or daylight saving time, **always adjust the LMT only!**

UTC time synchronisation is very important

Even though the clock electronics in newer PCs is quite precise, it can't guarantee that the UTC used by REFCON is correct and remains so. Therefore, time synchronisation is vital.

To avoid that this gets a manual duty to the users, we recommend that any REFCON system runs some kind of automatic synchronisation. When such a solution is found, your duty as a user is to react on alarms or other signs should it stop working.

Note that REFCON also supports automatic synchronisation of the LMT.

There are many possible automatic solutions, and the most adequate solution should have been chosen for your REFCON system.

However, basically there are two ways that external equipment can control the clock in REFCON:

- Via LAN, through the Windows clock on the Main PC.
- Through the REFCON Data Service.

Clock Configuration and Clock Setting

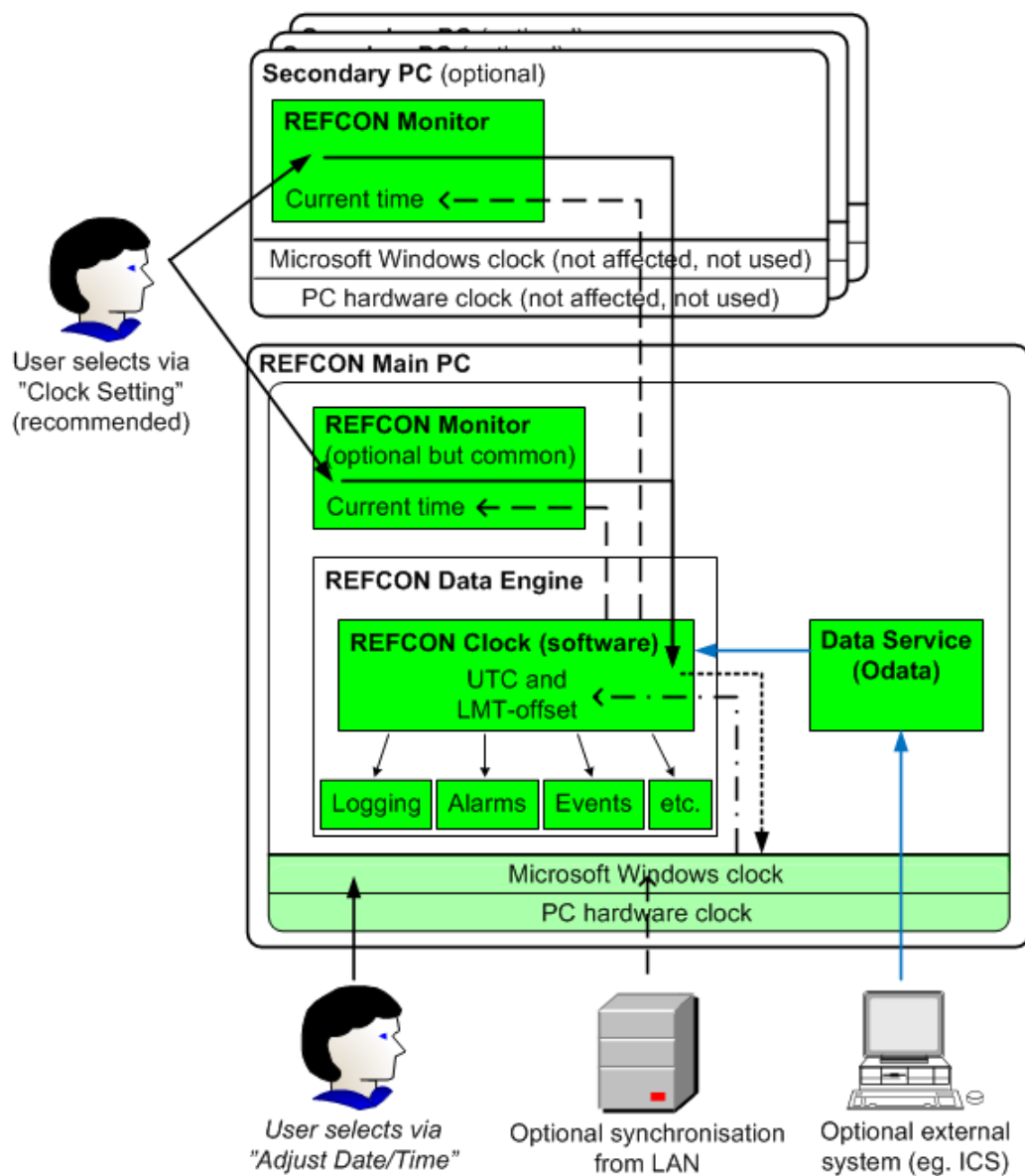
Please be aware of the two different activities, [Clock Configuration](#)^[39] and [Clock Setting](#)^[41].

Typically at commissioning time, the Clock Configuration is decided once for all (though it can later be changed). The Clock Configuration determines which of the connections in the below illustration are in fact active/valid. The Clock Configuration must match the chosen automatic time synchronisation solution, if any.

Clock Setting is, depending on the Clock Configuration, frequently done on ships sailing east-west, however, you would generally do it if the REFCON time is incorrect.

REFCON Clock System Overview

The following drawing should illustrate the essentials of the REFCON Clock System:



Symbols used in the illustration

The above drawing includes the following symbols:

Symbol	Meaning	Dependency
	Elements involved in or affected by the REFCON Clock System.	
	The external Windows/PC clock.	Used on Main PC only, and only if decided by Clock Configuration.
	User's Clock Setting.	If and where depends on Clock Configuration.
	Clock setting by external system via REFCON Data Service.	Only possible for a certain Clock Configuration.
	Time synchronisation.	



Symbol	Meaning	Dependency
----->	Setting Windows's clock on Main PC	Only if allowed by Clock Configuration.
· - · ->	Time synchronisation.	Only if allowed by Clock Configuration.
————>	Time stamping.	

Use of the Windows date and time

On any of the optional secondary PCs, the Windows operating system clock is **not** used nor affected. Thus, using Windows' time setting dialog on such a PC has **no** effect in REFCON.

The light green parts in above drawing illustrate the Windows operating system clock on the Main PC, being fully or partly used for deciding the REFCON clock. There are two main rules decided by the Clock Configuration:

- With main rule 1, the REFCON clock exactly reflects the Windows clock. Thus, using Windows' time setting dialog on the Main PC will make the REFCON clock change correspondingly. However, we recommend that you use the [Clock Setting](#)^[41] dialog in the REFCON Monitor in stead, if allowed by Clock Configuration.
- With main rule 2, the Windows clock is ignored. The PC hardware clock does, however, decide the rate of the REFCON clock.

2.6.1 Clock Configuration

Key subjects in this topic:

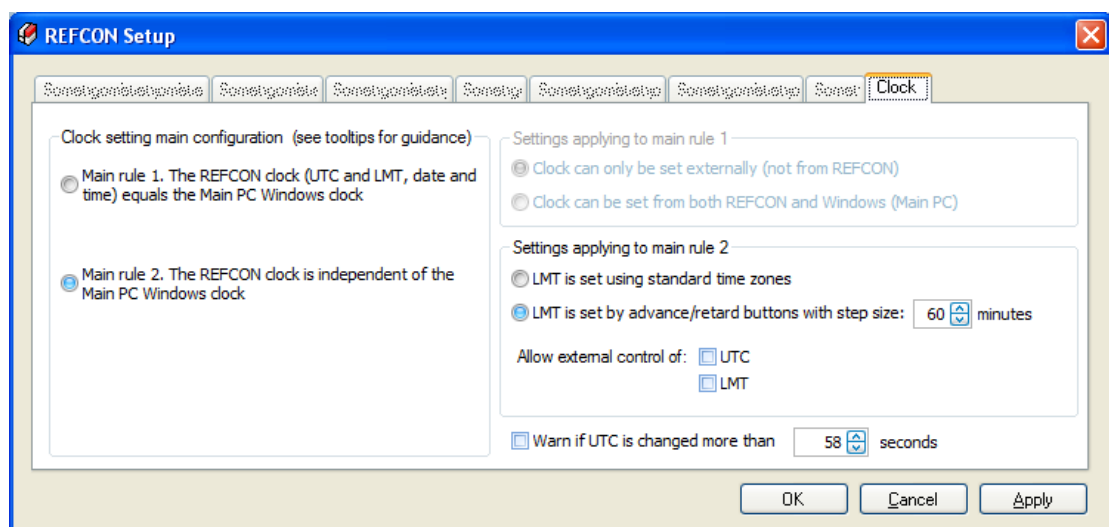
[Overview](#)^[39]
[Matching the chosen time synchronisation solution](#)^[40]
[Clock Configuration options](#)^[40]

Related topics:

[The REFCON Clock System](#)^[38]
[The REFCON Monitor](#)^[22]

Overview

The Clock Configuration is controlled through the "Clock" tab of the [REFCON Setup](#)^[330] dialog:



Note that within the Clock tab, tooltips are available with guiding instructions.

Remember to apply your changes.



Matching the chosen time synchronisation solution

It is essential that the clock configuration matches the chosen time synchronisation solution:

- If REFCON is synchronised via LAN (NTP or similar), select Main rule 1 with the option "Clock can only be set externally".
- If REFCON is synchronised via the REFCON Data Service, select Main rule 2 and make sure the relevant is checked: "Allow external control of UTC" and/or "Allow external control of LMT".

Clock Configuration options

The following table shows the guidelines on which options to select:

Main rule	Option	Guidelines
1: REFCON clock equals Main PC Windows clock.	Clock can only be set externally.	You need to select this combination if the REFCON Main PC is time synchronised via LAN (NTP or similar). In any other case, do not select this combination.
1: REFCON clock equals Main PC Windows clock.	Clock can be set from REFCON or Windows.	You should select this combination if the REFCON Main PC is not externally time synchronised, and it is important that REFCON time equals time in other applications on the REFCON Main PC. We do in this case recommend that you use REFCON's Clock Setting dialog to set the date and time rather than the Windows dialog. It is our experience that the latter can be a little difficult to use expediently.
2: REFCON clock is independent.	Either option; "Allow external control of UTC" and/or "Allow external control of LMT" checked.	If you want REFCON's clock to be decided by an external system, you need to select main rule 2 and make sure the relevant is checked: "Allow external control of UTC" and/or "Allow external control of LMT". Note that you can still set the clock manually, however, when a new synchro message arrives from the external system, your selection is overwritten.
2: REFCON clock is independent.	LMT set using time zones.	You should select this combination if you want to make sure the REFCON clock is not set unintentionally (important for data logging), and you do not want to use Advance and Retard buttons. This option lets you select LMT from a list of time zones. The time zone list is identical to the one found in Windows on the REFCON Main PC.
2: REFCON clock is independent.	LMT set by advance/retard buttons.	Select this option to have Advance and Retard buttons displayed with selected value. You can in this case also adjust the LMT Offset directly or set the time zone like above. Note that daylight saving time is in this case automatically applied only if selecting time zone.
	Warn if UTC is changed more than limit.	Logging functions appear most reliable when UTC runs continuously. A small adjustment can be



Main rule	Option	Guidelines
		necessary if the PC clock is not accurate, but a larger change could be unintentional.

2.6.2 Clock Setting

Key subjects in this topic:

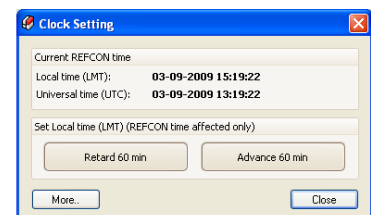
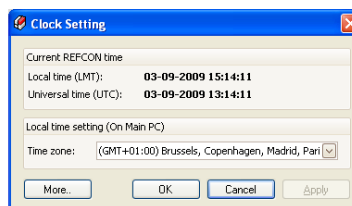
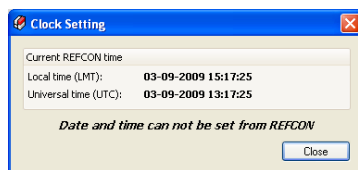
[Overview](#)^[41]
[Current REFCON time](#)^[41]
[Not changeable message](#)^[41]
[LMT synchronisation warning](#)^[42]
[Setting the time zone](#)^[42]
[Using Advance and Retard](#)^[42]
[Advance/Retard buttons and LMT](#)
[Offset in extended dialog](#)^[43]
[Setting the UTC](#)^[43]

Related topics:

[Clock Configuration](#)^[39]
[The REFCON Clock System](#)^[36]
[The REFCON Monitor](#)^[22]

Overview

The Clock Setting dialog box can be opened from the popup menu appearing when clicking the current time in the Top panel. The contents of the dialog box depend on the [Clock Configuration](#)^[39]. The primary variants are:



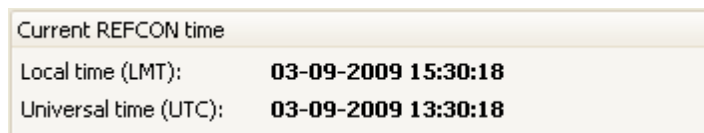
The dialog box clearly reflects the following principles:

- It is in the most cases permitted, and it can be a daily routine (on east <-> west going ships), to change the LMT, which is therefore immediately possible and easy.
- You should never, or only very rarely, change the UTC.
- The rarely used options can be displayed using the "More" button, if permitted.
- You are warned if the change you try to do does in some way conflict with the configuration or is not recommended.

The elements that can be found in the dialog box are described in the following.

Current REFCON time

At the top of the Clock Setting dialog box, the current LMT and UTC dates and times are shown for information:



It is not possible to make changes in this area.

Not changeable message

When, in the Clock Configuration, the main rule 1 has been selected in combination with "Clock can only be set externally", the Clock Setting dialog box is limited to the above mentioned current time and the following message:

Date and time can not be set from REFCON



In this case, the date and time is most likely externally time synchronised. If not so, you will have to make possible changes through the Windows' Adjust date/time dialog on the REFCON Main PC.

LMT synchronisation warning

When, according to the Clock Configuration, LMT is synchronised through the REFCON Data Service (Main rule 2, "Allow external control of LMT" checked), you are warned when opening the Clock Settings dialog box as follows:

***LMT is maybe controlled by an external source.
You should not change LMT here.***
*If, however, there is no external source, or it stops working,
you MAY click "More.." to change.*

As indicated, the Clock Settings dialog box can not check if the external source does *actually* control the LMT - it should do, but please make sure your self.

Setting the time zone

When the Clock Configuration does NOT read "Clock can only be set externally", you are able to decide the local time by selecting the time zone:

To change the time zone,

1. Select the appropriate in the drop-down list.
2. Select OK or Apply.

If the Clock Configuration main rule 1 has been selected, the header of the field reads "On Main PC" (as in above example), meaning that your change will affect the Windows clock on the REFCON Main PC.

If the Clock Configuration main rule 2 has been selected, the header of the field reads "REFCON time affected only", meaning that the Windows's clock is NOT affected.

Using Advance and Retard

When, in the Clock Configuration, the main rule 2 has been selected in combination with "LMT set by advance/retard buttons", the Clock Setting dialog box opens with two large buttons and a More button:

The number of minutes displayed on the buttons is selected in the Clock Configuration.

Clicking either of the large buttons will

- Advance or Retard the LMT the shown number of minutes.
- Close the dialog box.

Advancing and Retarding affect the REFCON time only, not the Windows clock.

Clicking the More button will change to an extended version of the Clock Setting dialog box. The Advance and the Retard button will at the same time slightly change behaviour as described below.



Advance/Retard buttons and LMT Offset in extended dialog

When the More button is selected as described above, the Advance/Retard field changes to the following:

Here, clicking either of the large buttons will temporarily change the LMT offset shown just below the buttons. You may alternatively type an LMT Offset or use the small arrow buttons to change it in steps of 5 minutes.

In any case you must select OK or Apply to activate your changes. Doing so affects the REFCON time only, not the Windows clock.

Setting the UTC

When the Clock Configuration does NOT read "Clock can only be set externally", you are able to decide the universal time by selecting the date and/or the time:

To change the UTC,

1. Select the date, maybe using the calendar available when clicking the down-arrow button.
2. Select the time by changing the hour, the minute and/or the second separately: Click the figure (eg. the minute), and type the desired value or click the small up and down arrows.
3. Select OK or Apply.

If the Clock Configuration main rule 1 has been selected, the header of the field reads "On Main PC" (as in above example), meaning that your change will affect the Windows clock on the REFCON Main PC.

If the Clock Configuration main rule 2 has been selected, the header of the field reads "REFCON time affected only", meaning that the Windows's clock is NOT affected.

2.7 Backup function

Key subjects in this topic:

[Overview](#)⁴³
[Backup chunks](#)⁴⁴
[The Backup dialog box](#)⁴⁴
[Backup procedure](#)⁴⁵
[Inspecting a backup](#)⁴⁶
[Restoring a backup](#)⁴⁶

Related topics:

[Configuration export\(/import\)](#)⁴⁶
[The REFCON Monitor](#)²²

Overview

We strongly recommend that you make backups regularly of system and log data.

If you have set up the [Automatic backup and restore](#)³²⁸ function during installation, you can normally rely on this and will not have to do manual backups.

All settings that you can decide in the REFCON Monitor as well as all container and event log files are as standard stored on the harddisk of the REFCON Main PC. Unless you have purchased some harddisk mirror solution, the harddisk should **not** be regarded as a reliable



long-time storage. Therefore, if you consider your settings and/or the container log files valuable, you must back up to an external media.

Note that the backup function is quite similar to the [Configuration Export](#)^[46] function.

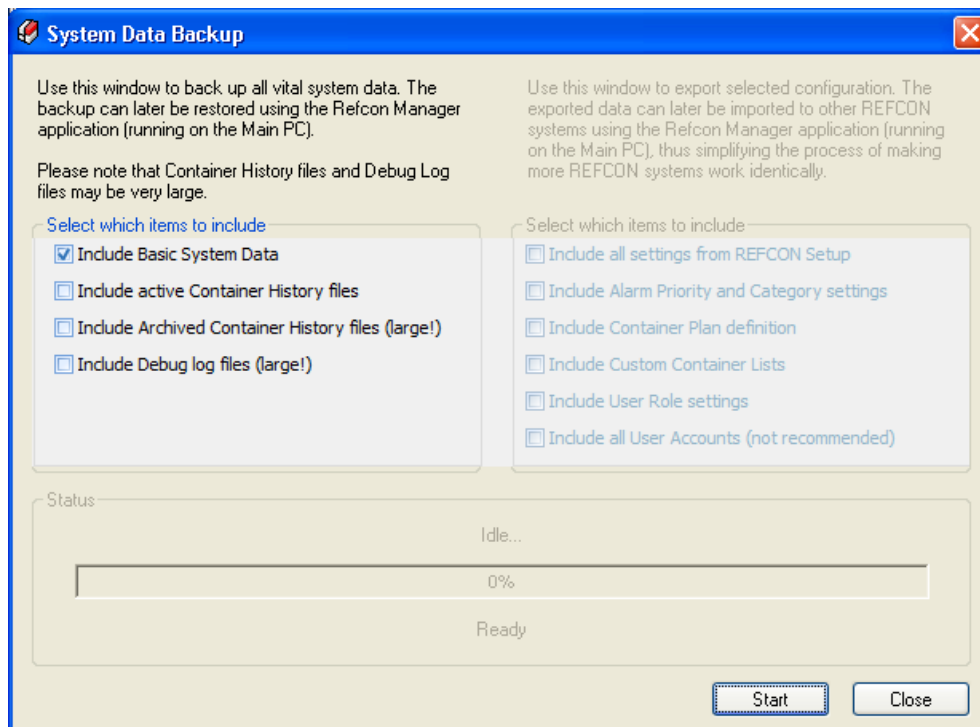
Backup chunks

The whole set of files that can be backed up are divided into the following chunks:

Chunk	Short	When to backup	Included information
Basic System Data	S	When changes have been made, or regularly.	• Installation settings. • Settings selected in REFCON Setup. • Customized Container List versions and the like. • User profiles. • Settings related to plugins. • Optionally other plugin-related files.
Active Container history files	L	Regularly, very often.	The trend log and event log files for every single container being currently monitored by REFCON.
Archived Container history files	La	According to your needs.	The trend log and event log files for containers having been monitored in the past. <i>Note that the size of this chunk can be giga-bytes, especially if you have a large amount of containers and/or have decided to keep the files for a long time.</i>
Debug log files	D	When requested / in case of troubles.	Logs that hopefully could help Copeland Transportation Solutions to solve problems / correct errors, or to improve the product.

The Backup dialog box

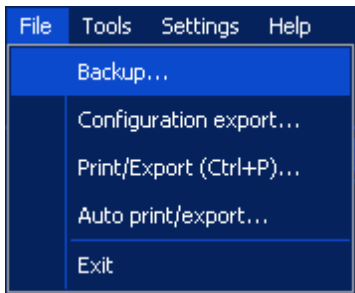
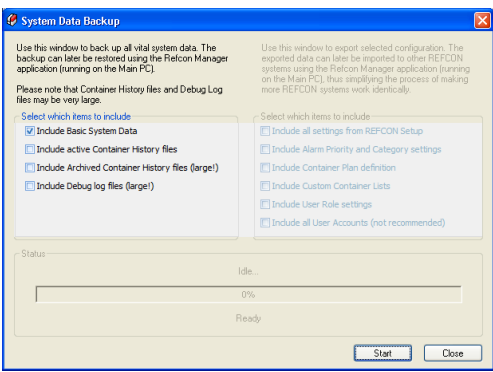
The Backup dialog box looks like this:



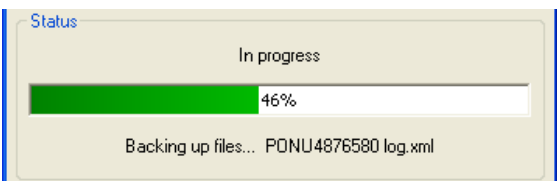
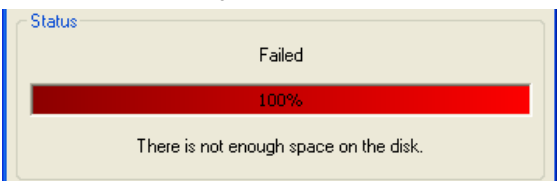
The dialog box has been designed with an active and an disabled half to illustrate the similarity with the [Configuration Export](#)^[46] function.

Backup procedure

To make a backup, proceed as follows:

Step	Action	Result
1	Select "Backup..." from the "File" menu. 	The Backup dialog opens. 
2	Select which items (= chunks) to include. <i>Note: You can combine more chunks in one backup, however, consider if this is a good idea, as the time scope of the chunks are different</i>	One or more checkboxes checked.
3	Select "Start".	A folder selection dialog opens.
4	Locate and select the folder where you want your backup to	The folder selection dialog closes, and the backup operation starts. You will observe the progress:



Step	Action	Result
	be created. <i>Note: Insert possible USB key or similar before selecting.</i>	
5	Await completion. <i>Note: First the backup zip-file is created on the REFCON Main PC harddisk. After this it is copied to the final destination, which may take quite some time.</i>	In case of troubles you are informed like this: 
6	If Completed/Success, remove the possible USB key or similar.	

Inspecting a backup

The backup is a single compressed file in zip format. It is automatically named as follows:
 "Installation name_yyyy-mm-dd-HHMMSSz_SLLaD.zip"

where

- "Installation name" is the official "name" of your REFCON system, as shown in the REFCON Monitor caption and in the About box.
- yyyy-mm-dd is the year-month-day when the file was created.
- HHMMSS is the hour minute second when the file was created.
- z indicates that the date and time is UTC.
- SLLaD (or a subset of these letters) indicate what chunks have been included, see the "Short" column [above](#) ^[47].

If you unpack and inspect the zip-file, you will see that it contains two folders:

- BackupFiles includes the data files, in the most cases in xml format.
- BackUpInfo includes information used when REFCON is to restore the backup.

It is **important** that you do not modify the zip-file, as this might make restore fail.

Restoring a backup

A backup can be restored using the [REFCON Manager](#) ^[355].

2.8 Configuration export(import)

Key subjects in this topic:

[Overview](#) ^[47]
[Configuration export chunks](#) ^[47]
[The Configuration export dialog box](#) ^[47]
[Configuration export procedure](#) ^[48]
[Inspecting a Configuration export file](#) ^[48]
[Importing a configuration file](#) ^[49]

Related topics:

[Backup function](#) ^[43]
[The REFCON Monitor](#) ^[22]



Overview

The main purpose of the Configuration export(/import) function is to allow copying of settings from one REFCON system to another. The similar [Backup Function](#)^[43] is not suitable for this as it contains data that should never be copied.

All relevant settings that you can decide in the REFCON Monitor can be copied.

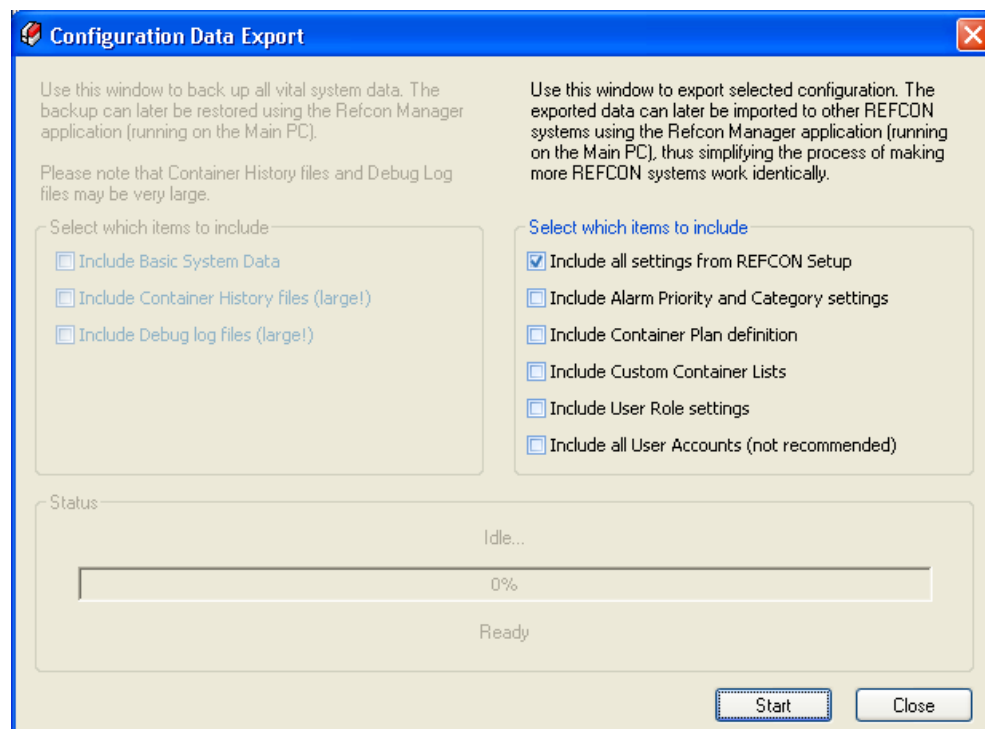
Configuration export chunks

The whole set of files that can be exported are divided into the following chunks:

Chunk	Short	Included information
Settings from REFCON Setup	R	All data selected in the REFCON Setup dialog box ^[330] , all tabs.
Alarm Priorities and Categories	A	The file AlarmPrioritiesAndCategories.xml - which can be edited externally, with greatest caution. <i>Note that alarm priorities are not used in this REFCON version.</i>
Container Plan	P	The one or two files defining the Container Plan ^[96] .
Custom Container Lists	C	All the possible unlocked Container Lists having been added ^[80] to your REFCON.
User Roles	Ur	All the User Roles ^[56] defined in your REFCON.
User Roles and User Accounts	Ua	All the User Roles and Accounts ^[56] defined in your REFCON. If you work with personal user accounting you should not copy this chunk.

The Configuration export dialog box

The Configuration export dialog box looks like this:

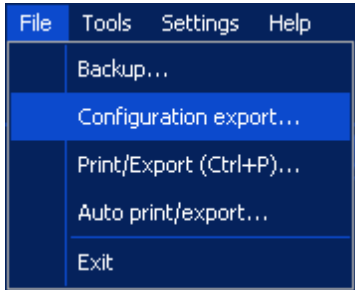
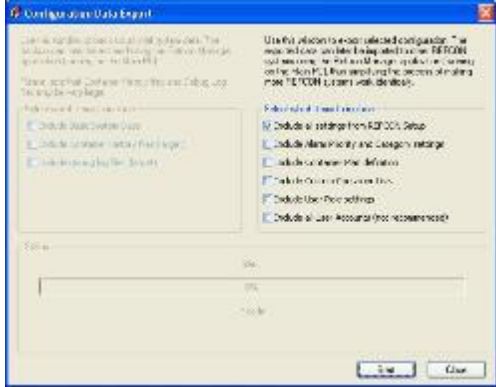
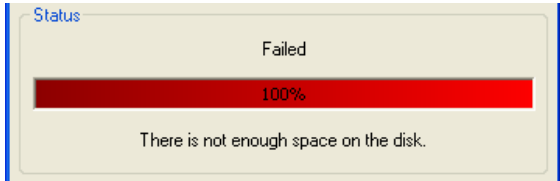


The dialog box has been designed with an active and an disabled half to illustrate the similarity with the [Backup](#)^[43] function.



Configuration export procedure

To make an export, proceed as follows:

Step	Action	Result
1	Select "Configuration export..." from the "File" menu. 	The Configuration export dialog opens. 
2	Select which items (= chunks) to include. <i>Note: You can combine more chunks in one export.</i>	One or more checkboxes checked.
3	Select "Start".	A folder selection dialog opens.
4	Locate and select the folder where you want your export to be created. <i>Note: Insert USB key or similar before selecting, if relevant.</i>	The folder selection dialog closes, and the export operation starts. Normally this goes quite fast so you will hardly observe the progress.
5	Await completion. <i>Note: First the export zip-file is created on the REFCON Main PC harddisk. After this it is copied to the final destination, which may take quite some time.</i>	In case of troubles you are informed like this: 
6	If Completed/Success, remove the USB key or similar, if relevant.	

Inspecting a configuration export file

The export is a single compressed file in zip format. It is automatically named as follows:

"Installation name_yyyy-mm-dd-HHMMSSz_RAPCUrUa.zip"

where

- "Installation name" is the official "name" of your REFCON system, as shown in the REFCON Monitor caption and in the About box.
- yyyy-mm-dd is the year-month-day when the file was created.
- HHMMSS is the hour minute second when the file was created.
- z indicates that the date and time is UTC.
- RAPCUrUa (or only some of these letters) indicate what chunks have been included, see the "Short" column [above](#) ⁴⁷.

If you unpack and inspect the zip-file, you will see that it contains two folders:



- BackupFiles includes the data files, in the most cases in xml format.
- BackUpInfo includes information used when REFCON is to import the configuration.

It is **important** that you do not modify the zip-file, as this might make import fail.

Importing a configuration file

An exported configuration can be imported using the REFCON Manager.

2.9 Selecting drive and folder on Main PC

Key subjects in this topic:

[Overview](#)^[49]
[How to edit a drive and path](#)^[50]
[The Choose Main PC folder dialog box](#)^[50]
[Manage network drives](#)^[50]
[Create new folder](#)^[51]
[Sharing a local folder](#)^[51]
[Defining a Windows user for sharing permission](#)^[52]
[Automatic re-establishment](#)^[52]
[Manual action needed to make sharing work](#)^[52]

Related topics:

[The REFCON Monitor](#)^[22]

Overview

Various functions in REFCON need to read from or write to a drive that is accessible from the REFCON Main PC. Keep in mind, however, that the set of drives and drive letters available to REFCON may very well differ from the ones available to the user in the Windows operating system.

Sometimes this is because the user works at a different location than the REFCON Main PC, but even though he works at the REFCON Main PC itself, the REFCON system maintains its own selection of available network and drive mappings which are not (necessarily) identical to those of the user's.

Example 1:

A user works at a remote machine with REFCON Monitor, and the REFCON Main PC is situated elsewhere.

- Clearly, the files and the folders of the user's C: drive is not the same as those of REFCON's (as the user works at a different machine than REFCON).
- Both the user's machine and the REFCON Main PC may (or may not) have a selection of mapped network drives. But there is no guarantee at all that they both map to the same network places (unless having deliberately been set up that way).

Example 2:

A user works at the REFCON Main PC and has a network drive E: available (in Windows). But – because the REFCON system maintains its own set of network drive mappings - REFCON may not be able to see that drive (or, REFCON's E: drive may point out a completely different network place).

When configuring REFCON in the REFCON Setup dialog, you often have to point out to REFCON where to look for or store different types of files (e.g. log files). You do so by browsing the file system in REFCON Monitor, but you must be aware that when doing so, *you will most often be looking at the REFCON Main PC's file system and drive mappings rather than your own!*



Anywhere in REFCON where this is the case, a light blue background colour is used instead of the normal white to indicate that you have left your own set of drive letters and network mappings and now use those of REFCON's.

How to edit a drive and path

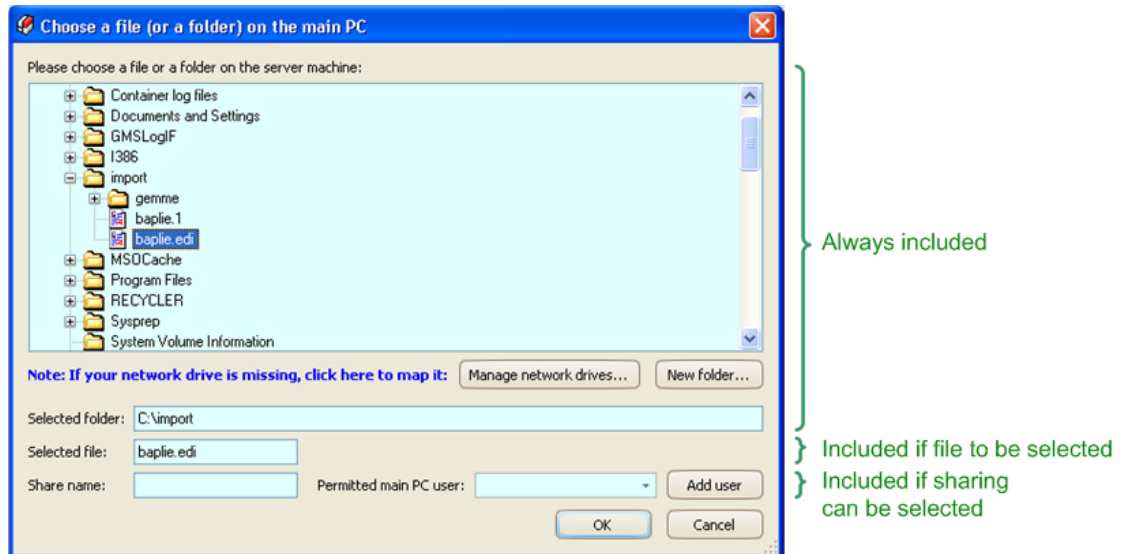
In relevant places you will find an Edit button, for example like this:



The textbox, with the light blue background, shows the previously selected path. To change it, select the Edit button.

The Choose Main PC folder dialog box

Having selected Edit, the following dialog box opens (note that it might take some time):



As indicated, the contents of the dialog box differs depending on the context.

In the upper part you see the drives and folders that are available to the [Data Engine](#)^[306] program on the REFCON Main PC.

If you need to select a network drive/folder that is not available, select [Manage network drives](#)^[50]. You may also create a [new folder](#)^[51].

Use the small + buttons to expand relevant parts of the folder structure.

Click the desired folder or, if relevant, the desired file to select it. You will see that the folder path and, if relevant, the file name appear in the respective textboxes.

If you are selecting a folder on the Main PC harddisk that is going to be accessed from an external computer, you need to [share it](#)^[51].

Having completed your changes, click OK to apply these and close the dialog box.

Manage network drives

To obtain access to a network drive/folder that is not already on the list, select the "Manage network drives" button. This opens the following dialog box:



Add/Remove network drive

Drive letter: **E:** READY

Drive path:
(e.g. \\ServerName\ShareName)

☒ Use log on credentials:

Credentials:

Username:
(use "username" or "domain\username" or "username@domain")

Password:

Domain:

Proceed as follows:

1. Select a vacant drive letter, for example F:.
2. Type the full path to the external computer's folder. As shown, start with two backslashes, then write the exact name of the external computer, and finally type the path on that computer. Example: "\\NameOfExternalComputer\REFCONfiles\PlanningData". Do **not** include any file name.
3. If the external computer is secured, check the "Use log on credentials" check box. If a domain is to be specified, either select this from the drop down list or prepend or append it to the username as mentioned in the dialog box. Type a valid user name and password. Note that if you type an invalid domain, user name or password, you must expect quite a long response time from the system before you are informed.
4. Select "Add connection". You do now automatically return to the [Choose a folder dialog](#)^[50], having the new drive selected.

In the "Manage network drives" dialog box you may alternatively:

- Simply inspect an existing connection - select a used drive letter, eg. F:, and see the corresponding path.
- Remove a connection - select a used drive letter, eg. F:, and click the "Remove connection" button.

Create new folder

To create a new folder,

1. Select the drive/folder where you want the new folder to occur, in the upper part of the [Choose a folder dialog](#)^[50].
2. Select the "New folder" button.
3. Type the name of the new folder and select OK.

If success you do automatically return to the [Choose a folder dialog](#)^[50], having the new folder selected. If not success, you are informed about this.

Sharing a local folder

If relevant in the context **and** you have selected a folder on the REFCON Main PC harddisk, you can define a share name, being required by an external computer to make use of the folder. You can additionally decide which Windows user on the REFCON Main PC to be permitted to access the share. It is possible to [define a Windows user](#)^[52] specifically for this purpose.



To share the folder, write a name in the "Share name" textbox. We recommend that you make it short and letters only.

If a share name is already defined you can

- change it by simply editing the name
- remove the share by deleting the existing name from the textbox.

Having defined a share name you should select a Windows user permitted to attach it from the "Permitted main PC user" drop down list.

Unfortunately, the setting of the folder sharing as described above does not always work fully automatically. You may have to follow the procedure [Manual action needed to make sharing work](#)^[52].

Defining a Windows user for sharing permission

If relevant in the context, you can define a single REFCON Main PC Windows user with limited rights. This user will be available at Windows logon like other Windows users.

If such a user has not already been defined through REFCON, select the "Add user" button. If a user is already defined, the button text is "Edit user". Clicking the button makes the following dialog box appear:



To create a user, type a valid Windows "User name". It is recommended that you add a password, which you have to type twice.

To change a user name, simply edit it.

To change the password, type the full new password twice.

To delete the existing user, delete the text in the "User name" field.

Automatic re-establishment

Each time the [Data Engine](#)^[306] is started, it checks if the folder, its possible sharing information and the possible Windows user are exactly as you have chosen. If not so, a background function attempts to re-establish it. This is in particular useful if the [Configuration Export/Import](#)^[46] function has been used to copy your settings from another REFCON system, as in this case the required folder, the possible Windows user and the possible share settings are automatically created.

Manual action needed to make sharing work

In order to have your selected folder(s), for example "C:\GmsLoglf", shared on the network, you must in addition to the above manually enable network sharing in Windows. Please proceed as follows:

1. In MyComputer, right click the folder (eg. "C:\GMSLoglf") and select Properties.
2. Open the pane Sharing.
3. Click the text "Network and Sharing Center"
4. Select the option "Turn on sharing so anyone with network access can read and write files in the Public folders".



5. Turn off password protected sharing.
6. Follow the same procedure for all folders to be shared.

2.10 The taskbar notify icon

Key subjects in this topic:

[Overview](#)^[53]
[The notify icon menu](#)^[53]

Related topics:

[The REFCON Monitor](#)^[22]

Overview

While the *REFCON Monitor* is running, you find it's *notify icon* in the taskbar, as standard, in the lower right corner of the screen:



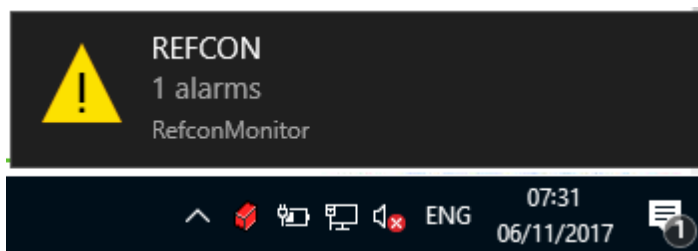
The idea of the *notify icon* is that you can use the PC for other purposes, the *REFCON Monitor* being not visible, and still be notified in case of alarms.

The icon is:

- Green when there is no alarms.
- Steady Red when there are one or more alarms, and all have been acknowledged.
- Flashing Red/Yellow when there are one or more unacknowledged alarms.

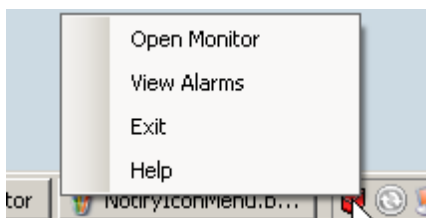
In case of alarm, the so-called "balloon" shows up, as indicated in the above figure, showing the alarm count. You may click the X button to remove the "balloon".

In case of Windows 10, the balloon may look like this:



The notify icon menu

Right-click the *notify icon* to have the menu displayed:



In the menu, select:

- "Open Monitor" to have the *REFCON Monitor* displayed.
- "View Alarms" to have the *REFCON Monitor* displayed with Alarm Summary open.
- "Exit" to close the *REFCON Monitor* and remove the *notify icon*.



- "Help" to display this page in the Online help.

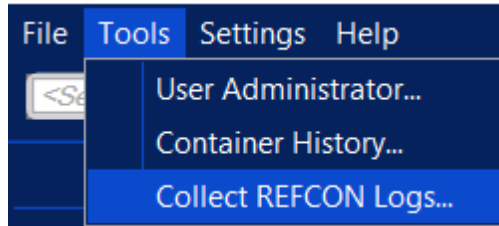
2.11 Collecting Refcon logs

Overview

This topic illustrates how to collect REFCON log files for analysis.

Collecting REFCON logs

From the REFCON Monitor navigate as shown below to collect all relevant REFCON logs. These logs are useful for troubleshooting issues:



User can choose the directory path to save the collected log files.

REFCON 6

Reefer Container Monitoring System

User Authentication



3 User Authentication

Key subjects in this topic:

[Overview](#) ^[56]

[Purpose of User Authentication](#) ^[56]

[Security levels](#) ^[56]

Related topics:

[User Authentication in daily use](#) ^[57]

[User Administration](#) ^[59]

[User Authentication Setup](#) ^[62]

Overview

User Authentication is the name of REFCON's security system.

Experience has shown that different customers and users have different views on security. Some customers want a high security level, while to others it is less important. As shown in the following, REFCON's User Authentication system can be set up to fit any of these viewpoints. As the starting point, the following description covers the standard level of security as recommended by Copeland LP.

In this REFCON version, the standard level of security has been upgraded, compared to older REFCON versions, to match newest IT standards.

Purpose of User Authentication

The purpose of User Authentication is to

- prevent unauthorized people from changing data;
- make it possible to log which operator made the change;
- prevent unauthorized access to container data.

It is therefore possible to

- assign personal User ID and Password to each individual REFCON user;
- assign specific rights to each User ID.

Each time important operations are done, the User ID that was used is logged.

Security levels

The User Authentication system can be configured to work at any of the following security levels:

Level		Title	Description
3		Standard security	It requires user ID and password to use REFCON. Any privileged action requires that the user ID is assigned to a user role permitting that action. When executed, the action is logged along with the user ID.
2		No passwords	Similar to level 3, however, passwords are not required.
1		No security	Any user (having access to the REFCON Monitor) can do everything (except security configuration), and it is not logged who does what.



3.1 User Authentication in daily use

Key subjects in this topic:

[Overview](#)^[57]
[Select password at first logon](#)^[57]
[Logging on to REFCON Monitor](#)^[57]
[Who is logged on?](#)^[58]
[User has insufficient rights](#)^[58]
[Automatic log off by timeout](#)^[59]

Related topics:

[User Authentication](#)^[56]

Overview

To use REFCON Monitor, you need to be and stay [logged on](#)^[57]. You can then do everything permitted by your assigned user role.

If you attempt to perform an action that you are not permitted to do, you are [prompted for user ID and password](#)^[58]. Normally you would cancel this prompt and ask a permitted user to do it for you.

Remember to close the REFCON Monitor when leaving the computer to make sure that no other user can do things in your name.

Select password at first logon

When you log on (see [Logging on to REFCON Monitor](#)^[57] below) using a user ID that was never used before, you enter the password that you received from the administrator and then select [OK]. After this you are prompted to select a different password:

You need to enter and re-enter a new password string containing

- At least 3 lower case letters a..z;
- At least 3 upper case letters A..Z;
- At least 3 digits 0..9;
- At least 3 special characters, please see suggestions in the dialog box.

Logging on to REFCON Monitor

Each time you start REFCON Monitor, you are prompted to log on:



REFCON Logon

User ID:

Password:

Security Notice

Use of the software is subject to Emerson's Terms (<https://climate.emerson.com/en-us/training-support/warranty>), including Terms of Use, Privacy Notice and Terms and Conditions.

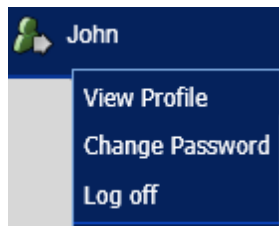
Please enter your user ID and password.

Who is logged on?

You can at any time see who is actually logged on to this REFCON Monitor in the upper left corner of the REFCON Monitor:



You may click the user indication to display the related menu:



- Select "View Profile" to display your role, description and more.
- Select "Change Password" to replace your existing password by another - works like above [Select password at first logon](#)⁵⁷.
- Select "Log off" to log off current user.

Note		At security level 1, the above indication is not visible.
------	--	---

User has insufficient rights

Being already logged on, if you attempt to perform an action for which you are not permitted according to your user role, you are prompted like this:

User has insufficient rights

User ID:

Password:

Permission required to:
Change how LMT can be set

The person symbol flashes, and you will have to type in a user ID with the required permission.



Below the person symbol it is indicated what permission is needed, if relevant. If your user account is not assigned that permission, you will not be able to log on.

Note		At security level 2, the Password entry is not visible - you just need to select the user ID and hit Enter/OK.
------	---	--

Automatic log off by timeout

If enabled in the configuration, the actual user is automatically logged off when he has done no privileged action for the decided time. If timeout is not enabled, users stay logged on until manually logged off.

For example, the timeout time could be set to 5 minutes. If you change setpoints on containers, once every 4 minutes, you keep logged on. But if you take a break and does nothing for 5 minutes, you are logged off.

3.2 User Administration

Key subjects in this topic: Overview ^[59] Example, User Accounts and Roles ^[59] The User Administrator dialog box ^[59] The User Account editor ^[60] The User Role Administrator dialog box ^[61] The User Role editor ^[61] User Administrator access ^[62]	Related topics: User Authentication ^[58]
---	---

Overview

User administration is to manage users, i.e., create, update and delete users. You do that in the *User Administrator* dialog boxes, see [User Administrator access](#)^[62].

You would normally create one **User Account** for each user. The User Account includes the unique user ID, the password, the assigned User Role and optionally a user name and a description.

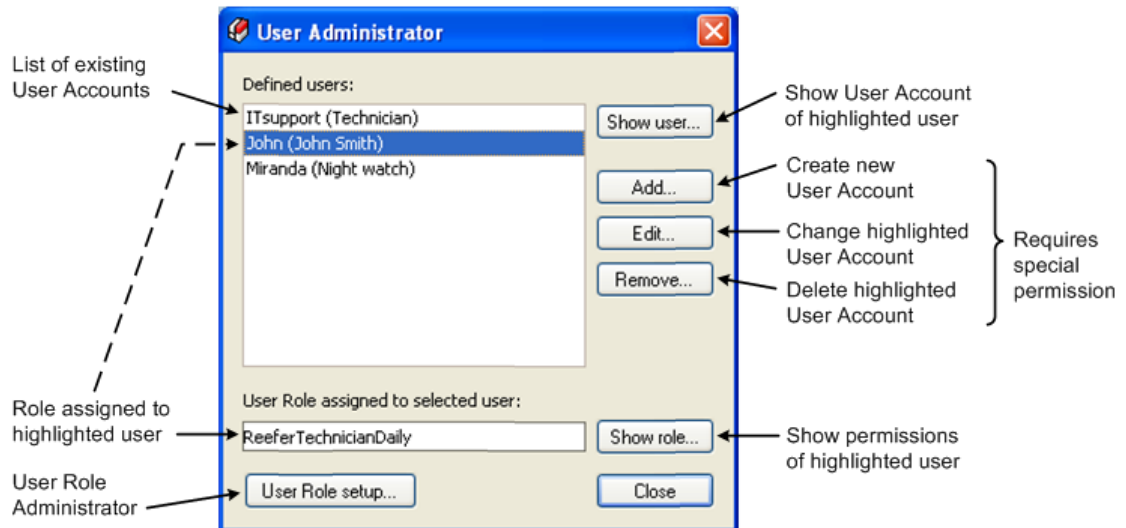
The permissions of the user is decided by the assigned **User Role**. The User Role is in fact a list of actions that users with this Role are allowed to do. **Example, User Accounts and Roles** Two persons, John and Miranda, are working on the same REFCON system as reefer electricians. The user administrator has created one User Role, named "ReeferTechnician", and two User Accounts with user IDs "John" and "Miranda". Both User Accounts are assigned to Role "ReeferTechnician", so John and Miranda have exactly the same permissions.

The User Administrator dialog box

You open the User Administrator from the "Tools" menu:

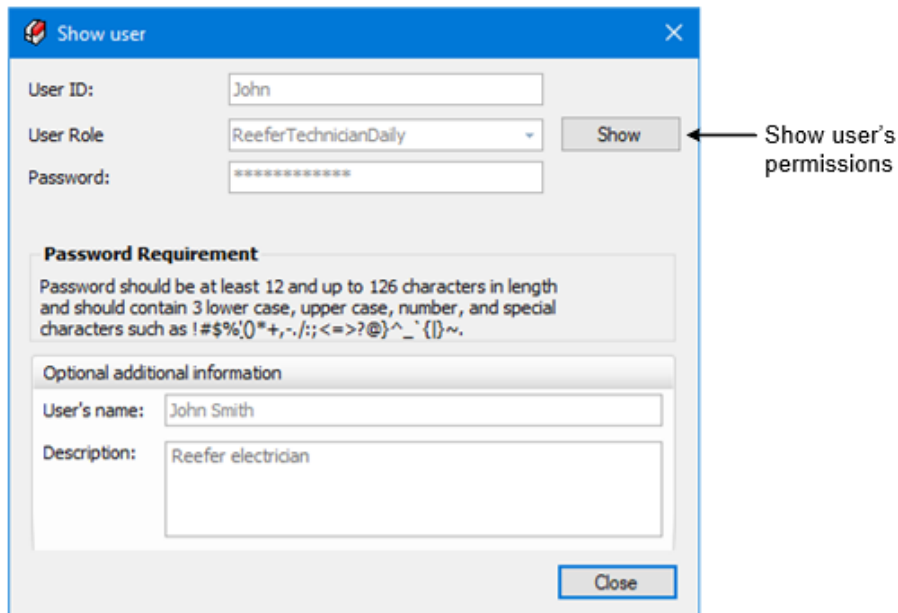


Clicking the option opens the following dialog box:



The User Account editor

Selecting either of the buttons "Show user", "Add" or "Edit" in the User Administrator opens the following dialog box:



In the above example, the dialog box is *showing* the user, in which case you can change nothing.

When creating a new user, all fields are empty when opening the dialog box. You must then

- Type a user ID. The string **must** be unique. Use alphanumeric characters only.
- Type a password. You must apply to the rules described in [Select password at first logon](#)^[57].
- Repeat/confirm the password.
- Select a User Role from the list.
- Optionally, type the User's name or similar - note that it is shown in parenthesis in the User Administrator.
- Optionally, type a Description.



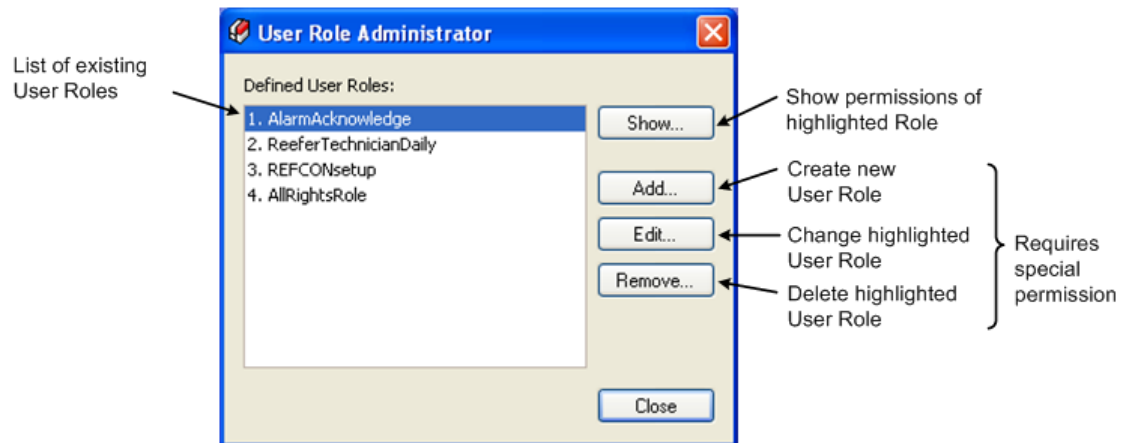
Note



At security level 2, the Password is optional. If there is any chance that you later enable passwords, you should select one at once - otherwise leave the field blank.

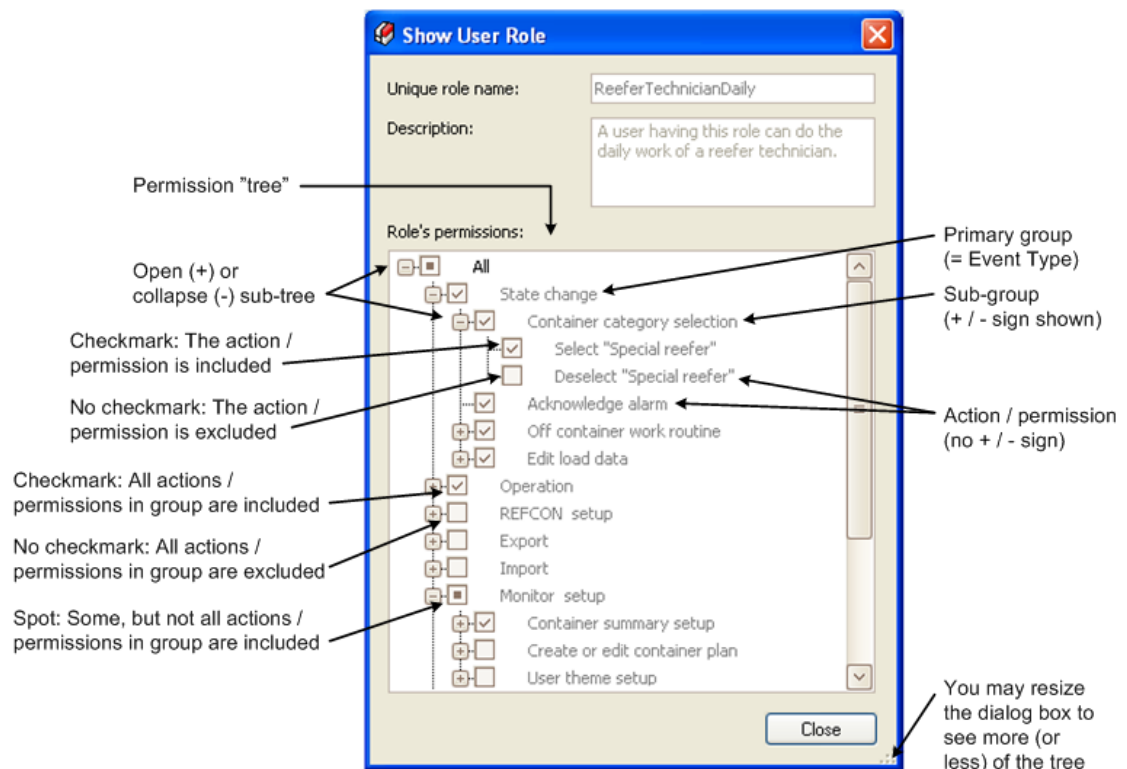
The User Role Administrator dialog box

Selecting "User Role setup" in the User Administrator opens the following dialog box:



The User Role editor

Selecting either of the buttons "Show", "Add" or "Edit" in the User Role Administrator opens the following dialog box:



In the above example, the dialog box is *showing* the role, in which case you can change nothing - but you can open or collapse parts of the tree.

When creating a new role, the two text fields are empty and all permissions in the tree are deselected when opening the dialog box. You must then

- Type a role name. The string **must** be unique. Use alphanumeric characters only.



- Optionally, type a Description.
- Select what groups / actions / permissions to be included.

User Administrator access

To open the User Administrator dialog box or the User Role Administrator dialog box, and when selecting "Add", "Edit" or "Remove" in either the User Administrator dialog box or the User Role Administrator dialog box, you are prompted for user ID and password. You need a user ID having the Administrators role assigned.

Note		At security level 2, no password is required.
------	---	---

3.3 User Authentication Setup

Key subjects in this topic:

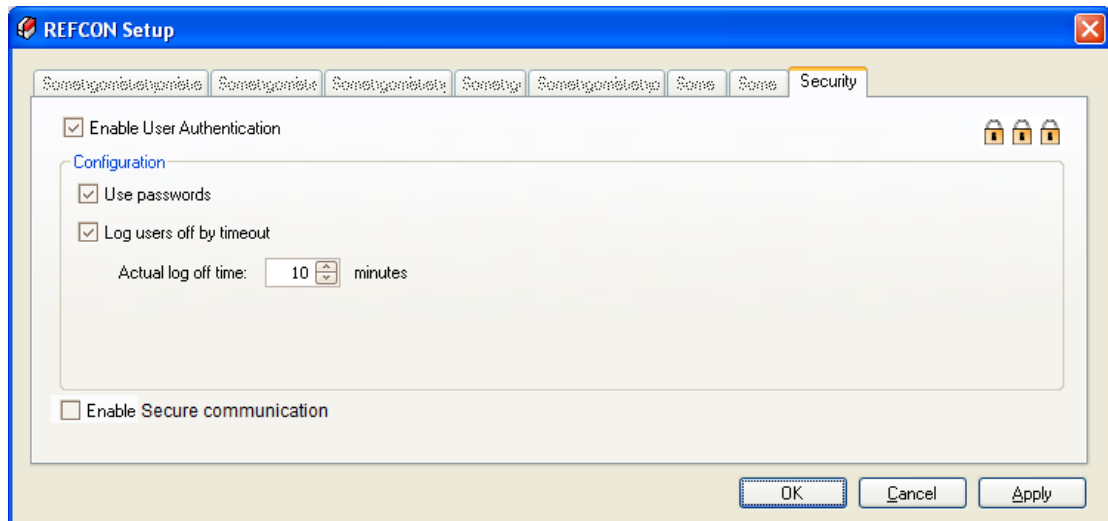
[Overview](#)^[62]
[User Authentication Setup access](#)^[63]
[Enabling Secure communication](#)^[63]

Related topics:

[User Authentication](#)^[56]

Overview

User Authentication is controlled through the "Security" tab of the [REFCON Setup](#)^[330] dialog:



The connections between the User Authentication configuration and the [security levels](#)^[56] are as follows:

When Security is configured as shown in above figure (all checkboxes selected), you operate at security level 3.	
If you deselect "Use passwords", you change to security level 2.	
If you deselect "Enable User Authentication", you change to security level 1.	



User Authentication Setup access

When trying to change anything under the Security tab illustrated above, you are prompted for user ID and password. You need a user ID having the Administrators role assigned.

Note		At security level 2, no password is required.
------	---	---

Enabling Secure communication

If you run one or more REFCON Monitors on computer(s) different from the REFCON Main PC, and you are operating on a vulnerable network, you may decide to enable secure communication.

This makes the data transferred between the Data Engine and the REFCON Monitor encrypted.

However, secure communication does reduce the communication speed, so if you do not strictly need the encryption, we recommend that you do not enable this.

REFCON 6

Reefer Container Monitoring System

The Container List

Transportation Solutions

COPELAND



4 The Container List

Key subjects in this topic:

[Overview](#)^[66]
[Standard lists, Pre-defined lists and User-defined lists](#)^[66]
[Colour codes within data fields](#)^[66]
[Bands and Columns](#)^[67]

Related topics:

[Using the Container List](#)^[67]
[Container List columns](#)^[69]
[Setting up the Container List](#)^[80]
[Setting up a Filter](#)^[84]
[How to use the Filter Builder](#)^[87]

Overview

The *Container List* is a list showing selected data for a number of reefer containers.

Each line in the Container List represents one container, having the container's data shown in columns. Because of the limited length of a line, only a smaller selection of the possible columns is displayed. Note that measurements appear with a [limited accuracy](#)^[142].

A list can either show all containers in REFCON's database, or it can have a filter applied that makes a subset shown.

The idea is that you work with a number of Container List versions, using at any time the one that is the most appropriate for your working duty. Each version has a unique combination of columns and a unique filter and thereby shows a specific part of the information available for each container and a specified subset of the containers.

We encourage you to make your own lists as described in [Setting up the Container List](#)^[80].

Standard lists, Pre-defined lists and User-defined lists

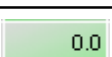
In REFCON you operate with three categories of Container Lists:

- **Standard lists** that occur in the Side panel and the menu in fixed positions and with fixed names and settings.
- **Pre-defined lists** that you can ["install"](#)^[81] and use if you wish.
- **User defined lists** that are created and designed by you or your colleagues.

Having "installed" a pre-defined list, it works exactly like a user-defined list. This is the whole idea: To make it easy for you to create your own lists.

Colour codes within data fields

The individual fields of the Container List are coloured as described in the following table:

Example	Colour	Explanation
	Grey	Field with no value or a value that can not get obsolete.
	Blue	The row (container, alarm etc.) is selected / highlighted.
	Green	The value is valid and not "obsolete", that is, it has been updated recently (see Data Obsolescence ^[332]).
	Orange	The data point is "obsolete" (see Data Obsolescence ^[332]). The shown value is the latest available, but you should not rely very much on it.

You will see that values get **bold** for a couple of seconds from time to another. This just indicates that the field is updated.



Bands and Columns

Because the amount of available columns is very high, the columns have been grouped into bands. The following figure shows a few examples:

Container List Example										
Standard Info			Manifest	Temperature				Cargo Probes		
Location	ContainerID	Alms	Outbound	Temp. SP Exp.	Temp SP	Supply	Return	USDA1	USDA2	USDA3
C1021.A	DEMO1000002	0		-8.0	-8.0	66.2	66.2	66.2	65.2	65.0
	DEMO1000003	0		7.0	7.0	31.3	32.5	31.3	32.7	33.0
C1022.B	DEMO1000004	0		-8.0	-8.0	77.3	78.6	78.9	80.3	79.2

The Container List version in the example includes 4 bands, named "Standard Info", "Manifest", "Temperature" and "Cargo Probes", with a total of 11 columns.

4.1 Using the Container List

Key subjects in this topic:

[Opening the Container List](#)^[67]
[Locked or Unlocked Container List](#)^[68]
[Operating the Container List](#)^[68]
[Operating the selected container\(s\)](#)^[68]
[Finding containers in the Container List](#)^[68]
[Making changes to a Container List](#)^[68]
[Selecting the order of the containers](#)^[68]
[Selecting the filter of the containers](#)^[69]
[Changing column widths](#)^[69]

Related topics:

[Container List columns](#)^[69]
[Setting up the Container List](#)^[80]
[The Container List](#)^[66]

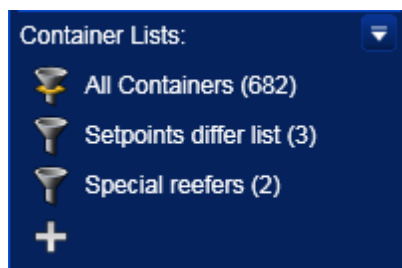
Opening the Container List

You would use the Container List to have an overview of an amount of containers. Being organised in a limited number of columns, the Container List shows a selected subset of the data for each container. So, before opening the Container List, you would probably think of:

- What subset of containers you need to deal with.
- What subset of data you are interested in.

You open the Container List by selecting it in the Side panel or, for certain specific list(s), in the [menu](#)^[25].

In the Side panel, Container Lists are available at the [Performance Bar](#)^[27], when relevant, and under the Container Lists heading:



Select a version to open the Container List.



Locked or Unlocked Container List

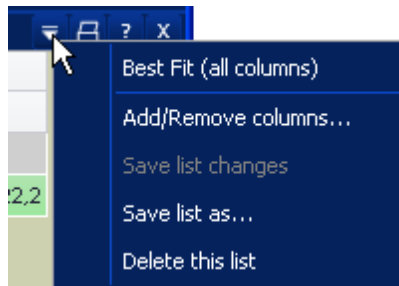
As can be seen in the above figure, two icons can occur next to the Container Lists headings:

Icon	Meaning
	The Container List version is locked : You can make certain temporary changes, but you can not save these, and you can not delete the version. You can not, even temporarily, change the filter.
	The Container List version is unlocked : You can make permanent changes, and you can delete the version.

If neither of these symbols are shown, the list is locked.

Operating the Container List

A specific set of commands is available allowing you to manipulate the Container List as such. You find these commands as part of the [Shortcut menu](#)^[34], but also you can use the options button:



Here, selecting "Best Fit.." will adjust the width of every single column so that it is just wide enough to show the actual content.

The other commands are explained in [Setting up the Container List](#)^[80].

Operating the selected container(s)

By right-clicking one container, or by selecting a number of containers and right-clicking one of them, you will have the Shortcut menu displayed. Please refer to [The Container Command section of the Shortcut menu](#)^[34] for details.

Finding containers in the Container List

If you need to locate one specific container in the Container List (as well as in any other picture), you might find the search function useful, please refer to [Container Quick Search](#)^[35].

If you are rather looking for a group of containers, you can use the filter functions described below.

Making changes to a Container List

Note that whenever you make any change to a Container List as described in the following, this is indicated by orange colours and symbols, see ["List changed" indication](#)^[26].

The orange indication means that the Container List is *temporarily* changed.

You can not save changes to a [locked](#)^[68] list. If you have made temporary changes to a locked list, they will disappear without warning when you open another picture. Note that you can, however, save the list with the temporary changes in a new name, using the "Save list as..." command.

Selecting the order of the containers

You can click any column header to change the sorting order of the containers. Having done so, you will see a small arrow symbol appear in the header:



If you click the same header once more, the arrow symbol flips to the opposite direction.

The arrow symbol indicates that now the containers are sorted alphabetically according to the appointed column. Only one column can be appointed at a time.

The meaning of the arrow symbol is:

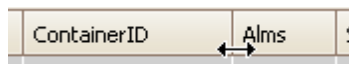
- When it points upward (like in above figure), the sorting order is ascending (A before B etc.).
- When it points downward, the sorting order is descending (B before A etc.).

Selecting the filter of the containers

You select which containers to include in the list by setting up the filter. Please refer to [Setting up a Filter](#)^[84] for instructions. Note that you can not change the filter of a locked list.

Changing column widths

You can make a column wider or narrower by dragging the lower part of the right column header edge:



In the example, the ContainerID column width is affected.

4.2 Container List columns

Key subjects in this topic:

[Overview](#)^[69]
[Column header descriptions](#)^[70]
[Abbreviations used in column headers](#)^[70]
[Alarm indication column](#)^[71]
[Descriptions of selected columns](#)^[71]
[Display temperature](#)^[72]
[Modem related columns](#)^[73]
[Communication State](#)^[73]
[Determining Comm State Should communicate](#)^[74]
[Selecting time ts](#)^[75]
[Comm States illustration, "normal" online case, Site State "Plugged on" supported](#)^[76]
[Comm States illustration, "normal" online case, other terminal](#)^[77]
[Comm States illustration, "normal" online case, ship with Planning data file transfer](#)^[78]

Related topics:

[Using the Container List](#)^[67]
[The Container List](#)^[66]

Overview

REFCON provides about 200 columns organised in about 17 [bands](#)^[67]. These can all be used in customised Container Lists as desired.

Note that

- Not all columns that origin from controllers are relevant to all controller types. If the controller does not provide that measure, the field will be empty.

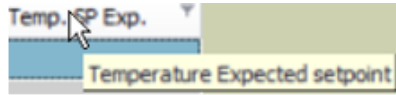


- Similarly, not all columns that origin from the Planning System are relevant to all Planning System interface protocols. If it is not included, the field will be empty.

Column header descriptions

When shown in list headers, column names can be abbreviated and therefore be difficult to understand.

Try moving the mouse cursor over the name in the list header, you will then normally get a tooltip that explains the abbreviation:



Abbreviations used in column headers

You will in general see the following abbreviations:

Abbreviation	Explanation
Abs	Absolute
Batt	Battery
Cap	Capacity
Comm	Communication
Comp	Compressor
Cond	Condenser
Ctrl	Controller
Defr	Defrost(ing)
Disch	Discharge
Econ	Economizer
Evap	Evaporator
Exp	Expected (= informed by Planning System or user)
Hum	Humidity
Interv	Interval
Msg	Message
Op	Operating
PD	Planning data
Press	Pressure
PTI	Pre-trip inspection
Rel	Relative
SP	Setpoint
Suct	Suction
SW	Software
Temp	Temperature



Abbreviation	Explanation
USDA	Used as synonym for cargo probe
Volt	Voltage

Alarm indication column

All versions include the narrow leftmost column with alarm indication:

Indication	Description
	No alarms on this container
 / 	There are one or more unacknowledged alarms on this container.
	There are one or more alarms on this container, they have all been acknowledged.

Descriptions of selected columns

The following table describes a few columns that could need a special explanation:

Column header	Explanation
Defrosting	Defrosting: A "D" is shown when the controller reports that it is being defrosting.
Special	An "S" is shown when the container is marked as Special Reefer.
Location	Shows the container's location as reported by the Planning System or typed in by the user. If (*) is appended to the location, it indicates that the container will be deleted as soon as all it's alarms have been acknowledged and it's Communication State ^[73] is no longer <i>Communicating</i> . See also Planning data import ^[182] .
Online	PCT communication status.  means that REFCON has connection to the container, or it had connection recently, while  means that it has not. Due to the way the power line communication works, "recently" in this context may mean 2 - 3 hours; therefore a container can stay "online" even quite some time after power off.
Comm State	Refer to Communication State ^[73] .
Monitored	This column has in fact been replaced by Comm State.  means that Comm State equals <i>Communicating</i> , while  represents any other Comm State.
Obsolete	 means that data once was received from controller, the one way or the other, but they are obsolete ^[332] , while  means that data are up-to-date or have never been received.



Column header	Explanation
OnlineDataOK	● means that data has been received through an online source (PCT, REFCON Wireless Hub ^[232] solution or similar) recently, while ● means that either data were never received through an online source, or they are obsolete.
OfflineDataOK	● means that data has been received through an offline source (REFCON Mobile ^[226] solution, RDC or similar) recently, while ● means that either data were never received through an offline source, or they are obsolete.
HasData	● means that data have been received from controller recently, while ● means that either data were never received or they are obsolete. In other words, HasData is ● if either OnlineDataOK or OfflineDataOK is ●.
Comm Start Time	The Communication Start time shows the date and time when REFCON received the very first "life sign" from the container, typically after arriving in terminal or on ship, but also after a longer period of no "life signs". You may use Comm Start Time as an approximation for power on time; but you need to be aware that Comm Start Time will always be some time after power on, in best case at least 15 seconds, in worst case several minutes or even more. "Life sign" means that some kind of communication from the modem on the container has been received by REFCON. First "life sign" is typically when REFCON receives the first part of the entire data package to be transferred; hence, the complete set of data for the container will normally appear in REFCON some time later.
Comm.Duration	Communication Duration is the time since above Comm Start Time; it is calculated every minute by REFCON.

In addition to the above, you may also see/use the special columns regarding [PCT Communication Statistics](#)^[346].

Display temperature

The "Display temp" data point / column is a copy of either the

- Return temperature when "Frozen", or the
- Supply temperature when "Chilled".

The actual temperature setpoint decides: It is "Frozen" when the setpoint is below or equal to the limit, and it is "Chilled" when it is above the limit.

User can select the limit in [REFCON Setup](#)^[330], under the Various tab. The limit is -10 deg.C as default.



Modem related columns

If you are using an [REFCON 6 extension](#)^[218], it is possible that more than one modem is attached to a container and communicating. In such case you may see semicolon-separated lists in the fields in the relevant modem columns, like in following example:

Modem			
Modem type	Modem HW rev	Modem serial	Modem SW ver
REFCON Portable Modem; RMM-W	042A; 1	11112232; 19280470	0.0.0-0S; 4.0.0_201050

In this example, two modems are communicating - a removable REFCON Portable Modem and a fixed RMM-W. As the modems are shown in this order in the "Modem type" field, you find data related to the REFCON Portable Modem *before* the semicolon in each relevant modem field, and data for RMM-W *after* the semicolon. So, in above example, the REFCON Portable Modem has serial number 11112232, while the RMM-W has 19280470.

Note that a possible communicating PCT modem will be shown in the separate "PCT Modem Info" column.

Communication State

The Comm State column exists both in the Container List and in the [Container Modems list](#)^[340].

In the Container List, Comm State indicates the status of the communication between the container controller and REFCON - no matter what modem is in play and what communication equipment is routing the data. However, the set of possible Comm States depend on this equipment.

The following time settings are involved when determining the Comm State:

Time setting	Explanation
t_1	Maximum time allowed for no "life signs" or no live data updates from container. The value of t_1 depends on the solution; for PCT it is 1 hour; for REFCON Portable Modem Solution, please refer to the description of this ^[219] .
t_{obs}	Customer decided data obsolescence time ^[332] for offline solutions.

The following table shows all the possible Comm States. For further explanation, please consult the [illustration below](#)^[69].

Please note that on your specific REFCON installation it is likely that some of these are never seen.

Priority	Comm State option	Explanation	Recommended action
1	Communicating	If your REFCON includes an online solution (PCT, REFCON Wireless Hub ^[232] solution or similar), <i>Communicating</i> is the desired, "normal" state for any container being on power.	None



Priority	Comm State option	Explanation	Recommended action
		<i>Communicating</i> means that REFCON has received controller live data from this container within last t_1 minutes through an online source. This state indicates that unless the container is powered off or similar, you can normally expect the container data to be automatically refreshed at intervals less than t_1.	
2	No controller	The modem on the container did report "no connection to controller" less than t_1 minutes ago. Note that there are cases where this Comm State for the container cannot be determined, as the modem cannot report the Container ID if it never had connection to the controller. Often you will in such case be able to locate a modem in Container Modems list reading <i>No controller</i>.	Check the plugs and cable between the modem and the controller.
3	Offline updated	If your REFCON includes an offline solution capable of transferring data from modems wirelessly (REFCON Mobile²²⁶ solution or similar), <i>Offline updated</i> is the desired, "normal" state for any container being on power. <i>Offline updated</i> means that REFCON has received controller data from this container within last t_{obs} minutes	None - but regular walk-bys are required to keep this state.



Priority	Comm State option	Explanation	Recommended action
		through an offline source that did automatically read data from the controller, that is, no data were manually entered, however, users need to walk-by to get data updated.	
4	Manually entered	REFCON has received controller data from this container within last t_{obs} minutes through an offline source (REFCON Mobile ²²⁶ solution, RDC or similar) where data were manually entered, fully or partly. Users need to walk-by to get data updated.	None - but regular walk-bys are required to keep this state.
5	Poor communication	REFCON has not received any complete live data update nor any indication of no connection to controller for this container within last t_1 minutes; however, REFCON has in fact registered some communication related to this container within last t_1 minutes.	If this state persists, check if there are any objects or signal noise that can affect the communication. If so, attempt to move objects / reduce signal noise.
6	Communication started	REFCON has registered some communication related to this container but has not yet received any complete live data update nor any indication of no connection to controller. Normally this state remains only for a short time, after which REFCON receives complete live data and then changes to	None



Priority	Comm State option	Explanation	Recommended action
		<i>Communicating</i> (or <i>No controller</i>). If REFCON keeps registering some communication but still no complete live data for t_1 minutes, the state changes to <i>Poor communication</i>. If, after change to <i>Communication started</i>, REFCON registers no communication at all for t_1 minutes, the state changes to <i>Not communicating</i>.	
7	Not communicating	REFCON has registered no communication related to this container for the past t_1 minutes after having seen communication before.	If power off was expected: None. Otherwise it <i>could</i> indicate unintended loss of power.
8	Obsoleted	This state is possible for offline solutions only (REFCON Mobile ²²⁶ , RDC or similar). If the state is <i>Offline updated</i> or <i>Manually entered</i> , and data was not updated for the time t_{obs} , Comm State gets <i>Obsoleted</i> , along with orange indications in the list.	Do/complete a walk-by with your hand-held device as soon as possible.
9	Should communicate	When a container has been in Comm State <i>Never communicated</i> (below) for some time without communication starts nor offline update is conducted, Comm State is changed to <i>Should communicate</i> to indicate to the user that something is/may be wrong. REFCON attempts to decide the change to	Check if • Container has power. • A modem, fixed or portable, has been mounted on the container, and that it indicates working all right.



Priority	Comm State option	Explanation	Recommended action
		Should communicate ^[77] in a smart way to avoid that user takes unnecessary action. Once the state has been any of the above, it will not return to <i>Should communicate</i> nor <i>Never communicated</i> .	
10	Never communicated	The container has been created due to data from Planning System "recently" and has not yet started communication nor been offline updated. This is a normal case when REFCON receives Planning data for a container before or at the moment power is plugged on, as after that it takes time before communication runs.	None

The Comm State options are listed in prioritized order. For example, if Comm State *Communicating* is determined, this generally overrules the lower prioritized options; similar, if *Poor communication* is determined, but *Communicating* was determined less than t_1 minutes ago, *Communicating* is kept.

Determining Comm State *Should communicate*

The state *Should communicate* means that, to the best of our knowledge, container has now had time to start communication and should therefore now have changed to one of the first 6 states in above table - but it has not.

Optimally, REFCON would be triggered exactly when the container was (claimed to be) powered on. Then it would start a timer, and if no communication within [ts minutes](#)^[78], it would change to *Should communicate*.

Unfortunately, in most cases, REFCON does not get such precise power on indication. The following describes how it works:

On ships

On ships, REFCON normally gets Planning data through imported files, Baplie format or other format. Normally the file transfer involves a manual effort.

To make sure you are free to transfer the Planning data file any time appropriate, the Comm State decision is based on the change from [Harbour mode to Sea mode](#)^[117], assuming that you will not transfer files while in Sea mode. Any container in Comm State *Never communicated* will change to *Should communicate* shortly after the switch to Sea mode.

If the ship has automatic online Planning data transfer it works like in a terminal...



In terminals,

- file transfer may be used as well, but here we have no Sea mode. Therefore, for the file transfer case, the switch to *Should communicate* will happen [ts minutes](#)^[78] after file transfer for all containers being new in the file.

Often some automatic Planning data transfer solution is running. Some solutions do support the [Site State](#)^[183] *Plugged on*, others do not.

- When Site State *Plugged on* is supported, we assume that when the Planning System reports *Plugged on* for a container, this does in fact indicate that the container has just been powered on. REFCON then allows [ts minutes](#)^[78] for communication to start; if not Comm State changes to *Should communicate*.
- When Site State *Plugged on* is **not** supported, we assume that when the Planning System reports any new container, this does in fact indicate that the container has just been powered on. REFCON then allows [ts minutes](#)^[78] for communication to start; if not, Comm State changes to *Should communicate*.

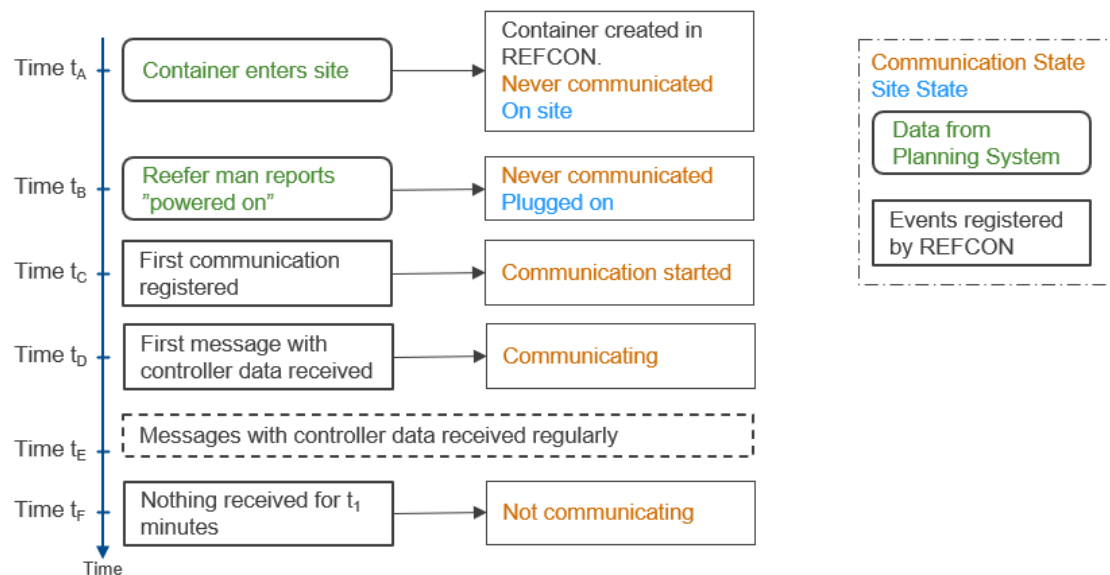
Selecting time ts

You can change the time **ts** in [REFCON Setup](#)^[330], Various tab - the value "Communication State Timeout".

You can choose any value between 5 and 300 minutes. The appropriate time depends on your working schedules.

Comm States illustration, "normal" online case, Site State "Plugged on" supported

If your REFCON includes an **online** solution (PCT, [REFCON Wireless Hub](#)^[232] solution or similar), and your Planning System specifically reports Site State *Plugged on*, the "normal" life cycle of a container can be illustrated as follows:



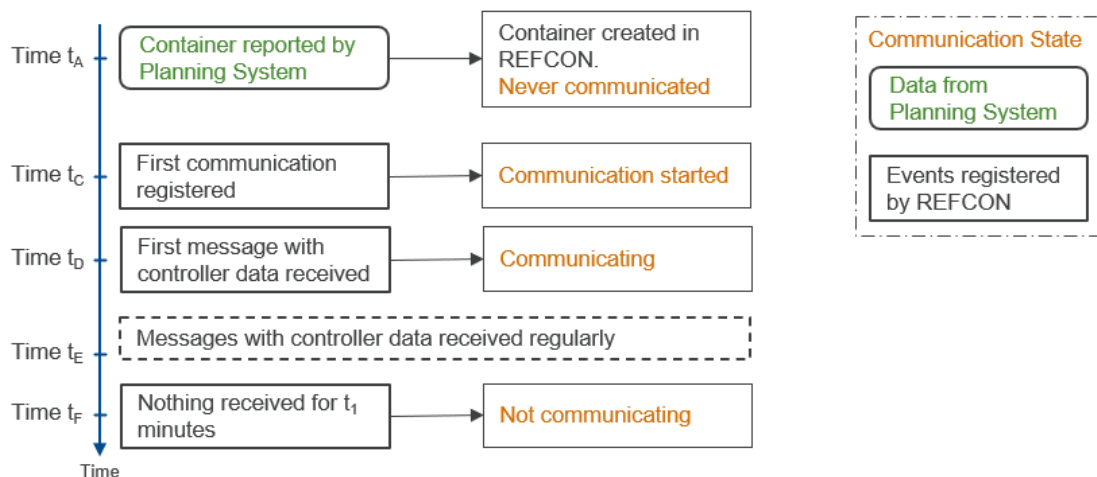
- The time span from t_A to t_B can be any duration - Comm State keeps *Never communicated* - unless communication does in fact start.
- The time span from t_B to t_C is normally short. If it exceeds t_2 minutes, Comm State changes to *Should communicate*.
- The time span from t_C to t_D is normally short. If it exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.



- After time t_D , messages with controller data will normally be received regularly. If the time span between two messages exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.
- After time t_E , the container is powered off. At time t_F , being t_1 minutes after t_E , Comm State changes to *Not communicating*. Hence, this happens up to t_1 minutes after power off.

Comm States illustration, "normal" online case, other terminal

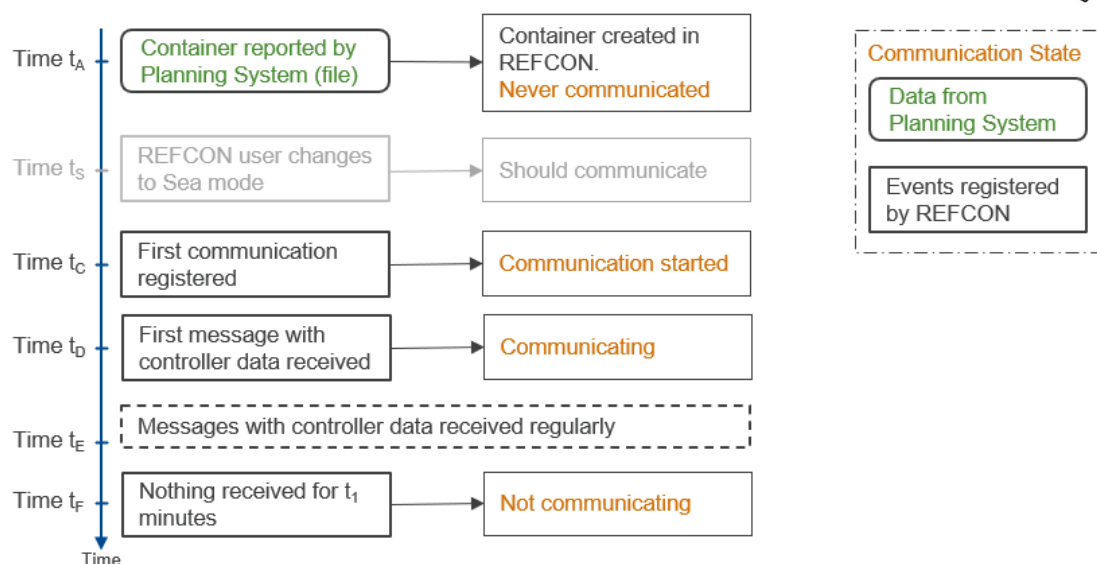
If your REFCON is in a container terminal and includes an **online** solution (PCT, [REFCON Wireless Hub](#)²³² solution or similar), and your Planning System does **not** specifically report Site State *Plugged on*, the "normal" life cycle of a container can be illustrated as follows:



- The time span from t_A to t_C is expected to be short. If it exceeds t_2 minutes, Comm State changes to *Should communicate*.
- The time span from t_C to t_D is normally short. If it exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.
- After time t_D , messages with controller data will normally be received regularly. If the time span between two messages exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.
- After time t_E , the container is powered off. At time t_F , being t_1 minutes after t_E , Comm State changes to *Not communicating*. Hence, this happens up to t_1 minutes after power off.

Comm States illustration, "normal" online case, ship with Planning data file transfer

If your REFCON is in a ship and includes an **online** solution (PCT, [REFCON Wireless Hub](#)²³² solution or similar), and Planning data are transferred via files, the "normal" life cycle of a container can be illustrated as follows:



- The time span from t_A to t_C can be any duration. While in Harbour mode, Comm State does not change to *Should communicate*.
- If/when REFCON user changes to Sea mode, containers with Comm State *Never communicated* are changed to *Should communicate*.
- The time span from t_C to t_D is normally short. If it exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.
- After time t_D , messages with controller data will normally be received regularly. If the time span between two messages exceeds t_1 minutes, Comm State changes to *Poor communication* or *Not communicating*.
- After time t_E , the container is powered off. At time t_F , being t_1 minutes after t_E , Comm State changes to *Not communicating*. Hence, this happens up to t_1 minutes after power off.

4.3 Setting up the Container List

Key subjects in this topic:

[Overview](#)^[80]
["Installing" a pre-defined Container List](#)^[81]
[How to create a Container List version](#)^[81]
[Selecting the name for your version](#)^[81]
[Selecting the order of the columns](#)^[81]
[Selecting the columns](#)^[82]
[Sorting the containers](#)^[83]

Related topics:

[Using the Container List](#)^[67]
[Setting up a Filter](#)^[84]
[How to use the Filter Builder](#)^[87]
[The Container List](#)^[66]

Overview

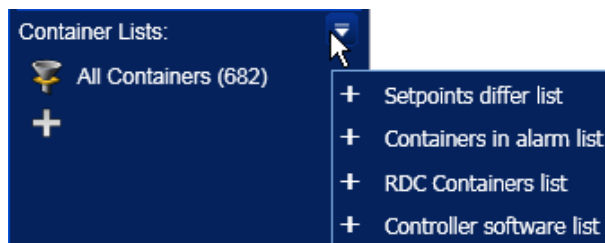
When REFCON 6 is installed the first time, it includes a limited number of standard and pre-defined Container List versions. All of these are **locked**^[68], meaning that you cannot change them.



The idea is, however, that you design your own user-defined Container List versions. If you from time to another need the same subset of containers or columns, follow the instructions below to create a version that fits your needs.

"Installing" a pre-defined Container List

Out of the box, REFCON 6 provides a number of pre-defined Container Lists that are found in the drop-down menu in the Side panel:



When you select one of the lists, it shows up in the Side panel as an unlocked list. Should you one day not need it anymore, simply delete it - you can always "install" it again.

How to create a Container List version

To create a new Container List version, proceed as follows:

Step	Action	Comments
0	Open the existing Container List version that fits best to your requirements.	This is in particular valuable if you are going to re-use or modify a complex list filter.
1	Click the + icon in the Side panel:  Alternative: In stead of the + icon you can use the "Save list as..." command. Alternative: "Install" a pre-defined list, see above ^[81] .	You should carefully write a name for your version ^[81] that clearly describes what is it's purpose, to you and other users.
2	Change the settings of the Container List as described below.	
3	Save the layout in any of the following ways: • From the Shortcut menu^[33] or the options menu^[68], select "Save list changes". • Or close the Container List and select Yes when asked whether to save.	If the version you are setting up is locked , the "Save list changes" option is greyed, and you are not asked when the Container List is closed.

Selecting the name for your version

When you create a new Container List version, it is important that you carefully select a name that clearly describes it's purpose. Remember that the name appears in the Side panel, along with others. To you and other users, the name will be the only means to decide which version to open.

The name that you select can only consist of letters, digits and spaces. Special characters are not allowed.

Selecting the order of the columns

You can drag the band headers or the column headers with the mouse to change the order that the bands and columns appear.

Dragging means:

1. Point to the header and press and keep the left mouse button.



2. While the left mouse button is pressed, move the mouse cursor to another header, and release the button.

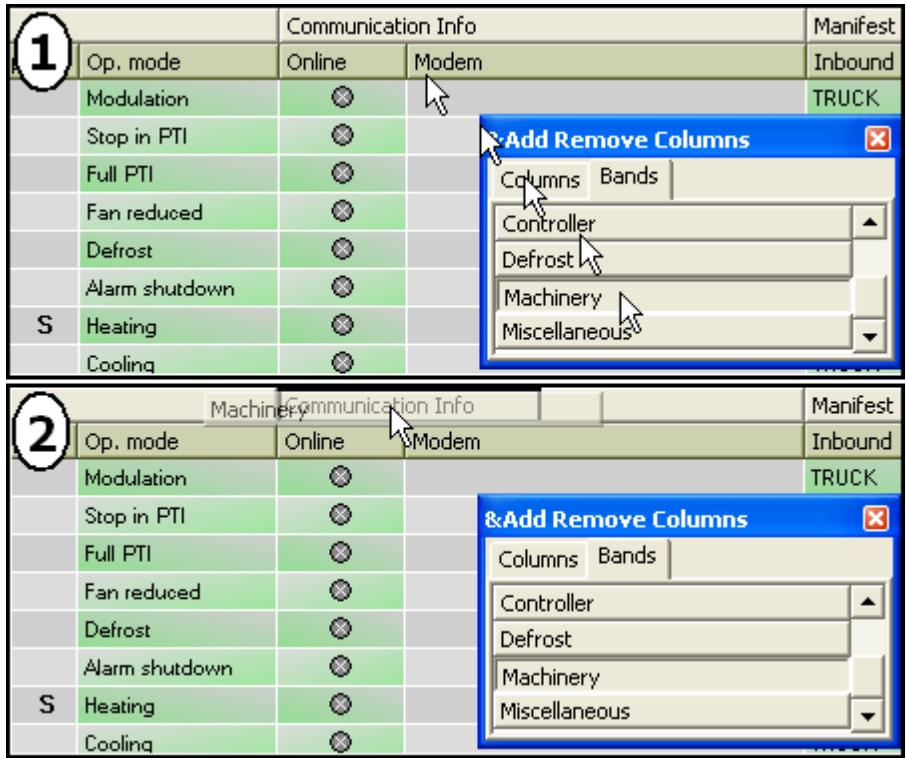
While dragging, you will observe a black rectangular frame within the appointed header. The frame is just visible for a short while. It appears in either of the ends of the header:

- If at the left end, the moved band or column will be placed left to the appointed column.
- If at the right end, the moved band or column will be placed right to the appointed column.

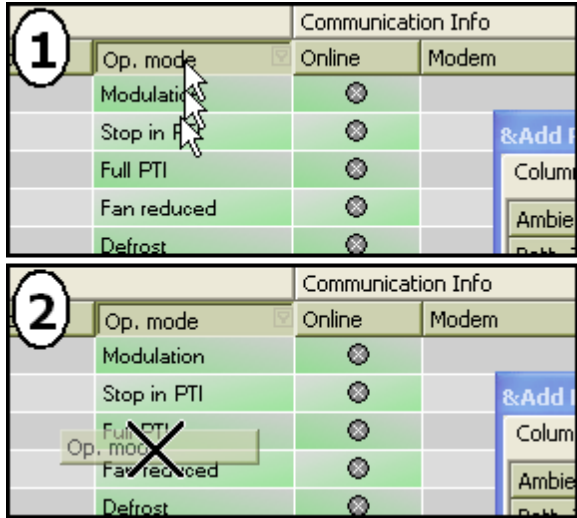
You will notice that you can't move a column outside its band.

Selecting the columns

You add or remove bands or columns from the Container List version as follows:

Step	Action
1	Right-click anywhere inside the Container List window, and select "Add/Remove Columns..." from the menu. Result: A small box named "Add Remove Columns" appears in the lower right corner of the Container List window. The box has two tab's, named "Columns" and "Bands". Under each tab, there will normally be a number of column and band headers, respectively.
2a	To add a band, locate it in the "Add Remove Columns" box under the "Bands" tab and drag it up upon the existing band header where you want it to be placed. See also Selecting the order of the columns ^[81].  A band can be placed anywhere among and besides the already displayed bands. Note that when adding a band, all the attached columns are automatically added as well, except for the possible columns being found under the Columns tab.
2b	To add a column that has been un-attached its band, locate it in the "Add Remove Columns" box under the "Columns" band, and drag it up upon the



Step	Action
	existing header where you want it to be placed. You can only add it to it's band, as clearly indicated. 
2c	To remove a band or column, drag it's header away from the headers.  When you have dragged it some way down, a cross appears upon the floating header - then release the mouse button, and the header has been removed. It automatically occurs in the "Add Remove Columns" box.

Sorting the containers

You can change the sorting order of the containers as described in [Selecting the order of the containers](#)⁶⁸.

You set up the filter as described in [Setting up a Filter](#)⁸⁴.



4.4 Setting up a Filter

Key subjects in this topic:

[Overview](#)^[84]
[Initially selecting a Filter](#)^[84]
[The Filter Status Bar](#)^[85]
[Customising the filter, method 1](#)^[85]
[Customising the filter, method 2](#)^[85]
[Customising the filter, method 3](#)^[87]

Related topics:

[Using the Container List](#)^[67]
[How to use the Filter Builder](#)^[87]
[The Container List](#)^[66]

Overview

You define a *filter* to specify a certain subset of all containers to be shown in the Container List. Filters can be applied to more columns at a time. Columns having filter applied are marked like this:



Filters can be rather simple or quite complex as you choose. There are several ways to define the filter, as described in the following.

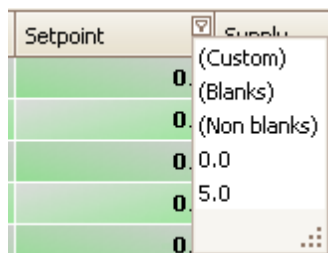
Initially selecting a Filter

When you are working with a Container List having no filter applied, the first step in defining a filter always is as follows. Note that you can also do this when a filter is active.

Move the mouse cursor onto the header of the column of interest (if there are more, just pick one for a beginning):



As soon as the mouse cursor moves over the header, it changes colour, and the small filter button gets visible. Click the button to open the list of filter options:



At the top of this list, you will see a number of options marked with parentheses. They can be:

Filter option	Explanation
(All)	Removes the filter regarding this column. After this, all containers are shown unless filtered by other column(s).
(Custom)	Selecting an advanced filter as described below.
(Blanks)	Do only show the container if the field in this column is empty.
(Non blanks)	Do only show the container if the field in this column is not empty.

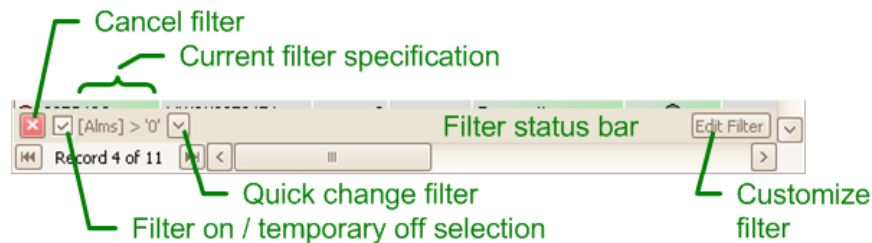
Below the parentheses options, all values that actually appear in the column are listed - in the above example, 0.0 and 5.0. When selecting one of these, only containers reading this specific



value in this column will be shown. For example, select 0.0 to show containers with setpoint zero.

The Filter Status Bar

Having selected a filter as described above, the Filter Status Bar appears near the bottom of the Container List:



From left to right, the Filter status bar includes:

Bar element	Description
"Cancel filter" button	Completely removes the filter. After this, all containers are shown.
"Filter on / temporary off" checkbox	Removing the checkmark makes the filter temporarily cancelled, which makes all containers shown. Checking anew re-invokes the filter.
"Current filter specification" field	Shows how the actual filter is defined.
"Quick change filter" button	If available the button leads to a list of related alternative filter specifications.
"Customize filter" button	Opens the Filter Builder ^[87] described below.

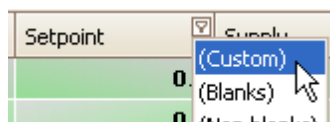
Customising the filter, method 1

One way to further develop the filter is to repeat the initial selection for another column, see [Initially selecting a Filter](#)^[84] above.

This way you achieve AND-logic, that is, for a container to be shown, it must meet the filter conditions for **all** columns where filters are applied. You will recognize the word "and" in the specification in the Filter status bar (and not the word "or").

Customising the filter, method 2

When you select "(Custom)" from the filter list,



the following dialog box opens:



← Selected column

← Criterion 1

← Criteria combination type

← Criterion 2

You will see that the selected column is indicated, in the example: "Setpoint".

It is possible to select one or two criteria. Each criterion is specified by a *type* and a *parameter string*.

Regarding the **parameter string**, note that:

- It is **not** case sensitive - you can use upper and lower case letters to wish.
- For some criterion types, the % character means "any character combination" - see examples below.

The available criterion types depend on the column type. All the possible criterion types are:

Criterion type	Explanation	Examples
equals	The column reading must exactly match the parameter string	"equals" "SoMu0123456" will display ID "SOMU0123456" only
does not equal	Display all column readings that do not match the parameter string	"does not equal" "SoMu0123456" will display all containers except the one with ID "SOMU0123456"
is greater than	• If numeric column: Display all column readings being greater than the parameter value. • If textual column: Display all column readings being higher in the alphabet than the parameter string	• "is greater than" "17" will display the value 18 and higher (or 17.0001 and higher) • "is greater than" "somu%" will display all IDs starting with "SOMU", "SONU", "TAAU" etc., but not "AAAU", "SOLU" etc.
is greater than or equal to	Same as "is greater than" but also including the exact match	"is greater than or equal to" "17" will display the value 17.0000 and higher
is less than	• If numeric column: Display all column readings being less than the parameter value. • If textual column: Display all column readings being lower in the alphabet than the parameter string	• "is less than" "17" will display the value 16 and smaller (or 16.9999 and smaller) • "is less than" "somu0123456" will display all IDs starting with "AAAU", "SOLU" etc. as well as "SOMU0123455", but not "SOMU0123456" and higher numbers or IDs starting with "SONU", "TAAU" etc.
is less than or equal to	Same as "is less than" but also including the exact match	"is less than or equal to" "17" will display the value 17.0000 and smaller



Criterion type	Explanation	Examples
blanks	Do only show the container if the field in this column is empty.	
non blanks	Do only show the container if the field in this column is not empty.	
like	Similar to "equal", but the % character can be used to hit more than one container	• "like" "SoMu0123456" will display ID "SOMU0123456" only • "like" "SOMu%" will display all IDs starting with "SOMU". • "like" "SOMu%20" will display all IDs starting with "SOMU" and ending with "20". • "like" "%77%" will display all IDs including the sequence of characters "77".
not like	The opposite of "like": Displays all containers that "like" would hide.	"not like" "%77%" will display all IDs that do not include the sequence of characters "77".

If you select two criteria, also choose the combination type:

- Select "And" if both criteria must be met.
- Select "Or" if one of the criteria or both must be met.

Examples:

- To have all containers with setpoint near zero selected, you could define criterion 1 as "is greater than" "-0.5", criterion 2 as "is less than" "0.5" and combination type "And".
- To have all containers in bay 02 and 06 displayed, you would choose the Location column and set criterion 1 to "like" "b02%", criterion 2 to "like" "b06%" and combination type "Or".

Customising the filter, method 3

When you select the "Edit Filter" button rightmost in the [Filter status bar](#)^[85], the advanced filter selection dialog box appears, see [How to use the Filter Builder](#)^[87].

4.5 How to use the Filter Builder

Key subjects in this topic:

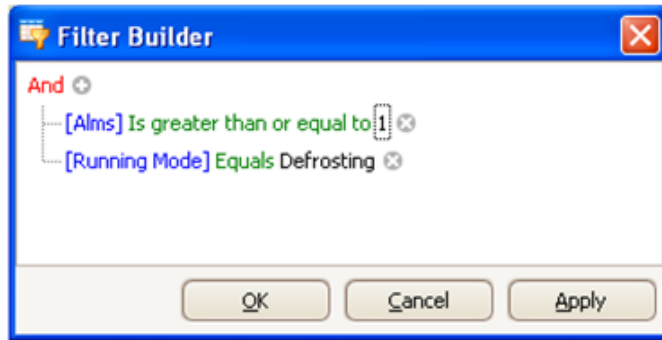
[Overview](#)^[87]
[Combination types](#)^[88]
[Column name selection](#)^[88]
[Criterion selection](#)^[89]
[Parameter selection](#)^[92]
[Other hints](#)^[92]
[Examples](#)^[92]

Related topics:

[Using the Container List](#)^[67]
[Setting up a Filter](#)^[84]
[The Container List](#)^[66]

Overview

When you select the "Edit Filter" button rightmost in the [Filter status bar](#)^[85], the Filter Builder appears:



Filter logic pane

Note that any element within the filter logic pane is click-able and changeable.

We believe that you will have to experiment a little to learn how to create advanced filters, but here are a few tips:

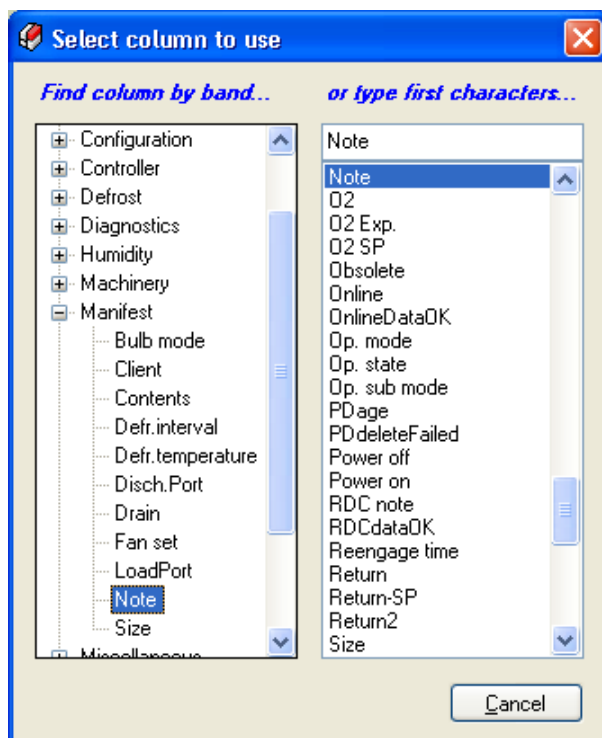
Combination types

Clicking "And" (or any of the following alternatives) displays a menu where you can select any of the following options:

Operator option	Explanation	Examples
 And	All related operands are "and'ed", that is, all conditions must be met.	([Alms] > 0) And ([Special] = Checked) => only containers that are marked Special and have alarms are selected.
 Or	The related operands are "or'ed", that is, one or more conditions must be met.	([Alms] > 0) Or ([Special] = Checked) => all containers that are marked Special or have alarms (or both) are selected.
 Not And	All related operands are "and'ed" and the result is inverted, that is, if all conditions are met the container is excluded.	Not (([Alms] > 0) And ([Special] = Checked)) => all containers that are not marked Special or do not have any alarms are selected.
 Not Or	All related operands are "or'ed" and the result is inverted, that is, if any condition is met the container is excluded.	Not (([Alms] > 0) Or ([Special] = Checked)) => all containers that are not marked Special and do not have any alarms are selected.
 Add Condition	Adds a condition row - same as clicking the addition symbol: 	There are two condition rows in above Filter Builder example.
 Add Group	Adds a new group of conditions.	Both advanced examples below have two groups.
 Remove Group	Removes a group of conditions.	

Column name selection

Clicking a column name (eg. [Alms]) opens a dialog box with all available columns (= container data fields), including those not displayed in the actual Container List:



As you see there are two lists side by side. You can use any of the lists as you wish.

You can use the mouse to select, or you can use the arrow keys, the tab key and the Enter key. In the following we assume you are using the mouse.

In the list to the left, all the columns are organised by **bands**⁶⁷. So, if you are looking for a column, and you know what band it belongs to, click the  next to the band name, and then select the column name.

In the list to the right, all columns are listed in alphabetical order. You can search this list by typing the first characters of the column name in the text box uppermost.

Criterion selection

Clicking "Equals" (or any of the following alternatives) displays a menu where you can select any of the following options:

Criterion	Explanation	Examples
= Equals	The column reading must exactly match the parameter string.	[Running Mode] = 'Defrost' => all containers in mode "defrost" are selected, but not containers in mode "defrosting".
≠ Does not equal	Display all column readings that do exactly not match the parameter string.	[Running Mode] <> 'Defrost' => all containers in mode "defrost" are not selected, but all others are, including those in mode "defrosting".
> Is greater than	• If numeric column: Display all column readings being greater than the parameter value. • If textual column: Display all column readings being higher	• [Alms] > 0 => all containers with alarms are selected. • [Running Mode] > 'Defrost' => all containers in modes coming after "defrost" when sorted alphabetically are



Criterion	Explanation	Examples
	in the alphabet than the parameter string.	selected - for example, "defrosting" is selected, but "defrost" is not.
➤ Is greater than or equal to	- If numeric column: Display all column readings being greater than or equal to the parameter value. - If textual column: Display all column readings being higher or equal in the alphabet than the parameter string.	[Alms] >= 1 => all containers with alarms are selected. [Running Mode] >= 'Defrost' => all containers in mode "defrost", and all modes coming after "defrost" when sorted alphabetically, are selected - for example, both "defrost" and "defrosting" are included.
< Is less than	- If numeric column: Display all column readings being less than the parameter value. - If textual column: Display all column readings being lower in the alphabet than the parameter string.	[Alms] < 1 => all containers without alarms are selected. [Running Mode] < 'Defrost' => all containers in modes coming before "defrost" when sorted alphabetically are selected - for example, "cooling" is selected, but "defrost" is not.
⚡ Is less than or equal to	- If numeric column: Display all column readings being less than or equal to the parameter value. - If textual column: Display all column readings being lower or equal in the alphabet than the parameter string.	[Alms] <= 1 => all containers with none or one alarm are selected. [Running Mode] <= 'Defrost' => all containers in mode "defrost", and all modes coming before "defrost" when sorted alphabetically, are selected - for example, "defrost" is included, but "defrosting" is not.
🌱 Is between	- If numeric column: Display all column readings being between the two parameter values. - If textual column: Display all column readings being between the two parameter strings in the alphabet.	[Alms] Between ('1', '4') => all containers with 2 or 3 alarms are selected. [Running Mode] Between ('Defrost', 'Remote') => all containers in modes coming after "defrost" and before "remote" when sorted alphabetically are selected - for example, "defrosting" is selected, but "defrost" and "remote" are not.
🚫 Is not between	If numeric column: Display all column readings being less than or equal to the smallest parameter value or greater	Not ([Alms] Between ('1', '4')) => all containers with 0, 1 or more than 3 alarms are selected.



Criterion	Explanation	Examples
	than or equal to the highest parameter value. If textual column: Display all column readings being not between the two parameter strings in the alphabet.	Not ([Running Mode] Between ('Defrost', 'Remote')) => all containers in modes coming before "defrost" and after "remote" when sorted alphabetically, as well as both "defrost" and "remote", are selected - for example, "defrosting" is not selected, but "cooling" and "unit off" are.
abc Contains	Display all column readings that includes the parameter string.	[Location] Like '%23%' => all containers where the exact substring "23" is found in the location are included, for example, 005R123A and 011R231B, but not 008R203A.
acb Does not contain	Display all column readings that do not include the parameter string.	[Location] Not Like '%23%' => all containers where the exact substring "23" is not found in the location are included, for example, 005R120A and 011R203B, but not 008R123A.
[a]b Begins with	Display all column readings that begin with the parameter string.	[Location] Like '00%' => all containers where the exact substring "00" is the beginning of the location are included, for example, 005R123A and 008R203A, but not 011R231B nor 123R001A.
b[c] Ends with	Display all column readings that end with the parameter string.	[Location] Like '%3A' => all containers where the exact substring "3A" is the end of the location are included, for example, 005R123A and 008R203A, but not 011R231B nor 123A001R.
a%c Is like	Similar to "equal", but the % character can be used to hit more than one container.	[Location] Like '005%A' => all containers where the location starts with the exact substring "005" and ends with "a" are included, for example, 005R123A and 005R203A, but not 005R123B nor 008R203A.
%c Is not like	Similar to "not equal", but the % character can be used to refine the specification.	[Location] Not Like '005%A' => all containers where the location does not start with the exact substring "005" and ends with "a" are included, for



Criterion	Explanation	Examples
		example, 008R203A and 005R123B, but not 005R123A.
☐ Is blank	Do only show the container if the field in this column is empty.	Not really a useful criterion!
☒ Is not blank	Do only show the container if the field in this column is not empty.	Not really a useful criterion!
☒ Is any of	Display all column readings that exactly match one of the parameters.	[Running Mode] In ('Defrost', 'Defrosting') => all containers in "defrost" or "defrosting" mode are selected.
☐ Is none of	Display all column readings that do not exactly match any of the parameters.	Not ([Running Mode] In ('Defrost', 'Defrosting')) => all containers not in "defrost" nor "defrosting" mode are selected

Parameter selection

Clicking the parameter field ("1" or "Defrosting" in above figure) lets you edit this.

Note that if the column is binary (on/off, yes/no etc.), repeating the click will walk through the options.

Other hints

Clicking an addition symbol adds a condition to the current node.

Clicking a deletion symbol removes the condition.

Examples

The following figure shows an example of how you can define a subset of locations (area 005R stack 23 or area 011R level B):

Standard Info					Communication Info	
Location	ContainerID	Alms	Special	Running Mode	Online	Routing
011R24B	MWCU5640114	0		Heat max		
011R22B	MWCU5708770	0	S	Alarm shutdown		
005R23A	MWCU6183291	0		Heat reduced		
005R23C	MWCU6230768	0		Heat reduced		
005R23B	MWCU6281282	0	S	Fan reduced		
011R23B	PONU2858289	0	S	Heat max		
011R21B	PONU2878393	0	S	Fan reduced		

Filter Builder

Or

And

[Location] Begins with 005R

[Location] Contains 23

And

[Location] Begins with 011R

[Location] Ends with B

OK Cancel Apply

[X] [Location] Like '005R%' And [Location] Like '%23%' Or [Location] Like '011R%' And [Location] Like '%B'

The following figure shows how you can combine certain locations and other parameters (it must be a special reefer located in area 005R with running mode either "defrost" or "fan reduced"):



Standard Info				
Location	ContainerID	Alms	Special	Running Mode
005R07B	CRLU1292931	0	S	Defrost
005R11B	MWCU6861151	0	S	Defrost
005R23B	MWCU6281282	0	S	Fan reduced

Filter Builder
[And] [Location] Begins with 005R [Special] Equals Checked [Or] [Running Mode] Equals Defrost [Running Mode] Equals Fan reduced
OK Cancel Apply

☒ [Location] Like '005R%' And [Special] = 'Checked' And ([Running Mode] = 'Defrost' Or [Running Mode] = 'Fan reduced')

REFCON 6

Reefer Container Monitoring System

The Container Plan

Transportation Solutions

COPELAND



5 The Container Plan

Key subjects in this topic:

[Overview](#)^[96]

[The Overview zoom level](#)^[96]

[The Detail zoom level](#)^[97]

[The Container Plan Tooltip](#)^[98]

Related topics:

[Using the Container Plan](#)^[99]

[Container Plan Legends](#)^[99]

[Container Plan Generator \(terminal version\)](#)^[100]

[Container Plan Generator \(ship version\)](#)^[102]

[Editing the Container Plan \(ships only\)](#)^[103]

Overview

The *Container Plan* is a system of graphical pictures showing the states of the reefer containers. It can be all containers, or a specified subset.

The Container Plan is build up of squares, called Container Items. Each Container Item represents one 40' container or one or two 20' containers. The Container Items are organised in the picture in a way that should associate to the physical locations of the containers. The picture layout is thus customised for the individual application.

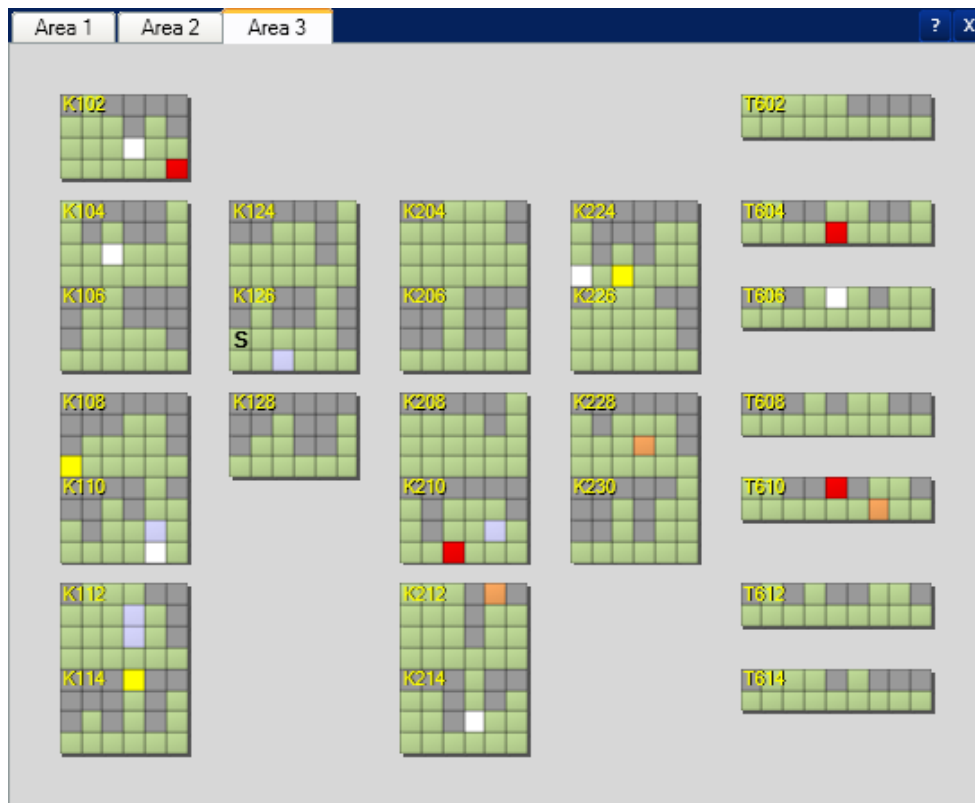
The Container Item indicates the status of the container at that specific location by colours, see [Container Plan Legends](#)^[99].

REFCON includes a generator/import tool that can be used to create or edit the Container Plan pictures, see [Container Plan Generator \(terminal version\)](#)^[100] and [Container Plan Generator \(ship version\)](#)^[102].

The Container Plan provides two zoom levels, as described below.

The Overview zoom level

The Overview zoom level aims at supervising all containers at the ship/terminal in one picture.



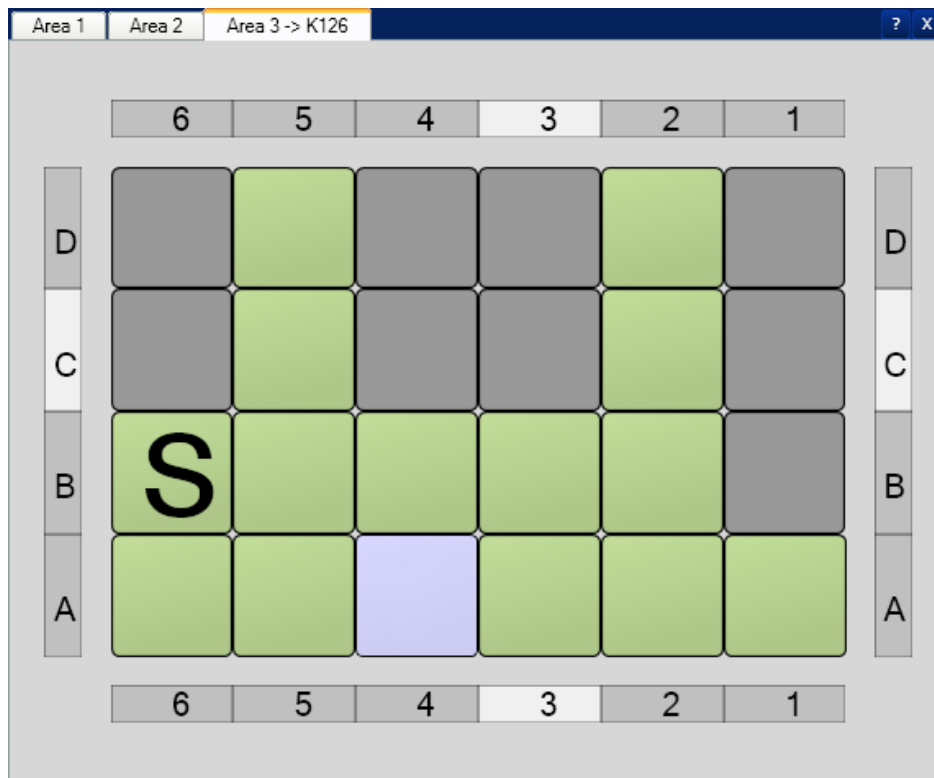
As indicated in above example, it is possible for the Container Plan designer to divide into two or more overviews - in this case three, named Area 1, 2 and 3. You can display one overview only at a time.

The overview automatically adapts to the available screen area. Hence, if you make the REFCON Monitor and/or the Container Plan window area smaller, the Container Items get smaller, and vice versa. The Container Plan designer would normally optimise the layout for the maximum window size.

Note: On ships you need to move the mouse over the bay to see the bay number.

The Detail zoom level

The Detail zoom level shows a decided segment of the overview. On a container ship, it would normally be a bay.



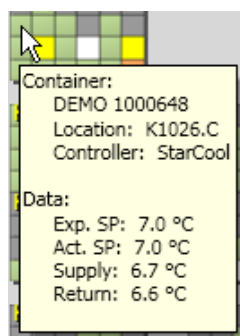
As shown in the above example, a system of co-ordinates appears in the Detail view. You will see that the axes dynamically indicates the position of the mouse cursor. Of course, also the Container Items are much larger.

The segment is identified next to the overview name at the top of the picture, eg. "Area 3 -> K126".

Apart from this, the information shown is the same as in the overview. The detail view automatically adapts to the available screen area, exactly like the overview.

The Container Plan Tooltip

When moving the mouse cursor over the Container Items, you will see that a Tooltip is displayed, showing primary data regarding the appointed container.



You will recognize the second line in the Tooltip as the container ID.

In case of one or more alarms on a container, these are listed in the Tooltip.

Note that measurements appear with a [limited accuracy](#)¹⁴².



5.1 Using the Container Plan

Key subjects in this topic:

[Opening the Container Plan](#)^[99]
[Operating the selected container](#)^[99]
[Zooming the Container Plan](#)^[99]

Related topics:

[Container Plan Legends](#)^[99]
[The Container Plan](#)^[98]

Opening the Container Plan

To open the Container Plan, click the item in the Side panel:



Operating the selected container

By right-clicking a container you will have the Shortcut menu displayed. Please refer to [The Container Command section of the Shortcut menu](#)^[34] for details.

Zooming the Container Plan

You can change between the [Overview zoom level](#)^[97] and the [Detail zoom level](#)^[97] in two ways:

- Use the scroll wheel on the mouse. If at Overview level, select a Detail section first (click it to have the yellow frame shown).
- Double-click - at Overview level, in the Detail section of interest; at Detail level, just somewhere within the Container Plan picture.

5.2 Container Plan Legends

Key subjects in this topic:

[Overview](#)^[99]
[Legends](#)^[99]

Related topics:

[Using the Container Plan](#)^[99]
[The Container Plan](#)^[98]

Overview

The Container Item, being the square representing one container, indicates the status of the container by colour and simple symbols, see [Legends](#)^[99].

Legends

The Container Item provides a set of [general legends](#)^[136]. The following table specifies the possible *additional* colour indications and symbols in the Container Item.

Symbo l	In words	Short description	Explanation
	Dark grey	No reefer container at location	The location is vacant, or it is occupied by a container that is unknown to REFCON.
	Divided in two	Two 20' containers at location	The foremost container is shown at the upper half of the square. Note that if there is only one 20' reefer container at the location, the other "half" being vacant or occupied by a dry reefer, the full square is used, in which case you can not



			graphically distinguish between a 20' and a 40' container.
	White	Never communicated	Any reefer container created by Planning System is initially Never communicated. When using Baplie files or similar, containers may stay white for a long time. With an online planning system a container should only stay white only for a few minutes.
	Light orange	Should communicate	Same as Never communicated, but REFCON core will switch from Never communicated to Should communicate when the Site State is "Plugged On" or user changes setting to "Sea Mode".
	Light grey	Not communicating	Reefer container turns Not communicating when all modems that has routed data from this container has changed to state Not communicating or are in state Should communicate.
	Blinking darker green	Communication started	Reefer container is in state Communication started when any related modem is Communicating but no data has yet been received.
	Dark green	Communicating	Container is sending data periodically.
	Dark blue	No controller	The modem is unable to communicate with the controller. A device alarm will been created.
	Light cyan	Poor communication	The modem is sending only partial messages. Review the hub placement or replace the modem on the reefer.
	Lighter green	Offline updated	Reefer container had controller data updated via Mobile, data for the container having been fully automatically received via BLE.
	Blueish green	Manually entered	Reefer container had controller data updated via RDC, or via Mobile where data for the container was fully or partly typed in by user
	Orange	Obsoleted	Reefer container had controller data updated via REFCON Mobile application, but is too long time ago (user decided obsoletion time)

5.3 Container Plan Generator (terminal version)

Key subjects in this topic:

[Overview](#) ¹⁰¹

[Procedure](#) ¹⁰¹

[Invoking the Container Plan Generator](#) ¹⁰¹

[Viewing the result](#) ¹⁰²

[Saving the Container Plan](#) ¹⁰²

Related topics:

[The Container Plan](#) ⁹⁸



Overview

This version of the Container Plan generator allows you to import a Container Plan designed in the separate tool "RefConfig".

Having successfully imported a new Container Plan and saved it, it replaces any possible existing Container Plan.

Procedure

To import a Container Plan, walk through the steps described below:

A. Make sure you have an XML-file exported from RefConfig.

B. [Invoke the Container Plan Generator](#)^[101].

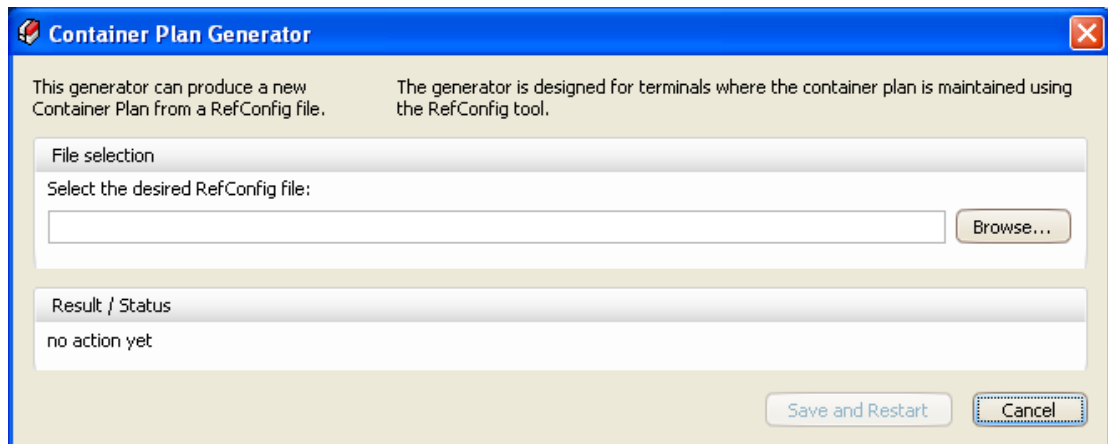
C. [View the result](#)^[101].

D. [Save the Container Plan](#)^[102].

Invoking the Container Plan Generator

To open the Container Plan Generator, select "Create Container Plan" from the "Settings" menu.

The following dialog box opens:



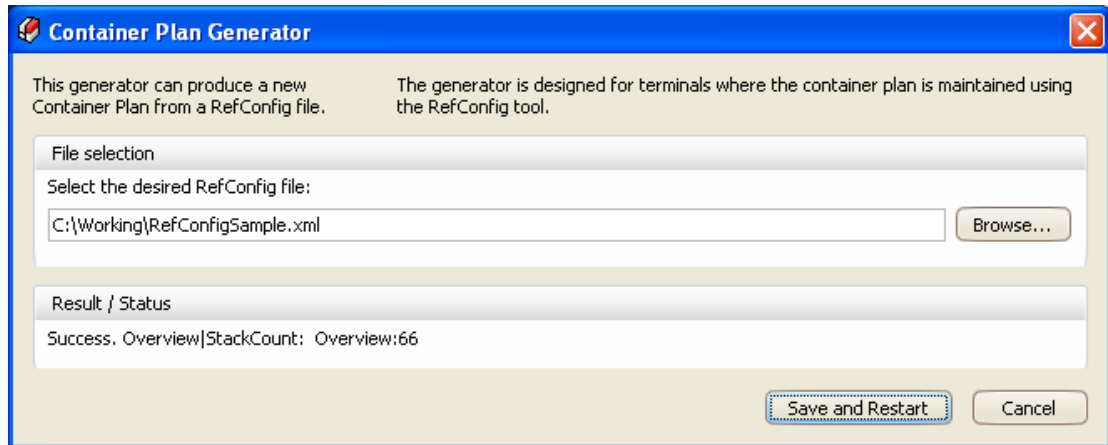
Type or paste the full path and filename into the textbox, or use the Browse function to locate it. Follow the result in the Result / Status area, see below.

Viewing the result

As soon as you have entered a valid RefConfig file path and name, the Generator immediately starts importing the Container Plan. This normally lasts just a few seconds.

Having completed, you can watch the result in the "Result / Status" area of the dialog box.

In case of **success**, the dialog box appears something like this:



In this example you observe:

- The "Result / Status" reads success, stating the number of overviews detected (66 in the example).
- The "Save and Restart" button has become enabled.

Saving the Container Plan

To complete your work, now select the "Save and Restart" button.

- "Save" means that the Container Plan you just generated will now replace a possible existing one.
- "Restart" means that the REFCON Monitor will restart to make the new Container Plan available.

5.4 Container Plan Generator (ship version)

Key subjects in this topic: Overview^[102] Procedure^[103] Invoking the Container Plan Generator^[103] Viewing the result^[103] Saving the Container Plan^[104] Enabled and disabled locations^[104] A container on a disabled location^[104] A container outside the basic bay^[105]	Related topics: Editing the Container Plan (ships only)^[105] The Container Plan^[96]
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Overview

This version of the Container Plan generator is designed for ships where the containers are organised strictly according to the ISO 9711-1:1990 standard, however, it also handles certain exceptions from this. Please examine the result if you have such exceptions.

The generator works with Baplie files, that is, files in the Baplie 1.5 or Baplie 2.0 format (specified by the SMDG group, see www.smdg.org).

All you have to do is to select a Baplie file including the actual locations on the ship. Preferably, the Baplie file should include all locations where you can have reefer containers, or if you wish, all locations at all. Your Planning System would probably be able to produce such a file. An incomplete Baplie file might also give a reasonable result.



Having successfully generated a new Container Plan and saved it, it replaces any possible existing Container Plan.

Procedure

To generate a new Container Plan, walk through the steps described below:

- A. [Invoke the Container Plan Generator](#)^[103].
- B. [View the result](#)^[103].
- C. [Save the Container Plan](#)^[104].
- D. [Optionally, edit the result](#)^[105].

Invoking the Container Plan Generator

To open the Container Plan Generator, select "Create Container Plan" from the "Settings" menu.

The following dialog box opens:

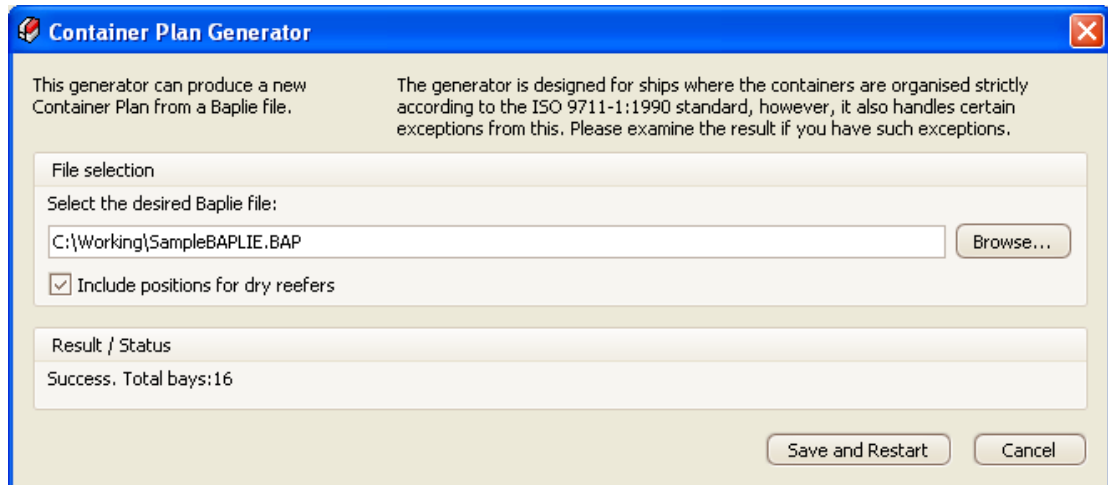
1. Type or paste the full path and filename into the textbox, or use the Browse function to locate it. Follow the result in the Result / Status area, see below.
2. You may, before or after step 1, select/deselect the option "Include positions for dry reefers". Many users would find the Container Plan more clear when dry reefers are left out.

Viewing the result

As soon as you have entered a valid Baplie file path and name in the above step 1, the Generator immediately starts calculating the Container Plan. This normally lasts just a few seconds.

Having completed, you can watch the result in the "Result / Status" area of the dialog box.

In case of **success**, the dialog box appears something like this:



In this example you observe:

- The "Result / Status" reads success, stating the number of bays detected (16 in the example).
- The "Save and Restart" button has become enabled.

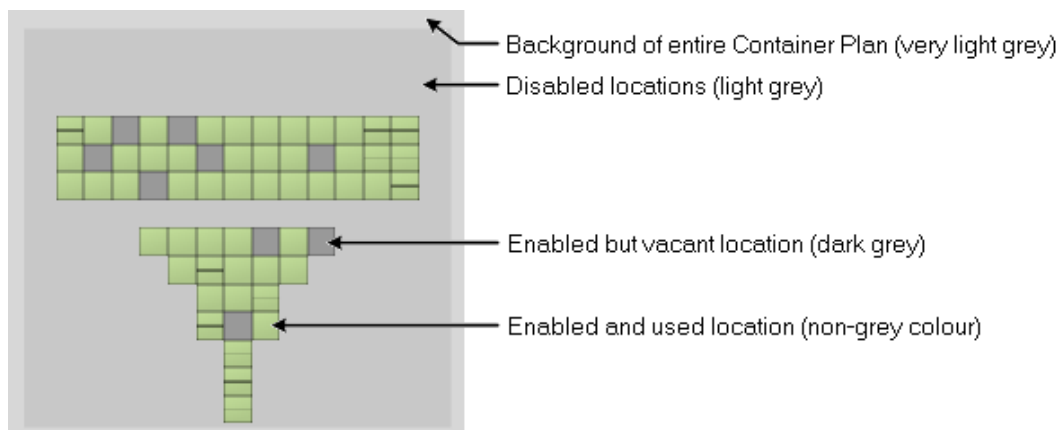
Saving the Container Plan

To complete your work, now select the "Save and Restart" button.

- "Save" means that the Container Plan you just generated will now replace a possible existing one.
- "Restart" means that the REFCON Monitor will restart to make the new Container Plan available.

Enabled and disabled locations

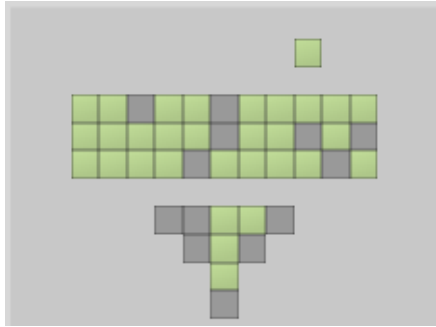
The Generator determines the overall maximum and minimum tier and row numbers and then constructs the **basic bay** to fit these. Then it constructs the specific individual bays in accordance with the specific locations defined in the Baplie file. The result (in the very most cases) is that a number of locations are disabled as indicated in the following example:



You can use the [Container Plan editor](#)¹⁰⁵ to enable or disable locations.

A container on a disabled location

If it happens that, after saving the Container Plan, a container appears at a disabled location, that location will be temporarily enabled.



When later the container is removed and deleted from REFCON, that location returns to the disabled state.

A container outside the basic bay

If it happens that a container appears at a location outside the basic bay area, it will not appear in the Container Plan. Still you can find it in the Container List, and alarms will be shown in the Alarm Summary.

5.5 Editing the Container Plan (ships only)

Key subjects in this topic:

[Overview](#)^[105]
[Exceptions](#)^[105]
[How to edit the Container Plan](#)^[105]
[The tooltip in edit mode](#)^[106]
[Enabling and disabling locations](#)^[106]

Related topics:

[Container Plan Generator \(ship version\)](#)^[102]
[The Container Plan](#)^[96]

Overview

On ships it is possible to manually modify the Container Plan layout by enabling or disabling locations within the existing bays, within the decided maximum size of these.

Exceptions

This editor has the following limitations:

- You can not add or remove bays.
- You can not add locations outside the defined maximum bay size.

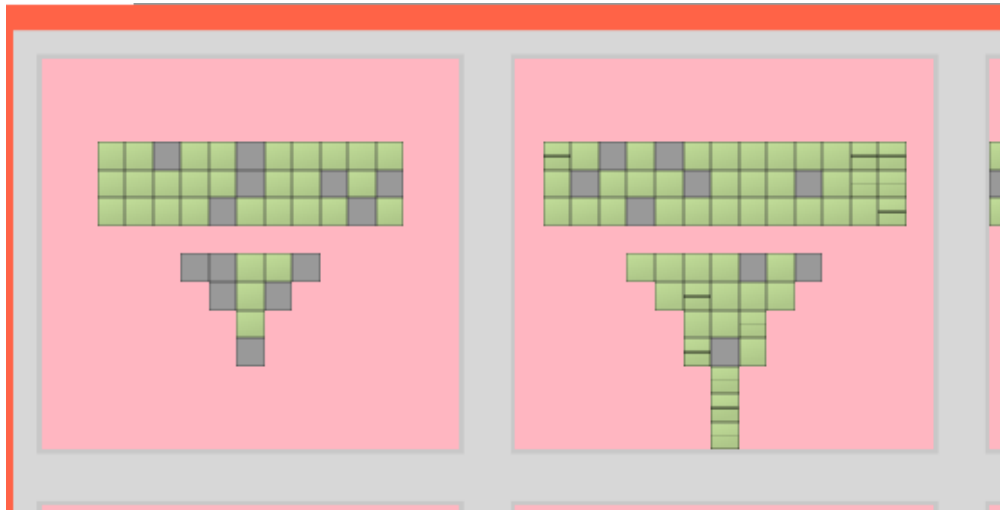
In both the above cases, if you need to do so, you will have to use the [Container Plan Generator](#)^[102] with an appropriate Baplie file.

How to edit the Container Plan

Invoke the Container Plan editor this way:

1. Open the Container Plan.
2. Select "Edit Container Plan" from the "Container Plan" menu.

You will now see (within a few seconds) that the frame of the Container Plan changes to orange, and the disabled locations in each bay change to pink:



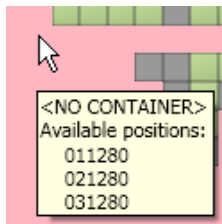
While these colours are observed, you are in edit mode. Below you find descriptions on editing details.

Having completed your editing:

3. Re-select "Edit Container Plan" from the "Container Plan" menu. The changes are saved (if you accept to do so), and you are back in normal mode.

The tooltip in edit mode

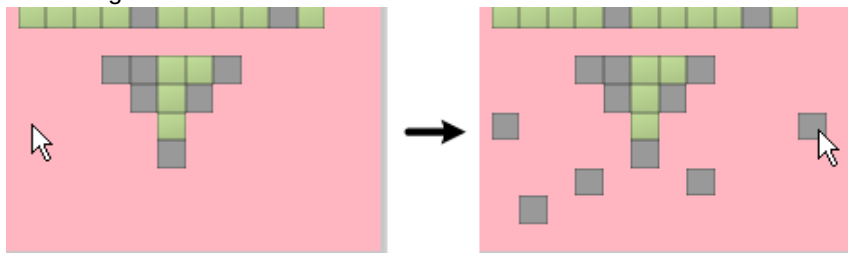
Normally, when moving the mouse over disabled locations within the bays, no tooltips occur. But when in edit mode, you will have a tooltip, making it easy to determine exactly where the mouse is positioned at the moment:



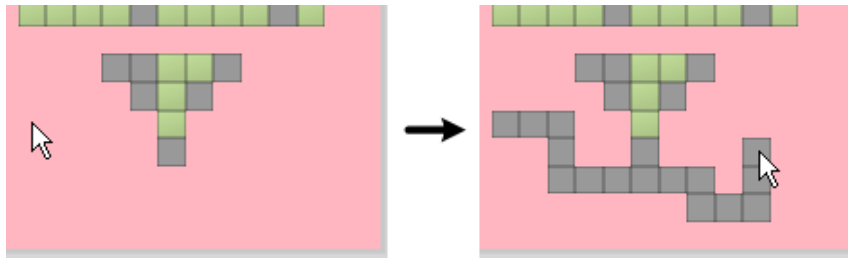
Enabling and disabling locations

You **enable** locations (the pink ones) by:

- Clicking individual disabled locations:

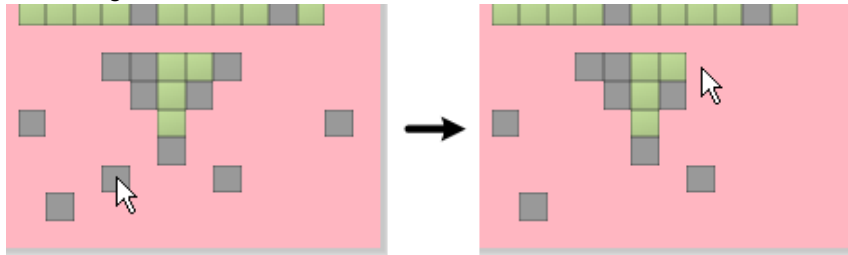


- By pressing the left mouse button on a disabled location and dragging to other disabled locations. Note that if the mouse passes enabled locations, these are not affected (because you started at a disabled location):

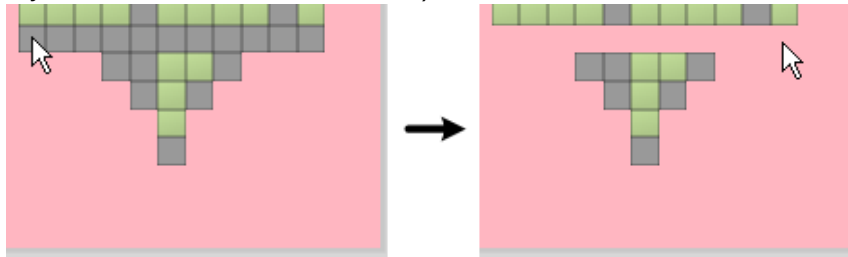


You **disable** locations (the non-pink ones) by:

- Clicking individual enabled locations:



- By pressing the left mouse button on an enabled location and dragging to other enabled locations. Note that if the mouse passes disabled locations, these are not affected (because you started at an enabled location):



REFCON 6

Reefer Container Monitoring System

The Alarm System



6 The Alarm System

Key subjects in this topic:

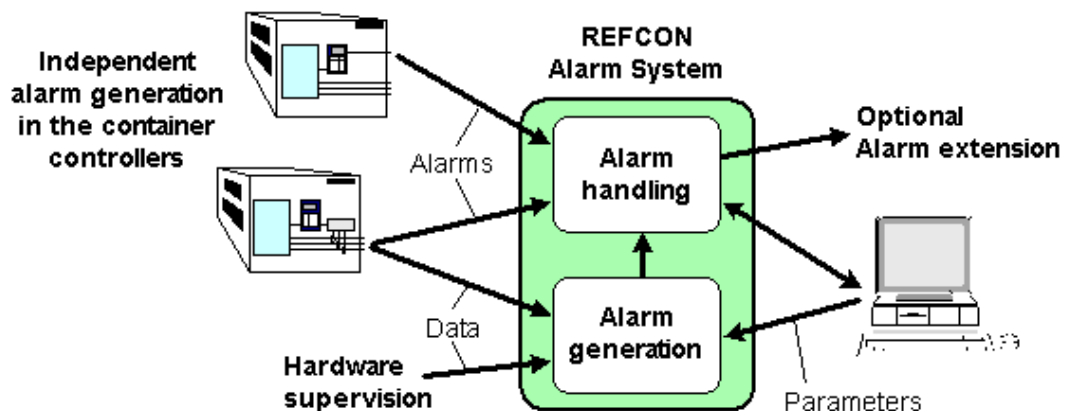
[Overview](#)^[110]
[Alarms generated by the controller](#)^[110]
[Reservation regarding alarm text strings](#)^[111]
[Alarms generated by REFCON](#)^[111]
[Alarm handling](#)^[111]
[Acknowledging a container's alarms](#)^[111]

Related topics:

[Alarm Summary](#)^[112]
[Alarm propagation](#)^[115]
[Special reefers](#)^[117]
[Container handling mode \(ships only\)](#)^[117]
[Temperature setpoint supervision](#)^[118]
[Possible loss of power alarm](#)^[119]
[Communication not started/stopped alarms](#)^[121]
[Alarms for missing or wrong ID](#)^[122]
[User generated alarms](#)^[124]
[Alarm Options setup](#)^[125]
[Power Alarm Options setup](#)^[127]

Overview

A vital part of REFCON is the *Alarm System*. Please look at the following sketch:



There are in fact three main parts:

- **Alarm generation** = the decision that an alarm must be given. Alarms are generated in the [container controller](#)^[110], in [REFCON 6 extensions](#)^[218] / [plugins](#)^[309] and in [REFCON](#)^[111].
- [Alarm handling](#)^[111] = maintaining a list of actual alarms and event logging.
- [Alarm propagation](#)^[115] = attracting your attention, or just showing on the screen that an alarm is present.

A number of tools are available at REFCON for your alarm handling. The primary tool is the [Alarm Summary](#)^[112] in which you can always inspect all the actual alarms.

Alarms generated by the controller

Reefer container controllers are independent alarm systems. Alarms can be inspected on the container – which you will have to do if the container is not connected to REFCON by PCT (PCT) or a relevant [REFCON 6 extension](#)^[218].

Each controller has a certain number of alarm points, each corresponding to an alarm reason. The specific list of possible alarms depends on the controller type.

In case of PCT or an online REFCON 6 extension, the states of all the alarm points are periodically transmitted to REFCON.

In REFCON, changes are detected, causing the Alarm Summary and Event Log to be updated and the alarm to be propagated, if relevant.



Reservation regarding alarm text strings

Please observe the following:

Copeland Transportation Solutions makes reservations regarding the alarm text strings attached to the controller generated alarms. Copeland Transportation Solutions **cannot** guarantee that these text strings are correct and clear.

Please **consult the controller manufacturer's manual** to be absolutely sure what is the correct alarm text.

The alarm text strings shown in REFCON were informed by the controller manufacturers. Copeland Transportation Solutions has made great efforts to ensure that each text string is true. Possible divergences could for example be due to controller changes that have not yet been updated in REFCON.

You should keep your REFCON system updated in order to ensure full compatibility with controller manufacturers' latest software, see [Understanding installation and update](#)^[314].

Alarms generated by REFCON

REFCON provides the following types of alarms:

- All REFCON hardware is supervised. In case of malfunction an alarm is released.
- All communication interfaces are supervised. If, for example, information that should be retrieved is not so, or information that should be transmitted can not be so, alarms are released.
- [User generated alarms](#)^[124].
- Various function alarms are described in relation to the respective functional descriptions.

REFCON will generate alarm on behalf of the container controller in the following cases where the controller would not itself have the necessary data to do the alarm decision:

- [Expected setpoint differs from actual setpoint](#)^[118].
- Modem does not communicate with the controller.
- [Container ID is missing or wrong](#)^[122].
- [Communication has stopped \(possible loss of power\)](#)^[119].
- [Communication not starting/stopping](#)^[121].

Alarm handling

Your main activities regarding alarms would be:

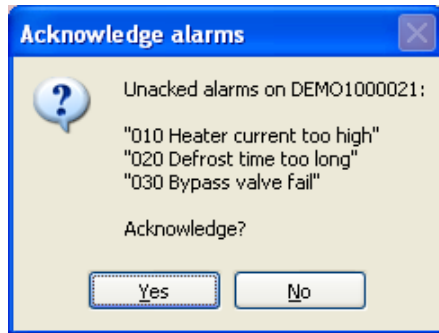
- Inspecting an [Alarm Summary](#)^[112], to find the reason for a new alarm and more. As regards container alarms, you may in stead inspect the optional Container Plan.
- Acknowledging alarms (see [Acknowledging a container's alarms](#)^[111] below and [Acknowledging alarms](#)^[114]).
- Taking action to solve the problem causing the alarm.
- [De-acknowledging](#)^[114] an alarm, if relevant.

Acknowledging a container's alarms

In addition to the Alarm Summary, you can acknowledge a container's alarms from the [Container Details](#)^[145].

Further you can acknowledge a container's alarms directly from pictures where the container occur, typically the [Container List](#)^[66] and the optional Container Plan.

To do so, right-click the container. In the [Shortcut menu](#)^[33], select "Acknowledge container alarms". This makes a container alarm overview appear:



Make sure to inspect this list of new alarms carefully. Then select Yes to acknowledge them all.

6.1 Alarm Summary

Key subjects in this topic: Overview ¹¹² Indication of temporary list changes ¹¹³ Opening the Alarm Summary ¹¹³ Acknowledge state indication ¹¹³ Acknowledging alarms ¹¹⁴ Selecting alarms for acknowledge ¹¹⁴ De-acknowledging an alarm ¹¹⁴ Sorting the alarms ¹¹⁴ Column specifications ¹¹⁴	Related topics: Alarm propagation ¹¹⁵ Special reefers ¹¹⁷ The Alarm System ¹¹⁰
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Overview

The *Alarm Summary* is a list showing all the pending alarms.

REFCON provides two Alarm Summary versions, one for container alarms and one for system alarms. The two versions work the same way, so only the container version is illustrated in the following.

Container Alarm Summary								
	Location	Origin	Text	Alarms	User	Time UTC	Time LMT	User's note
Special reefers having alarms (Count=2)								
	420484	MWCU6956200	018 Power supply phase error	1		2012-05-30 06:03:07	2012-05-30 08:03:07	Place of delivery: FRLEH Thermo King MP30
	020984	MEDU9023000	dAL90 - Future	3	John	2012-05-30 06:02:05	2012-05-30 08:02:05	Place of delivery: FRLEH Carrier Microlink 2/2
Containers having alarms (Count=9)								
	620886	LNXU9653900	dAL70 - Recorder Supply Temp. Out of ...	2		2012-05-30 06:18:52	2012-05-30 08:18:52	Place of delivery: FRLEH Carrier Microlink 2/2
	340986	MWCU5284121	Cable needs repair	1		2012-05-30 06:18:10	2012-05-30 08:18:10	Place of delivery: FRLEH John
	301486	MWCU5697300	caAL30 - Control Contactor Fuse Open ...	5		2012-05-30 06:18:04	2012-05-30 08:18:04	Place of delivery: FRLEH Carrier Microlink 2 E
	220882	MWCU5266600	AW301 RH short (other sensor)	2		2012-05-30 06:15:32	2012-05-30 08:15:32	Place of delivery: FRLEH StarCool RCCU5

As indicated, the Alarm Summary might be divided into sections. The individual section is shown only if there is one or more alarms in the related category:

1. Special reefers: Here you find all containers that are marked "Special" and have alarms (see [Special reefers](#) ¹¹⁷).
2. Containers: Here you find all non-special containers with alarms.
3. System alarms: Here you find all alarms that are not related to containers (see [Alarms generated by REFCON](#) ¹¹⁷).

Basically, each line in the Alarm Summary represents one alarm, however, if there are more than one alarm on the same container or the same system device, it is still just one line.

Please observe the Alarms column, counting the number of alarms on each container/device. In



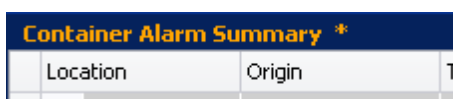
case of more alarms, the "Text" field relates to the newest alarm, while the indicator leftmost flashes if any of the alarms is not acknowledged. You can open a sub-list of the alarms by clicking the + sign leftmost.

The Alarm Summary includes a fixed number of columns. The contents of the columns are described in [Acknowledge state indication](#)^[113] and [Column specifications](#)^[114].

When opened, the Alarm Summary always includes all alarms in the relevant category, and it is always sorted chronologically, the newest alarm in each section being shown first. The sorting order can be changed as shown in [Sorting the alarms](#)^[114].

Indication of temporary list changes

It is possible to temporarily change the [Sorting of the alarms](#)^[114], the [selection of displayed columns](#)^[82], and more. As this can lead to serious mistakes for other users, the change is clearly indicated by orange colours of the list caption and more:

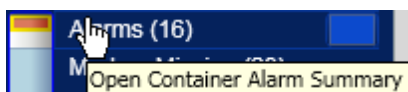


You revert to the standard settings by closing and re-opening the Alarm Summary or by clicking the "Reset sorting/filtering" button.

Opening the Alarm Summary

You can not open an Alarm Summary if there are no alarms in the relevant category. For example, if there is no pending system alarms, the System Alarm Summary can not be opened - but if it is already open, it stays so.

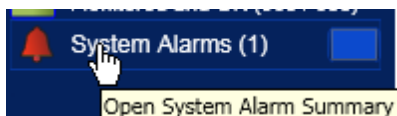
To open the **Container Alarm Summary**, click the "Alarms (nn)" text or the red/flashing field in the [Performance Bar](#)^[27]:



Note that the Alarms field is available even when the Side panel is minimized:



To open the **System Alarm Summary**, click the "System Alarms (nn)" text or the red/flashing bell icon below the [Performance Bar](#)^[27]:



Note that the System Alarms bell icon is available even when the Side panel is minimized:



Acknowledge state indication

The narrow leftmost column indicates the acknowledge state of each alarm:

Indication		Description
	Flashing	There are one or more unacknowledged alarms on this container.
	Steady	There are one or more alarms on this container, they have all been acknowledged.



Acknowledging alarms

Alarms can be acknowledged from the Alarm Summary as follows:

1. Select the alarm or the alarms to be acknowledged (see [Selecting alarms for acknowledge](#)^[114] below).
2. Either right-click the alarm (/ any of the selected alarms) and choose "Acknowledge selected alarms", or click the "Acknowledge selected alarm(s)" button upper right:



You can also acknowledge all alarms as follows:

- Either right-click any alarm and choose "Acknowledge all alarms",
- Or click the "Acknowledge all alarm(s)" button upper right:



Selecting alarms for acknowledge

Selecting specific alarms for acknowledge can be done as follows:

To select a single alarm,

1. Click the row and thereby make it highlighted.

To select a continuous series of alarms in the list,

1. Click the first alarm in the series.
2. Keep the Shift button pressed and click the last alarm in the series. You should now see the two clicked alarms and all in between being selected (highlighted).

To select chosen alarms in the list,

1. Click one of the alarms to be acknowledged.
2. Keep the Ctrl button pressed and click each of the other alarms to be acknowledged.

To select all alarms in the list,

1. Click any alarm to have the Alarm Summary "in focus".
2. Select the keyboard combination Ctrl+A.

De-acknowledging an alarm

An alarm that once has been acknowledged can be de-acknowledged from the Alarm Summary as follows:

1. Click the specific alarm of interest so that the row gets highlighted. Note that if you select a container or device with more alarms, you are warned (in step 2) that de-acknowledgement is not possible for a set of alarms.
2. Right-click the alarm and choose "De-acknowledge alarm".

De-acknowledgement means that alarm flashing and other [Alarm propagation](#)^[115] is restarted, but the original time of the alarm is not changed.

Sorting the alarms

When opened, the Alarm Summary always includes all alarms, and it is always sorted chronologically, the newest alarm in each section being shown first. It is possible to change the sorting, however, the change will be temporary as it can not be saved.

Changing the sorting is done exactly like in the Container List, please refer to [Selecting the sorting order of the containers](#)^[68] and [Selecting a sorting filter](#)^[69].

Column specifications

The Alarm Summary includes the columns described below. Note that the Source table-column shows where the information comes from.

Column header / abbreviation	Explanation	Source
---------------------------------	-------------	--------



Location	• If a container: It's location, if known. • If not a container: The location of the system device, if known.	Planning System or REFCON
Origin	• If a container: The ID (4 letters + 7 digits). • If not a container: The name of the system device.	Planning System / controller, or REFCON
Text	Description of the alarm cause.	Controller manufacturer / REFCON
Alarms	Number of alarms on this container / device.	Controller / REFCON
User	The user ID of the user who acknowledged the alarm.	REFCON
Time UTC and Time LMT	The time when the alarm occurred.	REFCON
User's note	The text written in the Note field of the container. The column is always empty for system alarms.	Planning System or user
Generated By	What unit did release the alarm. If it was a controller, the controller type is indicated. In case of a user generated alarm, the user ID is indicated.	Controller / REFCON
Action category (optional)	The alarm's category in the optional Email notification system.	REFCON

6.2 Alarm propagation

Key subjects in this topic:

[Overview](#) ¹¹⁵
[Alarm indication in the REFCON Monitor](#) ¹¹⁵
[Alarm indication with REFCON's Mini-status window](#) ¹¹⁶
[Alarm indication when the REFCON Monitor is not active](#) ¹¹⁶
[Alarm Extensions](#) ¹¹⁶

Related topics:

[The Mini-status window](#) ³¹⁴
[The Alarm System](#) ¹¹⁰

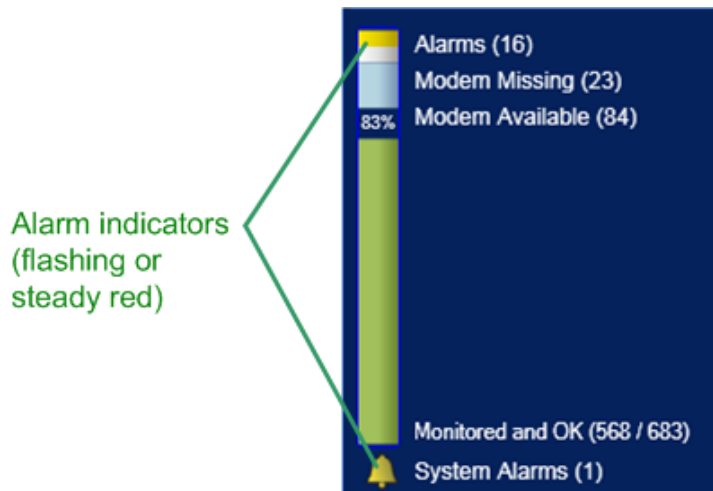
Overview

The purpose of the *Alarm propagation* is to attract your attention in case of alarm.

In the basic REFCON version, you need to watch the REFCON PC screen at intervals to discover new alarms. However, a number of Alarm Extension options are available that can alert you actively.

Alarm indication in the REFCON Monitor

When the REFCON Monitor is open and active, alarms are primarily indicated in the Side panel:



The indicators and the related texts show:

- The total number of container alarms (16 in the example) and system alarms (1 in the example). All possible alarm categories and priorities are included.
- If there are any unacknowledged alarm(s), the respective indicator is flashing.
- If there is no alarms in the category, the indicator is not shown.

Also, alarms are indicated in the Windows task bar, see [The taskbar notify icon](#)^[53].



Alarm indication with REFCON's Mini-status window

If you want the alarm indication to be visible at a distance, you may use the [Mini-status window](#)^[31].

Alarm indication when the REFCON Monitor is not active

If the REFCON Monitor is open but not active when an alarm occurs, you will see that the caption bar is flashing:



In the same situation, the taskbar field is flashing, keeping the taskbar visible even if auto-hidden:



Alarm Extensions

In general, the optional *Alarm Extension* functions more actively attract your attention in case of alarms. The current version of REFCON 6 includes the Alarm Extension functions listed below. Note that if the link does not work, it is because the particular option has not been enabled for your REFCON system.

- [Digital output signals](#)^[274] for external alarm system or alarm panel.
- [The Email Notification function](#)^[268].
- Possible plug-ins or external systems utilizing REFCON's [data-service interface](#)^[309].



6.3 Special reefers

Key subjects in this topic:

[Overview](#)^[117][Selecting "Special reefer"](#)^[117]

Related topics:

[The Container List](#)^[66][Alarm propagation](#)^[115][The Alarm System](#)^[110]

Overview

This version of REFCON includes the "Special Reefers" option, making it possible to change the alarming priority of individual containers. Normally, all container alarms belong to the Standard category. Selecting "Special Reefer" will move the container to the Special category.

This category change makes difference in the [Alarm Summary](#)^[112], as alarms are displayed in different sections.

"Special" is indicated by an "S" in the [Container Details](#)^[136] and in the optional Container Plan. It can also be included in versions of the [Container List](#)^[82].

Also, with [Alarm extensions](#)^[116], the idea is that alarms - depending on the configuration - can be distinguished externally, for example, on a main alarm system.

Selecting "Special reefer"

The alarm category of a container can be changed using the "Special" button in the [Container Details](#)^[143].

Alternatively you can proceed as follows:

Step	Action
1	Open the Container List ^[66] or the optional Container Plan and find the container.
2	Right-click the container.
3	From the Shortcut menu, • select "Mark special reefer" to change to the "Special" category; or • select "Unmark special reefer" to change to the "Standard" category.
4	In both cases, select OK to close the resulting dialog box.

6.4 Container handling mode (ships only)

Key subjects in this topic:

[Overview](#)^[117][Mode indication](#)^[118][Changing the mode](#)^[118]

Related topics:

[Possible loss of power alarm](#)^[119][Alarms for missing or wrong ID](#)^[122][The Alarm System](#)^[110]

Overview

This topic applies to container ships only.

When loading and discharging containers in a port (Harbour mode), containers are powered on and off rather randomly seen from a monitoring viewpoint. At the same time, the information received from the Planning System arrives asynchronously, which might result in containers that are temporarily [unplanned](#)^[183]. At the contrary, at sea (Sea mode) things are stable and predictable.



When a container is without power, REFCON loses its ability to supervise the container. After a while, REFCON will generate a *disconnect event* and, if in Sea mode, an *alarm*. In other words, if the handling mode is not set to Harbour mode before containers start to disconnect, there will be many "false" [loss-of-power-alarms](#) ^[119].

Similarly, if a container is connected before being informed by the Planning System, you will have a [wrong ID alarm](#) ^[122] if not in Harbour mode.

Mode indication

In the Top Panel upper left in the REFCON window, a field shows the selected handling mode by displaying either of the following:



Mode	Symbol	Description
Harbour	Bollard	Relevant alarms are suspended.
Sea	Propeller	Relevant alarms are enabled.

Changing the mode

To change the handling mode, click the mode indicator in the Top Panel.

In the drop-down menu that appears, select the new mode. If requested, enter user name and password.

6.5 Setpoint supervision

Key subjects in this topic:

[Overview](#) ^[118]

[How to act upon this alarm?](#) ^[119]

[Setpoint alarm parameters](#) ^[119]

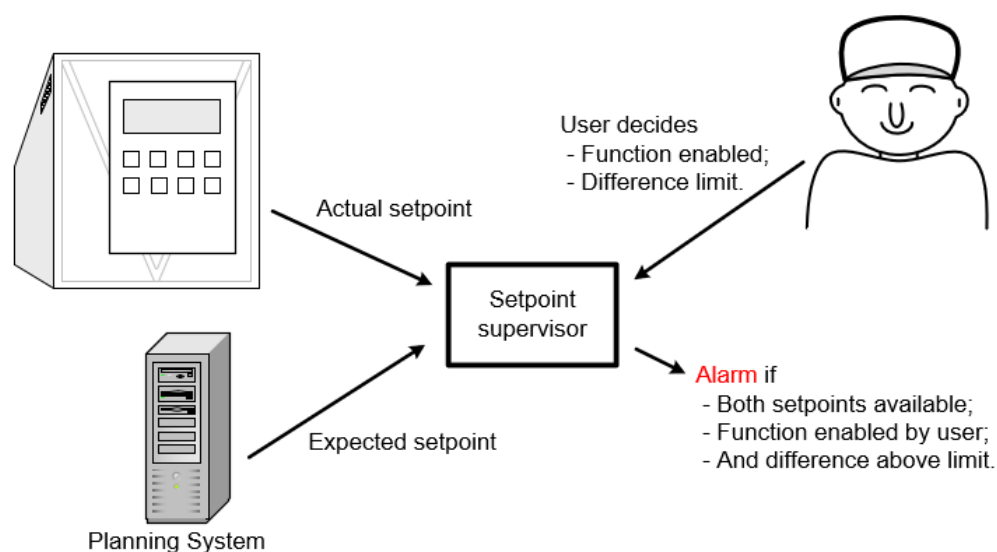
Related topics:

[Alarm Options setup](#) ^[125]

[The Alarm System](#) ^[110]

Overview

The Setpoint supervision function can be sketched as follows:





REFCON supervises all setpoints: Temperature, Humidity, CO2, O2 and Ventilation - but only if enabled, and only if **both** the actual and the expected setpoint are available.

When enabled, for all containers in the database, REFCON continuously supervises the actual setpoints, received from the container controllers, and compares them to the expected setpoints, received from the Planning System. If, for any specific container, the difference between the two is greater than "[Delta](#)"^[127], the "Actual and Expected setpoints differ" alarms is released.

Note that this is a very important alarm because it indicates that the cargo might be kept at wrong conditions.

How to act upon this alarm?

When the "Actual and Expected setpoints differ" alarm occurs, it is important that you carry out the following steps:

1. Find out if the expected setpoint is wrong, or the actual setpoint is wrong, or both.
2. If the expected setpoint is wrong, change this at the Planning System, and await the Planning data update - which you might have to trig manually.
3. If the actual setpoint is wrong, change this in the container controller. You can in many cases do this from the [Container Details](#)^[143], otherwise you need to visit the container.

Setpoint alarm parameters

You can change the Setpoint supervision parameters from the [REFCON Setup](#)^[127] dialog box.

6.6 Possible loss of power alarm

Key subjects in this topic:

[Overview](#)^[119]
[How the loss-of-power detection works](#)^[120]
[Limitations](#)^[120]
[How to avoid alarms on deliberate disconnects](#)^[120]
[Handling deliberate disconnects in terminals](#)^[120]
[Handling deliberate disconnects on ships](#)^[120]
[How to configure loss-of-power surveillance](#)^[121]

Related topics:

[Power Alarm Options setup](#)^[127]
[The Alarm System](#)^[110]

Overview

"Loss-of-power surveillance" is a REFCON feature that aims at detecting when a reefer container has broken down or has otherwise lost power in an unexpected way. If such a container is detected, REFCON releases an alarm showing: "**WARNING! Possible loss of power**", accompanied by the ID of the container in question. The user should then physically check the container as soon as possible.

As standard the following applies:

- On ships, this automatic surveillance is enabled when in [Sea mode](#)^[117] and disabled when in [Harbour mode](#)^[117].
- In terminals, this automatic surveillance is enabled per default.
- The user can change the alarm delay. A larger delay will reduce the risk of "false" alarms (see below).



- Because the feature may in certain cases produce "false" alarms (see below), the user can disable it.

How the loss-of-power detection works

When operating normally, a container equipped with a modem starts to communicate with REFCON shortly after it has been connected to power. It continues to regularly send readings to REFCON until it eventually is being disconnected in order to be moved away.

If for some reason the container loses power, the communication stops and does not start again by itself (unless the container is re-powered or repaired).

Whenever communication stops for more than the user decided **alarm delay** (being 3 hours as default), the loss-of-power alarm is released.

This alarm delay allows the container to recover from a temporary communication gap (due to noise, weak signals, and the like) without releasing the loss-of-power alarm. Also it allows a deliberate short disconnection.

Limitations

Because a loss-of-power alarm is really a "lengthy-loss-of-connection"-alarm, the following limitations apply:

- If a container was never connected to power or for some other reason never started communication, no alarm will ever occur for that container (since you can't lose a connection to a container that never had one!).
- If a container *for some other reason* loses the communication to REFCON for more than the alarm delay, the loss-of-power alarm will still be released. That is why the alarm text says "Possible loss of power". Reasons for communication stop while still on power may include: Weak signal, noisy connection, defect modem and others. REFCON can in no way distinguish between those reasons.

How to avoid alarms on deliberate disconnects

In general, the right way to avoid the alarm from a container that has been deliberately disconnected (in order to be moved away) is to have the container deleted from REFCON's database within the alarm delay time. This deletion is normally handled automatically by REFCON based on data received from the Planning System. It is therefore essential that the Planning System is kept up-to-date as regards what containers are supposed to be on power at any time.

Handling deliberate disconnects in terminals

On most container terminals, REFCON has a two-way online connection with the Planning System. In this case everything works automatically because the Planning System very quickly informs REFCON that a certain container has been disconnected, after which it is immediately deleted from REFCON's database.

If no two-way online connection with the Planning System is available, REFCON is updated via Planning data file transfers. At each file transfer, the Planning System produces a file that is then imported by REFCON. On every import, REFCON checks if all previously known containers are still present in the new file, and missing containers are automatically deleted. To avoid that deliberately disconnected containers will release unwanted loss-of-power alarms, it is essential that the file transfer is done frequently. For example, if the alarm delay is 3 hours, transfer should be done at least every 2 hours.

Handling deliberate disconnects on ships

On ships the procedure is a bit different: While at sea the set of connected containers remains the same throughout the sailing. Once in harbour, containers are connected and disconnected all the time, and finally, when leaving the harbour, REFCON is updated with the new/changed container load (by Planning data file transfer as described above).



This means that there may very well pass more than alarm delay time from a container is disconnected until it disappears from the REFCON database. To avoid unwanted loss-of-power alarms while in harbour, the automatic loss-of-power surveillance should be temporarily suspended by selecting [Harbour mode](#)^[118].

How to configure loss-of-power surveillance

Both enabling/disabling the surveillance and changing the alarm delay is done from the [REFCON Setup dialog box](#)^[125].

When disabling the surveillance (= removing the check mark), all already created loss-of-power alarms disappear (once they have been acknowledged) and no new loss-of-power alarms will be created.

When enabling the surveillance, all timing starts anew, meaning that at least the alarm delay will pass before any loss-of-power alarms will be created – even if communication has stopped hours ago.

Decreasing the alarm delay means quicker discovery of containers without power, but with the potential expense of more unwanted alarms.

Increasing the alarm delay generates less unwanted alarms, but also slows down the detection of containers losing power.

The optimal delay for your system depends on the quality of communication on your particular site. If communication quality is always good, you may choose a short delay (eg. 90 minutes), otherwise a longer delay should be chosen. You may have to experiment a bit to find the optimal value for your particular system.

6.7 Communication not started/stopped alarms

Key subjects in this topic:

[Overview](#)^[121]
[How the Communication not started alarm works](#)^[121]
[How the Communication not stopped alarm works](#)^[122]
[How to configure the above alarms](#)^[122]

Related topics:

[Power Alarm Options setup](#)^[127]
[The Alarm System](#)^[110]

Overview

When REFCON receives **detailed SiteState**^[183] information from the Planning System, user can enable either of the following alarms:

- [Communication not started alarm](#)^[121]
- [Communication not stopped alarm](#)^[122]

To avoid unwanted/"false" alarms, the alarms are only available when the SiteState changes to or from "Plugged On". For the majority of REFCON installations, such detailed SiteState information is not available, and you do therefore not see the alarms in the REFCON Setup dialog.

How the Communication not started alarm works

When the Reefer man reports that a container has been attached to power - meaning that the SiteState changes **to** "Plugged on" - a timer is started.

Normally, the container will start communicating within a short time. This proves that the power is in fact On, and that the optional [REFCON Portable Modem](#)^[222] has been attached. The timer



is canceled, and this alarm is then passive even if communication should unexpectedly stop (this is supported by the [Possible loss of power alarm](#)^[119]).

If the communication does not start within the user decided **alarm delay** (being 10 minutes as default), the Communication not started alarm is released within one minute. After this, as soon as the communication starts, the alarm is canceled (within one minute, once it has been acknowledged).

How the Communication not stopped alarm works

When the Reefer man reports that a container has been detached from power - meaning that the SiteState changes **from** "Plugged on" - a timer is started.

Normally, the container will stop communicating within some time. This indicates that the power is in fact Off. The timer is canceled, and this alarm is then passive even if communication should unexpectedly start anew.

If the communication does not stop within the user decided **alarm delay** (being 15 minutes as default), the Communication not stopped alarm is released within one minute. After this, as soon as the communication stops (based on timeout), the alarm is canceled (within one minute, once it has been acknowledged).

How to configure the above alarms

Both enabling/disabling these alarms and changing the alarm delays is done from the [REFCON Setup dialog box](#)^[127].

When disabling the alarm (= removing the check mark), all already created alarms of this type disappear (within one minute, once they have been acknowledged) and no new alarms of this type will be created.

When enabling the alarm, the alarm timing does not start until the SiteState changes and described above. For example, if the SiteState is already "Plugged on" when enabling the Communication not started alarm, the alarm will not appear even if communication has never started.

The optimal alarms delays for your system depends on the quality of communication on your particular site. If communication quality is always good, you may choose a short delay, otherwise you may need a longer delay. You may have to experiment a bit to find the optimal value for your particular system.

6.8 Alarms for missing or wrong ID

Key subjects in this topic:

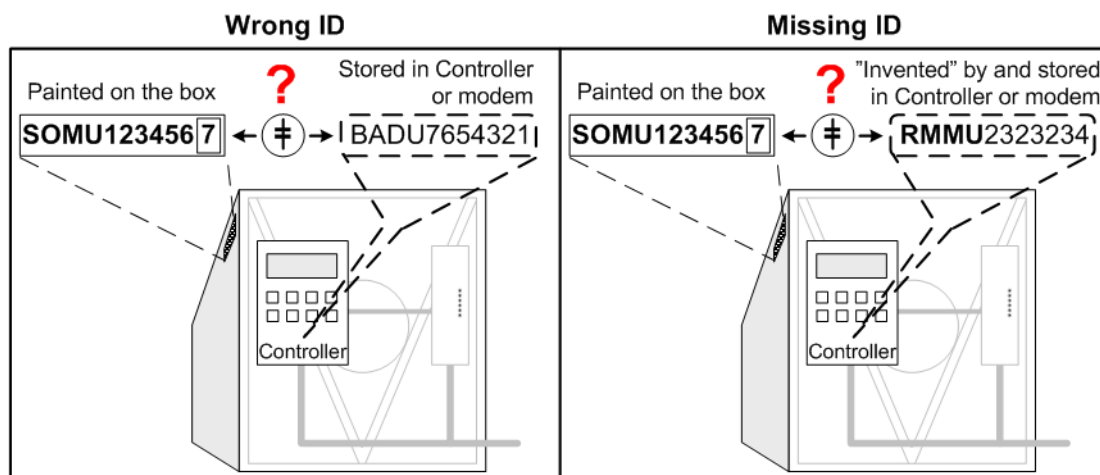
[Overview](#)^[122]
[How the Missing ID alarm works](#)^[123]
[How the Wrong ID alarm works](#)^[123]
[How to act upon these alarms?](#)^[123]
[Settings regarding missing or wrong ID alarms](#)^[123]

Related topics:

[Alarm Options setup](#)^[125]
[Modem Troubleshooting](#)^[278]
[The Alarm System](#)^[110]

Overview

It happens that a container has a mismatch between the ID painted on the box and the ID used by the power cable modem:



A *wrong ID* typically could occur because the controller has been exchanged, taken from another container, and was not re-programmed.

A *missing ID* would occur if a new controller did never have any ID programmed. In this case it uses a factory ID starting with RMMU, LOSU, LANG, RAND or MCID.

How the Missing ID alarm works

When a container is connected to the power cable transmission system, it runs through two steps:

1. It logs in. This is in fact just communication between REFCON and the *modem* on the container.
2. The controller is type detected, implying that communication is now established the whole way through.

There can be several minutes between the two steps.

In case of errors on the connection between the modem and the controller, the step 2 never occurs.

When a container logs in (step 1) using one of the factory IDs mentioned above (eg. RMMU), REFCON puts it on an observation list. If and only if step 2 does later occur, the Missing ID alarm is given immediately, but only if [enabled](#)¹²⁷.

How the Wrong ID alarm works

As REFCON can not actually *read* what is painted on the container box, the detection of a wrong ID is based on Planning data.

Assuming that REFCON has always a complete and updated list of the containers to expect received from the Planning System, the detection is simple: If a container starts communicating using an ID that is not on the Planning data list, the ID must be wrong.

In real life, the Planning data list might not be that perfect. In particular, when using file transfer, there might be some time lag. Therefore, the alarm text shown in the REFCON Monitor is "Container ID is wrong or Planning data missing".

On ships, the Wrong ID alarm is disabled while in [Harbour mode](#)¹¹⁷.

In terminals, user can decide a delay or even disable the Wrong ID alarm, see [below](#)¹²³.

How to act upon these alarms?

Please refer to the [Modem Troubleshooting](#)²⁷⁸ guide..

Settings regarding missing or wrong ID alarms

The Missing ID alarm can be disabled. It is enabled out-of-the-box.



On ships, there are no settings available for the Wrong ID alarm, as the disabling is handled by the [Container handling mode](#)^[117].

In terminals, user can

- Select an alarm delay. For example, if you normally update Planning data every 4 hours, you could select a delay of 4 hours to avoid unintended alarms.
- Disable the alarm. This is necessary if you intendedly operate with [unplanned](#)^[183] containers.

See also [Alarm Options setup](#)^[125].

6.9 User generated alarms

Key subjects in this topic:

[Overview](#)^[124]

[Creating a user alarm](#)^[124]

[Removing a user alarm](#)^[125]

Related topics:

[Container Details](#)^[143]

[The Alarm System](#)^[110]

Overview

You can create an alarm related to a specific container in two ways:

- By selecting the "User Alarm" button in the Container Details, see below.
- Using the optional RDC, please refer to the separate section of the online helpuser's manual.

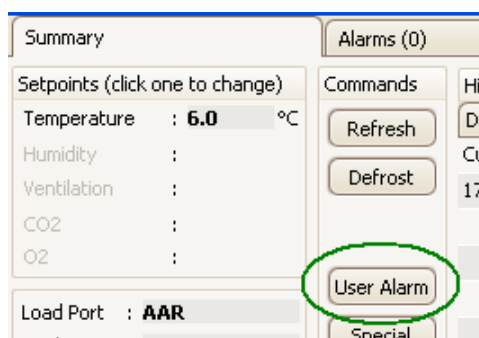
The alarm text you write appears - exactly as you write it - in the Alarm Summary and the Event Log:

Alarm Summary								✓	✕	📄	?
	Location	Origin	Text	Alarms	Initials	Time	Generated By				
🔴	Containers having alarms (Count=12)										
🔴	T7127.A	DEMO1001258	This is a user alarm example	1		08-03-2010 09:00:10	John				

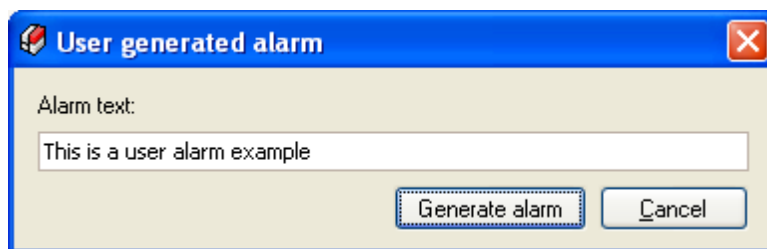
Event Log								📄	?	
	Event description	Event type	N...	Ol...	Unit	Item description	Scope	Trigger		
▶	This is a user alarm example	New alarm				DEMO1001258 T7127.A	Container	User	John	

Creating a user alarm

To create a user alarm, make sure the desired container is selected in the Container Details. Then click the "User Alarm" button under the Container Details' "Summary" tab:



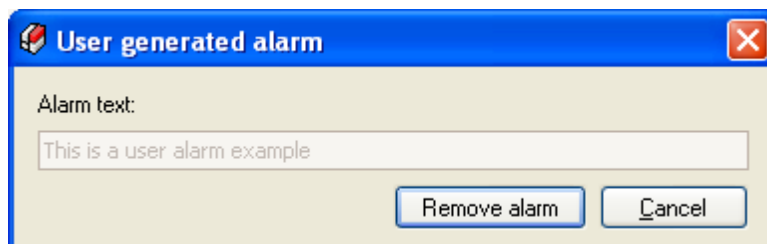
This opens the following dialog:



Type your alarm text and select "Generate alarm", and the alarm immediately appears.

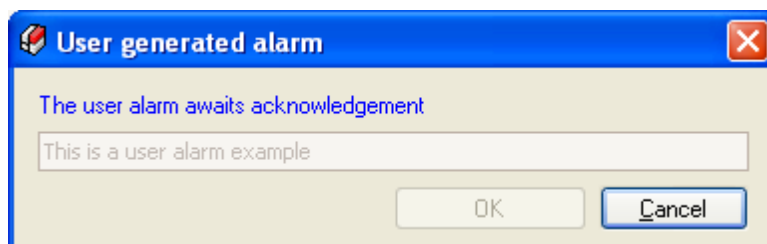
Removing a user alarm

To remove a user alarm, make sure the container with the alarm is selected in the Container Details. Then click the "User Alarm" button under the Container Details' "Summary" tab (see above). This opens the above dialog displaying the following:



Select "Remove alarm" - and the dialog closes.

If the alarm does **not** immediately disappear, it is because it has not been acknowledged. If you re-open the User alarm dialog, you will see this:



At the moment you acknowledge the alarm, it disappears.

6.10 Alarm Options setup

Key subjects in this topic:

[Overview](#)^[125]
[Remote operations parameters](#)^[126]
[GMS alarm parameters](#)^[126]
[Missing ID surveillance parameter](#)^[127]
[Wrong ID / missing planning data surveillance parameters](#)^[127]
[Setpoint difference alarm parameters](#)^[127]

Related topics:

[Temperature setpoint supervision](#)^[118]
[Alarms for missing or wrong ID](#)^[122]
[Remote operations](#)^[137]
[Power Alarm Options setup](#)^[127]
[The Alarm System](#)^[110]

Overview

The *Alarm Options* tab of the REFCON Setup dialog specifies data relevant to various alarms. For ships it looks like this:



In terminals there is one more option:

The "Email Notification" options are only visible when enabled in the license. For further description of this part, please refer to the separate "Email Notification" section.

The data fields are described below.

Remote operations parameters

This option is not shown if remote operations are not enabled in your [license](#)^[12].

In the first row of the *Alarm Options* tab you can enable or disable the alarm occurring when a [remote operation](#)^[137] fails.

A remote operation can fail in a number of ways. One of these is by timeout, the timeout time being decided in the field "Max. time to complete". Note that this setting is used no matter if the alarm is enabled or not.

GMS alarm parameters

In the second row of the *Alarm Options* tab you can enable or disable the alarm occurring should the [transfer of files to the Global Monitoring Server](#)^[176] (GMS) stop.

GMS files are produced once an hour. Preferably they are transferred immediately, however, in particular on ships it might be chosen to transfer only once or twice a day. Therefore you can decide a time limit - the number of hours you would allow the transfer to be delayed.



Missing ID surveillance parameter

Please refer to [How the Missing ID alarm works](#)^[123].

Wrong ID / missing planning data surveillance parameters

This option is only available in terminals. Please refer to [Settings regarding missing or wrong ID alarms](#)^[123].

Setpoint difference alarm parameters

In the right part of the *Alarm Options* tab you can enable or disable the various [setpoint supervision](#)^[118] functions. When the checkbox is checked, the function is enabled.

When enabled - as in the example in the above figure - you can further choose:

- The Delta value.

Please be aware that both REFCON and the container controllers may miscalculate Fahrenheit to/from Centigrade conversions by the last digit. To avoid that this causes alarms, the temperature Delta value should not be set to less than 0.2 degrees.

6.11 Power Alarm Options setup

Key subjects in this topic:

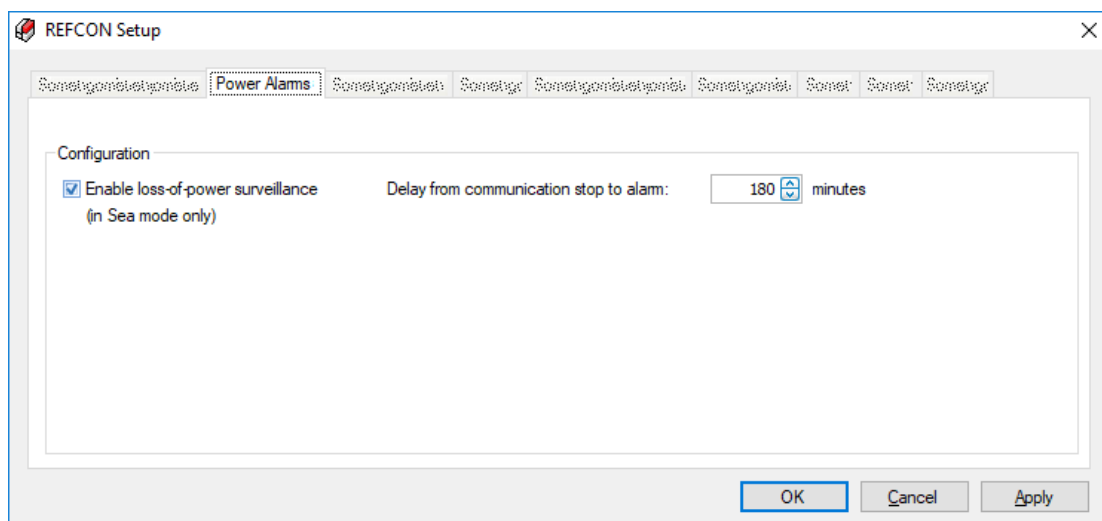
[Overview](#)^[127]
[Loss-of-power surveillance parameters](#)^[128]
[Communication not started / stopped alarm parameters](#)^[128]

Related topics:

[Possible loss of power alarm](#)^[119]
[Alarm Options setup](#)^[125]
[The Alarm System](#)^[110]

Overview

The *Power Alarms* tab of the REFCON Setup dialog specifies data relevant to various alarms. For ships it looks like this:



In terminals there can be two more options depending on the Planning System configuration:



REFCON Setup

Configuration

☒ Enable loss-of-power surveillance Delay from communication stop to alarm: 180 minutes

☒ Enable communication not started alarm Max time before communication must start: 10 minutes

☒ Enable communication not stopped alarm Max time before communication must stop: 15 minutes

OK Cancel Apply

The data fields are described below.

Loss-of-power surveillance parameters

Please refer to [How to configure loss-of-power surveillance](#)¹²¹.

Communication not started/stopped alarm parameters

Please refer to [How to configure communication not started/stopped alarms](#)¹²².

REFCON 6

Reefer Container Monitoring System

The Container Details

Transportation Solutions





7 The Container Details

Key subjects in this topic: Overview ¹³⁰ Locating the Container Details ¹³⁰ Container "Communicating" (Monitored) versus "Updated" ¹³¹ Container Connection Types ¹³¹	Related topics: Container Details common ¹³² The Container Item ¹³⁶ Remote operations ¹³⁷ Change Container ID operation ¹⁴¹ Measurement resolution ¹⁴² Sub-page Summary ¹⁴³ Sub-page Alarms ¹⁴⁵ Sub-page Operations ¹⁴⁶ Sub-page Data ¹⁴⁷ Sub-page Events ¹⁴⁸ Sub-page Communication ¹⁴⁸
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Overview

The *Container Details* allows you to work with one single specific reefer container. It includes all available information from and about the container, and it offers all possible commands and parameter changes.

The amount of available information and possibilities depends on the *connection type* and the *controller type*. The Container Details automatically adapts to both, that is, REFCON detects the type and adjusts the contents of the Container Details correspondingly.

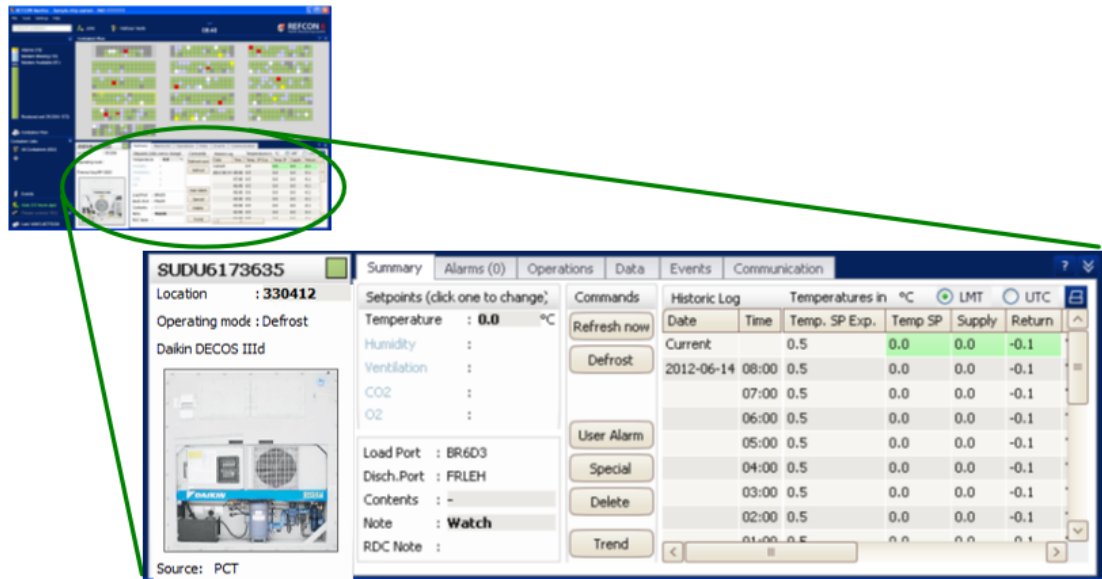
Controller manufacturers do, from time to time, release new controller types, and sooner or later they will connect to your REFCON. As the controller type was unknown at the time when your REFCON was installed, REFCON will categorize it as an *ISO compatible container* (if on PCT) or *Unknown* (if on [REFCON Extension](#) ²¹⁸). The *ISO 10368:1992(E)* standard decides a set of data that any controller communicating on the power cables must support. This is the set of data presented in the Container Details for an ISO compatible container.

On the contrary, when the controller type is known to REFCON, there normally is an extended data set available, being different from one type to another. REFCON supports the complete set, giving you the optimal means to handle the container.

Important note: You must think of the Container Details as a remote operating panel, similar to the one at the container. In this manualonline help, we are not able to tell you exactly what happens in the controller when you enter commands and parameter settings. If you are in doubt, please consult the controller manual.

Locating the Container Details

The Container Details is permanently located at the lower right of the REFCON Monitor:



Container "Communicating" (Monitored) versus "Updated"

It can be useful to understand how REFCON indicates the status of the communication path to the individual container.

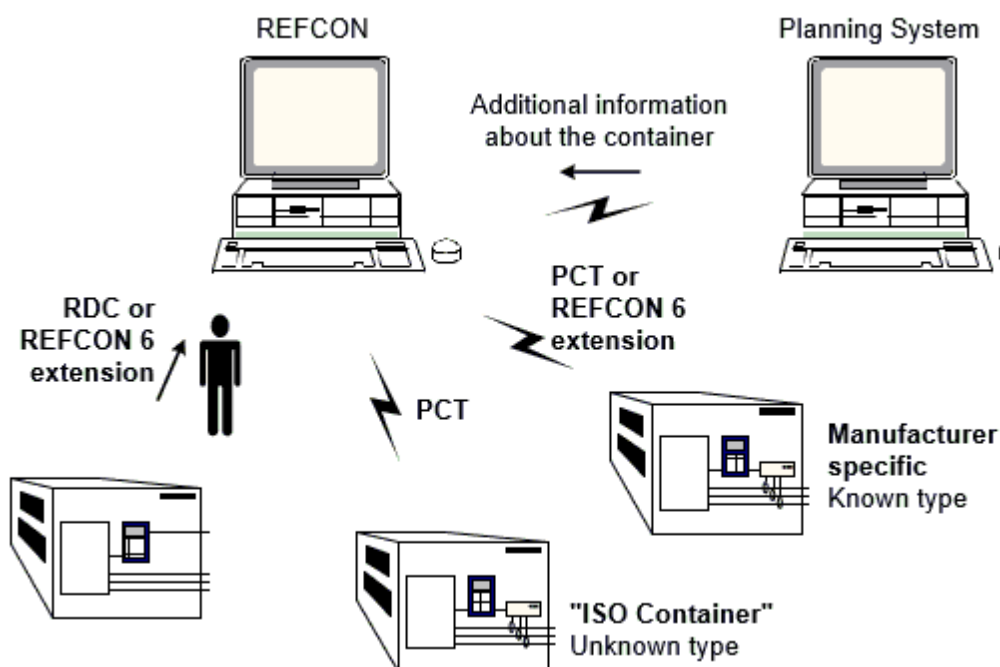
Mainly this is indicated by the [Communication State](#)^[73].

Three related terms are used:

Term	Explanation
Online	A container being "Online" means that the Power Cable Transmission system has had a contact to the PCT modem on the container "recently". If a container is "Online" and it has been updated within the last hour, it is also <i>Communicating</i> .
Communicating (Monitored)	When a container's Comm State is <i>Communicating</i> , it is also "Monitored". This state indicates that data were updated "recently" through a Source ^[134] that is presumed automatically updated at an interval of max 1 hour, and that REFCON has not later got any indication that it is no longer so. When Source is PCT, "Monitored" works like described above. For other Sources, normally "recently" is decided by a separate timeout setting. You will not be able to execute remote operations on a container that is not <i>Communicating</i> .
Updated	A container being "Updated" means that data were updated "recently" and are thus not obsolete ^[332] . An "Updated" container is either <i>Communicating</i> or <i>Offline updated</i> .

Container Connection Types

It is expedient to have the following connection types in mind:



From left to right, the potential amount of information increases.

RDC (option)	States and measured inputs, manually collected.
REFCON extension (option) with manual rounding	Depending on the extension, it may be a full data set - but the update frequency is depending on manual activity.
PCT, ISO	PCT running the basic ISO 10368:1992(E) protocol, transferring a certain minimum of information.
PCT or REFCON extension (option) with automatic communication, manufacturer specific	PCT running an extended ISO 10368:1992(E) protocol, transferring all available information, depending on the controller type. Or a REFCON extension running controller specific protocols.

For all connection types, there can be additional information available from the Planning System.

Note that in the Container Details, a container updated by RDC is called an "RDC-monitored container".

7.1 Container Details common

Key subjects in this topic:

[Minimising and restoring the Container Details](#) ¹³³
[The Identification area](#) ¹³³
[Source](#) ¹³⁴
[Using the tabs](#) ¹³⁵
[Editable fields in the Container Details](#) ¹³³

Related topics:

[The Container Details](#) ¹³⁰
[The Container Item](#) ¹³⁶
[Sub-page Summary](#) ¹⁴³
[Sub-page Alarms](#) ¹⁴⁵
[Sub-page Operations](#) ¹⁴⁶
[Sub-page Data](#) ¹⁴⁷
[Sub-page Events](#) ¹⁴⁸
[Sub-page Communication](#) ¹⁴⁸



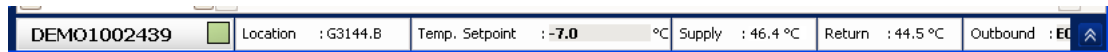
Minimising and restoring the Container Details

When opening the REFCON Monitor, the Container Details is found at the lower right in normal size, see [Locating the Container Details](#)^[130].

Click the down arrow button upper right to have the Container Details minimised:



This makes the Container Details appear as a single line bottommost:



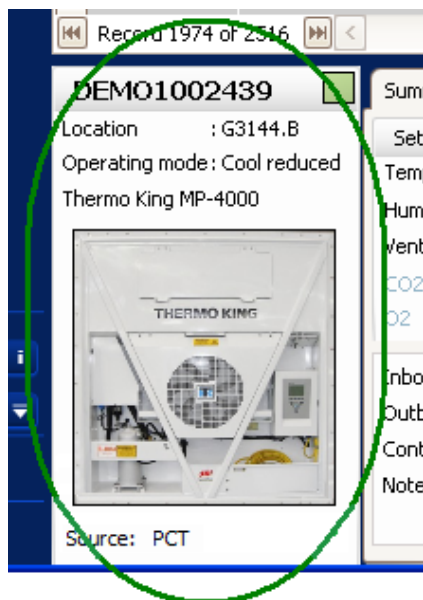
You will recognize the elements from the normal size Container Details, described in the following.

Click the up arrow button rightmost to return to the normal size.

Note that the up arrow button might cover parts of the information. To avoid this, resize the REFCON Monitor.

The Identification area

Leftmost in the Container Details window, the container being dealt with is identified:



- Uppermost, the container ID is shown.
- To the right of the container ID, a small [Container Item](#)^[136] square is shown, exactly as known from the optional Container Plan, thus indicating the main status of the container by colour and simple symbols.
- Below the container ID, the container's location according to the Planning System is shown. If the location originates from the Planning System, it can **not** be edited by the user, otherwise it is [editable](#)^[135].
- Next, the container controller's Operating mode is indicated, if known. Please refer to the controller manuals for explanations.
- In the lower part, the [connection type](#)^[131] and the controller type are indicated as described in the following table:

Condition	Result
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If the controller type is known (via PCT, REFCON 6 extension ^[218] or RMM+/RDC+)	Then the controller type is indicated. If a picture of the type is available, this is shown, otherwise a stylistic container is shown, see the following table.
If the controller type is read via PCT, but it is new to REFCON	Then you will read the type "ISO compatible container". A stylistic white container is shown, see the following table.
If the container communicates via PCT, but with no connection between modem and controller	Then you will read the type "Modem without controller". The stylistic light blue container is shown, see the following table.
If the container data is updated manually using the optional RDC	Then you will read the type "RDC-monitored container". The stylistic white or light blue container is shown, see the following table.
If the container is reported by the Planning System, but REFCON has no live data at all	Then you will read the type "Modem-missing container" or "Modem-available container" with corresponding stylistic white or light blue container shown, see the following table.

Three stylistic pictures can be found:

Picture	Meaning
	The controller type is detected, and data was received, but your version of REFCON was not prepared for this type, or it does not include a specific picture.
	This picture is shown in the following situations: - The container has logged in on the Power Cable Transmission, but it has not (yet) been type detected. Normally this is a temporary state, but if there is an error on the container like modem not connected to controller, it will be "permanent". - The controller type is detected, but your version of REFCON was not prepared for this type, or it does not include a specific picture. Data was not (yet) received.
	The container has not (yet) started communicating, and it is not known to have a modem.

Source

Below the picture discussed above, as well as in the Source column in [the Container List](#)^[66] and in [History](#)^[160], the data Source is indicated.

The Source will read "PCT", "RDC", "Fourpole" or a short name for a REFCON 6 extension.

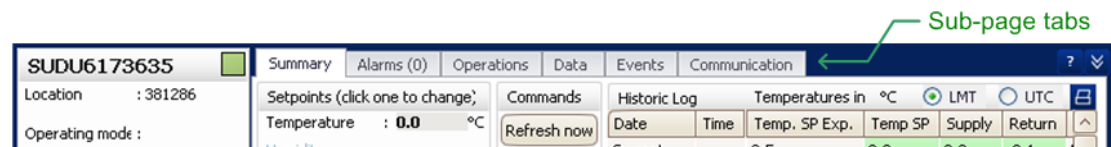


The real origin of data in any of these cases is primarily the container controller. The Source indicates how data were communicated to REFCON and thereby what you can expect regarding automatic updates, update frequency etc.

For example, the Source being "PCT" indicates full data set, automatic update at variable frequency, max interval 1 hour, and remote operations being possible; while "RDC" indicates limited data set, manual update, no remote operations.

Using the tabs

The Container Details consists of a number of sub-pages. You open a sub-page by clicking its tab:



The individual sub-pages are described in separate topic pages, see "Related topics" the Table of Contents.

Editable fields in the Container Details

In various places in various sub-pages, some data fields can (or maybe can) be edited by the user.

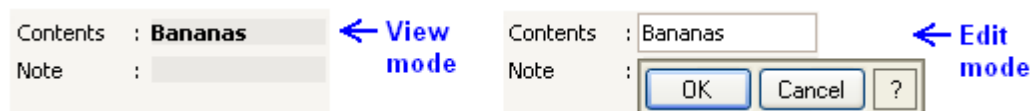
In contrary to a non-editable field, an editable field appears in bold characters on a slightly dark background. Also, when you move the mouse cursor over an editable field, a rectangle appears around the field.

Both editable and non-editable fields display a tool tip when the mouse cursor moves over. In particular, a non-editable field's tool tip will show why it is not editable, if not obvious.

To edit a field, click it. If it is not editable, nothing happens, except that the tool tip shows the reason.

Having clicked an editable field, you will be prompted for user ID and password if relevant, that is, if the User Authentication option is enabled and you are not already logged on with appropriate rights.

When the user ID/password is accepted, the field changes to a text box:



If it is a numeric field, the unit of measure is indicated, where relevant.

Now, simply change the reading and then:

- Press the Enter key or click the OK button to accept and activate your change. If relevant, your new value is validated. If it is not OK, you are informed, and the value is not activated. You will then have to edit anew and try again. Did you forget to observe the validation rules stated in the tool tip?
- Press the Esc key or click the Cancel button to revert to the existing value.

You may select the F1 key to obtain help. Also you may click the question mark button to have validation rules displayed.

Note: Having accepted your change, if the field represents data origin in a container controller, you will have to accept an extra time, see [Remote operations](#)¹³⁷. You may in this case observe that the field's value changes back to the "old" value, and then changes to the "new" value some time later, as it can take some time for the "new" value to be sent to the controller and then be read back to REFCON.



7.2 The Container Item

Key subjects in this topic:

[Overview](#) ¹³⁶

[Legends](#) ¹³⁶

Related topics:

[The Container Details](#) ¹³⁰

Overview

The Container Item, being the square indicating the container's status in the Container Details, and being the square representing one container in the optional Container Plan, indicates the status of the container by colour, see [Legends](#) ¹³⁶.

Legends

The following table specifies the possible colour indications and symbols in the Container Item.

Sym bol	In words	Short description	Explanation
	White	Never communicated / probably no modem	The container was included in Planning data but did not (yet) start communicating.
	Light blue	Communicating using an unexpected / default container ID	Typically the modem that has no connection to it's controller, or the container ID is recognized as the factory default, which cannot be used for a real reefer container.
	Green	Everything is OK	The container communicates by PCT or by a similar REFCON 6 extension ²¹⁸ and has recently been updated, and there is no alarms. It is " Monitored " ¹³¹ .
	Lighter green	Updated OK by RDC or similar	The container has recently been updated ¹³¹ by means of the optional RDC or via the REFCON Data Service, and there is no alarms. It may also be a darker green container that is no longer "Monitored".
	Flashing	New alarm(s)	There is one or more alarms on this container, and one or more of these has not been acknowledged.
	Red	Acknowledged alarm(s)	There is one or more alarms on this container, and they have all been acknowledged.
	Orange	Data obsolete	Primary data on this container are obsolete, that is, they have not been updated via PCT, the optional RDC or an optional REFCON 6 extension ²¹⁸ within decided time limit.
	An S, any colour	Special reefer	A user did mark this container "special".
	A filled blue-grey circle	Operation in progress	A user-demanded remote operation ¹³⁷ is awaiting completion.



Note that when the Container Item occurs in other contexts, there can be more legends than the above.

7.3 Remote operations

Key subjects in this topic: Overview ^[137] How a Remote operation is generally carried out ^[137] Indication of Remote operation in Side panel ^[139] Indication of Remote operation in Container Item ^[139] How a Remote operation generally works ^[139] The Remote operation failure alarm ^[140]	Related topics: Sub-page Summary ^[143] Sub-page Operations ^[146] The Container Details ^[130]
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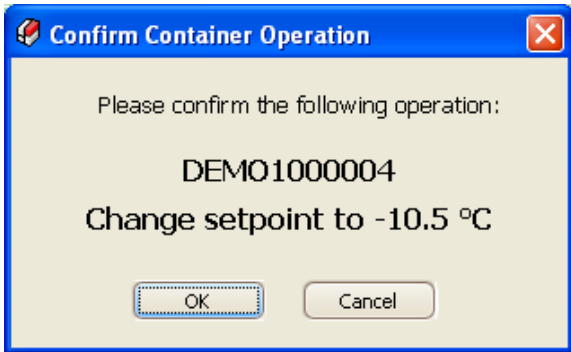
Overview

Remote operations - or short *operations* - is the common name for all commands and parameter selections that you can send to container controllers. Remote operations are possible on containers that communicate on PCT (Power Cable Transmission) or via a [REFCON 6 extension](#)^[218] being "presumed online" and supporting the operation. A few operations are common to all controller types, while others are available for certain types only.

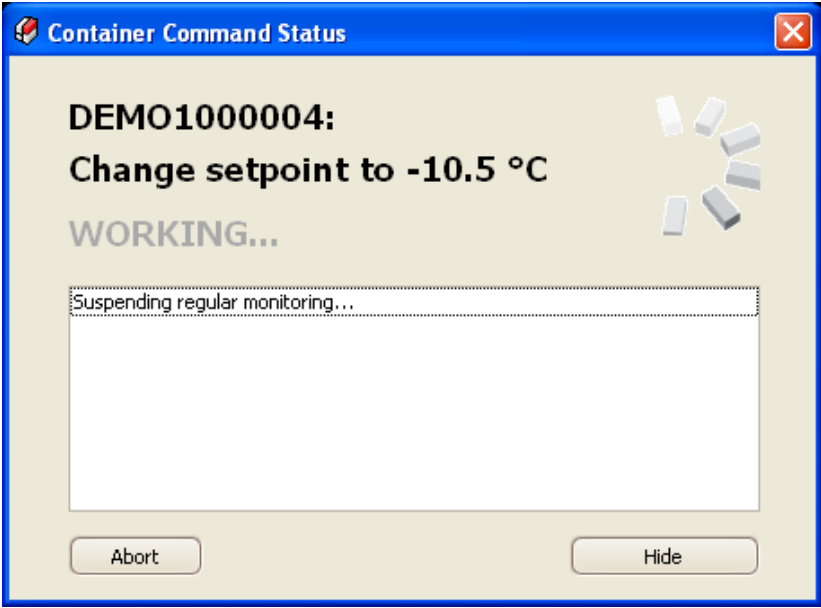
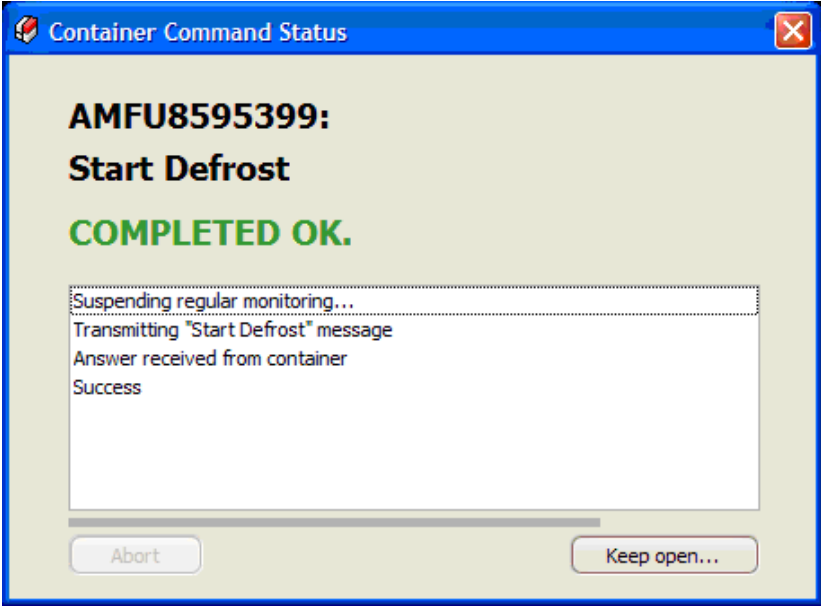
Remote operations are only possible if enabled in your [license](#)^[12].

How a Remote operation is generally carried out

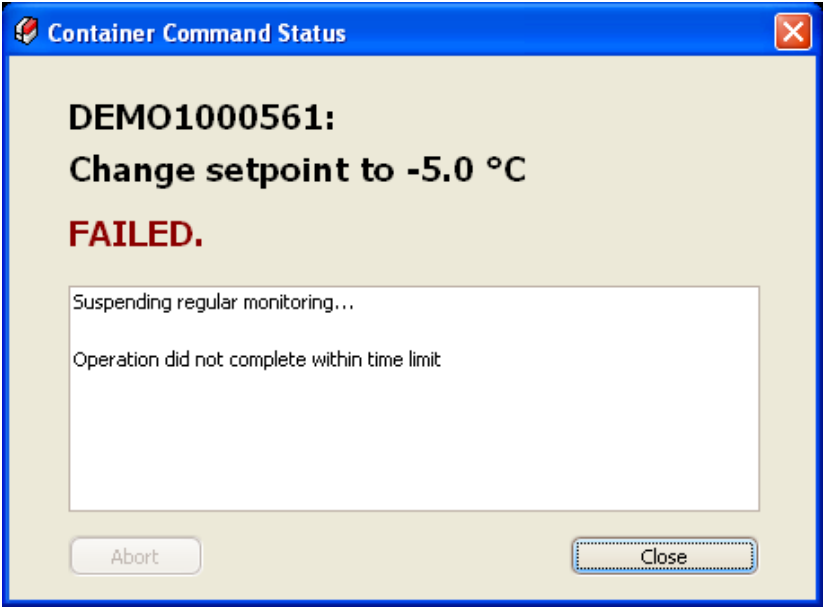
You initiate a Remote operation from the [sub-page Operations](#)^[146] or, for selected operations, the [sub-page Summary](#)^[143], see these for details. In general, having selected the parameter and clicked OK, or having clicked the command button, leads to a standardised number of dialog boxes:

Step	Action
1	If the User Authentication option is enabled, and you have presently insufficient rights to do the operation, you are asked for user name and password.
2	You are requested a confirmation, for example:  Select OK to proceed.
3	A progress dialog appears, showing how the operation proceeds.



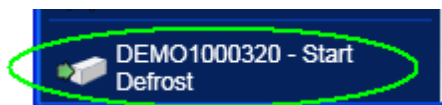
Step	Action
	 Refer to How a Remote operation generally works¹³⁹ for explanations to the messages that can appear in this dialog box. You may select the Hide button, which makes the dialog box disappear. It automatically shows up again should the operation fail.
4a	If/when the operation has completed successfully, the progress dialog shows this for a few seconds - if you did not close it in the mean time:  You may select the "Keep open" button to avoid that the dialog box is automatically closed, allowing you to inspect the messages.
4b	If the operation fails, the progress dialog stays open / re-opens, showing the error description. At the same time, an alarm occurs unless you have disabled this feature.



Step	Action
	 Refer to How a Remote operation generally works¹³⁹ for explanations to the possible error messages.

Indication of Remote operation in Side panel

While a Remote operation is being carried out, this is indicated lowermost in the Side panel:



If the operation is executed through a REFCON 6 extension, similar indication is shown just above the PCT indicator.

Indication of Remote operation in Container Item

While a Remote operation is being carried out, this is indicated in the [Container Item](#)¹³⁶:



How a Remote operation generally works

In the following table, the messages that can appear in the progress dialog (see above) are described:

Message	Explanation
Suspending regular monitoring... (PCT only)	The system is suspending its automatic container communication for a while. It does so because REFCON communicates with the specific container through a certain RMM Master. Most REFCON systems have more masters, of which only one can be active at a time, and it takes a while to change to the required.
Preparing telegram sequence... (PCT only)	The system is figuring out which telegrams to send to the container.



Message	Explanation
Communication failed	The system had to give up the operation because of communication problems with the container. You may try again a little later.
Container is reported offline	The system cannot currently communicate with the container.
Answer received from container	The system received an answer telegram from the container in reply to a previously sent request telegram.
Success	The whole container operation was carried out successfully.
Exceeded max. retries	The system gave up completing the requested container operation (because of communication problems).
Operation did not complete within time limit	The system gave up completing the requested container operation because it did not complete within the time selected in the REFCON Setup dialog.
Aborted	The container operation was aborted by the user.
Transmitting "xxxx xxxx" message	The actually selected command or parameter is sent to the container, the "xxxx xxxx" being replaced by the "name" of the operation, e.g. "Start Defrost".
The controller accepted the operation.	The container has understood the request and now starts carrying out the requested operation.
Container answer had wrong size (PCT only)	The container has replied to a request with a telegram that has a different length than expected by the system. <i>Note: This does not mean that the requested operation has failed.</i>
Reply was invalid - retransmitting... (PCT only)	The system had difficulties communicating with the container.
Transmitting telegram no. 3 (PCT only)	The system is sending telegram no. 3 in the sequence ("Refresh" operation).
Received reply (PCT only)	The system received a reply to the most recent sent telegram ("Refresh" operation).
No reply (PCT only)	The system did not receive an expected reply from the container within time.

Note that certain special messages might occur for the [Change Container ID operation](#)^[141] and when the operation is executed through a [REFCON 6 extension](#)^[218].

The Remote operation failure alarm

As standard, if/when an operation fails, an alarm is released. The benefit is that you don't have to wait for the operation to complete - consider it successful unless you get the alarm.

However, if you do not want this alarm, you can disable it in the [REFCON Setup](#)^[126] dialog. This is, of course, also where you re-enable it.



7.4 Change Container ID operation

Key subjects in this topic:

[Overview](#)^[141]
[Greatest caution needed](#)^[141]
[Communication key change](#)^[141]
[How to change the ID](#)^[141]
[What happens when changing the ID](#)^[141]
[Unwanted situations](#)^[142]

Related topics:

[Remote operations](#)^[137]
[Sub-page Operations \(optional\)](#)^[146]
[The Container Details](#)^[130]

Overview

To some degree, remote change of Container ID works like the other [remote operations](#)^[137], however, it requires special attention and works somewhat differently as described in the following.

Greatest caution needed

Most controller types accept the Change Container ID command.

You should, however, be strongly aware that it requires the **utmost care** to use this remote command. This is because the container ID is the key in the communication with the container. Should you, by accident, change the ID of a container to an ID other than the one printed on the container box, you will have what we call a [ghost container](#)^[286] and thereby a monitoring problem.

Communication key change

Notice that when the ID is changed in the controller, it works as if the container was disconnected and another was connected. This is, to REFCON, a complicated change that takes some time and with some risk of failure.

Please be aware that we talk about *one* container that had the **old ID** before and now is going to have the **new ID**.

In general, various combinations of modem types and controller types act in one of two ways when remotely changing the container ID:

- It responds to the command by the old ID, and then after a few seconds changes to the new ID.
- It changes to the new ID immediately without having responded by the old ID.

Because of the possible second behaviour, REFCON acts on the "no response" situation by assuming that the command actually went well. Just like the "response=OK" situation, REFCON proceeds by checking if the "new ID" has shown up.

How to change the ID

To change an ID, proceed as follows:

- Select the container going to have a new ID. Make **absolutely sure** that it is the right one shown in the Container Details.
- Click the command button and then enter the new ID. Be **absolutely sure** that you type the correct 4 letters and 7 digits. Then select OK.

What happens when changing the ID

What exactly happens when executing a change of ID though a [REFCON 6 extension](#)^[218] cannot be described here. Please refer to the extension documentation.

When executed through PCT the following happens:



When user has confirmed the new ID, things proceed as a standard [remote operation](#)^[137], however, with some additions:

1. When REFCON receives the OK message from the container, still using *old ID*, **or** when it gets the "no response" time-out, it expects the container to start communicating by the new ID within less than a minute.
2. REFCON waits for a while and then tries to contact the container using *new ID*. If this fails, it tries again a few times.
3. When (and if) the OK message is received from the container, using the *new ID*, a new record is created in REFCON's database with the *new ID* as key. All data, except log files, are moved from the record with the *old ID* to the new record. Then the record with the *old ID* is deleted.
4. The [Refresh now](#)^[146] command is initiated to make sure that the new record is up-to-date.

Unwanted situations

In case of any fault or time-out during this process (steps 1 - 3) you end up in an unwanted situation that can probably be described by one of the following statements:

- Though responding OK, the controller did *not* change the ID. You recognize this situation by a) the record with the *new ID* does not appear, and b) the record with the *old ID* persists and keeps updating (you could try the [Refresh now](#)^[146] command to check). You may try again, but probably you will have to walk to the container and change the ID directly in the controller.
- The controller changed the ID, but it did not start communicating using the *new ID* before time-out. REFCON concludes that the ID did not change, and things appear like the above bullet except that the *old ID* record is no longer updated. Most likely a record with the *new ID* shows up within an hour, as the container is (re-)discovered by the standard PCT functionality. Then you will have two records representing the same container, one with *old ID* and one with *new ID*. You may delete the one with *old ID*.

7.5 Measurement resolution

Key subjects in this topic:

[Overview](#)^[142]
[Temperature resolution example](#)^[142]

Related topics:

[The Container Details](#)^[130]

Overview

Any measurement that originates from a container controller and is shown on the REFCON Monitor has a limited accuracy. The sensors and the controller operate with certain resolutions, and after this the communication system will in some cases further reduce the accuracy in order to improve communication performance.

Temperature resolution example

For specific controller types and modem types, cargo temperatures are logged with following resolutions depending on temperature intervals:

Range	Resolution
Below -1.71°C	1.0 °C resolution
From -1.71°C to 3.39°C	0.05 °C resolution
Above 3.39°C	0.25 °C resolution



7.6 Sub-page Summary

Key subjects in this topic:

[Overview](#)^[143][The Setpoint area](#)^[143][The Planning data area](#)^[143][The Commands area](#)^[143][The Historic Log](#)^[144]

Related topics:

[Container Details common](#)^[132][Remote operations](#)^[137][The Container Details](#)^[130]

Overview

The Summary is the common / front page of the Container Details.

The screenshot displays the 'Summary' tab for container SUDU6173635. It includes a location (381286), operating mode (Cool reduced), and a Thermo King MP-3000 unit. The 'Setpoints' section shows Temperature at 0.0 °C, Humidity, Ventilation, CO2, and O2. The 'Commands' section has buttons for Refresh now, Defrost, User Alarm, Special, Delete, and Trend. The 'Historic Log' table shows data for 2012-06-14, including Date, Time, Temp, SP Exp., Temp SP, Supply, and Return.

Date	Time	Temp	SP Exp.	Temp SP	Supply	Return
Current		0.5		0.0	0.0	-0.1
2012-06-14	08:00	0.5		0.0	0.0	-0.1
	07:00	0.5		0.0	0.0	-0.1
	06:00	0.5		0.0	0.0	-0.1
	05:00	0.5		0.0	0.0	-0.1
	04:00	0.5		0.0	0.0	-0.1
	03:00	0.5		0.0	0.0	-0.1
	02:00	0.5		0.0	0.0	-0.1
	01:00	0.5		0.0	0.0	-0.1

The Summary includes the information and options that, according to feedback from users, are generally the most important and/or the most frequently used.

The Setpoint area

Next to the [Identification area](#)^[133] uppermost, the setpoints available for the specific controller type are listed. The first one is always the Temperature setpoint. The following "rules" apply to each of the setpoints:

- If the actual setpoint is read from the controller, this is shown.
- Otherwise, if the expected setpoint is received from the Planning System, this is shown. An (E) is appended to the text.
- If both actual and expected setpoint are known, these are indicated in the tool tip.
- If both actual and expected setpoint are known, and they differ by more than the [delta value](#)^[118], the value is shown in red colour.
- If the actual setpoint is known (and therefore shown), and the controller allows remote changing it, the field is [editable](#)^[135]. Editing it results in a [remote operation](#)^[137]. Note that if the field is **not** editable, the tool tip will indicate why.
- If the expected setpoint is shown, the field is not editable.
- If neither setpoint is known, the value field is blank, and the text is greyed.

The Planning data area

Below the Setpoints area, selected, available Planning data are shown (not including data shown elsewhere).

If it is a terminal, the Inbound and Outbound data are shown. If it is a ship, the Load Port and Discharge Port are shown in stead.

The following "rule" applies to each of the Planning data fields: If the value is received from the Planning System or otherwise is automatically generated, it is **not** [editable](#)^[135].

The Commands area

In the centre part of the Summary sub-page, you see a column of buttons.



The very first button is "Refresh now". This command works only for a container communicating via PCT. It suspends the PCT data polling for a short while, sending an update request immediately to this container's controller. As a result all live data on this specific container should be updated within a short time.

After this you find one or more Operations buttons ("Defrost" in above example), that is, commands applying directly to the container controller. Only the most frequently used commands are included here - other commands may be available on the [Sub-page Operations](#)^[146]. Note that the Operations buttons are not available if [remote operations](#)^[137] are not enabled in your [license](#)^[12].

If the operation is not supported by the container controller or is for some other reason not available, the button is greyed, and the tool tip indicates the reason.

Below the Operations buttons, the following command buttons are found:

Command	Action
User Alarm	You can use this command to have an alarm released on this container with an alarm text you decide, please refer to User generated alarms ^[124] . Also, if you have generated an alarm via the optional RDC, you can use this command to remove it, if desired.
Special	If the container is not marked "Special reefer", this button appears "normal" - otherwise it appears "pressed down". See Special reefers ^[117] .
Delete	See Container Shortcut menu ^[34] .
Trend	This button opens the Container Trend ^[166] window displaying data for the actual container.

The Historic Log

Rightmost in the Summary sub-page, the Historic Log is found. It combines the most recent readings in the first row and periodically logged data in the succeeding rows.

At the moment a new periodic logging is present (once per hour if not disabled by the user), the Historic Log list automatically updates: The new recordings are shown in the second row, and the first row (named "Current") gets empty. After this, when new measures are received over the PCT (or by other means), these appears in the first row with green background colour. Until the next periodic log occurs, the green fields keep updating - you will see this indicated by short time bold characters. In other words, current values are only shown if they are newer than the latest periodic logging.

Note that if the [Source](#)^[134] is "RDC" or a similar manual extension, data are immediately logged when transferred from the handheld device. This means that the green "Current" line will never be visible in this case (or only for a fraction of a second).

Note that measurements appear with a [limited accuracy](#)^[142].

The Date and Time columns show the times of the recordings as LMT or UTC as indicated and decided upper right. Simply click the option buttons to change.

You can [print/export](#)^[152] the historic log by selecting the print button upper right:





7.7 Sub-page Alarms

Key subjects in this topic:

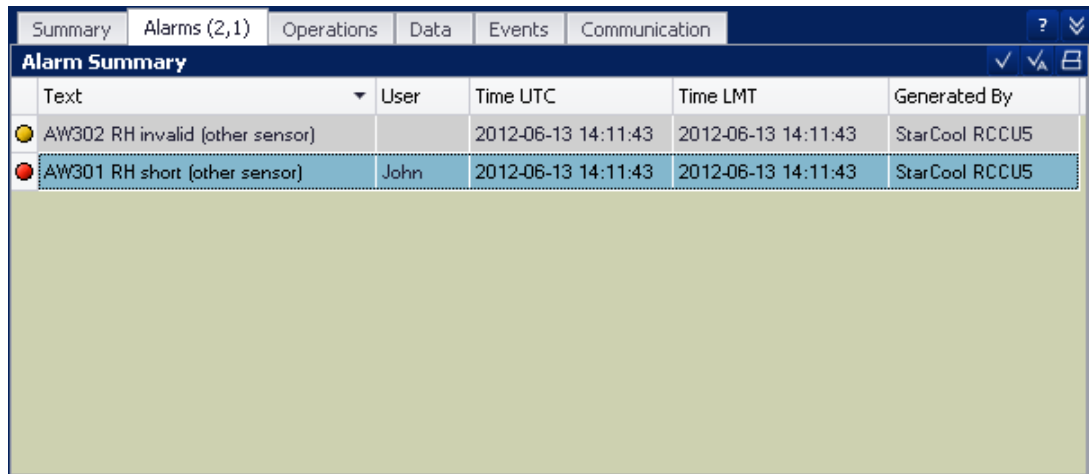
[Overview](#)^[145]
[Columns in the Container Alarm Summary](#)^[145]

Related topics:

[Container Details common](#)^[132]
[The Container Details](#)^[130]

Overview

The Alarms sub-page shows the list of present alarms on this container.



Summary	Alarms (2,1)	Operations	Data	Events	Communication
Alarm Summary					
Text	User	Time UTC	Time LMT	Generated By	
AW302 RH invalid [other sensor]		2012-06-13 14:11:43	2012-06-13 14:11:43	StarCool RCCU5	
AW301 RH short [other sensor]	John	2012-06-13 14:11:43	2012-06-13 14:11:43	StarCool RCCU5	

If no alarms, the list is empty. The alarm's acknowledge status is indicated exactly like in the main [Alarm Summary](#)^[113]. The alarms can be [acknowledged](#)^[114] individually or in groups using the small acknowledge buttons upper right.

Note that within the tab header



the number of alarms on this container is indicated, for example,

- (0) means there are no alarms.
- (2,1) means there are 2 alarms, and 1 of these has not been acknowledged.
- (2) means there are 2 alarms that have all been acknowledged.

You can [print/export](#)^[152] the container alarms summary by selecting the print button upper right:



Columns in the Container Alarm Summary

The columns in this list is the relevant subset of the columns in the main Alarm Summary. Please refer to [Column specifications](#)^[114] for explanations.



7.8 Sub-page Operations (optional)

Key subjects in this topic:

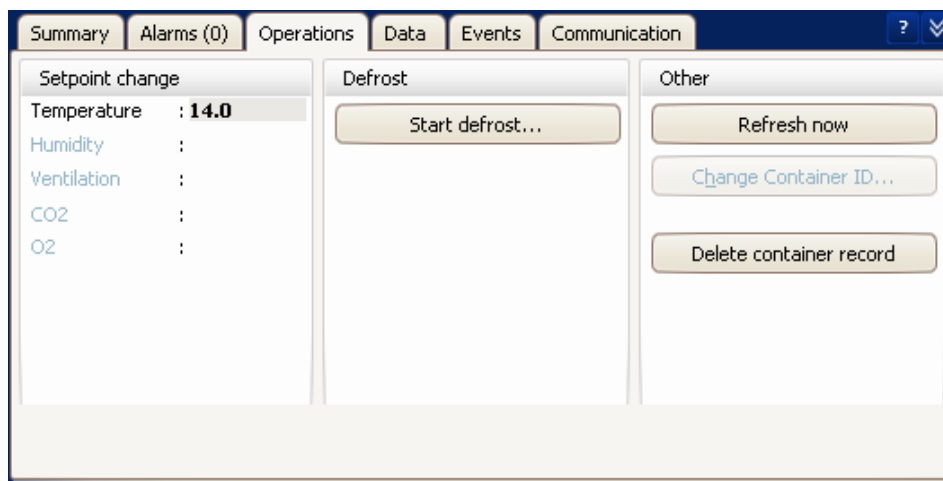
[Overview](#)^[146]
[Changing a setpoint](#)^[146]
[Start defrost](#)^[146]
[Refresh now](#)^[146]
[Change Container ID](#)^[147]
[Delete container record](#)^[147]

Related topics:

[Remote operations](#)^[137]
[Change Container ID operation](#)^[147]
[Container Details common](#)^[132]
[The Container Details](#)^[130]

Overview

When Remote container operations are enabled in the [license file](#)^[12], you find the Operations sub-page in the Container Details:



Within the Operations sub-page you find all the possible container commands, of which you recognize some from the [Summary page](#)^[143].

If a certain command is not supported by the specific container, or it is for another reason not available, it is disabled (greyed) - like "Change Container ID" in above example. Move the mouse cursor over the disabled command to have a tool tip show you the specific reason.

Changing a setpoint

As standard all controller types accept the temperature setpoint change command. A few controllers accept other setpoint commands as well. No matter the setpoint type, you proceed as follows:

1. Change the setpoint value shown in bold (14.0 in above example). While doing so, an OK button is available just below the input field - select this to activate your change.
2. After this, things proceed as a standard [remote operation](#)^[137].

Note that if you select a value different from the expected setpoint, a [setpoints differ](#)^[118] alarm will occur, if enabled.

Start defrost

When you select "Start defrost", things proceed as a standard [remote operation](#)^[137].

Note that even if the remote operation results in "Success", you can not conclude that defrosting has actually been started - this is a decision made by the controller.

Refresh now

This command is described under [The Commands area](#)^[145].



Change Container ID

The command is described in [Change Container ID operation](#)^[141].

Delete container record

The Delete command is described under [Container Shortcut menu](#)^[34].

7.9 Sub-page Data

Key subjects in this topic:

[Overview](#)^[147]

[Data field descriptions](#)^[147]

[Data field colours](#)^[147]

Related topics:

[Container Details common](#)^[132]

[The Container Details](#)^[130]

Overview

The Data sub-page shows all current data related to the actual container, except alarms and communication-related data.

Summary

Alarms (1,1)

Operations

Data

Events

Communication

Planning data

Controller measurements, states etc

Controller settings, other data

DataPoint	Value	DataPoint	Value	DataPoint	Value
Contents	E0	Ambient	0.3 °C	ACT mode	False
First PlanData UTC	2019-11-08 13:47:26	Batt. Temp.	0.5 °C	All SP match	False
Inbound	BLI833	Batt. Volt.	0.6 V	ASC mode	True
Location	350R01/02B	Capacity	0.2 %	Controller ID	Str 128
Note	E0	Cmpr.Runn	False	Controller text	Str 127
Outbound	TRUCK	CO2	1.0 %	Ctrl serial	Str 25
PDdeleteFailed	False	CO2 SP	1.2 %	Ctrl SW version	?
Planned	True	Comp. Temp.	1.9 °C	CtrlType	Daikin DECOS IIIj
Temp. SP Sup	14.0 °C	Comp. Temp.	1.5 °C	Def. interval	00:05:30

Note that measurements appear with a [limited accuracy](#)^[142].

As shown, the data has been divided into three tables:

- Planning data: Data origin from the Planning System or entered by the user.
- Controller measurements and states: Dynamic data origin from the container.
- Controller settings, other data: Static data and trip settings from the controller, and miscellaneous data if any.

You can [print/export](#)^[152] each of these tables individually by selecting the print button upper right:



Data field descriptions

Some of the data field descriptions are abbreviated rather much. To get the full description, move the mouse cursor over the text.

Supply	-16.6 °C
Temp SP	-15.0 °C
Supply temperature measured	

Data field colours

The fields in the Value column are coloured exactly like in the [Container List](#)^[66].



7.10 Sub-page Events

Key subjects in this topic:

[Overview](#)^[148]
[Comments to Container Event Log](#)^[148]
[Log](#)^[148]

Related topics:

[Container Details common](#)^[132]
[The Container Details](#)^[130]

Overview

The Events sub-page shows the list of events relevant to this container origin from the present cargo voyage as described in [Event logging](#)^[171].

Summary Alarms (0) Operations Data Events Communication										
Container Event Log										
Event description	Event type	New value	Old value	Unit	Item description	Scope	Triggering	Trigger	UTC time	
Select temperature setpoint delta	REFCON Setup	0.5	0.4	°C	REFCON Setup Setpoint difference alarming	System	User	Son	2011-05-19	
Temperature out of range	Acknowledge				MSWU1033746 020010	Container	User	John	2011-05-17	
Actual and expected setpoints differ	New alarm				MSWU1033746 020010	Container	System	REFCON	2011-05-17	
Temperature out of range	New alarm				MSWU1033746 020010	Container	User	John	2011-05-17	
Container data record creation	Creation				MSWU1033746	Container	System	Reefer Con	2011-05-17	
Container data record deletion	Deletion				MSWU1033746 020010	Container	System	Reefer Con	2011-05-17	
Container data record creation	Creation				MSWU1033746	Container	System	Reefer Con	2011-05-17	
Actual and expected setpoints differ	New alarm				MSWU1033746 020010	Container	System	REFCON	2011-04-29	

The list is similar to the general event list, see [Event log](#)^[171].

You can [print/export](#)^[152] the container event log by selecting the print button upper right:



Comments to Container Event Log

Referring to the example above, note that

- A few general events are logged to all container event log files because they are considered directly relevant to each container. An example is the "temperature setpoint delta" changed in the first event in the figure. You will see that the "Scope" is *not* "Container" for these events.
- All events with "Scope" = "Container" and "Item Description" showing the specific container ID are logged for this container only (plus the general event list).
- When the location of the container is known, this is added to the "Item description". You will see that when the container database record is created, causing the "Creation" event, the location is never shown.

7.11 Sub-page Communication

Key subjects in this topic:

[Overview](#)^[148]
[Displaying multiple modems](#)^[149]
[Data field descriptions](#)^[149]
[Data field colours](#)^[149]

Related topics:

[Container Details common](#)^[132]
[The Container Details](#)^[130]

Overview

The Communication sub-page shows all data related to the communication between REFCON and the actual container. To the right, all modem-related data are listed, while to the left, other communication-related data are shown.



Summary	Alarms (0)	Operations	Data	Events	Communication
Communication data			Modem data		
DataPoint	Value		DataPoint	Value	
LastUpdatedExt	2018-11-12 10:05:12		Ctrl.Conn.	True	
Monitored	True		Modem HW rev	1.1	
Source	WirelessHub		Modem serial	FB0837	
			Modem SW ver	1.0.1-4	
			Modem type	REFCON Portable Modem	
			On PCT	False	

You can [print/export](#)^[152] each of the lists by selecting the print button upper right:



Displaying multiple modems

If you are using an [REFCON 6 extension](#)^[218], it is possible that more than one modem is attached to a container and communicating. In such case you may see semicolon-separated lists in the modem fields, like in following example:

Modem HW rev	042A; 1
Modem serial	11112232; 19280470
Modem SW ver	0.0.0-0S; 4.0.0_201050
Modem type	REFCON Portable Modem; RMM-W

In this example, two modems are communicating - a removable REFCON Portable Modem and a fixed RMM-W. As the modems are shown in this order in the "Modem type" field, you find data related to the REFCON Portable Modem *before* the semicolon in each relevant modem field, and data for RMM-W *after* the semicolon. So, in above example, the REFCON Portable Modem has serial number 11112232, while the RMM-W has 19280470.

Note that a possible communicating PCT modem will be shown in the separate "PCT Modem Info" field.

Data field descriptions

Some of the data field descriptions are abbreviated rather much. To get the full description, move the mouse cursor over the text. See [example here](#)^[147].

Data field colours

The fields in the Value column are coloured exactly like in the [Container List](#)^[66].

REFCON 6

Reefer Container Monitoring System

Print and Export



8 Print and Export

Key subjects in this topic:

[Overview](#)¹⁵²
[Print/export on demand procedure](#)¹⁵²

Related topics:

[The Preview](#)¹⁵³
[Automatic Print/Export](#)¹⁵⁴

Overview

In REFCON, "printing" and "data export" is combined in two common "print" functions.

"Printing" means sending the selected picture to a printer, which can either be a part of the REFCON system or a network printer.

"Exporting" means creating a file and saving it on disk or attaching it to an email. The file can be PDF or other formats.

You have two options:

- Print/Export on demand, being described below.
- Automatic, timed print/export, see [Automatic Print/Export](#)¹⁵⁴.

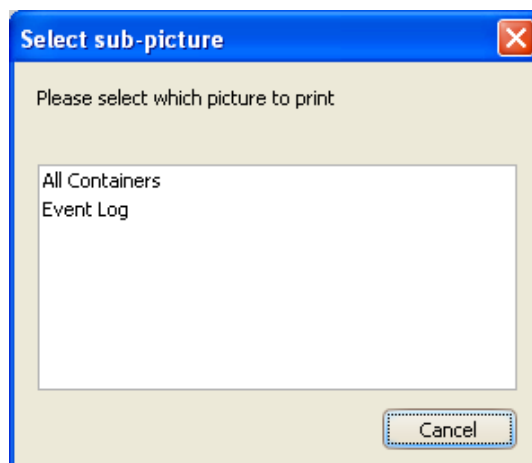
Print/export on demand procedure

To have a picture printed or exported, you always proceed as follows:

1. Make sure the picture of interest is open.
2. Select
 - the keyboard combination "Ctrl"+"P", or
 - "Print" from the "File" menu, or
 - the Print button, which is only available if the picture can be printed:



3. If you did select "Ctrl"+"P" or "Print" from the "File" menu, and there is more than one printable picture open, you must choose which to print:



Simply click the one to print.

4. Wait for the preview window to open.
5. In the preview window, select your option as described in [The Preview](#)¹⁵³.



8.1 The Preview

Key subjects in this topic:

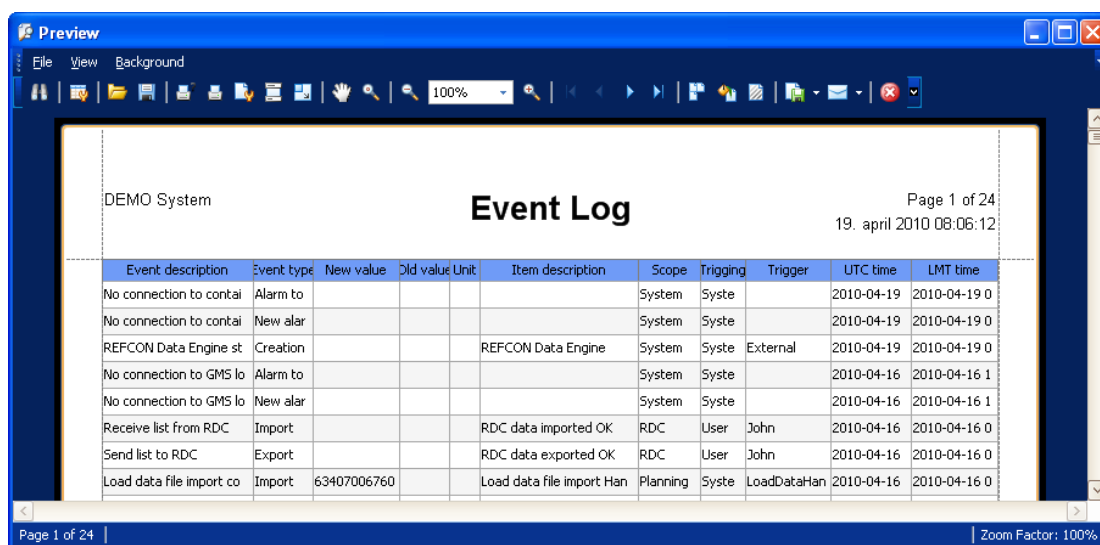
[Overview](#)¹⁵³
[Print the Preview](#)¹⁵³
[Export the Preview](#)¹⁵³
[Manipulate the Preview](#)¹⁵⁴

Related topics:

[Print and Export](#)¹⁵²

Overview

When opened, the Preview shows the active picture as it would look if printed in colours on A4 paper portrait:



In the preview itself you will notice:

- Upper left, the REFCON application is indicated, exactly like in the REFCON Monitor's [Caption bar](#)²²¹ ("DEMO System" in the example).
- Upper centre, the picture name is shown in large font.
- If a list, all columns from the screen picture are included. The widths of the columns are adjusted to the available paper size, which might mean that text is truncated.

Print the Preview

To have the preview sent to a printer, use one of the following options:

Toolbar button	Menu option	Keyboard shortcut	Result
 Print...	File -> Print...	"Ctrl"+"P"	The standard print dialog opens, allowing you to select a printer etc.
 Quick Print	File -> Print		The preview is immediately printed on the default printer.

Export the Preview

To have the preview exported to a file, use one of the following options:



Toolbar button	Menu option	Keyboard shortcut	Result
 Export Document	File -> Export Document...		Displays a list of the available file formats. Selecting one will display one or more dialog boxes where you decide properties for the file export.
 Send via E-mail	File -> Send via E-mail...		Displays a list of the available file formats. Selecting one will display one or more dialog boxes where you decide properties for the E-mail and the attachment.

Manipulate the Preview

To manipulate the preview before printing or exporting it, you can use one of the following options among others:

Toolbar button	Menu option	Keyboard shortcut	Result
 Page Setup	File -> Page Setup...		Opens the standard print setup dialog box, where you can change the orientation of the paper, among others. In the case that text is truncated, changing to "landscape" might be a solution.

8.2 Automatic Print/Export

Key subjects in this topic:

[Overview](#) ¹⁵⁴
[The Automatic Print/Export Administrator window](#) ¹⁵⁴
[The Automatic Print/Export Create/Edit dialog box](#) ¹⁵⁵
[Automatic Print/Export job data fields](#) ¹⁵⁷
[The background scheduler](#) ¹⁵⁸

Related topics:

[Print and Export](#) ¹⁵²

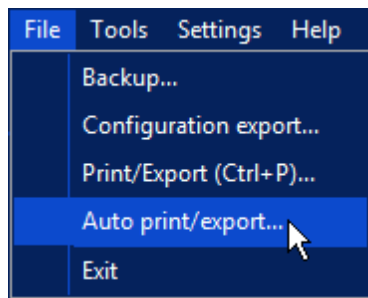
Overview

This function allows the user to have any Container List or the Alarm Summary automatically printed and/or exported at selected intervals and time points. The user can set up any number of such printing jobs.

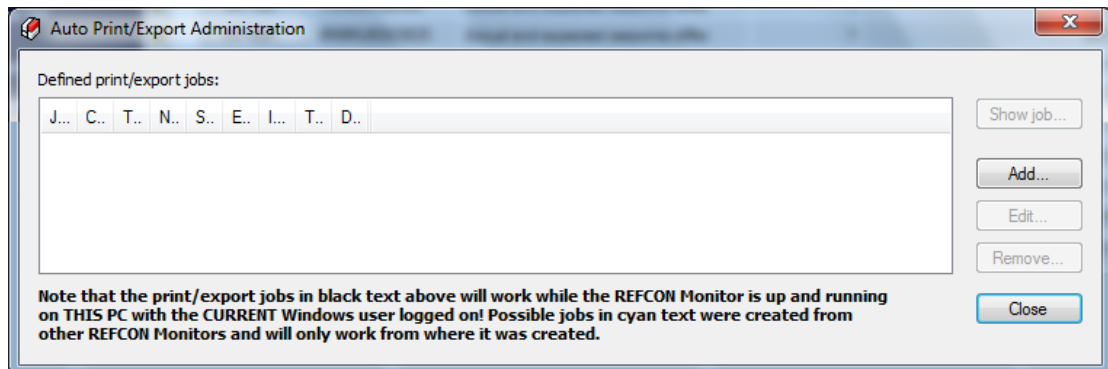
The jobs decided by a user on a specific PC with a specific Windows logon is separate from any other jobs, and they work only while that PC is running AND that Windows user is logged on AND the REFCON Monitor is running. This is because on different PCs and to different Windows users it can differ what printers and network drives are available.

The Automatic Print/Export Administrator window

To set up automatic print/export jobs, select "Auto print/export..." from the File menu:

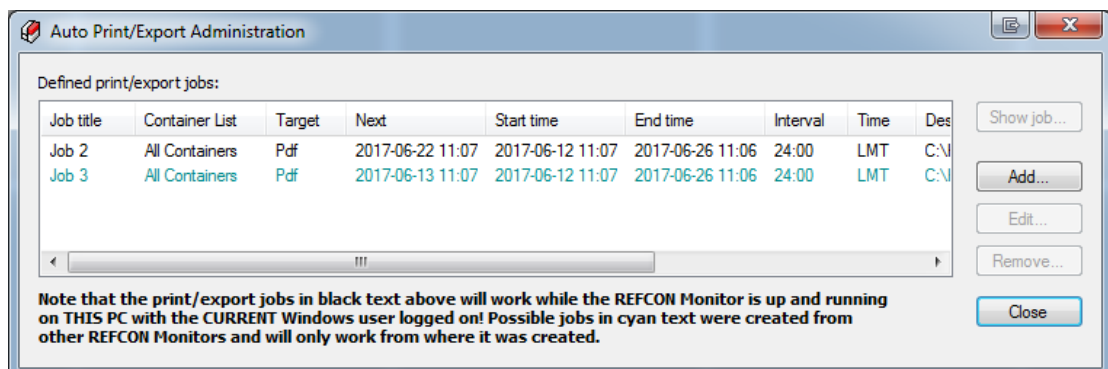


This opens the Administrator window. While no jobs have been created, it looks like this:



Here you would select the Add button to create your job(s).

When one or more jobs are defined, they appear like this:



Here "Job 2" is defined by you / the current Windows user on this PC, while "Job 3", as indicated by different colour, was defined by some other Windows user and/or on some other PC.

The most columns you will recognize from the Create/Edit dialog box described below, however, the "Next" column shows when the job is scheduled to run the next time - being identical to the Start time when it is the first time.

From the Administrator window you can, in addition to create new jobs, select an existing job and click the

- "Show job..." button to view the job settings. *This command is only available for local jobs (listed in black colour).*
- "Edit..." button to change a job. *This command is only available for local jobs (listed in black colour).*
- "Remove..." button to delete a job. *Before removing a job created by some other Windows user / on some other PC (listed in cyan colour), well, maybe you should ask first?*

The Automatic Print/Export Create/Edit dialog box

Selecting "Add..." from the Automatic Print/Export Administrator window opens the following editor:



Create new print/export job

Enter a unique job title:

Select the Container List:

Select the job target type:

Select timing type

☒ Once ☐ Recurring limited ☐ Recurring unlimited

Select job start date: and time: ☒ LMT ☐ UTC

Select job end date: and time:

Select time interval: hours minutes

Test now OK Close

You will see that the content is dynamic as items get visible/invisible or enabled/disabled according to your selections. Here is an example of a completed job definition:

Create new print/export job

Enter a unique job title:

Select the Container List:

Select the job target type:

Select file naming method

☐ Selected fixed ☒ Selected + \'-ymmddN\' ☐ \'-ymmddN\'

Select file folder: Browse...

Select file name: -ymmddN.Xls

Select timing type

☐ Once ☐ Recurring limited ☒ Recurring unlimited

Select job start date: and time: ☒ LMT ☐ UTC

Select job end date: and time:

Select time interval: hours minutes

Test now OK Cancel

All the possible fields and selections are described below.

When you have entered or changed data, you can at any time select the Cancel button to discard your changes.



Probably you would rather select the "Test now" or the "OK" button. In both cases the selected data are validated. If not valid you are warned and informed about what is wrong.

Assuming valid data,

- "Test now..." runs the job as if it was decided to run now, that is, the selected list is immediately printed or exported. Note that the operation can take some time. When finished you are informed about the result, however, of course you would normally check the produced paper or file.
- "OK" saves the job for further handling by the [background scheduler](#)¹⁵⁸.

Automatic Print/Export job data fields

In the Automatic Print/Export Administrator and the Automatic Print/Export editor, the following data fields are or can be available:

Field name	Explanation / options
Job title	Aims at making the purpose of the job obvious to you and your colleagues. You decide the text, but it is required to be at least 4 characters.
Container List	Select any item in the drop-down list, that is, all locked and customer decided lists in the Side Panel, the Alarm Summary and the PCT Communication Statistics list.
Job target type	Select Printer or a file format. The available file formats are: • Pdf - being viewable in Adobe Reader, and being ready to print. • Csv - comma separated values - being readable by certain programs including Microsoft Excel. • Txt - text format - being useful in special cases. • Xls - Microsoft Excel 2003 format. • Xlsx - Microsoft Excel 2007 format.
Selected printer (only if target is Printer)	Select any item in the drop-down list.
File folder (only if target is not Printer)	Use Browse to locate the desired folder, which can be on the local hddisk or on a network drive. You also can select a removable drive, however, are you sure it keeps available?
File naming method/file name (only if target is not Printer)	Select one of the options: • "Selected fixed": The exported file always has the same name. It is overwritten at each job-run. • "Selected + '-yymmddN'": The file is given a name composed of two parts; first the text you select in the "file name" field (for example "Morning report"; second the year (2 digits), month and day followed by a serial number (resulting example, third job that day: "Morning report-1105133.pdf"). • "-yymmddN'": The file is given a name composed of the year (2 digits), month and day followed by a serial number (for example, third job that day: "1105133.pdf").
Paper orientation (only if target is Printer or Pdf)	Select Portrait or Landscape. With many columns in the list, landscape normally gives the best result.
Timing type and time selections	Select one of the options: • "Once" - the job runs at the selected Start date and time and is then deleted. • "Recurring limited" - the job starts at the selected Start date and time. It repeats at the selected time interval. At the selected End date and time the job is deleted.



Field name	Explanation / options
	• "Recurring unlimited" - the job starts at the selected Start date and time. It repeats at the selected time interval. It keeps repeating until you remove it.
LMT or UTC	Select if the date and time(s), you select, must be according to LMT or UTC.

The background scheduler

The job(s) being listed in the [Automatic Print/Export Administrator window](#)¹⁵⁴ are checked a few times a minute by a background scheduler running in the specific REFCON Monitor.

When a job is due, it is executed. The execution causes the REFCON Monitor to "freeze" for some time, depending on the length of the list and the type of target. When the job has completed, the REFCON Monitor is again at your disposal.

If two jobs are due at the same time, they will run one at a time, that is, the second job is delayed for the time it takes the first to complete.

If the REFCON Monitor is not running at a job's due time, the job will execute as soon as the REFCON Monitor has been started (within half a minute or so); however, if the job is more than one hour overdue, the user will be asked if the job is to execute.

If the job can for any reason not be completed successfully, an alarm is released. The alarm text explains the reason.

REFCON 6

Reefer Container Monitoring System

Historic data logging



9 Historic data logging

Key subjects in this topic:

[Overview](#)^[160]
[Logging parameters](#)^[160]
[Log file philosophy](#)^[160]
[Deciding the log file location](#)^[161]
[Deciding the "new cargo" limit](#)^[161]
[Deciding the archiving duration](#)^[161]

Related topics:

[Periodic logging](#)^[161]
[Using the Container History menu](#)^[163]
[Container trend](#)^[166]
[Historic Log](#)^[169]
[Event logging](#)^[171]
[Event Log](#)^[171]

Overview

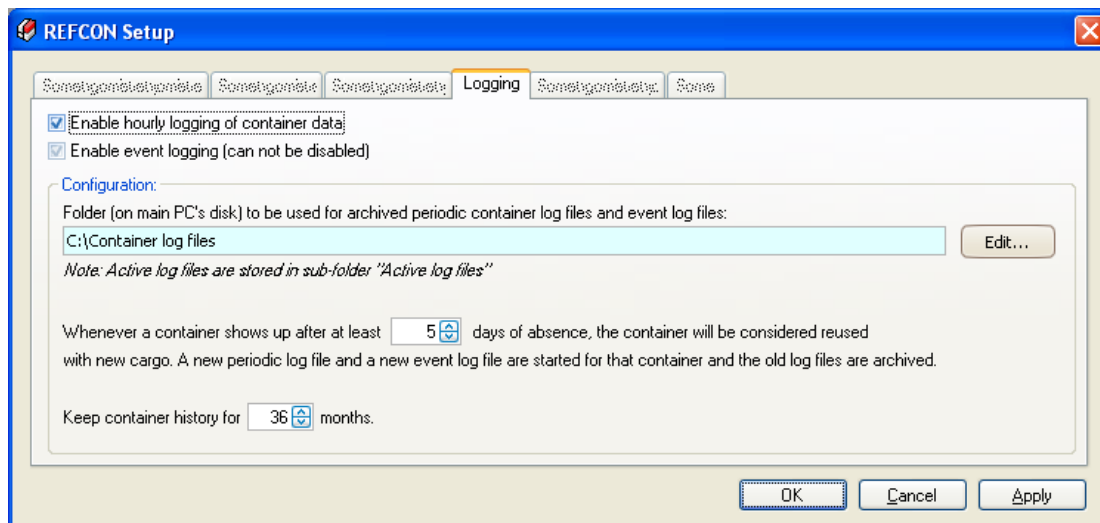
REFCON includes two primary logging functions, making it possible for daily users and service technicians to view historic trends and sequences of events.

The [periodic data logging](#)^[161] function records all available *live data*, that is, controller measurements and primary status indicators. We sometimes refer to this as *temperature logging*, as temperatures are important in this context, however, also other measurements are recorded.

The [Event Log](#)^[171] function records all important events triggered by users, container controllers, the REFCON system itself or any external source.

Logging parameters

Under the *Logging* tab of the [REFCON Setup](#)^[330] dialog box, you can decide the parameters for periodic logging and event logging:



Unchecking "Enable hourly logging of container data" will stop REFCON's periodic logging. Data are then no longer recorded, but you can still use the [Container Trend](#)^[166] or the [Historic Log](#)^[169] to view the logs from before the stop.

When enabled, the automatic logging function runs according to the Configuration parameters that you decide.

Log file philosophy

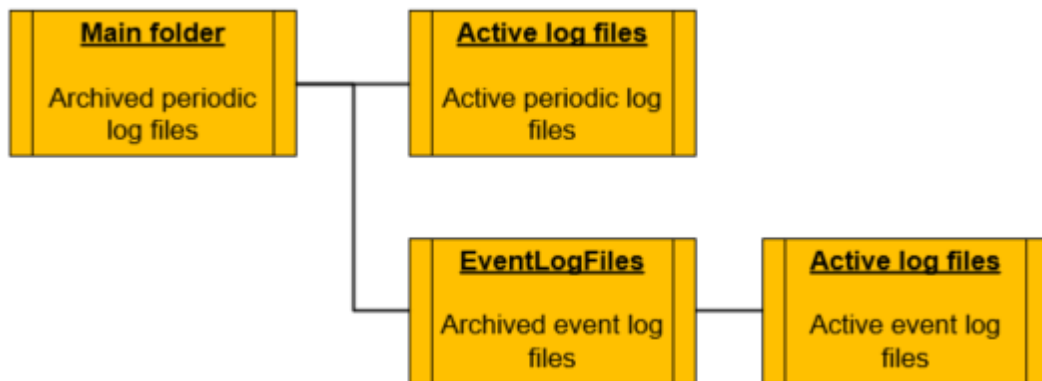
To promote important factors like data safety, easy backup and high performance, REFCON's periodic log files and container event log files are organised according to the following philosophy:



- Logs for different containers are stored in separate files. The file name includes the container ID.
- REFCON attempts to make it one periodic log file and one event log file for each container journey, that is, when the container has been loaded with new cargo, new files are started, and the old are archived. Unfortunately, REFCON can not specifically detect that the cargo is new. In stead, you select a time limit: If the container is disconnected from REFCON for more than the selected number of days, it is considered a new journey.
- When archiving a log file, it's name is extended to also include the date range of the included logs. For example, if the container was continuously connected to REFCON from August 29th to September 4th, you will recognize this time span in the file name - along with the container ID.
- To avoid that the haddisk runs full, archived log files are automatically deleted after your selected number of months.

Deciding the log file location

The log files are stored in the folder decided in the light blue textbox and automatically created sub-folders:



To change the folder, select the Edit button and proceed as described in [Selecting drive and folder on Main PC](#)^[49]. Note that you can only select a folder on a haddisk situated in the REFCON Main PC (fixed drive), **not** on an external disk or a network drive.

Please **be aware** that when changing the folder, all the existing log files are moved from the old folder to the new one, unless you deselect this. If you selected another drive, this file removal might take quite some time. Apart from this, it takes place in the background, and REFCON keeps working, so you normally should not worry.

Deciding the "new cargo" limit

Regarding the second bullet in above philosophy, you select the limit in the field labeled "Whenever a container shows up after at least". The maximum is 14 days.

Deciding the archiving duration

Regarding the fourth bullet in above philosophy, you select the archiving duration in the field labeled "Keep container history for". The maximum is 120 months. If the haddisk is small you should minimize the duration.

9.1 Periodic logging

Key subjects in this topic:

[Overview](#)^[162]

[How to use periodic logging](#)^[162]

[How periodic logging works](#)^[162]

Related topics:

[Historic data logging](#)^[160]

[Container Trend](#)^[166]

[Historic Log](#)^[169]



Overview

Being independent units, most reefer containers do their own data logging. What exactly is logged depends on the controller type, but typically, events are logged as they occur, and temperatures and other measurements are logged periodically.

What makes the containers own logging a little inconvenient is that it takes some manual work and some time to retrieve it. It might therefore be helpful for you to use one of the supplementary logging options.

One option is to take advantage of the Global Monitoring Server (GMS). Your REFCON system should be sending data to the GMS regularly as described in [Data Transfer for Global Monitoring](#)^[176]. When doing so, you can use the internet to view the logs from your containers.

The other option is that you can have REFCON do it's own data logging, based on the live data that are periodically read via the PCT or the other possible data sources.

How to use periodic logging

You use REFCON's periodic logging as follows:

- In the [REFCON Setup dialog](#)^[160], you decide about the automatic logging - if you want it to be active, the file location, and some time parameters.
- In the [Container Trend](#)^[166] window, you watch the trend curves resulting from the logging.
- In the Historic Log of the [Container Details](#)^[144] as well as in the [Historic Log](#)^[169] window, you watch the logged data as a table.

How periodic logging works

The periodic logging function records a set of live data for each active container. The specific data set depends on the controller type and version.

The individual data point is included only if it is *fresh* and *valid*.

- A data point being *fresh* means that it is not [obsolete](#)^[332] at the time of recording. You decide the timeout settings as this regards.
- A data point could be **not valid** due to, for example, a sensor error.

Depending on your REFCON system configuration, you can be working with **online** live data providers (PCT, REFCON Wireless Hub solution) and **offline** live data providers (REFCON Mobile solution, RDC), or a combination of the two.

Online data

REFCON logs live data from an online provider once an hour, starting every full hour (UTC time).

If, at the time of logging, a certain data point has not changed since it was last logged (one hour earlier), it is still recorded in the log file if still *fresh*, however, the time stamp shows when it was actually updated, so viewing the log reveals that it is old data.

Offline data

REFCON logs live data from an offline provider once the data has been transferred to REFCON.

The periodic logging supports import of "historic" data. If recorded data are imported (for example from the REFCON Mobile), REFCON imports and logs all data newer than the latest existing log, while older data are rejected.

Example: The supply temperature was last logged at 7.00. Historic data are imported containing the supply temperature values at 6.15, 7.00, 8.00 and 8.35. REFCON ignores the values from 6.15 and 7.00 and inserts the values from 8.00 and 8.35 in the relevant container log file. Note that this way there can be more or less than one log an hour.



9.2 Using the Container History menu

Key subjects in this topic:

[Overview](#)^[163]

[How to select](#)^[164]

[Choosing file / time period](#)^[166]

[Displaying event log](#)^[166]

Related topics:

[Historic data logging](#)^[160]

Overview

The *Container History* menu entry allows you to select a specific container and have its periodic log data displayed graphically (see [Container Trend](#)^[166]) or as a list (see [Container Details](#)^[144] or [Historic Log](#)^[169]).

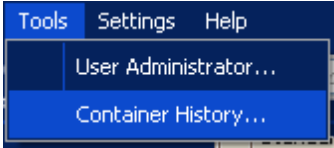
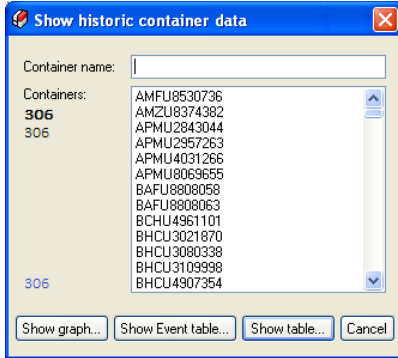
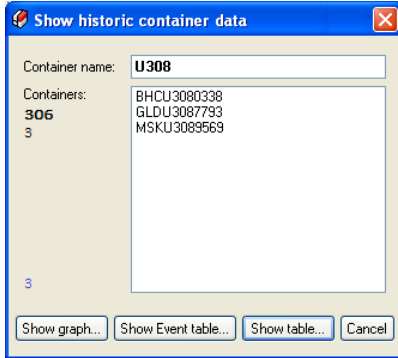
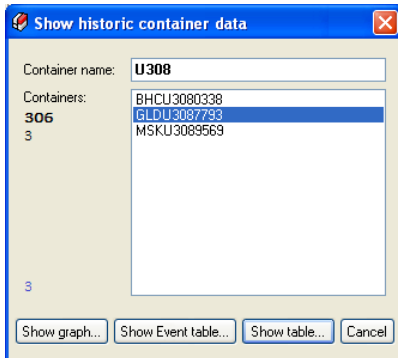
Also it allows you to inspect event log files.

If you want to inspect archived log files, this is the only way.

**How to select**

To select a container and have it's logged data displayed, proceed as follows:

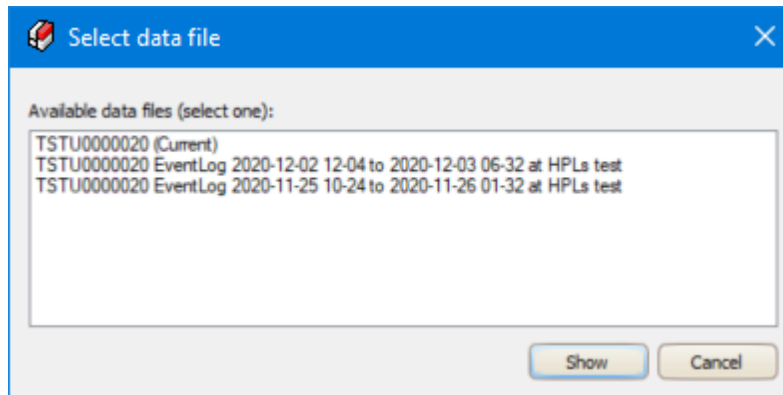


Step	Action	Result	Description
1	Open the "Show historic container data" dialog by selecting "Container History" from the Tools menu: 		The "Containers" list displays all containers for which active or archived log files exist in the selected folder^[160]. When opened you see identical numbers, 306 in the example. If you have thousands of container IDs you may see that it takes a little time for all 3 figures to update. The first number, in bold, is the number of container IDs in the list.
2	Type a search string (in the "Container name" field).		The "Containers" list changes as you type, still showing all container IDs including the typed string. As shown in the example, this might very well be a string in the middle of the ID. You will see that the second and the third number change as you type, showing the number of "hits" of your search. These two counts should be identical, however, the blue may be delayed.
3	Select the desired container by • Clicking it in the "Containers" list, or • Using the Down and Up keys.		The container is highlighted. Note: If you double-click the container, it will work as if you first single-click and then select [Show table...].
4	Select [Show graph...], [Show Event table...] or [Show table...].	The dialog closes. Then one of the following happens: 1. If only one log file exists for the container, the Container	



Choosing file / time period

Typically if a certain container has visited your site several times, probably with different cargo and some time interval, both an active log file and a number of archived log files can exist. In such case you will see an intermediate window like this:



If the container is currently at your site, you will normally find its container ID followed by (Current) as the first entry in the list.

After this the archived files follow. For each file the relevant time period is a part of the file name.

Double-click the file you want, or use arrow keys and Enter key. The window of the type selected in above step 4 will open.

Displaying event log

When you choose to view an event log, a window open with columns and data as described in [Event Log](#)^[171].

The layout of the window, and the way you operate it, is similar to the [Historic Log](#)^[169].

9.3 Container Trend

Key subjects in this topic: Overview^[166] Opening the Container Trend^[166] Understanding the Container Trend^[167] Understanding the dots^[167] Zooming^[168] Scrolling^[168] Selecting data points^[169] Viewing old log files^[169]	Related topics: Periodic logging^[161] Historic Log^[169] Historic data logging^[160]
--	---

Overview

The *Container Trend* is a window showing historical data related to one specific container in a co-ordinate system. As standard, values from a certain number of selected data points are shown, depending on the type of the container controller. When opening the window, approximately the last 24 hours are selected, and the range of the Y-axis matches the values found within that time period. You can manipulate the appearance of the trend as described below.

Opening the Container Trend

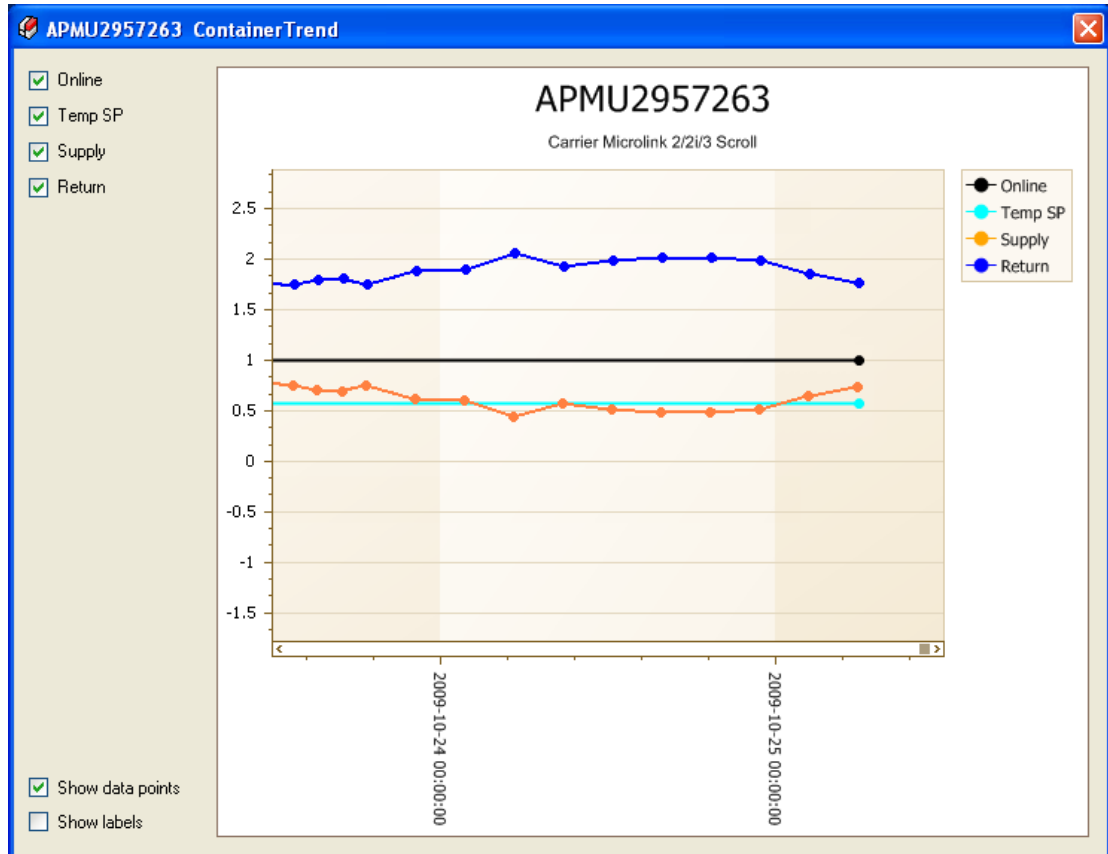
There are three ways to open a Container Trend:



1. Click the "Trend" button in the [Container Details](#)^[143].
2. Use the [Container history menu option](#)^[163].
3. If you are not sure about the container ID, and the container is still in REFCON's database, you can instead open the [Container List](#)^[66] or the optional [Container Plan](#)^[96] and find the container of interest. Right-click the container and select *Show historic data (graph)* from the [Shortcut menu](#)^[33] to have the graphical window opened.

Understanding the Container trend

When just opened, the Container trend appears like this, showing data from the latest 24 hours or the selected range:

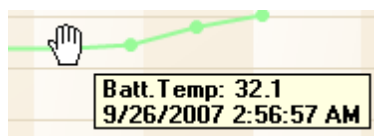


- In the caption of the window, and in large font above the co-ordinate system, the affected container's ID is shown.
- Upper left, all available data points have been enabled - all checkboxes are checked.
- Lower left, it has been chosen to *show data points* - the dots on the curves - but not to *show labels* - no value readings at each dot.
- Upper right, the colours used for each data point are indicated.
- Below the X-axis, the UTC log times are indicated.

Understanding the dots

When *Show data points* is checked lower left, dots are shown on each curve, as in the above example.

Moving the mouse cursor on top of a dot will make it's data shown in a tooltip:





In this example, logging was typically done at 3:00 AM UTC. The indicated time is about 3 minutes before that. This is because the indicated time shows when the data was received from the container controller. If data was received more than once between two loggings, always the newest data is used.

If **no** data was received between two loggings, **no** dot will appear. In the Container trend example above, this is true for the "Online" data point and the "Setpoint" data point - neither of them have changed or been updated since at least 27 hours ago. Note that neither the humidity (top green curve) has changed in the example, but humidity data have been received from the container controller, and therefore dots appear.

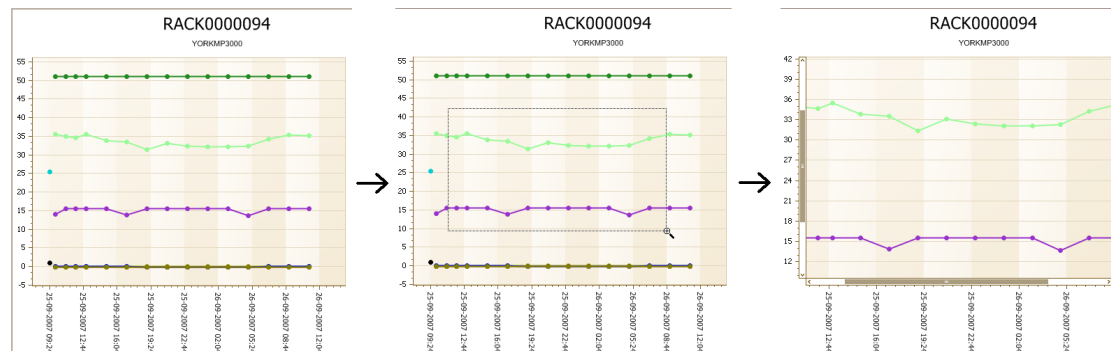
Note that measurements appear with a [limited accuracy](#)^[142].

It may occur that dots are "missing" in between. A typical reason is [data obsolescence](#)^[332].

Zooming

You may wish to have a closer look by zooming in or a better survey by zooming out. Both can be done in the following ways:

1. You can use the scroll wheel on your mouse to zoom in (= scroll up) or out (= scroll down).
2. You can keep the **Shift** button (on the keyboard) down and then **drag the mouse** over a part of the curve area to zoom in:



3. You can keep the **Shift** button down and then **click the mouse** inside the co-ordinate system to zoom in by a pre-defined step - both axes are affected.
4. You can keep the **Alt** button on the keyboard down and then **click the mouse** inside the co-ordinate system to zoom **out** by a pre-defined step - both axes are affected.

Note that in both zoom directions there are limits, so you can't just keep zooming.

You will see that when using the scroll wheel or when zooming in steps (3 and 4 above), both axes are zoomed simultaneously.

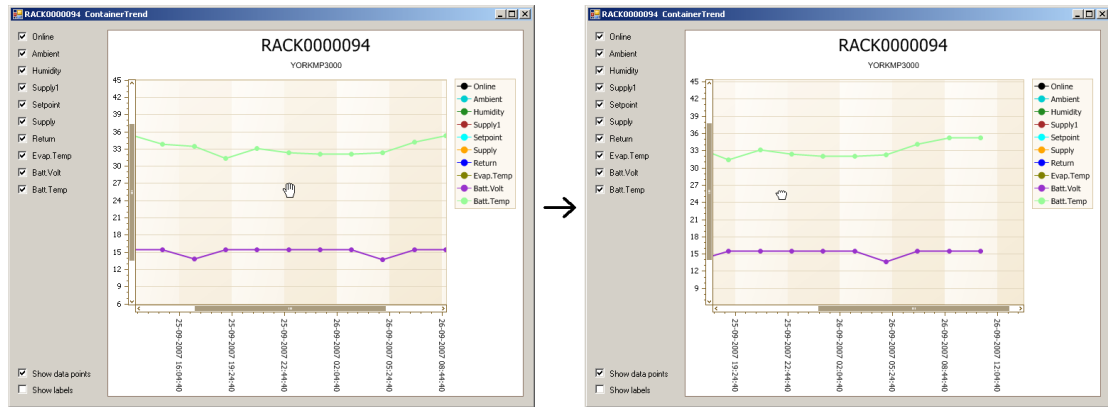
Scrolling

Unless you have zoomed out to the limit, scroll bars occur within the co-ordinate system, horizontal and/or vertical. In such case you can:

- scroll horizontally to change to another duration of time
- scroll vertically to change the span of the Y-axis.

You can scroll in the following ways:

1. Move the mouse cursor inside the co-ordinate system and then drag in any direction:



2. Drag one of the the scroll boxes, or click the scroll bar arrows or the base like a standard Windows scroll bar:



Selecting data points

All data points are selected when opening the Container trend window. You can remove a curve by deselecting the data point's checkbox upper left. When doing so, the curve disappears from the co-ordinate system and from the legend list upper right.

Viewing old log files

Even if the container has been discharged and removed from REFCON's database, and even if the periodic logging would have been stopped, there might still be "old" log files available. You can view these exactly like described above; however, if the container is no longer in REFCON's database, you need it's container ID, as you will have to use the access method **1** (see [Opening Container Trend](#)^[166] above).

9.4 Historic Log

Key subjects in this topic:

[Overview](#)^[169]
[Opening the Historic Log](#)^[170]
[Changing the time base](#)^[170]
[Scrolling the list](#)^[170]

Related topics:

[Periodic logging](#)^[167]
[Container Trend](#)^[166]
[Historic data logging](#)^[160]

Overview

The Historic Log is a list showing the periodically logged historic data, that is, the same data as shown in the [Container Trend](#)^[166]. The Historic Log is a part of the Summary tab of the [Container Details](#)^[143] picture, but also it is available as an independent window:



Recent container data						
DEMO1000587						
Switch to UTC						
Date (LMT)	Time (LMT)	SP Exp. (°C)	Setpoint (°C)	Supply (°C)	Return (°C)	Online
08-09-2008	09:00	-10.0	-10.0	-11.9	-10.4	
	08:00	-10.0	-10.0	-9.8	-9.2	
	07:00	-10.0	-10.0	-11.2	-9.3	
	06:00	-10.0	-10.0	-10.2	-10.4	
	05:00	-10.0	-10.0	-11.9	-10.4	
	04:00	-10.0	-10.0	-10.2	-9.9	
	03:00	-10.0	-10.0	-10.8	-9.5	
	02:00	-10.0	-10.0	-11.6	-9.3	
	01:00	-10.0	-10.0	-8.2	-10.9	
07-09-2008	00:00	-10.0	-10.0	-9.8	-9.2	
	23:00	-10.0	-10.0	-11.2	-9.3	
	22:00	-10.0	-10.0	-10.2	-10.4	
	21:00	-10.0	-10.0	-11.9	-10.4	
	20:00	-10.0	-10.0	-10.2	-9.9	
	19:00	-10.0	-10.0	-10.8	-9.5	
	18:00	-10.0	-10.0	-11.6	-9.3	
	17:00	-10.0	-10.0	-8.2	-10.9	
	16:00	-10.0	-10.0	-8.0	-9.4	
	15:00	-10.0	-10.0	-9.6	-9.2	
	14:00	-10.0	-10.0	-10.2	-10.4	

Each line in the Historic Log represents one sample of each of the available measures.

Note that measurements appear with a [limited accuracy](#)^[142].

The Historic Log includes one column for each available measure, depending on the type of the container controller.

The Historic Log includes all logs available for the specific container. It is always sorted chronologically, the newest sample being shown first.

Opening the Historic Log

There are two ways to open a Historic Log:

1. Use the [Container history menu option](#)^[163].
2. If you are not sure about the container ID, and the container is still in REFCON's database, you can instead open the [Container List](#)^[66] or the [Container Plan](#)^[96] and find the container of interest. Right-click the container and select *Show historic data (tabular view)* from the [Shortcut menu](#)^[33] to have the list window opened.

Changing the time base

Use the button upper right in the window to change between LMT and UTC.

Scrolling the list

To scroll the list, you can use:

- The vertical scrollbar.
- The "Page Up" and the "Page Down" keys.
- The key-combinations "Ctrl"+"Home" (= move to top) and "Ctrl"+"End" (= move to bottom).



9.5 Event logging

Key subjects in this topic:

[Overview](#)^[171]
[How to use event logging](#)^[171]
[Event log storage housekeeping](#)^[171]

Related topics:

[Event Log](#)^[171]
[Historic data logging](#)^[160]

Overview

The Event Log function records all important events detected by REFCON. An event is considered "important" if it influences the cargo in the containers or the "permanent" behaviour of the REFCON system, or similar. The events can have all kinds of origins, such as users, controllers and REFCON itself.

There are in fact two event logs:

- The *general event log* contains all the newest events, including system events and container events. One single general event log file exists.
- The *container event log* contains all the events relevant to a specific container during a specific journey. There is one file for each container, plus possible archived files.

How to use event logging

You use REFCON's event logging as follows:

- In the [REFCON Setup dialog](#)^[160], you decide about the event logging file location and archiving parameters.
- In the [Event Log](#)^[171] window, you watch the events found in the general event log. You can use standard list facilities to search and sort the events.
- In the [Container Details, Events tab](#)^[148], you watch the events found in the specific container event log.

Event log storage housekeeping

In this version of REFCON 6, the general event log storage is limited to 10000 events. When the storage is full, a new event automatically makes the oldest event removed from the storage.

The individual container log file is literally unlimited. It is archived, and a new started, according to the [log file philosophy](#)^[160].

9.6 Event Log

Key subjects in this topic:

[Overview](#)^[171]
[Opening the Event Log](#)^[172]
[Number of records](#)^[172]
[Freezing the list](#)^[172]
[Sorting the events](#)^[172]
[Column specifications](#)^[173]
[Event types](#)^[173]
[Event scopes](#)^[174]
[Triggering types](#)^[174]

Related topics:

[Event logging](#)^[171]
[Historic data logging](#)^[160]

Overview

The Event Log is a list showing the most recent events as defined in [Event logging overview](#)^[171].



Event Log										
Event description	Event type	New value	Old value	Unit	Item description	Scope	Triggering	Trigger	UTC time	LMT time
Device never connected	Alarm to Normal				RMM Master (1,5)	System	System	REFCON	12-03-2010 10:28:35	12-03-2010 11:28:35
010 Heater current too high	New alarm				DEMO1002526 (location unk...	Container	Container	Thermo King...	12-03-2010 10:28:32	12-03-2010 11:28:32
020 Defrost time too long	New alarm				DEMO1002526 (location unk...	Container	Container	Thermo King...	12-03-2010 10:28:32	12-03-2010 11:28:32
030 Bypass valve fail	New alarm				DEMO1002526 (location unk...	Container	Container	Thermo King...	12-03-2010 10:28:32	12-03-2010 11:28:32
Delete list layout	Monitor Setup	Ny			Container Summary	Monitor	User	John	12-03-2010 09:40:37	12-03-2010 10:40:37

Records in list: **1015** No. of logs to show: 1000 Freeze (stop updating)

Each line in the Event Log represents one event.

The Event Log includes a fixed number of columns. The contents of the columns are described in [Column specifications](#)^[173].

When opened, the Event Log shows the newest events, a pre-decided number. It is always sorted chronologically, the newest event being shown first. The sorting order can be changed as shown in [Sorting the events](#)^[172].

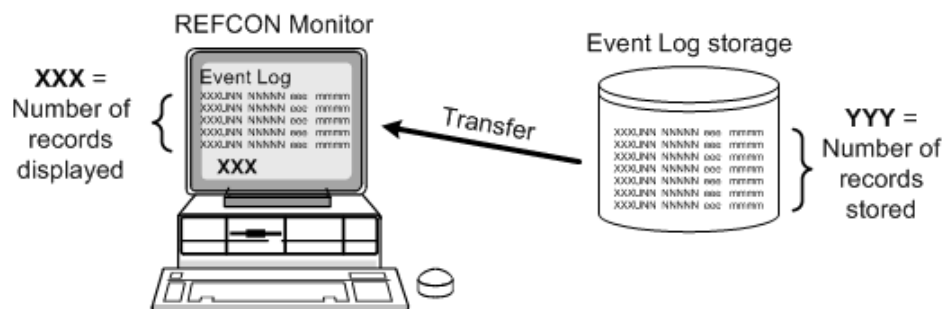
Opening the Event Log

To open the Event Log, click the Events item in the Side panel:



Number of records

In the bar at the bottom of the Event Log, the field "Records in list" shows the number of events XXX that are actually included in the displayed list.



As indicated above, XXX might be smaller than the number YYY of records stored in the event logging storage. This is because the transfer of logs takes some time, and to make the Event Log open quickly, only the newest 1000 events are transferred immediately.

As new events occur while the Event Log picture is open, you will see that the number XXX grows, as there is no reason to remove events once they have been transferred.

If you need to view older events, you can change the setting "No. of logs to show". A change will take effect within a few seconds. Note that if you change to a smaller number, it will have no effect.

Freezing the list

In the bar at the bottom of the Event Log, you may click the "Freeze (stop updating)" button. Doing so makes the button text change to "Unfreeze (resume updating)".

While "frozen", new events will not be added to the displayed list. This could allow you to study a set of events without being disturbed by the list scrolling and the like.

Sorting the events

When opened, the Event Log always shows all events, and it is always sorted chronologically, the newest event being shown first. It is possible to change the sorting, however, the change will be temporary as it can not be saved.

Changing the sorting is done exactly like in the Container List, please refer to [Selecting the sorting order of the containers](#)^[68] and [Selecting a sorting filter](#)^[69].



Column specifications

The Event Log includes the columns described below.

Column header / abbreviation	Explanation
Event description	Together with the following four fields, the Event description shows what happened.
Event type	The category of what happened, see Event types ^[173] .
New value	If relevant, the value/setting that is now active.
Old value	If relevant, the value/setting that was active before the event occurred.
Unit	If relevant, the unit of measure of the New and the Old value.
Item description	Together with the Scope, the Item description shows WHOM or WHAT was affected by the event. If a container: The ID (4 letters + 7 digits) and the location (if known), otherwise the name of the system item.
Scope	The category of whom or what was affected by the event, see Event scopes ^[174] .
Triggering	The category of who or what caused the event, see Triggering types ^[174] .
Trigger	If "Triggering" = User: The user ID of the user doing the action. Otherwise: The hardware or software unit that triggered the event.
UTC time	The UTC time when the event occurred.
LMT time	The LMT time when the event occurred.

Event types

The "Event type" column can show the following types.

Event type	Explanation
Operation	User did send a command or parameter change to a container controller.
Monitor Setup	User did change the behaviour of the REFCON Monitor or one of its pictures or functions.
REFCON Setup	User did change the behaviour of the REFCON system.
Installation	A change occurred during installation.
Export	Data was exported to an external system.
Import	Data was imported from an external system.
State change	The state of a specific function was changed.
Work action	User did perform a standard action as a part of the normal job.
Parameter change	A parameter for a specific function was changed.
Creation	A database object was created, eg. a container object.
Deletion	An object was removed from the database, eg. a container object.
New alarm	An alarm occurred.



Event type	Explanation
Alarm to Normal	The cause of an alarm returned to the normal state. If the alarm had been acknowledged, it disappeared at the same time.
Acknowledge	User did acknowledge an alarm. If it had gone normal, the alarm disappeared at the same time.
Other	An event not covered by the above.

Event scopes

The "Scope" column can show the following types.

Scope type	Explanation
Container	The event affected a specific container, defined by the Item description.
System	The event affected the REFCON system.
Monitor	The event affected the REFCON Monitor.
RDC	The event affected the optional RDC unit.
Planning system	The event affected the planning system, in the optional case that REFCON sends data to this.
Other	A scope not covered by the above.

Triggering types

The "Triggering" column can show the following types.

Triggering type	Explanation
User	The event was caused by a user.
Container	The event was caused by a specific container controller or similar.
Planning system	The event was caused by the planning system.
External	The event was caused by another external system.
System	The event was caused by the REFCON system.
RDC	The event was caused by the optional RDC.

REFCON 6

Reefer Container Monitoring System

Data Transfer for Global Monitoring

Transportation Solutions

COPELAND



10 Data Transfer for Global Monitoring

Key subjects in this topic:

[Background](#)^[176]

[When is your interaction](#)

[required?](#)^[176]

[The Data Transfer Link](#)^[176]

[Checking if your transfer works](#)

[manually](#)^[176]

Related topics:

[GMS Setup](#)^[177]

[Trouble shooting the GMS](#)

[connection](#)^[179]

Background

The Global Monitoring concept makes it possible for container owners and other interests to establish automated and centralized remote monitoring of the reefer containers, covering the whole journey.

The central part of the concept is the **Global Monitoring Server (GMS)**. The GMS is an internet based database that regularly receives data from most REFCON systems worldwide about the connected reefer containers. Also it receives data directly from containers equipped with the RMM+ modem, the RMM-W modem or similar, and from other data sources.

The highest efforts are done to keep the data on the GMS *private*, that is, they can only be accessed by people with legal rights to read them.

Using standard internet PCs, interested parties with legal rights can log in to <http://gms.myreefer.com/ProActTransport> and inspect data. The value of this highly depends on the regularity and the coverage.

When is your interaction required?

If you are a **daily user** of REFCON, you should pay attention to GMS in the following cases:

Case	Your action	Reference
A system alarm regarding GMS appears.	Identify and solve the problem.	Trouble shooting the GMS connection ^[179]

If you are a commissioning responsible, a service technician or similar, you are probably the one being responsible for making the data transfer work automatically or for preparing the manual transfer. In such case, please proceed below.

The Data Transfer Link

For the Global Monitoring concept to work, it is vital that the data transfer link between your REFCON system and the GMS is established and runs well. Data transfer as standard takes place by attaching these files to emails or to a cloud hosted API service.

If the REFCON Main PC has access to the internet, the built-in functionality of REFCON can be used, by selecting any option but the first in [GMS Setup](#)^[177]. When the REFCON Main PC is offline, data is buffered in the system to be sent via email or to the API when the has access to the internet. It is recommended to keep REFCON Main PC connected to the internet constantly in order to synchronize data to the GMS system in real time.

Checking if your transfer works manually

You should normally have been instructed if your manual interaction is required. Further, if data are not received by GMS for a longer time, we would normally contact you to find the reason.



10.1 GMS Setup

Key subjects in this topic:

[Overview](#)¹⁷⁷
[Method for exporting data](#)¹⁷⁷
[Sync via API](#)¹⁷⁸
[Sync interval](#)¹⁷⁸
[Email Server settings](#)¹⁷⁷
[GMS alarm settings](#)¹⁷⁸

Related topics:

[Data Transfer for Global Monitoring](#)¹⁷⁸

Overview

The GMS data transfer is controlled through the GMS tab of the [REFCON Setup](#)³³⁰ dialog box:

Method for exporting data

In the major lower part of the dialog box, you select the exporting method. In the example shown above, there are only two options; however, at your REFCON system there might be a third option as well, as described here:

Method option	Explanation
Deliver by email	Selecting this option enables the "Email Server" tab ¹⁷⁷ shows up in the REFCON Setup ³³⁰ dialog box, if not already visible. Select the email recipient as " logif@reefers-online.com ". Also ensure that the Email Server settings are correct.
Sync via GMS API	Selecting this option is recommend if Secondary PC has consistent connection to the internet in order to export data to the GMS system.
Store in local folder	Selecting this option will allow to collect files files on a user specified file folder.

Email Server settings

The "Email Server" tab is available in the [REFCON Setup](#)³³⁰ dialog box when REFCON is to sent emails for GMS or other purpose:



You need once for all to specify your mail account:

- The **Mail server** is the address of your Internet company's SMTP server.
- The **Port no.** on the mail server.
- The **Security** information fields must be filled in if required by that SMTP server. In some cases these fields must be left blank.
- The **Sender address** is two strings before and after the @ in the email address that will occur in the "From" field when the user receives the email. Note that many mail servers pay restrictions to the "From" domain (string after @), in order to avoid spam mails. The string before the @ is automatically created based on the [installation name](#)^[12].

Having completed and applied the settings you may select "Send Test Email" to check that your data work.

Sync via API

Select the 'Sync via API' method to ensure container data is automatically exported and stored in GMS system. This option will maintaining an up-to-date and unified view of your information in the GMS container data repository. By choosing this, you enable a seamless and secure data flow, making sure container data is consistently synchronized and accessible in the GMS system. The email support to export data will be discontinued in a future release.

Store in local folder

Select this option to store container data in a local folder. User can select the file folder which must have write permissions.

Sync Interval

Users can specify the frequency of exporting data to the GMS system. The default interval is set at 60 minutes, with a minimum value of 60 minutes.

GMS alarm settings

In the ["Alarm Options" tab](#)^[125] in the [REFCON Setup](#)^[330] dialog box, you can enable the "alarm when GMS file sending stops" to check whether your automatic solution is working.



10.2 Trouble shooting the GMS connection

Key subjects in this topic:

[Overview](#) ¹⁷⁹

[System alarm: REFCON](#)

[Communication Service: Device](#)

[went offline](#) ¹⁷⁹

[System alarm: "Failed to connect to connect to the GMS API. Contact support if GMS data is missing"](#) ¹⁷⁹

[System alarm: "Database storage has reached an unhealthy state. Add storage space now to prevent failure"](#) ¹⁷⁹

[System alarm: "Database storage has reached a degraded state. Add storage space soon to prevent errors"](#) ¹⁸⁰

[System alarm: "Jobs Health Check Degraded, one or more jobs have failed. Contact support"](#) ¹⁸⁰

Related topics:

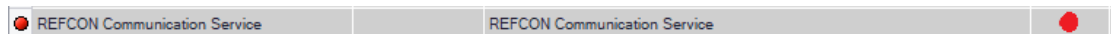
[Data Transfer for Global Monitoring](#) ¹⁷⁶

Overview

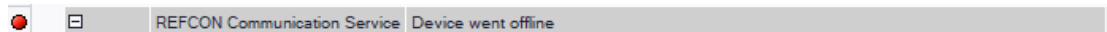
Should an alarm occur that includes the abbreviation GMS in the alarm text, please take action as described below.

System alarm: REFCON Communication Service: Device went offline

Indication in [System Devices List](#) ³³³:



Indication in [System Alarm Summary](#) ¹¹²:



Remedy:

- Restart the REFCON Communication Service from Windows Task Manager by right clicking on the service.
- If this does not solve the issue, restart REFCON Main PC.
- Contact Copeland support if problem persists.

System alarm: "Failed to connect to connect to the GMS API. Contact support if GMS data is missing"

Remedy:

- Verify that REFCON Main PC has connection to the internet.
- Verify that REFCON Main PC HTTP port 80 is open in the firewall.
- Contact Copeland support if problem persists.

System alarm: "Database storage has reached an unhealthy state. Add storage space now to prevent failure"

Remedy:

- Add more disk space to REFCON Main PC.
- Clean up disk space.
- Contact Copeland support if problem persists.



System alarm: "Database storage has reached a degraded state. Add storage space now to prevent failure"

Remedy:

- Add more disk space to REFCON Main PC.
- Clean up disk space.
- Contact Copeland support if problem persists.

System alarm: "Jobs Health Check Degraded, one or more jobs have failed. Contact support"

Remedy:

- Restart REFCON Main PC.
- Contact Copeland support if problem persists.

REFCON 6

Reefer Container Monitoring System

Planning data import



11 Planning data import

Key subjects in this topic:

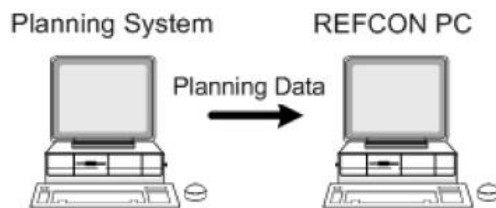
[What is the Planning System?](#)^[182]
[What is Planning data?](#)^[182]
[Why are Planning data important?](#)^[182]
[What is "Planned" and "Unplanned" containers?](#)^[183]
[What is "SiteState"?](#)^[183]
[Possible interface types](#)^[183]
[Understanding the Data Flow](#)^[183]
[Working schedule flexibility](#)^[184]
[When are containers deleted?](#)^[184]
[About "Delete-candidates"](#)^[184]

Related topics:

[The Planning data element in the Side panel](#)^[185]
[Planning data file transfer settings](#)^[187]
[File reading rules](#)^[189]
[Manual Planning data file transfer](#)^[192]
[Planning data file transfer information](#)^[193]
[CSV format editor](#)^[197]

What is the Planning System?

The majority of container companies are using computers - the vessel's load calculator or the terminal's administrative system - to manage the positioning of the containers, and more. We generally refer to such a computer as the *Planning System*. As standard, manual or automatic data transfer should be implemented:



Optionally it is possible to additionally implement customized data transfer in the opposite direction, from REFCON to Planning System.

What is Planning data?

In most cases, the Planning System stores

- the complete list of reefer containers found in the ship/terminal right now
- and, for each container,
- the container ID
- if it is a reefer container
- the expected temperature setpoint
- and other information.

We call this information *Planning data*, opposed to the live data available from the container controllers.

Why are Planning data important?

It is a great advantage to have the Planning data transferred from the Planning System to REFCON. In particular, the following data are valuable:

- The presence of a container, as it allows the user to identify containers that are not updated and thereby potentially are without power.
- The container location, as it allows REFCON to show the containers graphically, see the optional [Container Plan](#)^[96].
- The expected setpoint, as it allows REFCON to check the setpoint actually running in the container controller, see [Actual and Expected setpoints differ](#)^[118].

Other information can be valuable as well for specific purposes.



What is "Planned" and "Unplanned" containers?

We denote a container as **planned** if its data record includes valid Planning data, that is, if information about the container has been received from the Planning System.

We denote a container as **unplanned** if it is not *planned*, that is, if REFCON did **not** receive information about the container from the Planning System. Such a container will always have *live data*, as the data record was created because the container was discovered by the Power Cable Transmission system or by [REFCON 6 extensions](#)^[218].

What is "SiteState"?

The SiteState is the container's state according to the Planning System.

With some of the below mentioned interface types, including file transfer, only two states are possible:

- "On site" means that the container is *planned* (see above).
- "Unplanned" or empty means that the container is *unplanned*.

With certain online solutions, more detailed SiteState information is available. Please refer to the interface user manual for specific information. This manual is normally available from the [Planning data element in the Side panel](#)^[185].

If SiteState is not empty and not reading "Unplanned", the container is generally considered "Planned".

Possible interface types

REFCON includes the following interface options:

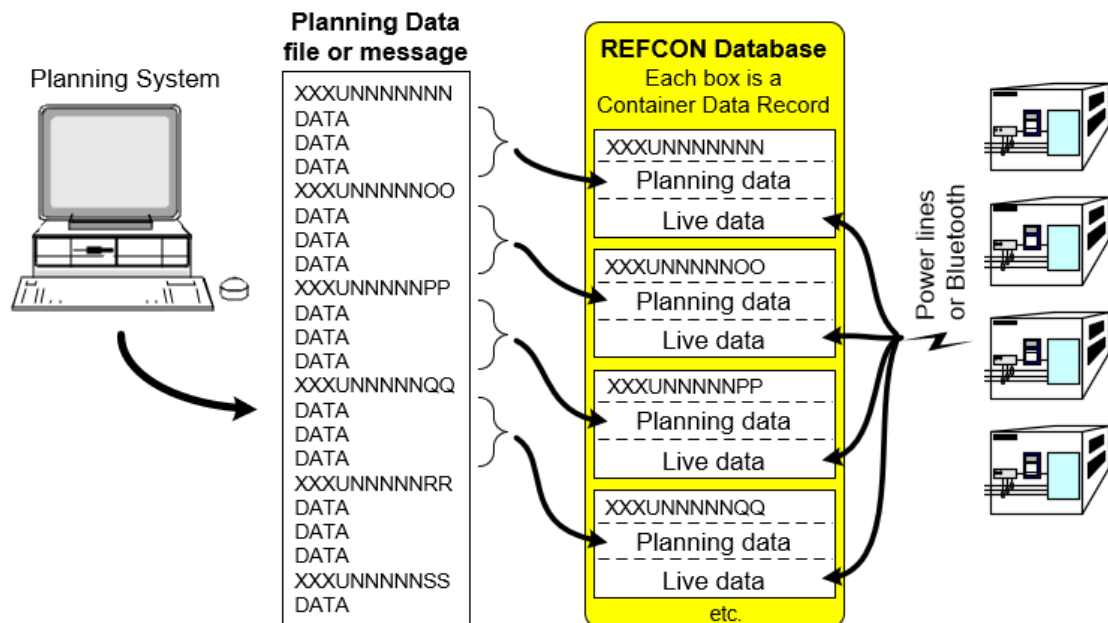
- [Manual Planning data file reading](#)^[192], via removable disk or LAN.
- Optional [Automatic Planning data file reading](#)^[187], via LAN.
- Optional [2-way automatic online interface](#)^[202], via LAN.
- Optional [external application or dedicated plugin](#)^[214] delivered by Copeland Transportation Solutions, interfacing through the [REFCON Data Service](#)^[309].

We strongly recommend that you always select an automatic interface option.

Only the two file transfer types are described in this topic and its sub-topics.

Understanding the Data Flow

The following figure illustrates how data enter the Container Data Record:





- There is one Container Data Record for each container known to REFCON – no matter if it is
- reported by both Planning data and live data, or by only one of them;
 - communicating or not.

Container Data Records are dynamically created and deleted. For example, if a container appears in Planning data and has not been known to REFCON before, a new record is created, and the Planning data is inserted.

Selected data in any Container Data Record can be inspected in the [Container Details](#)^[130] and the [Container List](#)^[66].

Working schedule flexibility

REFCON is highly flexible and allows you to transfer the Planning data file at any time you wish and as often as you wish:

- If the container ID is not known to REFCON in advance, a new Container Data Record is created.
- If the container ID is known to REFCON in advance, it is checked if Planning data have changed. If so, they are updated in the existing Container Data Record.
- If a container ID is no longer in the Planning data File, the record is conditionally deleted as described [below](#)^[184].

In particular, on a ship, you are supposed to transfer Planning data at least once in each port. You can do it

- when arriving to the port, before starting discharging and loading;
- after leaving the port;
- or any time in between.

In the first mentioned case it is likely that a number of containers will keep communicating for hours, maybe even for days, after having disappeared from the Planning data file. These containers are denoted "delete-candidates", as described below. As REFCON assumes it important to keep supervision as long as possible, the delete-candidates are not deleted until they actually stop communicating. And should such a container log in anew (within 3 days), it retains its data, as it could be a temporary plug out.

In the second mentioned case it is likely that you will get a number of unplanned containers while the file has not yet been transferred. These are fully supervised, but the user will not know their locations. Therefore we recommend the first mentioned schedule or a combination.

When are containers deleted?

It is essential that container data records are removed from REFCON's database when they are no longer in use. As standard this is done automatically, and you should normally never use the available manual delete commands.

Planned containers are deleted on basis of the received Planning data as follows:

- If the optional [2-way automatic online interface](#)^[202] or an optional external application or dedicated plugin is running, REFCON receives a specific delete command from the Planning System on every single container. When such a command is received, the container is immediately and unconditionally deleted from the database, and it is canceled in the Power Cable Transmission system. Should this be a fault, and the container is in fact still connected to power, it will start communicating again after up to a couple of hours, and will then appear as an *unplanned* container.
- If Planning data is received via file transfer (manual or automatic), a container is noted as a "delete-candidate" if it is no longer found in the file, see below.

Unplanned containers are deleted on timeout, see [Container Deletion parameters](#)^[331].

About "Delete-candidates"

A *delete-candidate* is a container that is no longer found in the Planning data file.



A delete-candidate is indicated in the REFCON Monitor by having the (*) symbol appended to it's location. For example, if the location is "140184", it now reads "140184(*)" on the screen. It does no longer appear in the Container Plan, but keeps available in Container Lists and the Container Details.

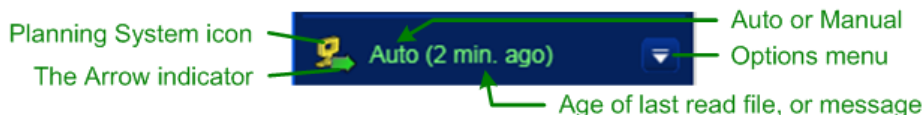
A delete-candidate is automatically deleted as soon as it meets the "delete-conditions". The "delete-conditions" are that the container must be not monitored (that is, it is no longer communicating or has never been) and it must not have any unacknowledged alarms. In other words, if a container is monitored or has one or more unacknowledged alarms at the moment the Planning data file is read, it's deletion is postponed, and it stays in the database.

11.1 The Planning data element in the Side panel

Key subjects in this topic: Overview ^[185] The Arrow indicator ^[185] The text ^[186] Possible selections ^[186]	Related topics: Planning data file transfer settings ^[187] Manual Planning data File transfer ^[192] Planning data file transfer information ^[193] The Planning System Communication Interface (optional) ^[202] Planning data import ^[182]
--	--

Overview

Lower left in the REFCON Monitor you find the Planning data element:



In general, the element

- indicates the overall Planning System interface setting
- indicates the Planning data import status
- gives access to a number of options.

Note that if the optional Planning System Communication Interface is enabled, or similar, the element can occur slightly different than described below. Please refer to the respective parts of the online help manual.

The Arrow indicator

The Arrow occurs as follows:

Arrow occurrence	Description
Ponting to the right only 	Planning data import works by file import, auto or manual.
Pointing left and right 	A two-way Planning System Communication Interface is enabled.
Green, steady arrow 	Auto is selected.
Flashing red/white 	Unacknowledged alarm on Planning System interface.
White, steady arrow 	Manual is selected.
Flashing green/white 	Manual is selected, and a new Planning data file is found.



The text

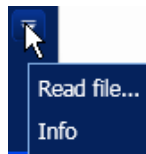
The text occurs as follows:

Text occurrence	Description
Reading "Auto"	The Planning System interface runs in automatic mode. Note that you still might have to take manual action in the Planning System.
Reading "Manual"	The Planning System interface runs in manual mode. You need to take action each time Planning data needs to be updated.
Time duration in parentheses	Time since a Planning data file was last imported, automatically or manually.
Message in parentheses	If you read "data error", the last read file was rejected. Other messages may occur.
Green text	The Planning System interface runs in automatic mode and works fine.
White text	The Planning System interface runs in manual mode, no problems detected.
Flashing red/yellow	Unacknowledged alarm on Planning System interface.
Steady red	Acknowledged alarm on Planning System interface, problem has not been solved.

Possible selections

You can at any time, no matter the mode, click the icon or the text to open the "Read planning data file" dialog box, see [Manual Planning data File transfer](#)¹⁹².

Click the down-arrow button  to display the options menu. The contents of this can vary. The minimum menu is as follows:



Here,

- selecting "Read file..." works like clicking the icon or the text of the element, opening the ["Read planning data file"](#)¹⁹² dialog box;
- selecting "Info" opens the [Planning System Interface Information](#)¹⁹³ dialog box.



11.2 Planning data file transfer settings

Key subjects in this topic:

[Overview](#)¹⁸⁷
[Automatic Planning data file transfer](#)¹⁸⁷
[Planning data file transfer routing](#)¹⁸⁷
[Setting up Planning data file transfer](#)¹⁸⁸
[Automatic \(re-\)creation of folder and share](#)¹⁸⁹
[File import error alarming options](#)¹⁸⁹
[Establishing Online Connection](#)¹⁸⁹

Related topics:

[File reading rules](#)¹⁸⁹
[CSV format editor](#)¹⁹⁷
[Planning data import](#)¹⁸²

Overview

The settings regarding Planning data file transfer comprises

- Selection of automatic file transfer, and settings for this.
- Decisions regarding error handling in the Planning data files.
- Alarm settings regarding file errors and rejection.

Automatic Planning data file transfer

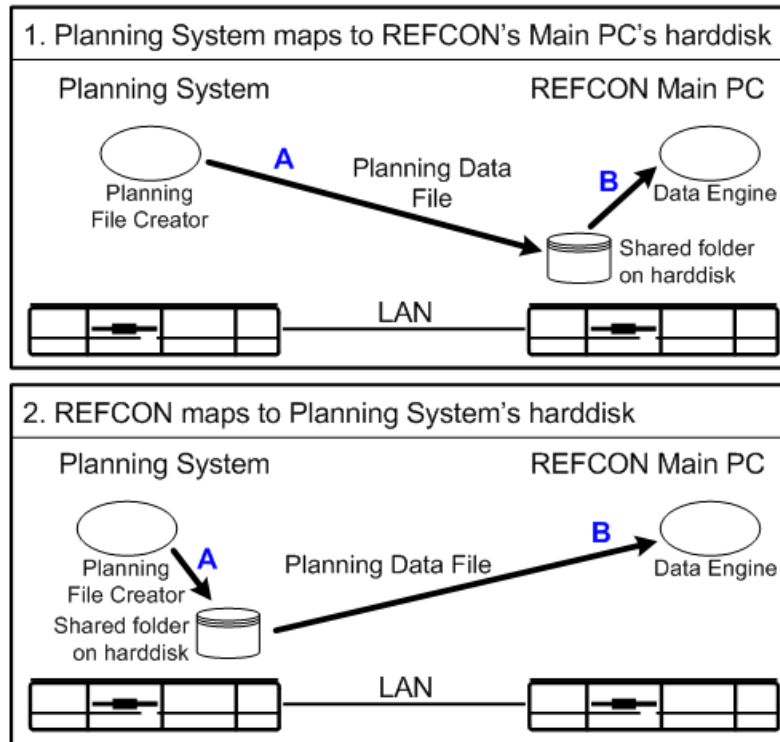
When *Automatic Planning data File Transfer* is enabled, REFCON will periodically look for a file with a specific name in a selected file location. Please be aware that at the Planning System, the file still has to be generated, manually or automatically.

At a certain (small) time interval, REFCON reads the file and checks for changes.

You can **not** select automatic file transfer if any other [automatic planning system interface](#)¹⁸³ is licensed. When not so, we recommend that you always use the Automatic Planning data File transfer, if possible (requires network connection).

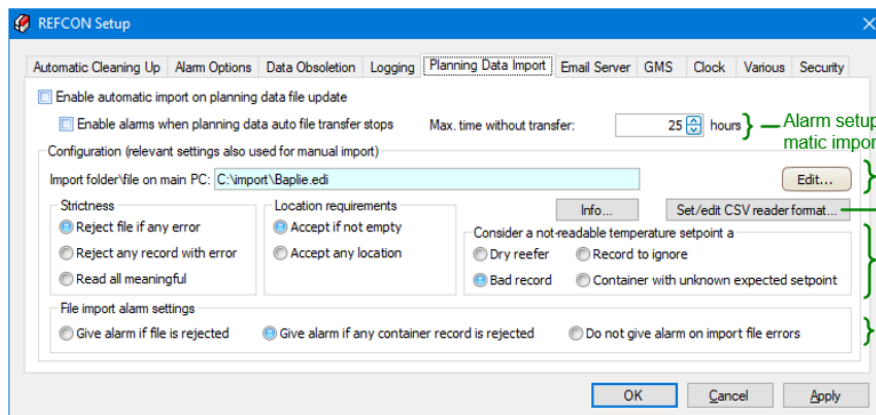
Planning data file transfer routing

When in auto, the [Data Engine](#)³⁰⁶ reads the Planning data file from the Planning System in one of the following ways:



Setting up Planning data file transfer

Planning data file transfer is controlled through the "Planning data Import" tab of the [REFCON Setup](#) ³³⁰ dialog:



Not available if online solution is licensed

Alarm setup, automatic import only

Automatic import settings

File format custom setup

File reading rules setup
Used for both automatic and manual import

Alarming rule setup
Used for both automatic and manual import

To have the automatic file transfer run, make sure the "Enable automatic import..." check-box is checked. Note: This is not possible if any other [automatic planning system interface](#) ¹⁸³ is licensed.

You further need to decide as follows:

- In the "Import folder/file on main PC" field, the selected path for automatic import is shown. To change it, select the Edit button and proceed as described in [Selecting drive and folder on Main PC](#) ⁴⁹. If file transfer routing option 1 (see [above](#) ¹⁸⁷) is used, you should see the "Shared as" field next to the Edit button.
- If a file format is used that is not immediately supported by REFCON, use the "Set/edit CSV reader format" button to define the appropriate reader, see [CSV format editor](#) ¹⁹⁷.
- Select the [File reading rules](#) ¹⁸⁹.
- Select the [File import error alarming option](#) ¹⁸⁹.



- Check the "Enable alarms..." check-box to get an alarm if the automatic Planning data file transfer stops. Carefully select the time limit.

Remember to Apply your changes.

While experimenting with your settings you may wish to select the "Info..." button to inspect the result for the latest file transfer, see [Planning data file transfer information](#)^[193].

Automatic (re-)creation of folder and share

Using the Edit button mentioned above, you can not only point to an existing folder, but you can create a new folder, select sharing and even define a Windows user to access the share.

Each time the [Data Engine](#)^[306] is started, it checks if the folder and it's possible sharing information is exactly as you have chosen. If not so, a background function attempts to re-establish it. This is in particular useful if the [Configuration Export/Import](#)^[46] function has been used to copy your settings from another REFCON system, as in this case the required folder, the possible Windows user and the possible share settings are automatically created.

File import error alarming options

You have the following alarming possibilities in case of errors in a Planning data import file, no matter if you are importing manually or automatically:

Option	Description
Give alarm if file is rejected	If REFCON dismisses the file as a whole, thus using no data at all from the file, the alarm is released. In contrary, if just some data are used, maybe despite of errors, no alarm occurs.
Give alarm if any container record is rejected	If there is one or more errors in the file with the result that data from one or more containers are totally ignored, the alarm is released. This also occurs if the whole file is rejected.
Do not give alarm	No matter if the file is fully or partly rejected, no alarm is released. Still the event is logged in such case.

Establishing Online Connection

Below you find a little information necessary to establish the Local Area Network connection between the Planning System and REFCON.

REFCON is running under Microsoft Windows, and is supporting all communication protocols that Windows is supporting. As network driver, Microsoft's TCP/IP is used. This means that it is not important which operating system the Planning System is using, as long as it can exchange data using TCP/IP and supports file sharing with the Windows version of the REFCON Main PC.

The communication hardware is standard Ethernet.

11.3 File reading rules

Key subjects in this topic:

[Overview](#)^[196]

[User selected rules](#)^[196]

[Requirements to the single container record](#)^[196]

[Requirements to the import file](#)^[197]

Related topics:

[CSV format editor](#)^[197]

[Planning data file transfer settings](#)^[187]

[Planning data import](#)^[182]



Overview

The ideal situation is that all Planning data files are error free, that is, the Planning System produces files that are always fully readable by REFCON. Though this ideal situation is also the real situation in many cases, experience has shown that there are often "errors" in the files. An "error" could, for instance, be caused by a simple typing error in the Planning System (or in the manifest imported to the Planning System).

Further, REFCON supports a number of file formats, including the [customised CSV format](#)^[197]. On the Planning System side it must be decided, once for all or repeated, which format to use. REFCON automatically detects the format, which in case of errors can be difficult or even impossible.

In the following it is explained how REFCON handles file errors. The handling is based on a set of rules. Some of these rules are selected by the user.

User selected rules

In the [REFCON Setup](#)^[188] you find the rule settings:

How these rules apply is explained below.

Requirements to the single container record

Any Planning data file is a list of containers. For each container there is a set of data - this is what we call a container record.

A specific container record can be in one (and only one) of the following "states":

State	Description
OK	The container ID is OK, and all required data fields are found and fulfill the conditions.
Partly OK	The container ID is OK, and at least one other required data field is OK, but other required fields are not.
Bad	A record is considered "Bad" in any of the following cases: • The record is wrong format. • The container ID is missing or incorrect. • None of the required data fields are OK. • The temperature setpoint is not-readable and the "Bad record" rule option has been chosen.
Useless	Like state "OK", but the temperature setpoint is not-readable and the "Record to ignore" rule option has been chosen.

The following data fields are or can be required:

Data field	Conditions
Container ID	The container ID is accepted if (and only if) it consists of exactly 4 letters followed by exactly 7 digits. <i>For example, if the check digit (= digit 7) has been omitted in the Planning System, the record is considered Bad.</i>
Location	Decided by the user selected rule "Location requirements":



Data field	Conditions
	• If "Accept if not empty" is selected, it is required that a location string is found, and this string is not allowed to be spaces only. If this condition is not met, the container record can not be considered OK. • If "Accept any location" is selected, location is not a required field. It is, however, used if a non-empty string is read, and the record is used.
Temperature setpoint	If a readable decimal value is found, it is range checked: • If the value is within the range -99°C to 99°C, the field is OK, and the record is then considered OK if the above conditions are met. • If the value is outside this range, it is considered not readable and is then handled as follows: If the setpoint is not found or not readable, the handling is decided by the user selected rule "Consider a not-readable temperature setpoint a": • "Dry reefer": The field is OK. If the record is used, the container is created in REFCON's database without any expected setpoint. • "Bad record": The entire record is considered bad. • "Record to ignore": The entire record is considered not existing. • "Container with unknown expected setpoint": The field is OK. If the record is used, the container is created in REFCON's database with an expected setpoint having the value "unknown".

Requirements to the import file

REFCON has a number of *format readers*, one for each of the supported file formats. Each time a Planning data import file is read, the format readers are tried one by one. The outcome of a trial is

- Fully applicable - at least one container record was identified, and all container records were OK (as defined above).
- Partly applicable - at least one container record was identified that was at least Partly OK, but also there was at least one error.
- Not applicable - neither of the above.

In practice, all format readers but one will be Not applicable.

If all format readers are Not applicable, the file is rejected.

Normally one format reader is Fully or Partly applicable. If it is Fully applicable, data are immediately read into REFCON's database.

If the format reader is Partly applicable, the handling depends on the user selected "Strictness":

Strictness	If file is Partly applicable:
Reject file if any error	The file is rejected.
Reject any record with error	If there are one or more records that are OK (as defined above), these records are used, and the file is accepted, otherwise the file is rejected. All records that are not fully OK are ignored.
Read all meaningful	If there are one or more records that are OK or Partly OK (as defined above), the useful parts of these records are used, and the file is accepted, otherwise the file is rejected. All records that are not OK or Partly OK are ignored.



11.4 Manual Planning data file transfer

Key subjects in this topic:

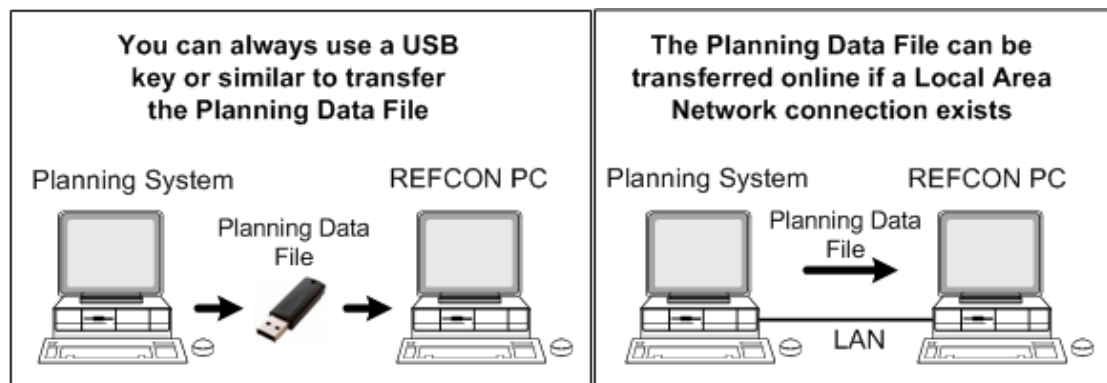
[Overview](#)^[192]
[File transfer procedure](#)^[192]
[Using the Planning data import dialog](#)^[193]

Related topics:

[CSV format editor](#)^[197]
[Planning data file transfer settings](#)^[187]
[File reading rules](#)^[189]
[Planning data file transfer information](#)^[195]
[Planning data import](#)^[182]

Overview

The Planning data file must be transferred to the REFCON. If [automatic transfer](#)^[187] is not enabled, you need to do it manually. You can use a USB key or, if available at your specific REFCON system, the LAN network connection:



File transfer procedure

To have the Planning data transferred to REFCON, you must proceed as follows:

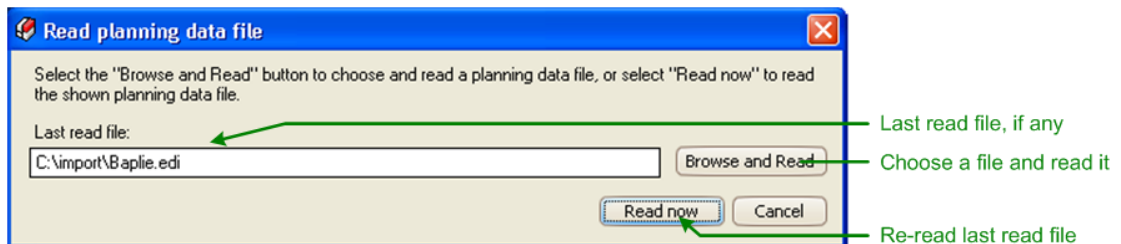
Step	Action	Comments
0	Decide the format of the transferred file (normally this is done once for all).	The file must be in the decided format, which in the most cases is called Baplie 1.5 or Baplie 2.0 (specified by the SMDG group, see www.smdg.org). Also it can be a comma-separated file (csv) in a customized format, which must be configured before you can proceed, see CSV format editor ^[197] .
1	At the Planning System, execute the command that stores the Planning data as a file.	The file must be stored at an appropriate location, which can be a USB key or a network folder, accessible from both REFCON and the Planning System.
→	If you have enabled the "Automatic Planning data File Transfer" function, don't do more.	Shortly after step 1, the file is automatically imported, see Automatic Planning data File Transfer ^[187] .
2	If using a USB key, move it to the REFCON Monitor PC.	That is, the REFCON Monitor PC where you are working.



Step	Action	Comments
3	Open the Planning data import dialog.	See The Planning data element in the Side panel ^[186] .
4	Select the file stored in step 1 and read it.	See Using the Planning data import Dialog ^[193] . The last used location and filename appears automatically.
5	If you have any reason to doubt that the data transfer was successful, inspect the Status.	Does the number of updated containers match your expectation? See Viewing the Import Status ^[193] .

Using the Planning data import dialog

The Planning data import dialog is used to retrieve a Planning data file manually:



If you previous have read a Planning data file, and a file with same name (but maybe newer) still exists in the same folder, it is indicated in the "Last read file" field. You can in this case simply select the "Read now" button to re-read it immediately.

If no such file exists, or you want to read another file, select the "Browse and Read" button and use the standard Windows file-open dialog box to locate the file on the harddisk (of the PC you are working at), a USB key or on the network.

Having selected the file, the file import immediately starts. You will see the import progress being indicated in the [The Planning data element in the Side panel](#)^[185].

11.5 Planning data file transfer information

Key subjects in this topic:

[Overview](#)^[193]
[Understanding the Import Status](#)^[194]
[File rejection](#)^[195]
[Why rejected?](#)^[195]
[Example of file with a few errors](#)^[195]

Related topics:

[Planning data file transfer settings](#)^[187]
[File reading rules](#)^[189]
[Planning data import](#)^[182]

Overview

You can at any time select the Info command from the [options menu](#)^[186] or the [REFCON Setup](#)^[188] to watch the result of the latest Planning data file import:



Planning System Interface Information

File Import

Status

Automatic import enabled!

Last imported file..... C:\import\Baple.edi

Last import at..... 2012-06-21 12:27 UTC

Imported.....	552	Updated.....	20
Delete-candidates.....	496	Rejected due to error.....	0
Not imported.....	0	Dry reefers included.....	

Strictness

☒ Reject file if any error

☐ Reject any record with error

☐ Read all meaningful

Location requirements

☒ Accept if not empty

☐ Accept any location

Consider a not-readable temperature setpoint a

☐ Dry reefer

☒ Bad record

☐ Record to ignore

☐ Container with unknown expected setpoint

Manual import alarm settings

☐ Give alarm if file is rejected

☒ Give alarm if any container record is rejected

☐ Do not give alarm

Setup... **Open REFCON Setup** Close

The "Status" area shows the result of the latest import. The fields are explained below, see [Understanding the Import Status](#)^[194]. If the file import went wrong, this is clearly indicated, see [File rejection](#)^[195].

Understanding the Import Status

The Status information shown in the Planning System Interface Information dialog box relates to the latest automatic or manual file import and can be interpreted as follows:

Field name	Description
Last imported at	The UTC date and time (year-month-day hour:second) when an import file was last successfully read. If the last file was rejected, this is indicated in stead, and a button appears allowing you to inspect the import log, see below.
Imported	The number of containers that were identified as new reefers and therefore caused container records to be created, including dry reefers if so decided.
Updated	The number of reefer containers that did already exist in REFCON's database. If data were changed, the corresponding records were updated.
Delete-candidates	The number of containers that were no longer in the file and therefore were deleted if the conditions ^[184] were met.
Rejected due to error	The number of unreadable container records in the file.
Not imported	If the "Record to ignore" rule option is selected, this count shows the number of not used records (typically dry reefers) identified in the file; otherwise this count is 0.
Dry reefers included	Yes or No according to your rule selection.



File rejection

If and when a Planning data import file is rejected (according to the [File reading rules](#)^[189]), the following things happen:

- An alarm is released, if so selected by the user.
- The event is logged.
- In the [Planning System Interface Information](#)^[193] dialog box, the situation is indicated as follows:

Status

Last import at.....	Last file rejected		
Imported.....	0	Updated.....	0
Delete-candidates.....		Rejected due to error.....	0
Not imported.....		Dry reefers included.....	

Last file rejected due to errors

Show log..

Here you can select "Show log" to inspect the import log as explained in the following.

Why rejected?

As explained under [File reading rules](#)^[191], generally there are two possible reasons why a file is rejected:

- either because **all** format readers are Not applicable - which is displayed as follows in the log viewer:

Planning data file import log

File name: C:\import\baplie.edi

Used reader: No suitable reader found

Total no. of records: 0 No. of error-free records: 0 No. of records with useful data: 0

Error report (first max. 5 error records):

REFCON has no reader matching the selected import file

Top of imported file (for your information):

Close

- or because the relevant format reader is Partly applicable but rejected due to the user selected rules.

Example of file with a few errors

In this example, a sample file in Baplie format was read, in which two errors were found. Due to highest Strictness, the file was rejected. The log showed the situation like this:



Planning data file import log

File name: C:\import\baplie.edi

Used reader: BaplieReader

Total no. of records: 12 No. of error-free records: 10 No. of records with useful data: 0

Error report (first max. 5 error records):

Error when applying Baplie reader to file

Errors detected in following record
 LOC+147+0060682::5'FTX+AAA+++Place of delivery:
 FRLEH'MEA+WT++KGM:28600'LOC+6+BR6D3'LOC+12+FRLEH'RFF+ET:0020682'EQD+CN+MWCU6747431+4532+++5'NA
 D+CA+MAE:172:20'
 Error description: Unable to locate temperature setpoint ("TMP+2+")

Top of imported file (for your information):

UNB+UNOA:1+Lica M□rsk+Unknown+060326:1036+0603261036+++++MSK'
 UNH+0603261036+BAPLIE:1:911:UN:SMDG15'
 BGM++LOADSTAR+9'
 DTM+137:0603261036:201'
 TDT+20+0517++OWAY2:103::Lica M□rsk:DK++275:172:20'
 LOC+5+E5ALR'
 DTM+178:0603261036:201'

Close

At the top of the log viewer, the File name of the tested file is shown for your information.

The "Used reader" shows the name of the identified Partly applicable [format reader](#)^[191].

Below this three counters are shown:

- The "Total no. of records" shows the number of records in the file, of which at least one has errors. Often this count would match the current total number of reefer containers according to the Planning System - is it so?
- The "No. of error-free records" shows the number of container records that were considered fully OK according to the [rules](#)^[190].
- The difference between the above two counts is the number of records with errors. The "No. of records with useful data" shows how many of these were considered Partly OK according to the [rules](#)^[190]. Of course, error-free records are certainly useful, too, but these are not included in the third counter.

Below the counters you see two scroll-able text fields.

The second text field, named "Top of imported file", simply is a copy of the first many lines of the file in question, for your recognition.

The first text field, named "Error report", attempts to visualise the errors. The list has been limited to max. 5 errors - if there are more you would hardly inspect all the details, anyway.

For all format readers but the Baplie reader, the Error report is relatively clear and easy to survey, because each record is one line in the file.

But in our example, the Baplie format was used, in which each record is several lines with a very special syntax. Let's try reading it anyway:

Location identifier

Location data field

Start of log

Container ID

Error report (first max. 5 error records):

Error when applying Baplie reader to file

Start of error record

Errors detected in following record

LOC+147+0060682::5'FTX+AAA+++Place of delivery:
 FRLEH'MEA+WT++KGM:28600'LOC+6+BR6D3'LOC+12+FRLEH'RFF+ET:0020682'EQD+CN+MWCU6747431+4532+++5'NA
 D+CA+MAE:172:20'
 Error description: Unable to locate temperature setpoint ("TMP+2+")

Explanation of the error

The Baplie code (missing in this case)



Here is one more example:

Location identifier

Location data field

Start of error record

The error: Container ID has only 6 digits

Errors detected in following record

LOC+147+0100182+5'FTX+AAA+++Place of delivery: FRLEH'MEA+WT++KGM:28600'TMP+2+-18.U:CEL'LOC+6+BR6D3'LOC+12+FRLEH'RFF+ET:0020682'EQD+CN+MWCU747837+4532+++5'NAD+CA+MAE:172:20'

Error description: Missing or invalid container id after 'EQD+CN+' - Actual text: EQD+CN+MWCU747837+4532+++5

Explanation of the error

The Baplie code

11.6 CSV format editor

Key subjects in this topic: Overview ¹⁹⁷ Opening the CSV format editor ¹⁹⁷ Automatic format detection ¹⁹⁷ Example of fully configured CSV format editor ¹⁹⁷ How to configure a CSV format ¹⁹⁸ Understanding the parts of the CSV format editor ¹⁹⁹ Columns in Mapping grid ²⁰⁰	Related topics: Planning data file transfer settings ¹⁸⁷ Planning data import ¹⁸²
--	--

Overview

Depending on the type of the Planning System, it might be expedient that the Planning data file is in a proprietary comma-separated format. In such case, REFCON needs to know the exact specifications of the format, for which purpose you need to use the *CSV format editor*.

A CSV-formatted Planning data file is supposed to include one line for each container. The line includes a number of data source columns, separated by tab characters, commas, semi-colons or another selected character (despite of the name "comma-separated"). So what you mainly configure in the CSV format editor is: what is the separator character, also called the delimiter, and what is the meaning of each individual data source column.

Opening the CSV format editor

To open the editor, click the "Set/edit CSV reader format..." button in the REFCON Setup dialog, see [Setting up Automatic Planning data File Transfer](#) ¹⁸⁸.

Automatic format detection

Each time REFCON reads a Planning data file, it automatically attempts to recognize the format. In principle, it tries the available known formats one by one and checks if the parsing of the file makes sense.

In particular, when checking a CSV format, the following rules apply:

- A container line is only valid if all data source columns are parsed correctly, that is, the number of separator characters is as specified, and the values in between are reasonable (but not necessarily valid).
- If more than 50% of the container lines are somehow in-valid, the file-type identification fails, and then none of the values in the file are imported (unless another format succeeds).
- Note that every single valid data source column in a container line is imported, i.e. a single in-valid value does not result in skipping all the other container line-values.

Example of fully configured CSV format editor

The following diagram shows a fully configured format:



CSV format editor: Guessed format for navis.txt *

Name : Save

Test file : Select file...

Delimiter : Header lines : Footer lines : Encoding : Guess?

Data source column numbers Preview

1	2	3	4	5	6
Equipment ID	Current Pos...	Type	Temp°C	Outbound ...	Inbound Carrier
BCHU1865172	008R25A	4532	-24,0°C	TRUCK	BLI833
CRLU1111566	010R27A	4532	-20,0°C	TRUCK	TRUCK
CRLU1116362	009R20B	4532	-24,0°C	TRUCK	BLI815

Mapping of data source column to target data point

Source column	Search	Replace	DataPoint name	DataPoint Unit	DataPoint Kind	DataPoint Type
1			* Container ID *			System.String
4 °C			Temperature Expected setpoint	Celcius	Temperature	System.Double
2			Location	None	None	System.String
6			Inbound	None	None	System.String
5			Outbound	None	None	System.String

Preview import Preview

* Container ID *	Location	Container size	Temperature Expected setpoint	Outbound	Inbound	Note (free text)
BCHU1865172	008R25A	4532	-24	TRUCK	BLI833	
CRLU1111566	010R27A	4532	-20	TRUCK	TRUCK	
CRLU1116362	009R20B	4532	-24	TRUCK	BLI815	

How to configure a CSV format

Assuming that no format has been configured in advance, we recommend that you proceed as follows:

Step	Action	Result
1	Click "Select file..." and open a (sample) Planning data file of the decided format.	Path and file name occurs in Test file field
2	Click "Guess?"	- Name field indicates what was done. - Delimiter and succeeding field shows the determined format.
3	If either Delimiter, Header/Footer lines or Encoding is not as expected, make appropriate changes.	Reflects your changes.
4	Click either of the "Preview" buttons.	- Grid 1 (the uppermost table) shows the parsed content of the test file. - If any header or footer lines exist, errors are probably indicated. Ignore these if the lines are in fact headers/footers. - Grid 3 shows a preliminary mapping result, which is most likely useless.
5	Make changes to the Source column of Grid 2, using the drop-down lists. Also maybe make changes to the Search and/or Replace columns.	Grid 2 rows mapped to relevant grid 1 columns.



Step	Action	Result
6	Click either of the "Preview" buttons.	Grid 3 should now reflect data from the file positioned at appropriate data points (you may find the column header tooltip useful to see the full name of the data point).
7	Repeat steps 3 - 5 as required if any errors are indicated in grid 3, or the result in grid 3 is not as expected.	

Understanding the parts of the CSV format editor

Referring to the above figure, the following tables explain the individual fields of the editor:

Option	Option description
Name	The Name has no direct impact on the logic, but it is an appropriate place to capture a reference back to a defined export-list and possible format-version information.
Test file	The file is used during the format definition, and to test and verify that files can be parsed.
Delimiter	The separator character used between data source columns.
Header and footer lines	The number of lines to exclude/ignore during every import.
Encoding	Must be selected, to handle old DOS-text-encoding-formats like codepage 850 and special characters like Turkish 'i' or Danish 'ø' etc.

Button	Button description
Save	Saves the changes. An un-saved change is indicated in the form's title bar by a trailing *.
Save → Delete	Deletes the format, and no custom CSV-format parsing will be attempted by REFCON.
Select file...	Opens a file-selection dialog for the Test file.
Guess?	Tries to determine the format of the selected Test file, i.e. the delimiter, and also if possible the header lines, footer lines, and the encoding. Please note that it is only a guess, so you must verify the choices your self. In most cases the guess must be corrected!
Preview	Updates all the grids according to the selected options, i.e. with the parsed content of the Test file.

Grid (= table)	Button description
1. Data source column numbers	Shows all the delimited lines of the Test file, i.e. unfiltered content that is only tokenized according to the selected delimiter.
2. Mapping of data source column to target data point	Shows the possible target data points for the import, i.e. container id, expected set points, and text fields. The columns of this grid are described below.



Grid (= table)	Button description
3. Preview import	Shows the resulting import, with the chosen data source columns to be mapped, with the input values searched and replaced and potential error-indications, like trying to parse a free text column as a type of double. Every failed container line-parsing is indicated at the start of the row and, if possible, at the failing cell like this: 

Columns in Mapping grid

The columns in the grid "Mapping of data source column to target data point" can be described as follows:

Column	Description
Source column	The values in this column map to the column numbers in the "Data source column numbers" grid. Choose from the drop-down list to change.
Search and Replace	Is a very simple transformation option, which helps to change the input value before it is stored in REFCON. For example it can trim away a trailing "°C" before the decimal value is parsed into REFCON as a set point.
DataPoint columns	All the columns starting with "DataPoint..." are just information to help you choose, e.g. the right Search and Replace values, and that you do not import Fahrenheit degrees into a data point expecting Celsius degrees.

REFCON 6

Reefer Container Monitoring System

The Planning System Communication Interface

Transportation Solutions

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12 The Planning System Communication Interface

Key subjects in this topic:

[Overview](#)^[202]
[Purposes of the Planning System Interface](#)^[202]
[Communication status](#)^[202]
[The options menu](#)^[203]
[The Re-read all command](#)^[203]
[The Send all command](#)^[203]
[Overview of transferred data](#)^[203]
[Data transferred to REFCON](#)^[203]
[Data transferred from REFCON](#)^[204]
[Planning System Interface failure alarm](#)^[204]

Related topics:

[Planning System Interface Setup](#)^[204]
[Trouble-shooting the interface](#)^[207]
[Planning System Interface Live Monitor](#)^[210]

Overview

Your This REFCON system might include includes the *Planning System Communication Interface* option. The option is relevant in container terminals and similar places where reefer containers frequently come and go (in contrary to container ships).

To make the Planning System Communication Interface work, the Planning System and the REFCON Main PC must be connected by an ethernet LAN:



When properly connected and software configured, the interface runs automatically. You will be warned by an alarm should the connection fail, please refer to [Trouble-shooting the interface](#)^[207].

Purposes of the Planning System Interface

The basic idea of the Planning System Communication Interface is to automatically keep data updated on both sides:

- The Planning System continuously gets live data from the containers.
- REFCON continuously gets the data necessary to check the containers and to display them graphically (Container Plan).

Please refer to the Planning System for instructions on how to send the below messages. Be aware if it runs automatically, or you have to do things manually.

On the REFCON side, data transfer runs automatically, however, you can trig a complete re-transfer of data as described below.

Communication status

In the Side panel you can at any time see the status of the Planning System Communication Interface:



In the left example, the interface runs well - to the right it does not so.

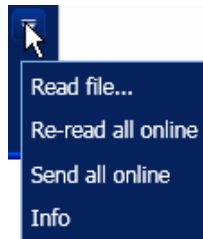


Note that if the Planning System Interface is not enabled, [Planning data Import](#)^[182] status is shown in stead.

You will notice that the arrow in the icon points both ways only when the Planning System Interface is enabled.

The options menu

Click the  button to have the options menu displayed:



Here,

- The "Read file..." can be used to import a [Planning data file manually](#)^[192]. This might be useful should the automatic interface fail.
- The ["Re-read all online"](#)^[203] and the ["Send all online"](#)^[203] commands are described below.
- The "Info" command displays the [Planning System Interface Live Monitor](#)^[210].

The Re-read all command

Though data are transferred automatically from the Planning System to REFCON whenever changes occur, you can trig a full re-transfer by selecting the "Re-read all online" command in the [options menu](#)^[203].

Having entered the command, and accepted the warning, a request is sent from REFCON to the Planning System, which should then start re-sending data for all [planned](#)^[183] containers, one by one. This may take a few minutes to complete.

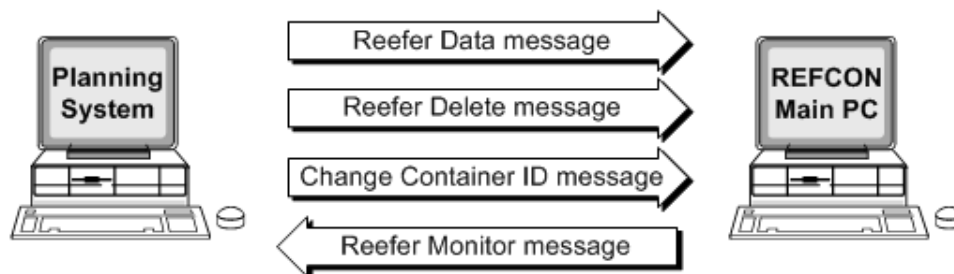
The Send all command

Use the "Send all online" command in the [options menu](#)^[203] to trig that Reefer Monitor messages (see [below](#)^[204]) are sent to the Planning System for every single reefer container in REFCON's database.

Note that normally this is not necessary if [Timer based data transmission](#)^[205] has been selected.

Overview of transferred data

The Planning System Communication Interface transfers data from the Planning System to REFCON and vice versa. The primary data messages are:



Data transferred to REFCON

From the Planning System, the following information can be sent:



Name	Description of main contents
Reefer Data message	General container information, including ID, location and expected setpoint.
Reefer Delete message	Command to remove a certain container from the REFCON database.
Change Container ID message	Command to change the ID of a reported container. Note that this will never make any change in the container controller – the message is only used to change the ID in the REFCON database, and it only works if no communication is or has been running with a container with the "old" ID.
Request message	Request for Reefer Monitor message (see below).

Data transferred from REFCON

From REFCON, the following information can be sent:

Name	Description of main contents
Reefer Monitor message	Online data for specific containers, including alarm status, temperature readings and actual setpoints. REFCON sends this message in the following situations: • In case of a new alarm on the specific container. Only one message is sent (though of course, if more containers have alarms simultaneously, it will be more messages). • Periodically if so decided by the REFCON user (timer based transmission^[205]). One message for every single container in REFCON's database is sent, up to 8 times a day. • As response to the request message (mentioned above) - one response message for each request message.
Request message	Request for Reefer Data message (see above).

Planning System Interface failure alarm

Should the connection to the Planning System fail, an alarm is released.

12.1 Planning System Interface Setup

Key subjects in this topic:

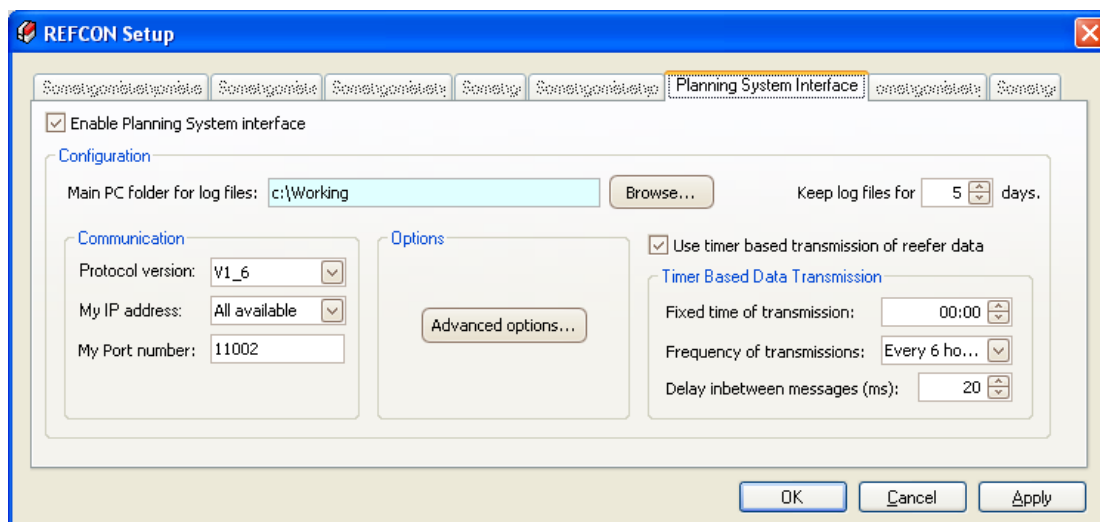
[Overview](#)^[204]
[Preparing the Planning System](#)^[205]
[Communication setup](#)^[205]
[Timer based data transmission](#)^[205]
[Transmission log files](#)^[206]
[Advanced options](#)^[206]

Related topics:

[Planning System Interface Live Monitor](#)^[210]
[The Planning System Communication Interface](#)^[202]

Overview

The Planning System Communication Interface is controlled through the "Planning System Interface" tab of the [REFCON Setup](#)^[330] dialog:



To have the Planning System Communication Interface work, first of all make sure the "Enable" checkbox is checked. Also decide the configuration as described in the following.

Preparing the Planning System

The Planning System must be prepared for this interface. The separate *Planning System Interface Programmers Guide* describes how to do this preparation. The Guide is available on www.myREFCON.com, but it requires a password that can be requested through the email-address shown under [Contact Information](#).

Communication setup

These three settings are vital to make the communication work:

Setting	Explanation
Protocol version	What protocol version to select is normally agreed between the parties when the REFCON system was planned. V1_6 is the newest protocol. If a wrong protocol is selected here, the Planning System and REFCON is likely to misunderstand the data from each other.
My IP address	Here you would normally select "All available", however, if there is more than one LAN card (internally or externally) in the REFCON Main PC, it might be preferred to select a specific card.
My Port number	The Port number is agreed between the parties. If the selected Port number does not match the number used by the Planning System, the communication will not work.

Timer based data transmission

The Reefer Monitor message is automatically sent in case of a new alarm, but not in any other case, such as measured data changing. If you want such data periodically updated in the Planning System, check "Use timer based transmission of reefer data" and select the parameters:

Setting	Explanation
Fixed time of transmission	Here you select when to have the data transmitted. For example, if a new crew is meeting at 7 o'clock, you might select the fixed time 6.45 to provide fresh data (note that the total transmission may take some time).
Frequency of transmissions	Here you select how often the data are transmitted. For example, selecting "Every 6 hours" and the fixed time 6.45 would start the transmission at 0.45,



Setting	Explanation
	6.45, 12.45 and 18.45.
Delay between messages	Normally leave this value as it is, however, if you experience troubles (like data for some containers are not updated in the Planning System), try to increase the value. Be careful not to select a very high value. For example, if you select 500 milliseconds, and you have 2000 reefer containers, the total transmission time will be at least 1000 seconds - such a long time would hardly be expedient.

Transmission log files

REFCON logs all messages transmitted on the interface, in both directions. You may select the parameters:

Setting	Explanation
Main PC folder for log files	The log files are stored on the REFCON Main PC's harddisk. You can select the specific folder by clicking Browse, see Selecting drive and folder on Main PC ⁴⁹ .
Keep log files for...	The purpose of the log files is trouble shooting. While everything runs well, they are needless and just occupies space on the harddisk. But when one day troubles should occur, you will need the log files covering a time range starting before things went wrong. The log files are as default kept for 5 days, leaving you 4 days to discover and act on the troubles. If you think 4 days is not enough, you may increase the value.

Advanced options

Selecting "Advanced options" opens the following self-explaining dialog box:



Advanced options for Planning System Interface

Send options

- ☒ **Send REEFER_MONITOR messages on new container alarms**
If this box is checked, a REEFER_MONITOR message is sent to the Planning System every time a new container alarm occurs.
- ☐ **Do not send REEFER_MONITOR messages for modem-offline containers**
If this box is checked, REEFER_MONITOR messages are never sent for containers which are currently offline (but do have modems).
- ☐ **Do not send REEFER_MONITOR messages for no-modem containers**
If this box is checked, REEFER_MONITOR messages are never sent for containers not known to have a modem.

REEFER_MONITOR message time stamping

Use ☐ UTC ☒ LMT

Receive options

- ☐ **Don't accept REEFER_DATA messages with blank or unknown locations**
If this box is checked, incoming REEFER_DATA messages are ignored if the YARD_LOCATION field of the message does not match any location defined in the container plan (this also applies if the field is blank).

REEFER_DELETE handling

- ☒ **Immediate deletion**
The specified container is deleted from REFCON immediately and unconditionally upon reception of the REEFER_DELETE message.
- ☐ **Delayed deletion (not possible yet)**
REFCON awaits the specified container to go offline and have all its alarms acknowledged before it is deleted.

OK Cancel

12.2 Troubleshooting the interface

Key subjects in this topic:

[Overview](#)^[207]
[Starting Up](#)^[208]
[If Communication Stops](#)^[208]
[Not All Old Containers Are Deleted](#)^[208]
[Change Container ID does not work](#)^[209]
[Request for Reefer Monitor message does not work](#)^[210]

Related topics:

[The Planning System Communication Interface](#)^[202]

Overview

The Planning System Communication Interface is a vital connection. Should it stop working, it is important that you take action. Here is a few hints.



Starting Up

When starting up the first time, or starting up after a break e.g. for service, the communication to and from the Planning System must be checked. The check must be done to be sure that the interface is working all right.

The following can be done for checking the communication line:

1. Check that the communication is enabled, in the [Setup dialog box](#)^[204].
2. Inspect the [Planning System Interface Live Monitor](#)^[210]. Does it show OK upper right?
3. Try sending a message from the Planning System. You should be able to identify the message received by REFCON in the Live Monitor.

Checking that messages can be sent to the Planning System:

You may use the "[User Alarm](#)^[124]" function to create an alarm on a container. If enabled, this will cause a message to be sent to the Planning System. The [Planning System Interface Live Monitor](#)^[210] will show that the message has been sent, however, you also need to check that the message is received at the Planning System.

In both above cases, if the messages do not work, please check the Planning System, the LAN connection and the [setup](#)^[204].

If Communication Stops

If the connection to the Planning System has been working well, and it is then disabled or fails, and this situation remains for a long time, you will have to take care of the deletion of the old containers in REFCON. Otherwise the REFCON database will be filled up with old containers. The deleting must be done manually from the Container List. Note that containers having been reported by the Planning System are [planned](#)^[183] and will therefore not be deleted by the [automatic clean-up function](#)^[331].

Please note; in the situation that the connection is disabled by user's choice, **no alarm** will appear.

Not All Old Containers Are Deleted

It will happen that containers communicating with REFCON over the power cables do have container IDs different from the ones expected by the Planning System. The result of such mismatches is that the Reefer Delete message doesn't work on the specific containers. Such containers are [unplanned](#)^[183] and will therefore be deleted by the [automatic clean-up function](#)^[331].

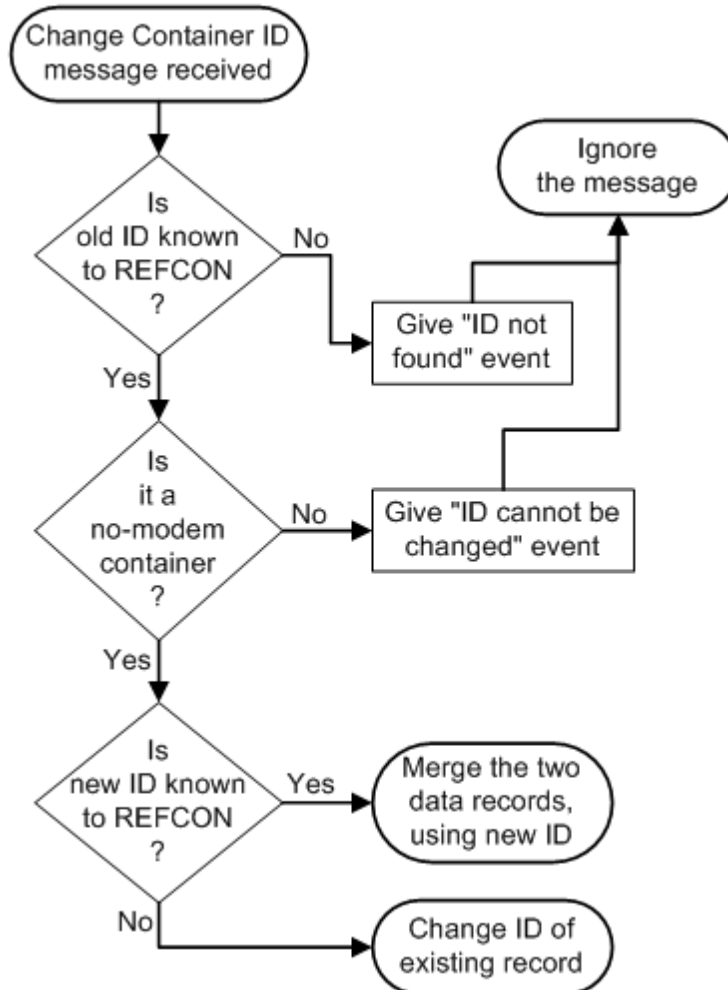
If the ID mismatch is due to a fault on the container, this should be corrected. Please refer to the description of "ghost containers".



Change Container ID does not work

The "Change Container ID" message is sent from the Planning System to REFCON, one message for each container. Note that this will **never** make any change in the container controller – the message is only used to change the ID in the REFCON database, and it only works for containers that did never communicate on the power lines.

When received, the message is treated as follows:

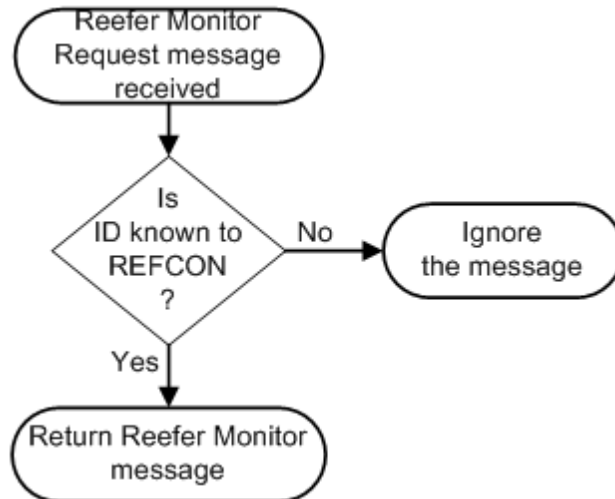




Request for Reefer Monitor message does not work

The Request for Reefer Monitor Data message is sent from the Planning System to REFCON, one message for each container. REFCON is expected to respond with the Reefer Monitor message.

The message is treated as follows:



12.3 Planning System Interface Live Monitor

Key subjects in this topic:

[Overview](#) ^[210]

[Status indication](#) ^[211]

[Show or Hide alive-messages](#) ^[211]

[The list of messages](#) ^[211]

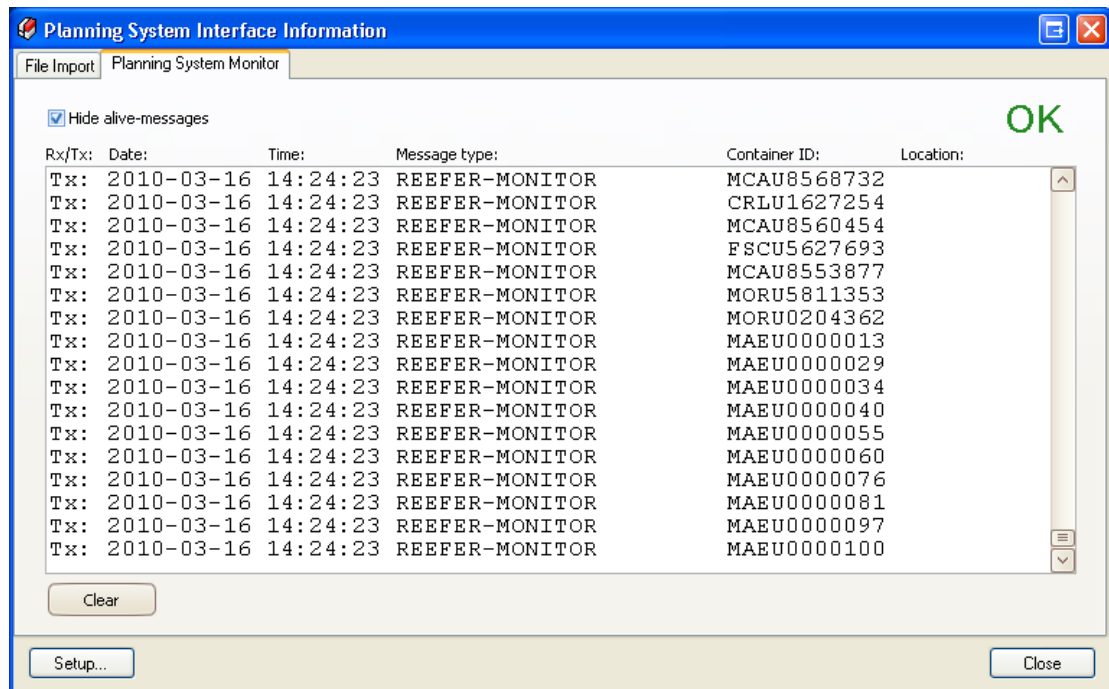
Related topics:

[The Planning System](#)

[Communication Interface](#) ^[202]

Overview

The *Planning System Interface Live Monitor* is found under the "Planning System Monitor" tab of the Planning System Interface Information dialog available from the [options menu](#) ^[203].



The Live Monitor displays one line for each message being transmitted on the interface. The Live Monitor is **not** able to display messages that were transmitted **before** the Live Monitor was opened - for such purpose you will have to inspect the log files.

Status indication

Upper right, the Live Monitor indicates the status of the communication:

Indication text	Explanation
WAIT...	Is shown right after opening the Live Monitor. Please wait until this indication changes before using the Monitor.
TIME OUT	The interface communication does not run - something is wrong.
OK	The interface communication is running.

Show or Hide alive-messages

If you uncheck the checkbox upper left, the Live Monitor will also show the non-data messages, like this:

```
Tx: <Alive>
Rx: <Alive>
```

Normally it is unnecessary to have these messages displayed, as the status OK upper right exactly means that the alive-messages run as intended.

The list of messages

The major part of the Live Monitor displays one line for each message. The columns are:

Column header	Explanation
Rx/Tx	"R" is short for Received (by REFCON): A message sent from the Planning System to REFCON. "T" is short for Transmitted (by REFCON): A message sent from REFCON to the Planning System.



Column header	Explanation
Date	The date indication stamped in the message, if any.
Time	The time indication stamped in the message, if any.
Message type	The type of the message, as described in Overview of transferred data 203 .
Container ID	The container ID stamped in the message, if any.
Location	The container location stamped in the message, if any.

You may use the Clear button to remove everything in the list.

REFCON 6

Reefer Container Monitoring System

Automatic Planning System interface

Transportation Solutions





13 Automatic Planning System interface

Key subjects in this topic:

[Overview](#)^[214]
[Purposes of the Planning System Interface](#)^[214]
[Documentation](#)^[215]
[Communication status](#)^[215]
[The options menu](#)^[215]
[Planning System Interface failure alarm](#)^[215]

Related topics:

[Planning data import](#)^[182]
[The Planning System](#)
[Communication Interface](#)^[202]

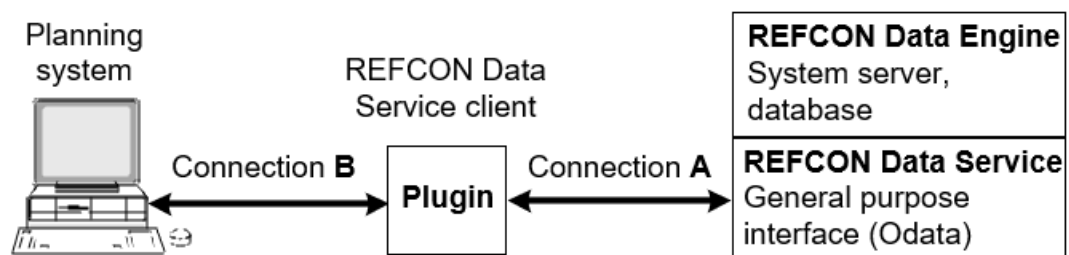
Overview

Your This REFCON system might include includes an automatic Planning System interface solution based on a [REFCON Data Service client](#)^[309] (**plugin**) interfacing between the Planning System and REFCON. An automatic solution is highly valuable in container terminals and similar places where reefer containers frequently come and go.

To make the solution work, the Planning System and the REFCON Main PC must be connected by an ethernet LAN:



The [REFCON Data Service client](#)^[309] may be implemented as an integrated function in the Planning System, as a solution provided by the customer or by third party vendor. If not so, it will be a plugin acting as middleware between the two systems:



When properly connected and software configured, the interface runs automatically. You will be warned by an alarm should either of the connections fail.

Purposes of the Planning System Interface

The basic idea of the interface is to automatically keep data updated in REFCON or on both sides depending on the decided solution:

- If decided, the Planning System continuously gets selected live data from the containers.
- REFCON continuously gets the data necessary to check the containers and to display them graphically (Container Plan).

Please refer to the Planning System for instructions on how to interact with REFCON on that side. Be aware if it runs automatically, or you have to do things manually.



Documentation

Please refer to the documentation provided with the specific plugin for detailed information about the interface: How it works, what data are transferred, settings, troubleshooting etc.

As standard, this documentation is available from the [Planning data options menu](#)^[215].

Communication status

In the Side panel you can at any time see the status of the Planning System Interface:



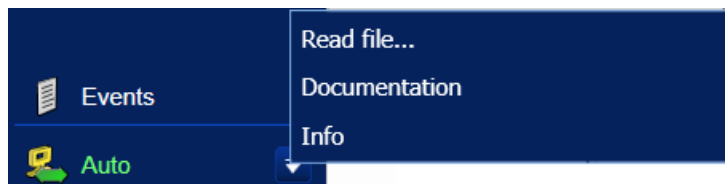
In the left example, the interface runs well - to the right it does not so.

Note that if no automatic Planning System Interface is defined, [Planning data Import](#)^[182] status is shown in stead.

You will notice that the arrow in the icon points both ways for any automatic Planning System Interface even if data transfer from REFCON to Planning System has not been decided.

The options menu

Click the  button to have the options menu displayed. The contents of the menu are partly decided by the [REFCON Data Service](#)^[309] client. As minimum, the following items are available as standard:.



Here,

- The "Read file..." can be used to import a [Planning data file manually](#)^[192]. This might be useful should the automatic interface fail.
- The "Documentation" option opens the manual provided by the plugin. Please refer to this for descriptions of the possible other parts of the options menu.
- The "Info" command displays the [file import status window](#)^[194].

Planning System Interface failure alarm

Should the connection to the Planning System fail, an alarm is released.

REFCON 6

Reefer Container Monitoring System

REFCON 6 extensions

Transportation Solutions





14 REFCON 6 extensions

Key subjects in this topic:

[Overview](#) ²¹⁸

[What is a REFCON 6 extension?](#) ²¹⁸

[Indicator for live data from](#)

[extensions](#) ²¹⁸

Related topics:

[The REFCON Portable Modem](#)

[Solution](#) ²¹⁹

Overview

REFCON 6 is a reefer container monitoring system that originally consisted of the Power Cable Transmission system, a database with various interfaces, and the REFCON Monitor.

In latest REFCON versions, *REFCON 6 extensions* have been introduced.

What is a REFCON 6 extension?

A REFCON 6 extension is a separate package that extends the functionality of REFCON 6. A package always includes one or more software module(s), and can also include hardware. Any extension includes one or more [plugins](#) ³⁰⁹ that runs the communication with REFCON 6.

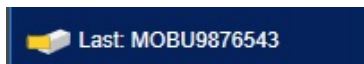
An extension typically provides new ways of communication with containers.

The plugin(s) and all the possible hardware devices are normally shown in REFCON 6, as system devices (when working as parts of the REFCON system) or as modems (when attached to individual containers).

The user guidance for the extension [REFCON Portable Modem Solution](#) ²¹⁹ is included here. Future extensions are likely to come with separate user manuals (PDF) that can be found by right-clicking the respective plugin in the System Devices list.

Indicator for live data from extensions

Lower left in the REFCON Monitor you may find the following indicator:



The purpose of this element is solely to indicate that the live data source is alive, similar to the PCT live indicator just below it.

Each time a container is updated via any REFCON 6 extension, the ID of the container is shown. As the frequency of updates can be high, the shown ID will stay displayed for minimum 1 second - which means that maybe not all individual container IDs are seen.

Note that while a [container remote operation](#) ¹³⁷ is taking place through an extension, the indicator shows the status of this in stead.

If no live data REFCON 6 extension is found, or no live data have yet arrived from such source, this indicator is not shown.



14.1 The REFCON Portable Modem Solution

Key subjects in this topic:

[Overview](#) ^[219]
[The REFCON Portable Modem](#) ^[219]
[Communication solutions](#) ^[220]
[Working schedule](#) ^[220]
[Understanding REFCON Portable Modem power on cases](#) ^[221]
[Understanding the REFCON Portable Modem power on and communication start schedule](#) ^[221]

Related topics:

[REFCON 6 extensions](#) ^[218]
[The REFCON Portable Modem](#) ^[222]
[The REFCON Mobile](#) ^[226]
[The REFCON Wireless Hub](#) ^[232]

Overview

The REFCON Portable Modem Solution is an extension to the REFCON system, introducing the [REFCON Portable Modem](#) ^[222].

It aims at 100% remote monitoring coverage even when

- some reefer containers are not equipped with communication modems
- or do have modems being unable to interface with REFCON 6.

The REFCON Portable Modem communicates wirelessly through Bluetooth with REFCON 6 through either the REFCON Mobile or the REFCON Wireless Hub. Note that the REFCON Portable Modem Solution also supports certain types of modems permanently mounted on containers, such as the RMM-W.

The REFCON Portable Modem

The fundamental part of the REFCON Portable Modem Solution is the [REFCON Portable Modem](#) ^[222].



REFCON Portable Modems are small wireless communication devices that you attach to the log-retrieval port of the containers. The REFCON Portable Modem supports all controller types from Carrier, Daikin, Star Cool and Thermo King.

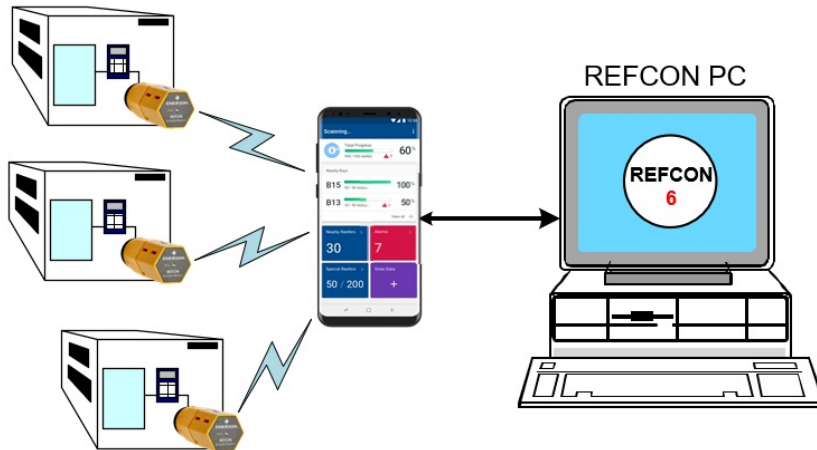




Communication solutions

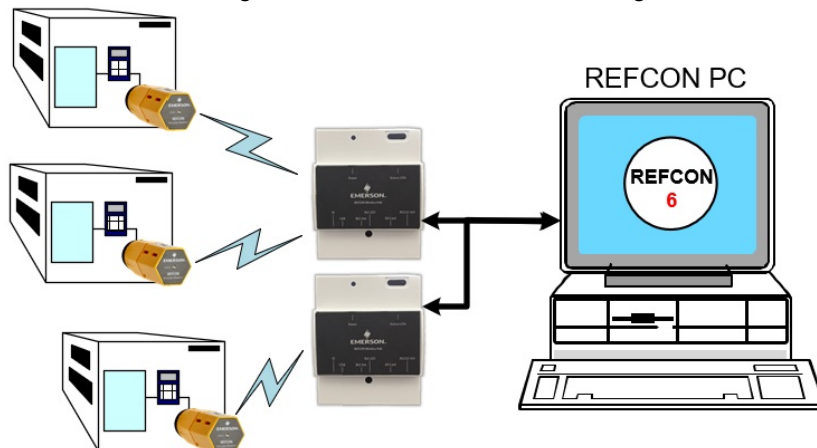
The controller data obtained by the REFCON Portable Modems can be forwarded to REFCON 6 in two ways, of which you may have one or both available:

- The REFCON Mobile and REFCON Mobile Plugin solution utilizes one or more smart phones, running the REFCON Mobile app.



With this solution user needs to carry the smart phone around the container site and collect the container data wirelessly, and then connect to the computer running the REFCON Mobile Plugin to have data transferred to REFCON 6.

- The REFCON Wireless Hub solution utilizes a number of mounted REFCON Wireless Hub devices, covering the decided container area as regards the wireless signals.



By this solution, container data are automatically forwarded to REFCON 6 periodically, through the hidden REFCON Wireless Hub Plugin.

Working schedule

To utilize the REFCON Portable Modem Solution you will need to run through the following steps whenever relevant:

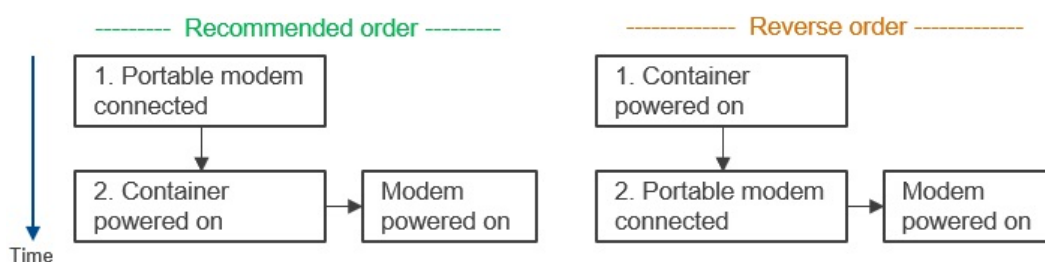
Step	Action	Comments
1	When new reefer containers arrive / are loaded, mount ²²³ REFCON Portable Modems where required.	You may mount on all reefer containers, or only those that do not communicate with REFCON 6 by other means, for example Power Cable Transmission.
2a	If REFCON Wireless Hubs are in operation, wait for some 15 minutes after mounting	Please expect that it takes some time for the REFCON Wireless Hubs to detect



Step	Action	Comments
	modem / powering on the container.	the new containers and to fetch and transfer their data.
2b	If the REFCON Mobile smart phone app is used, run the working schedule relevant to this.	See the REFCON Mobile description ^[226] .
3	Open an appropriate container list ^[66] in REFCON 6 including the Comm State ^[73] column - we recommend the <i>Attention required</i> pre-defined list. Optimally, all containers are communicating and updated as expected, in which case the list should be empty. Take action on containers in the list not updating as expected - see recommendations in the Communication States table ^[73] . • Are containers on power? • Are REFCON Portable Modems mounted? • Are their Status LEDs^[224] blinking green?	See also Troubleshooting ^[225] .
4	Maybe open a list including all containers and check if they are all updated with the expected set of data (measurements etc).	
5	When any reefer container is about to leave / be discharged, dismount ^[223] the REFCON Portable Modems in time.	Maybe when disconnecting power anyway.

Understanding REFCON Portable Modem power-on cases

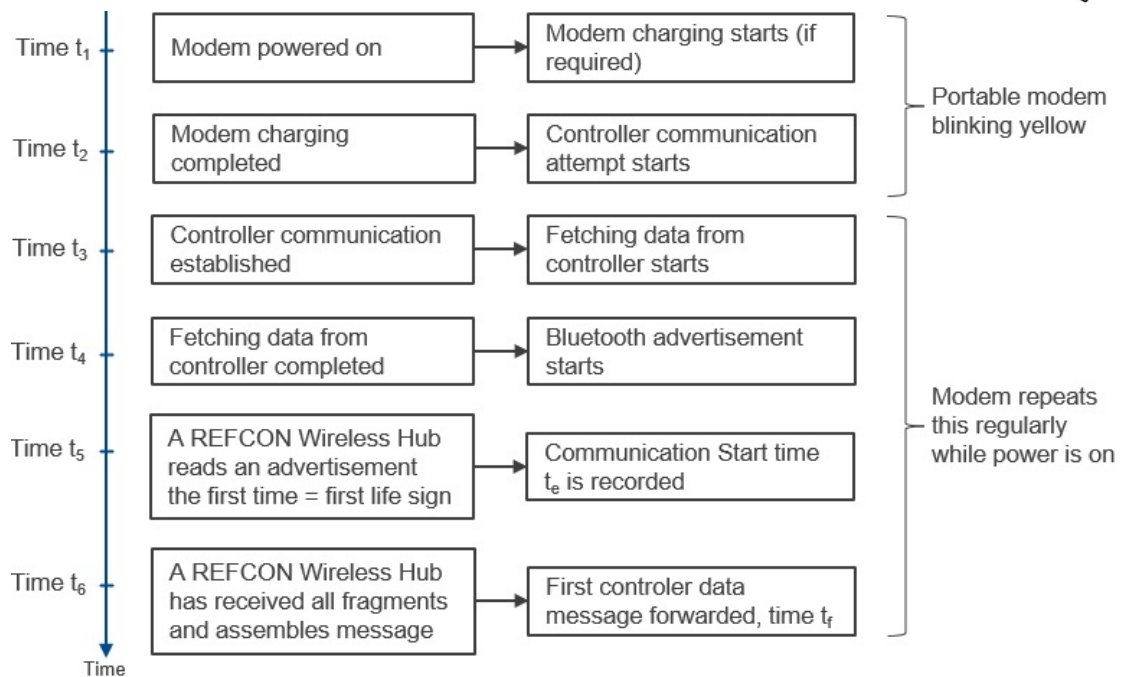
This is just to mention the two practical options:



- In the left case, reefer man first plugs on the modem and then plugs on power to the container. In this case, modem's power on time is same as container's power on time. This is the recommended order.
- In the right case, reefer man first plugs on power to the container and then plugs on the modem, any time later. There is no way the modem can know about the time between the two events.

Understanding the REFCON Portable Modem power-on and communication start schedule

When your REFCON installation includes the [REFCON Wireless Hub solution](#)^[232], it can be useful to know about the activities being carried out from modem is powered on until REFCON receives first controller data message:



- The time span from t_1 to t_2 is near zero if the modem was used recently, as then it is pre-charged; or it can be a minute or more for a modem having not been mounted for long.
- The time span from t_2 to t_3 is depending on the controller type - can be 5 - 20 seconds.
- The time span from t_3 to t_4 also depends on the controller type - typically about 10 seconds.
- The time span from t_4 to t_5 is depending on Bluetooth communication conditions; in best case it is below 1 second, but in practice it will most often be more, even a minute or more.
- The time span from t_4 to t_6 similarly is depending on Bluetooth communication conditions.

The REFCON Wireless Hub (or REFCON Mobile) must read some 12 - 20 *different* advertisements, all repeated by the modem in turn at 6 - 10 seconds interval. Best case is typically some 15 seconds, average more like a minute.

Having advertised same data for about 2 minutes, the schedule is repeated. If no REFCON Wireless Hub (nor REFCON Mobile) managed to assemble the message, it was lost.

14.2 The REFCON Portable Modem

Key subjects in this topic:

[Overview](#) ^[222]
[Mounting](#) ^[223]
[Dismounting](#) ^[223]
[Fixed installation](#) ^[223]
[Status LED](#) ^[224]
[Modem check flowchart](#) ^[225]
[Troubleshooting](#) ^[225]
[Disposal](#) ^[225]
[Accessories](#) ^[226]

Related topics:

[The REFCON Portable Modem Solution](#) ^[219]

Overview

The REFCON Portable Modem is a small wireless communication device that you attach to the Service Port (also called log-retrieval connector) of the reefer container. It can be connected directly or through an extension cable.

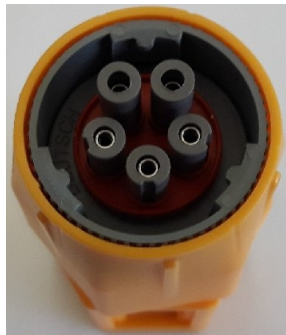


A strong benefit of the REFCON Portable Modem is that it never needs any battery charge or replacement. This is possible because it uses extremely little power. For example, you will see the Status LED blinking very shortly and being off for rather long time in between simply to save energy.

Mounting

Before mounting, make sure the REFCON Portable Modem is in the Unlocked position:

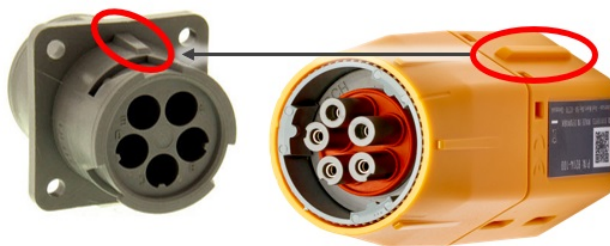
Unlocked:



Locked:



To mount the device, hold the Connector Part in one hand. Make sure the Lock marker is aligned with the corresponding marker on the log-retrieval port connector:



Press the REFCON Portable Modem against the port while turning the Connector Part clockwise until click.

Dismounting

To dismount the REFCON Portable Modem, turn the Connector Part counter clock-wise while gently drawing it from the port.

Fixed installation

The REFCON Portable Modem can be installed using zip ties or similar products to fixate the modem on or near the container.



This solution requires an additional extension cable to connect to the log-retrieval port. Please contact Copeland Transportation Solutions for further information.

Status LED

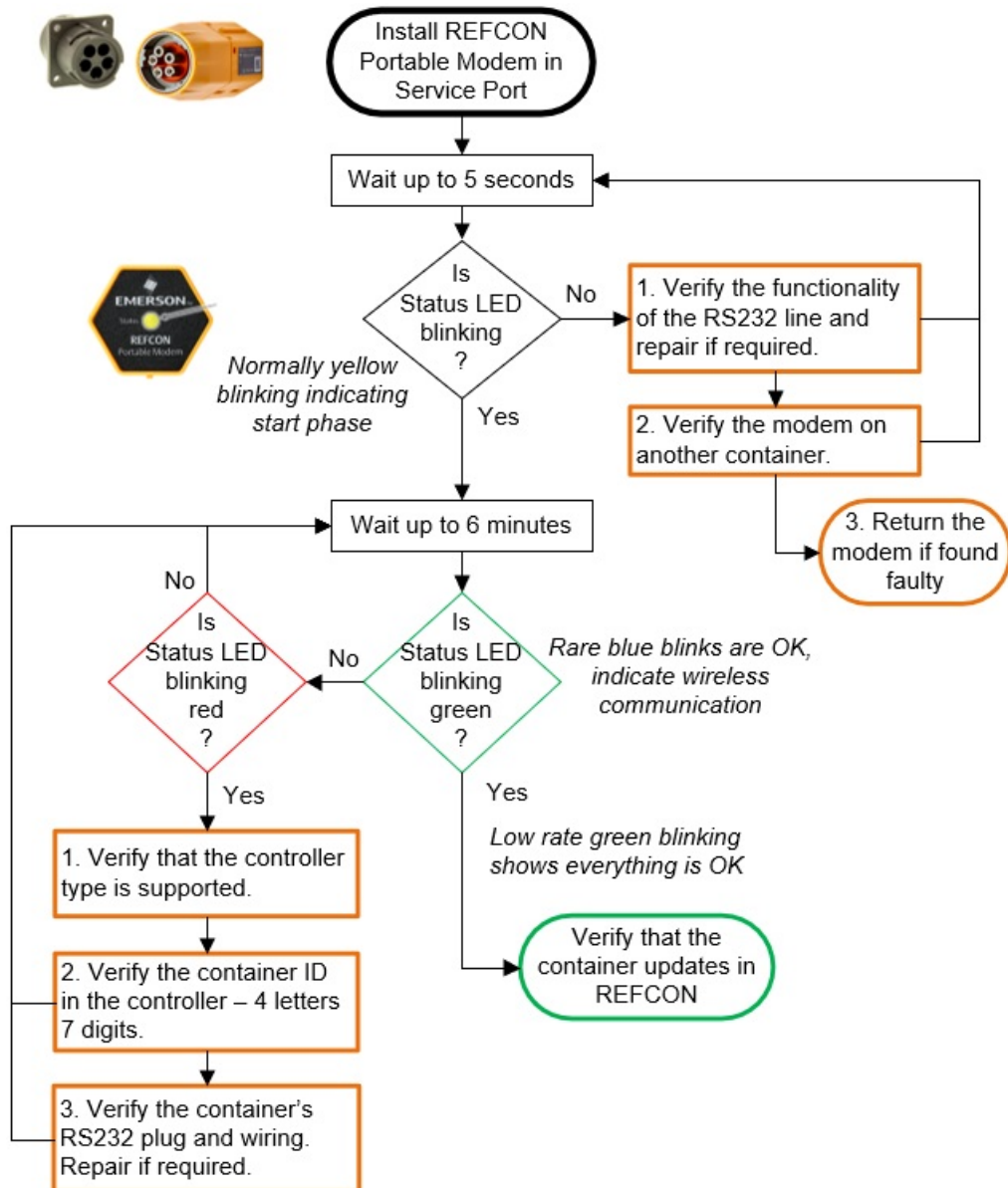
The Status multi-colour LED provides feedback to the user.



LED behaviour	Status	Comments
Yellow blink	Starting up / Connecting to controller	After connection it can take a few seconds before blinking starts. Yellow blink can last up to a couple of minutes.
Green blink	Controller found / Collecting reefer data	This is the normal state - everything is fine.
Blue blink	Transmitting data over Bluetooth	Expect this blink colour to occur quite rarely and for short duration.
Red blink	Error: Unknown controller / Faulty connection	Please refer to the troubleshooting ²²⁵ advices.



Modem check flowchart



Troubleshooting

If the Status LED on the REFCON Portable Modem is blinking in red colour, please try the following steps:

1. Dismount, wait until the red blinking stops, and then mount anew. Wait for yellow blinking to stop.
2. If now again red blink, dismount, wait until the red blinking stops, and then mount on another container - preferably one you have seen work recently.

Disposal

Should you decide to dispose of the device, please follow your local standard rules for disposing/recycling electrical equipment.



Accessories

A range of accessories is available for the REFCON Portable Modem, including protection and fixation solutions. Please contact Copeland Transportation Solutions for information.

14.3 The REFCON Mobile

Key subjects in this topic:

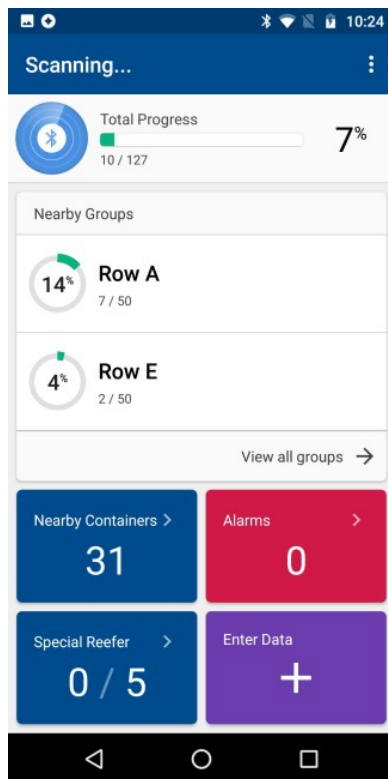
[Overview](#) ²²⁶
[Working schedule](#) ²²⁶
[Understanding the visit list](#) ²³⁰
[Filtering the visit list](#) ²³⁰
[First start-up](#) ²³¹
[Understanding the REFCON Mobile main screen](#) ²³¹

Related topics:

[The REFCON Portable Modem Solution](#) ²¹⁹
[The REFCON Portable Modem](#) ²²²
[The REFCON Mobile Plugin](#) ²³²

Overview

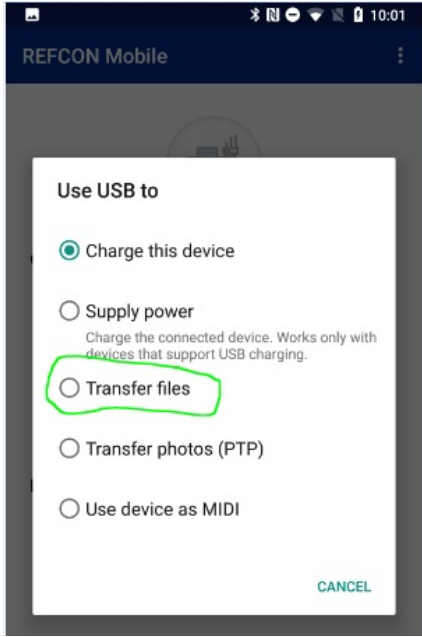
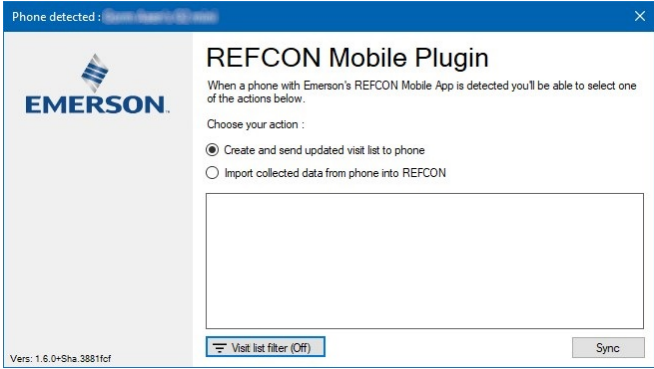
The REFCON Mobile is an app for Android smart phones used to collect data from all reefer containers having a REFCON Portable Modem mounted. Further it is capable of retrieving data from other modem types supporting same Bluetooth protocol, such as the RMM-W. Also it offers a solution for manually recording data from containers without a compatible modem.



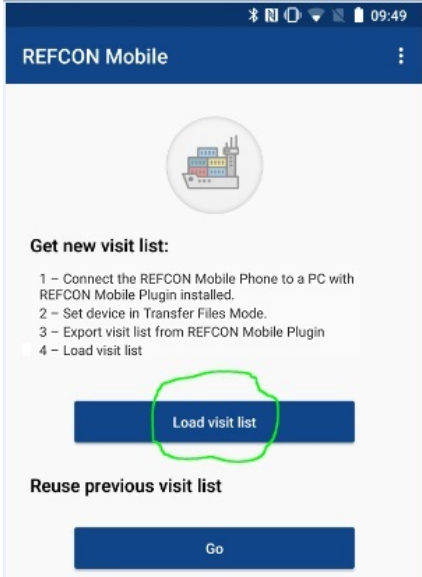
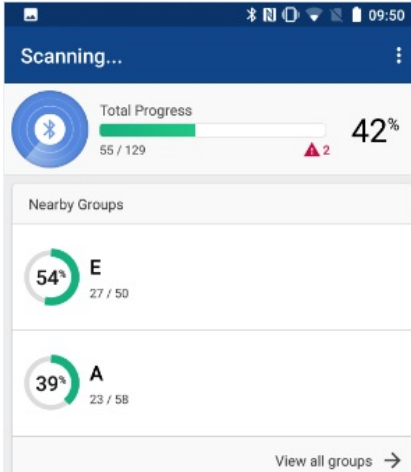
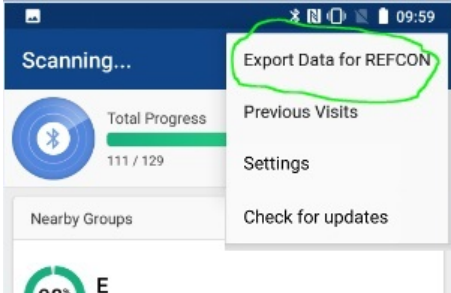
Working schedule

The REFCON Mobile working schedule is a part of the [overall schedule](#) ²²⁰. Assuming that REFCON Portable Modems have been mounted where required, the following rounding schedule should be performed periodically:

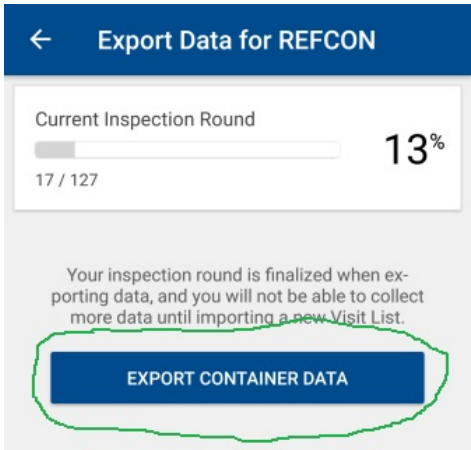
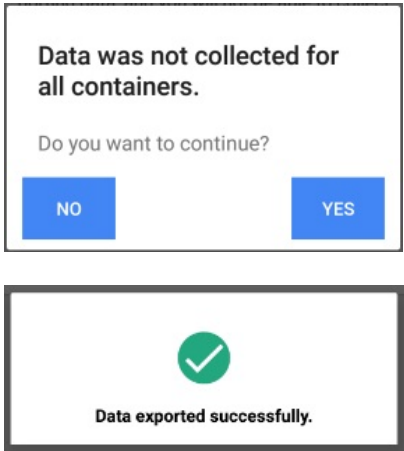
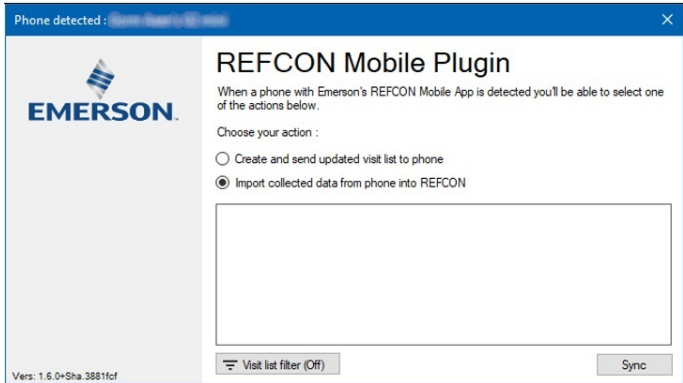


Step / Action	Comments / Examples
1. If a manual task, update Planning data in REFCON.	As the current list of containers on site should be the basis for the visit list ²³⁰ .
2. Connect the smart phone to a computer running the REFCON Mobile Plugin ²³² , using a USB cable.	This must be the beginning of any rounding, including the very first (see below).
3. Set the smart phone's USB interface into "Transfer Files Mode" <i>Note: This step can vary from device to device, please refer to the manual of the mobile device.</i>	This is due to standard security functionality in Android. Normally this step needs to be completed each time the USB cable is connected. 
After this, the REFCON Mobile Plugin should pop up automatically.	
4a. Maybe select a filter for your visit list.	See Filtering the visit list ²³⁰ below.
4b. Export the "visit list" by choosing the first option and then selecting the [Sync] button.	A number of files are transferred to the phone.
5. Open REFCON Mobile on phone.	



Step / Action	Comments / Examples
6. Load the visit list from the REFCON Mobile app.	Screen shows progress. 
7. Move around the site / ship getting close enough to all the containers on the visit list.	The wireless signal has a range of several hundreds of meters in an open area, while with many containers and other metal constructions the range will vary.  If 100% is never reached, it may be because REFCON Portable Modems have not been mounted on all containers, or are failing. You may enter data manually to reach the 100%.
8. When you have completed, export data to a file on the phone.	From the menu upper right: 



Step / Action	Comments / Examples
9. In the screen opening, select EXPORT CONTAINER DATA	
9a. If you, like in above example, did not collect data for all containers on the visit list, select YES to export anyway, or NO to be able to continue the inspection round.	
10. Return to a computer running the REFCON Mobile Plugin ²³² . Connect the smart phone to the computer by USB, and repeat step 2.	
11. From the REFCON Mobile Plugin, retrieve data from the phone by choosing second option and then selecting the [Sync] button.	 This will transfer the collected data from the mobile device into REFCON 6.
12. Check the REFCON Monitor and see that data were updated.	



Understanding the visit list

The visit list is primarily a list of all containers to be visited during the pending inspection round. As standard a visit list includes all containers with status "planned" in REFCON at the moment. "Planned" means that the container has a valid location origin from the Planning System. You may *filter* the list as described [below](#) to have only a subset of the "planned" containers included.

For each container the visit list includes:

- The container ID and location.
- A sorting index and an "area" decided by the "walk route".
- Other data, such as known alarms and monitoring status.

Filtering the visit list

Select the [Visitlist filter] button at the bottom of the REFCON Mobile Plugin window to work with the visit list filter.



This opens the "Visitlist Filter" dialog box, if relevant with the last selected filter preloaded:

The filter consists of a number of main conditions, represented by the checkboxes to the left. When a checkbox is selected, this main condition and its subconditions / parameters are taken into account when deciding the visit list.

When a main condition is checked, its subconditions / parameters are displayed to the right. Here you finalize the specification of your filter.

Having completed your selections, you can

- Select [Close] to apply your filter.
- Select [Options] -> [Save filter] to save your filter to disk.
- Select [Preview] to inspect the list of containers resulting from the current filter selection.
- Select X (upper right) to close the window without applying any changes.

You may alternatively select [Options] -> [Load filter] to have last saved filter displayed. Note that if you have more REFCON Mobile Plugin applications running, it is one common saved filter - which can therefore be shared by all users.

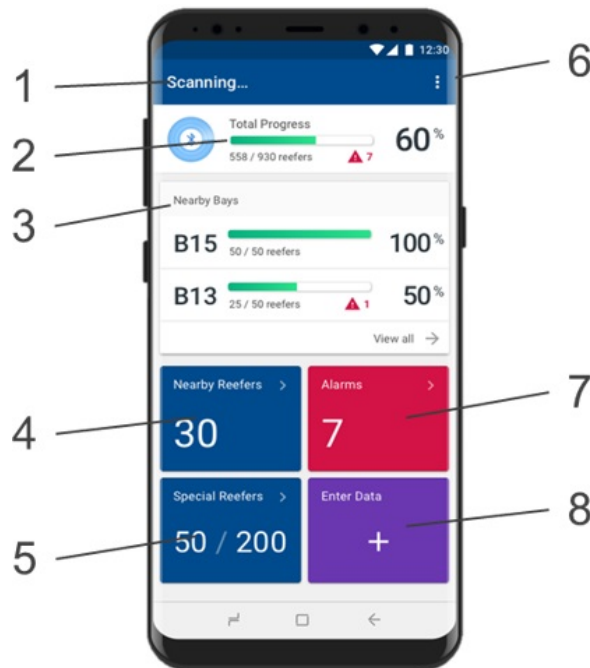


First start-up

To start using REFCON Mobile you need to run through the above [Working schedule](#)²²⁶ from step 1. The app will not work without a visit list and other information retrieved from REFCON 6 the first time.

Understanding the REFCON Mobile main screen

You find the following elements in the REFCON Mobile main screen:



Part	Description
1	Bluetooth communication status.
2	Total progress of the data collection. Shows the percentage of the reefer containers in the visit list that have been successfully monitored during this rounding.
3	The "Nearby Areas" - or, on a ship, "Nearby Bays" - shows the progress in the two nearest reefer areas as defined by the visit list.
4	"Nearby Reefers" shows the number of currently sensed REFCON Portable Modems within wireless signal range.
5	"Special Reefers" shows the monitoring progress for the containers marked as special reefers in REFCON 6.
6	Menu.
7	"Alarms" show the number of containers that have one or more alarms according to data retrieved during the current rounding.
8	"Enter Data" allows the user to enter information manually, even for containers that are not on the visit list.



14.3.1 The REFCON Mobile Plugin

Key subjects in this topic:[Overview](#) ²³²**Related topics:**[The REFCON Portable Modem](#)[Solution](#) ²¹⁹[The REFCON Mobile](#) ²²⁶

Overview

The *REFCON Mobile Plugin* is a small application that can be installed and run on any Windows computer connected by LAN to the REFCON 6. You can run it on more computers if appropriate. As standard, REFCON Monitor offers you an easy way to install the REFCON Mobile Plugin.

The REFCON Mobile Plugin provides the interface between the REFCON Mobile and REFCON 6. Please refer to [The REFCON Mobile](#) ²²⁶ for user guidance.

14.4 The REFCON Wireless Hub

Key subjects in this topic:[Overview](#) ²³²[Disposal](#) ²³³**Related topics:**[The REFCON Portable Modem](#)[Solution](#) ²¹⁹[The REFCON Wireless Hub](#)[Plugin](#) ²³³

Overview

The REFCON Wireless Hub is a device that allows REFCON 6 to communicate wirelessly with reefer containers having REFCON Portable Modems mounted. Further it is capable of retrieving data from other modem types supporting same Bluetooth protocol, such as the RMM-W.



Normally a suitable amount of REFCON Wireless Hubs are permanently installed around the ship or terminal so that all container locations are covered regarding the wireless signals.

The REFCON Wireless Hub is connected to REFCON 6 through LAN, WiFi or Bluetooth. Keep it permanently on power, and it will automatically run its communication tasks needing no attention.

Disposal

Should you decide to dispose of the device, please follow your local standard rules for disposing/recycling electrical equipment.

14.4.1 The REFCON Wireless Hub Plugin

Key subjects in this topic:

[Overview](#) ²³⁴

[Connections](#) ²³⁴

[Installation and configuration](#) ²³⁴

[Changing REFCON Wireless Hub](#)

[Plugin settings](#) ²³⁴

Related topics:

[The REFCON Portable Modem](#)

[Solution](#) ²¹⁹

[The REFCON Wireless Hub](#) ²³²



Overview

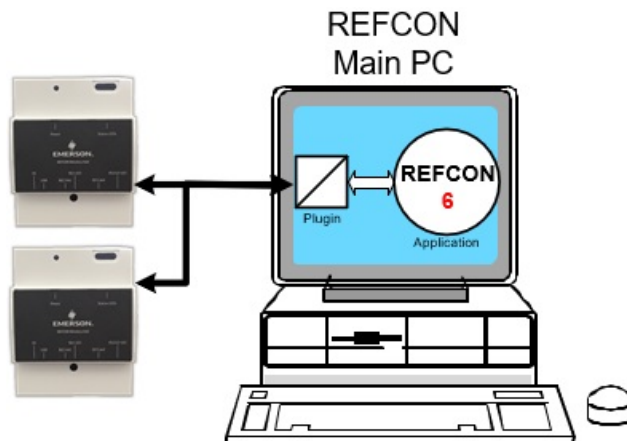
The *REFCON Wireless Hub Plugin* is a part of the REFCON Portable Modem Solution extension to REFCON 6. It takes care of the communication between REFCON 6 and the REFCON Portable Modems through the REFCON Wireless Hubs.

The REFCON Wireless Hub Plugin runs as a Windows Service on the computer and is therefore normally not visible on the Windows screen. It runs permanently and does normally require no user attention. It even installs along with REFCON 6 as standard.

Nevertheless there are situations where you, as system responsible user, may need to handle the plugin. Please find information and advice in the following.

Connections

The REFCON Wireless Hub Plugin is as standard running on the REFCON Main PC. It connects through wired LAN or WIFI to all the REFCON Wireless Hubs:

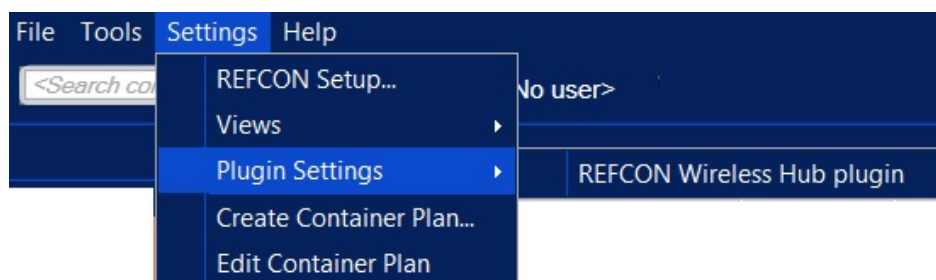


Installation and configuration

As standard the REFCON Wireless Hub Plugin is automatically installed as a part of the REFCON 6 installation. A default configuration is installed as well, which will normally be suitable. All this requires that the installation files are found in the used [REFCON System Bundle](#) ³¹⁴.

Changing REFCON Wireless Hub Plugin settings

It is possible to change a number of settings controlling the behavior of the REFCON Wireless Hub Plugin. To see or edit these settings, select the plugin from the Settings menu:



This opens the [Settings Editor](#) ³³⁸.

In the Settings Editor, please refer to the tooltips of the individual settings for explanations.

REFCON 6

Reefer Container Monitoring System

REFCON Data Collector

Transportation Solutions





15 REFCON Data Collector

Key subjects in this topic:

[Overview](#) ²³⁶

[The hand-held computer: The RDC unit](#) ²³⁶

[Recommended inspection schedule](#) ²³⁶

[Why Use the RDC?](#) ²³⁷

[RDC-monitored containers](#) ²³⁸

Related topics:

[Operating RDC from REFCON](#) ²³⁸

Overview

The RDC (REFCON Data Collector) allows systematic data collection from reefer containers that are not communicating on the power cables. Also it allows you to check the setpoints of the containers.

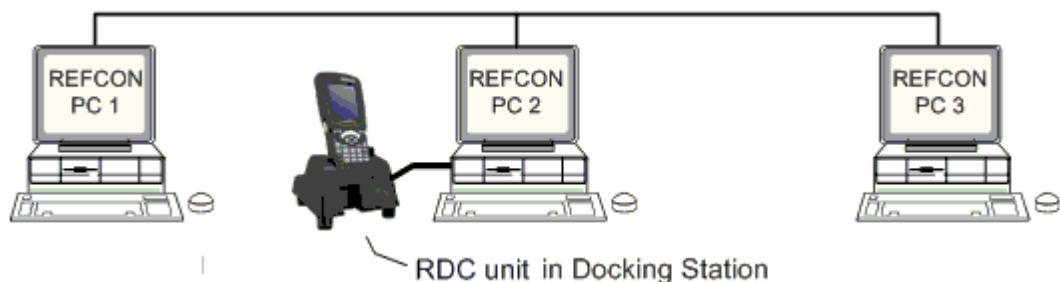
To take advantage of the RDC, you need to do regular *inspections*, where you carry the [RDC unit](#) ²³⁶ and walk to the selected containers. Before an inspection, you should [prepare](#) ²⁴⁷ the RDC unit by deciding exactly which containers to visit (the *selection*), and in which order to visit them (the *walk route*).

The RDC software has been and will be released in various versions. It is installed and updated independently of the REFCON installation, using the *Handheld Software Commander*. Please find the RDC operating manual on the PC where Handheld Software Commander is installed, being the REFCON Monitor PC where you interact with the RDC.

The hand-held computer: The RDC unit

Inspections are done by means of a hand-held computer equipped with RDC software. We generally refer to this as the *RDC unit*.

The RDC unit needs to be connected to REFCON for data transfer. Normally this takes place through a docking station connected to a USB port of the REFCON PC. If you have more than one REFCON PC, any of them can be used.



Important note: You must import RDC data and prepare the RDC unit from the REFCON PC to which it is connected. In the above example, you can operate the RDC from PC 2, but not from PC 1 or PC 3 (unless you move the docking station).

Also important: On the same PC (PC 2 in above example), the Handheld Software Commander must be installed and running. The Handheld Software Commander helps you installing the RDC software and keeping it up-to-date.

Recommended inspection schedule

Your normal data collection and setpoint-checking schedule would look like this:



Step	Action	Comments
1	Receive container Planning data from the Planning System.	To be able to prepare the RDC with the correct list of containers and their locations.
2	Await that all relevant containers have had time to start communication .	With RDC you will <i>normally</i> only deal with containers that do not communicate on the power lines - but this is your decision.
3	Place the RDC unit in the docking station , being connected to REFCON.	Note that if you have more than one REFCON PC, the RDC unit is connected to a specific one of them.
4	Select which containers to include in the preparation list (= the <i>selection</i>).	You do this from the RDC Preparation Selection ²⁴⁷ dialog box on that particular PC.
5	Select the order in which you intend to visit the containers (= the <i>walk route</i>).	This requires one or more walk routes to be specified once for all. Normally this is done at commissioning time.
6	Prepare the RDC , that is, send the sorted preparation list to the RDC unit.	You may observe file transfer progress on both REFCON and the RDC unit.
7	Do your manual inspection round .	That is, carry the RDC unit to the containers, in the order stated in the display.
8a	For each container, check the setpoints .	Compare controller's setpoints with the expected, and change the first mentioned on the controller if necessary.
8b	Also for each container, get it's runtime values (wait or type).	Data is retrieved automatically if your RDC and the container are equipped with matching wireless technology; otherwise you need to type in setpoints, primary temperatures, alarm state and maybe other measures on the RDC.
9	Make sure you have covered all the selected containers.	If your preparation list is complete: Are they all green?
10	Return and place the RDC unit in the docking station.	After this, wait a few seconds for the RDC unit and the PC to connect.
11	Select the "Import data file" command on the PC.	Note that only containers with updated data are affected, and only if they meet the conditions for RDC-monitored containers ²³⁸ .

Usually you would repeat the inspection periodically. If no new Planning data, you start succeeding inspections at above step 4 or, if same selection and walk route, at step 6.

Why Use the RDC?

Normally, most containers communicate with REFCON over the power cables (PCT), however, it will happen that a number of containers don't. This can be because of defects, or simply because the containers are not equipped with power cable modems.

Thus, containers that do not communicate will have to be supervised manually, to some extent. With the RDC this manual inspection gets easier, and it allows data to be integrated in REFCON and on the Global Monitoring Server.

The main advantage is that the RDC shows you the list of containers that you need to visit in the order you have chosen. This can considerably reduce your manual inspection time.



Another advantage is that data from the non-communicating containers get into REFCON, making these containers work similar to any communicating container. Of course this will depend on your inspection interval.

RDC-monitored containers

When you import data from the RDC unit to the PC, REFCON checks the **conditions** mentioned below. We refer to containers that fulfill these conditions as the *RDC-monitored containers*.

The following conditions must be fulfilled for a container to be RDC-monitored:

- The container has never been communicating via the PCT, or data from the container are obsolete, or the container is communicating, but data could not be retrieved (most likely because of an error on the container).

If you during your manual inspection have recorded setpoints and/or measures for a container that, at the time of import, does **not** meet the above conditions, these data will be ignored and thus **not** imported to REFCON. Note that if you, for the same container, also have changed the Note or the textual Alarm field, these data **will** be imported, as these do not conflict with PCT data.

15.1 Operating RDC from REFCON

Key subjects in this topic:

[Overview](#)^[238]
[The RDC Side panel element](#)^[238]
[The main schedule](#)^[239]
[Exceptions](#)^[243]
[RDC Side panel element details](#)^[244]
[The RDC Side panel element extension](#)^[245]
[The Options Menu](#)^[245]
[The RDC status viewer](#)^[245]
[Removing an RDC alarm from REFCON](#)^[247]

Related topics:

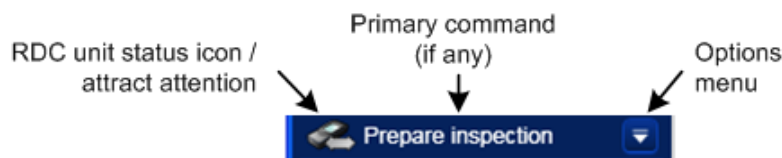
[RDC Preparation Selection](#)^[247]
[How the Default Walk Route is organised](#)^[250]
[RDC+ introduction](#)^[236]

Overview

If REFCON is configured to include the RDC, any REFCON Monitor will include the RDC element in the Side panel as well as the pictures and functionality described in the following.

The RDC Side panel element

The key to the user's handling of the RDC from the PC is the RDC Side panel element:



By very simple means, the RDC Side panel element always:

- Informs you about the actual situation (as regards the RDC).
- Provides the most relevant command available in the situation.
- Provides access to alternatives and detailed status information.

Below, you find further information about [RDC Side panel element details](#)^[244], [The RDC Side panel element extension](#)^[245], [The Options Menu](#)^[245] and [The RDC status viewer](#)^[245].

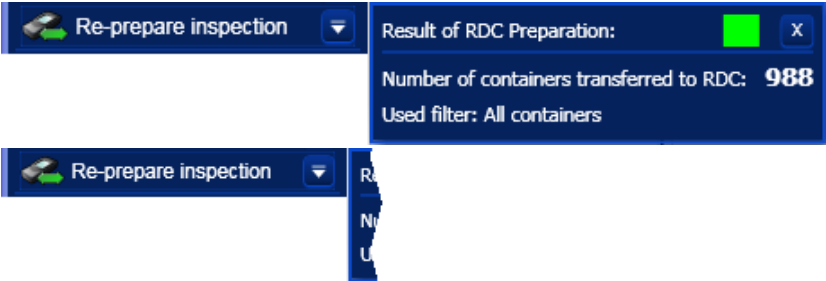


The main schedule

When everything works as expected, a manual inspection runs through the following steps:

Pre-condition	Step	Procedur e
	1. Prepare the RDC.	Click the element to have containers selected and sorted according to the last saved options (= quick preparation). Choose "Prepare inspection (select)" from the Options menu to open the RDC Preparation



Pre-condition	Step	Procedure
		on Selection dialog ^[247] to adjust the options. In both cases, the resulting list is sent to the RDC.
	2. Do your manual inspection.	1. Disconnect the RDC unit. 2. Carry out

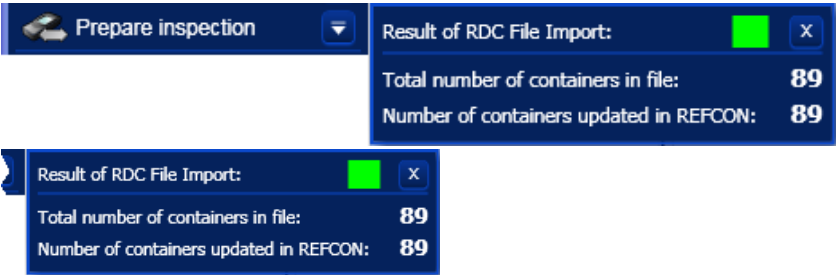


Pre-condition	Step	Procedure
		o u r m a n u a l i n s p e c t i o n r o u n d .
 Please connect RDC 	3. Connect the RDC unit.	Plug it into the docking when you have completed your inspection round. Wait for a while.
 Import data file 	4. Import changed data.	Click the element to have the



Pre-condition	Step	Procedure
		data recorded during your inspection imported into REF CON's database. Choose "Ignore data file" from the Options menu to skip the import. In this case, the file with changes is still found in the RDC unit, meaning that you



Pre-condition	Step	Procedure
		may disconnect the unit <i>without</i> a new preparation and add more changes.
	5. Watch the import result.	The square should be green, and the two counters should match the number of containers that you did visit.

Exceptions

It might occur that things do not quite appear as described above. The most common exceptions occur when importing the container data file. Such exceptions will make the RDC Side panel element look something like this:



Notice the yellow square and the third counter. The following exceptions are the most common:

- One or more containers have been disconnected since the RDC unit was prepared, and since you recorded their data. Usually you would know, of course, but it *could* be an unintended disconnect, maybe?
- One or more containers have started communicating on the PCT since the RDC unit was prepared. In this case, if you have made changes to the Note or the Alarm field of any such container, these data are imported all right, but setpoint and temperature data are not. Even if you have changed nothing but the Note and/or the textual Alarm, it will count as a "container that was not or only partly updated".

If it is even worse than this, for example the data file being not readable, the square will be red.

RDC Side panel element details

The RDC status icon can appear as follows:

Icon appearance	Explanation
	No RDC unit is connected to this PC.
 (light gray arrow)	An RDC unit is connected, awaiting preparation for next inspection.
 /  (flashing)	An RDC unit is connected, data are ready for import.
 (green arrow)	An RDC unit is connected, it has been prepared for next inspection and is awaiting your disconnection.

The command text appears as follows:

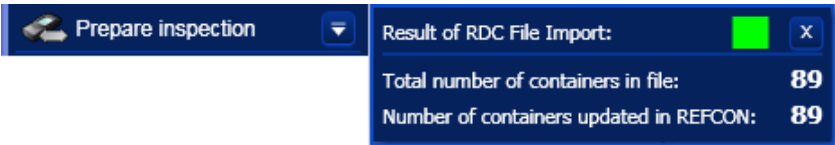
Text / appearance	Explanation
Please connect RDC	No RDC unit is connected to this PC. No primary command is available as you can do nothing.
Connecting	An RDC unit has just been connected, it's status and possible data files are being retrieved - please wait for a while.
Import data file	An RDC unit is connected, data are ready for import. Click the field to start the import, or select alternative from Options menu.
Importing file	An RDC unit is connected, data are being imported - please wait for a while.
Prepare inspection	An RDC unit is connected, awaiting preparation for next inspection. Click the field to prepare with standard configuration, or select alternative from Options menu.
Re-prepare inspection	An RDC unit is connected, it has been prepared for next inspection and is awaiting your disconnection. Still you have the option to overwrite the already prepared data file.
Being prepared	An RDC unit is connected, the preparation data file is being created - please wait for a while.



Text / appearance	Explanation
Being transferred	An RDC unit is connected, the preparation data file is being sent to the RDC unit - please wait for a while.

The RDC Side panel element extension

Each time a primary RDC operation has completed, the RDC Side panel element extension appears to the right of the RDC Side panel element itself.

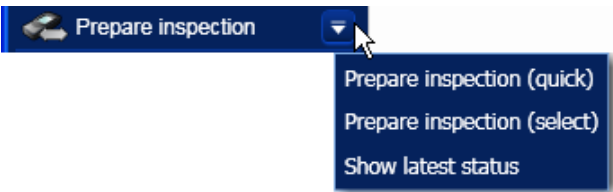


The extension disappears automatically in case of certain events, for example, when the RDC unit is disconnected or user selects the Options menu. If none of these events occur, the extension fades away after a short time. You can remove it by clicking the small X button.

For further information, use the [RDC status viewer](#)²⁴⁵.

The Options Menu

In general, the Options Menu provides the alternatives to the primary command in the actual situation:

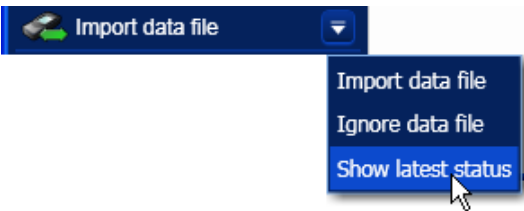


The first option in the menu is identical to the primary command. After this follows one or more alternatives, if any.

As the last option you find the [RDC status viewer](#)²⁴⁵ described below. If this is the only option in the menu, the viewer opens immediately when selecting the Options menu button.

The RDC status viewer

To open the RDC status viewer, select it in the Options menu:



If the "Show latest status" option is the only one in the menu, the viewer opens immediately when selecting the Options menu button.

Note that **the viewer always displays a snapshot!** Should status change while the viewer is open, this will not be reflected - you must close and reopen the viewer to have it updated.

The RDC status viewer contains two tabs, "Status latest Import" and "Status latest Preparation". If the latest primary operation was a preparation, the latter tab automatically opens.

With the "Status latest Import" tab open, the viewer looks like this:



- Actual RDC unit status →
- Time and responsible of latest import →
- Tabs →
- Result of latest import →
- Time of latest import →
- Statistics of latest import — {
- Details / explanations about not-updated containers, if any — {

RDC Status

Awaiting RDC to be connected.

Last RDC data import: **16 hours ago** by John

☒ Status latest Import
 ☐ Status latest Preparation

Latest import was completed with warnings

RDC data imported at: 16-03-2010 15:07:34 (16 hours ago)

Statistics:

Description	Count
Total number of containers in file:	85
Number of containers updated in REFCON:	84
Number of changed containers that were not or only partly updated in REFCON:	1

Problem report:

Container ID	Reason	File extract
BMOU9009926	Container was deleted	BMOU9009926;241112;4.5C;48.6C;5...

Close

With the "Status latest Preparation" tab open, the viewer looks like this:

- Same as above — {
- Result of latest preparation — {
- Time and responsible of latest preparation →
- Filter used for latest preparation →
- Walk route used for latest preparation →
- Click to show latest preparation list →

RDC Status

RDC is connected. Awaiting users preparation for inspection.

Last RDC data import: **16 hours ago** by John

☐ Status latest Import
 ☒ Status latest Preparation

Latest preparation was successful

483 containers were sent to RDC

RDC prepared for inspection at: 17-03-2010 07:36:16 by John

Used filter:
Decided by Cold treatment

Used walk route:
Sorted alphabetically by ID

Show list of containers

Close

Note: The "Used filter" is a description of the selections done in the [RDC Preparation Selection](#) dialog. In the above example, the filter selected is the Container List "Cold treatment".

Clicking "Show list of containers" opens the following dialog:

Container IDs

The following containers were sent to the RDC in the shown order at latest preparation:

- BMOU9005005
- BMOU9011965
- BMOU9012004
- BMOU9012426
- CRLU1605201
- CRLU1605337
- CRLU1606673

No. of containers: **483**

Close



Here, the selected containers appear in the selected walk route order.

Removing an RDC alarm from REFCON

An "RDC alarm" is an alarm generated by the user by selecting it from the standard alarms list or by typing in an alarm text at the RDC unit during an inspection.

Normally you would check the same container during later inspections. When observing that now the alarm condition is no longer present, you would delete the alarm at the RDC unit and thereby have it removed from REFCON.

There are, however, situations where it is more convenient to remove the RDC alarm from REFCON. You can do this by selecting the "User Alarm" button in the [Container Details](#)^[143]. You will see that the normal User Alarm dialog box has been extended with an "Alarms created from RDC" section:

In the section, the two "types" of alarms can occur:

- There can be any number of "Alarm(s) selected from list". To remove one, click it in the list and then select the "Remove" button. If an alarm is grayed in the list, it has already been removed, but awaits acknowledgement.
- There can be one "Typed alarm text"; in the above example, the words "This is a free text alarm" were typed in at the RDC unit. Select the "Remove" button to have it removed without using the RDC unit. You will have to acknowledge the alarm before it actually disappears.

15.1.1 RDC Preparation Selection

Key subjects in this topic:

[Overview](#)^[248]
[The RDC Preparation Selection dialog](#)^[248]
[Using filter, list or both](#)^[249]
[Selecting the filter](#)^[249]
[Selecting an RDC list](#)^[249]
[Using a selected Container List](#)^[250]
[Defining the Sorting order \(walk route\)](#)^[250]

Related topics:

[RDC-monitored containers](#)^[238]
[Event logging](#)^[171]
[How the Default Walk Route is organised](#)^[250]
[Creating a custom walk route](#)^[253]
[Operating RDC from REFCON](#)^[238]
[RDC+ introduction](#)^[238]



Overview

Preparation Selection means making the RDC unit ready for your next inspection round. You select which containers to include in the RDC list and the sorting order (walk route).

Note that it is possible to include non-RDC-monitored containers, that is, containers that are or starts communicating. Such containers receive data via the power lines, and therefore the data you may collect using the RDC unit will be ignored - of course, only where conflicts are detected.

The RDC Preparation Selection dialog

You make your choices in the following dialog:

RDC preparation selections

Define how containers are selected

☒ Use specified filter only
☐ Use filter and list
☐ Use specific list only
☐ Use selected Container List

☒ Containers that never communicated
☐ Containers with alarms
☒ Include containers missing data
☐ Special reefers
☐ Exclude containers w. unknown locations due to planning data
☒ Exclude "ghost containers" (no planning data) and dry reefers
☐ All containers

Please generate RDC list from a Container List Show

Select Container List version to decide RDC list:
 Cold treatment

Warning: RDC recorded setpoint and temperatures will be ignored if the container communicates when the RDC is returned

Define how containers are sorted (walk route)

☒ Alphabetical by container ID
☐ As defined by default walk route
☐ As defined by walk route loaded from following file:

Save as file
Browse Show

Prepare (and save as "Quick") Prepare (without saving) Cancel

In the upper part you select which containers to include in the preparation list. In the lower part you select the sorting order (walk route).

Having made your choices, select one of the Prepare buttons to have the preparation list generated and, if not empty, sent to the RDC unit:

- The left Prepare button also saves your choices so that you can very easily repeat it the next time you are going to prepare.
- The right Prepare button makes your choices used only this time.

Selecting Cancel will dismiss your possible choices, and the RDC unit is not prepared.



Using filter, list or both

To the left in the upper part of the dialog, the first three radio buttons decide whether to Use filter, Use list or both. To use list you have to generate a list from the Container List (see below).

If you select "Use filter and list", the result will limit to containers that both are on the list and meet the filter conditions.

Selecting the filter

In the frame to the right of the "Use specified filter only" radio button, you can set up a filter deciding which containers to include in the preparation list:

Choice	Result
Containers that never communicated	The result depends on the "Include containers missing data" checkbox: • If not checked, the RDC list will include all containers reported by the Planning System that have not (yet) started communicating. If the container has communicated, even if it was just for a while, it is not included. • If checked, the RDC list will include the same containers as if not checked plus all containers for which setpoint and temperature data are not known or are obsolete. Note that this selection corresponds to what we call RDC-monitored containers ²³⁸.
Containers with alarms	The RDC list will include all containers with alarms, no matter if the alarm origins from the container controller, from REFCON or from the user, and no matter if the container is communicating or not.
Special reefers	The RDC list will include all containers marked as Special reefers, no matter all other conditions of the container.
All containers	The RDC list will include all containers unconditionally.

You can refine the filter by selecting either of the checkboxes:

Choice	Result
Exclude containers with unknown locations due to Planning data	When this checkbox is checked, containers that were included in the Planning data but without a location are excluded from the resulting RDC list. In other words, containers that pass the other filter definitions are only included in the list if either they have Planning data-specified locations or they have no Planning data at all.
Exclude "ghost containers" (with no Planning data) and dry reefers	When this checkbox is checked, containers without Planning data and containers with no expected temperature setpoints are excluded from the resulting RDC list.

Selecting an RDC list

You can pick out a specific collection of containers for the RDC as follows:

1. Open (or switch to) the Container List (any version).
2. Highlight the containers of interest.
3. From the Shortcut menu, select "Make RDC list...".

What happens next depends:

- If the RDC unit **is** connected, and the RDC Preparation Selection dialog is **not** open, this will automatically open, and "Use specific list only" will be checked.
- If the RDC unit **is** connected, and the RDC Preparation Selection dialog is already **open**, this will be updated, and "Use specific list only" will be checked, ready to use the list.



- If the RDC unit is **not** connected, you will get a message that you need to connect it before you can use the list.

Using a selected Container List

As described in [Setting up the Container List](#)^[80], it is possible to define Container List versions with built-in filters. Such a filter can be anything from simple to very advanced, thus selecting exactly the subset of containers that you want. Before proceeding, you might have to create a Container List version if the appropriate one does not exist. REMEMBER to SAVE it before using it.

If you select the fourth radio button upper left, named "Use selected Container List", and choose the desired Container List version from the list, the RDC list will be identical to the actual list of containers in that version, at the moment you select "Prepare...". Next time you prepare, the filter is applied again, meaning that the list of containers might have changed.

Defining the Sorting order (walk route)

In the lower part of the RDC Preparation Selection dialog you decide the sorting order of the selected containers. You have three options:

Choice	Result
Alphabetical by container ID	All the containers are sorted by ID, meaning that the locations are not taken into account. This is your only option if you have defined no walk route.
As defined by default walk route	A walk route is basically a list of locations. All containers with locations defined in the walk route list are sorted according to this, while containers without defined locations or with locations not defined in the walk route are sorted alphabetically by ID and appended to the sorted list. The default walk route would be automatically created when generating the Container Plan, if this option is enabled, please refer to How the Default Walk Route is organised^[250].
As defined by walk route loaded from file (see procedure below)	As alternative to the default walk route, you can define your own as described below, see Defining a custom walk route ^[253] .

15.1.2 How the Default Walk Route is organised

Key subjects in this topic:

[Overview](#)^[250]
[How the default walk route for a terminal is organised](#)^[251]
[How the default walk route for a ship is organised](#)^[252]

Related topics:

[RDC Preparation Selection](#)^[247]
[Operating RDC from REFCON](#)^[238]
[REFCON Data Collector](#)^[236]

Overview

The default RDC walk route is created on basis of the optional Container Plan. If no such is included, the following description is not relevant.



How the default walk route for a terminal is organised

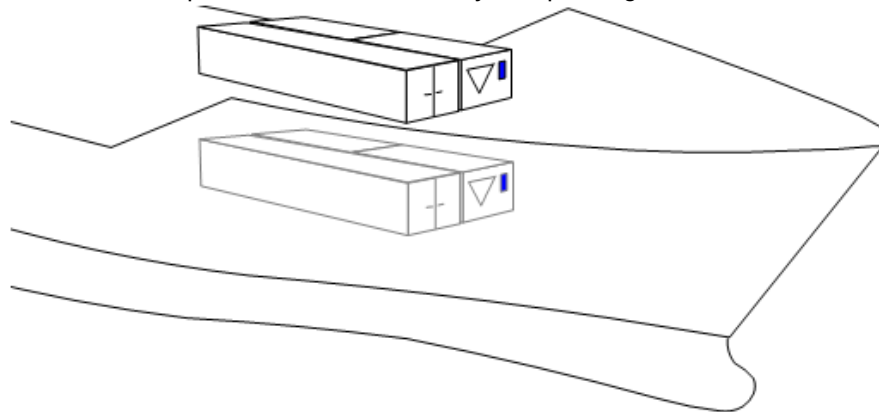
As there is no common standard for the organisation of the locations in terminals, the default walk route for terminals is one area, named "Div", where all locations appear in kind of random order. The only real value of this default walk route is to be the basis for your custom walk route, please refer to [Creating a custom walk route](#)^[253].

How the default walk route for a ship is organised

The calculation of the default walk route for a ship is based on the assumption that the ship's containers are organised according to the ISO 9711-1:1990 standard, however, it also handles certain minor exceptions from this.

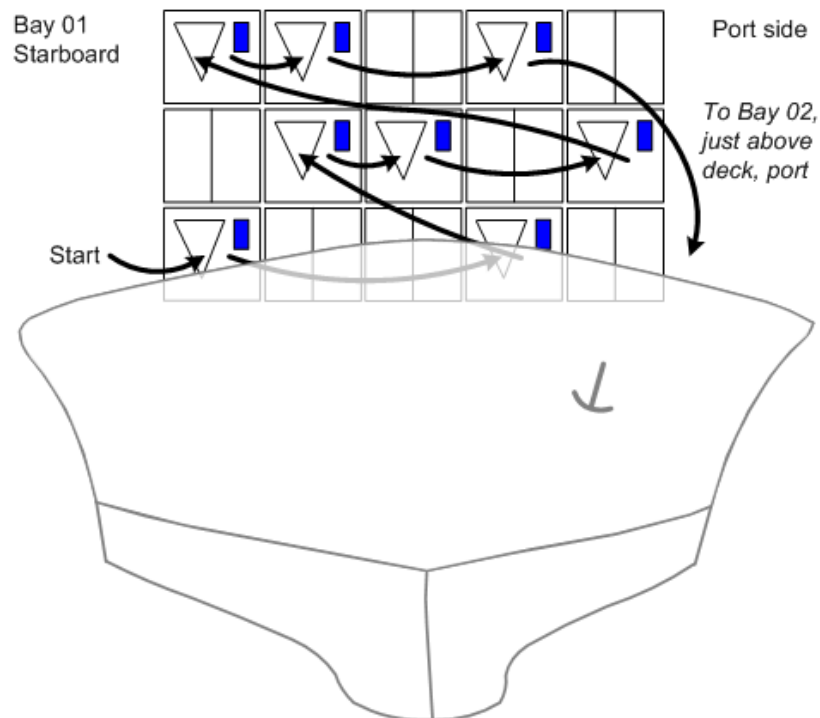
The order of the locations in the default walk route is decided by the following rules/assumptions:

- All 40' containers are placed with the machinery end pointing astern.

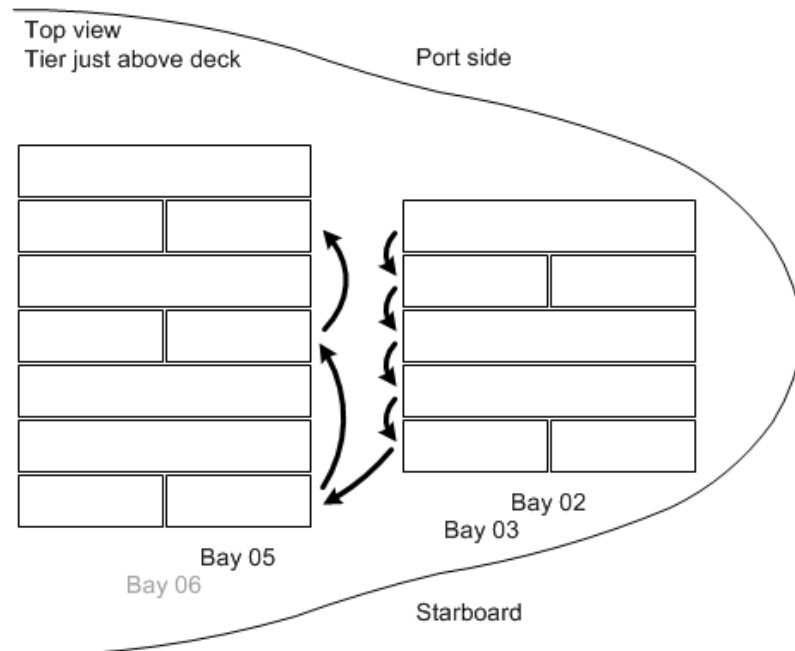


For example, a 40' container in bay 06 and a 20' container in bay 07 will have the controller displays accessible side by side, while a 20' container in bay 05 will have the controller display at the other end (ahead) of the 40' bay.

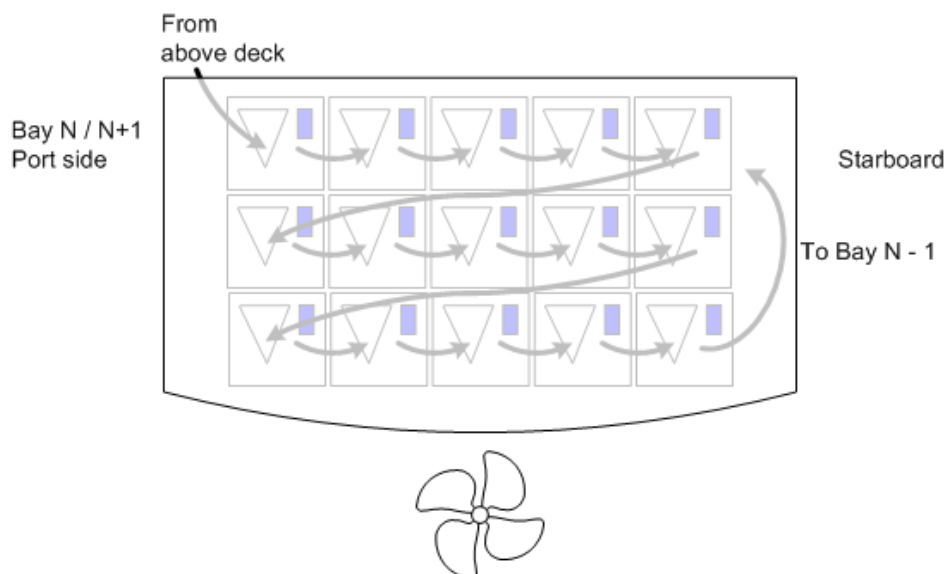
- The walk route starts ahead, on deck.
- You inspect all containers on deck first, and then all containers below deck.
- Far most ahead you might have 20' containers in bay 01. You inspect these first, starting just above deck in starboard side. Having finished the first tier, you continue at the tier just above, starting again in starboard side. Etc.



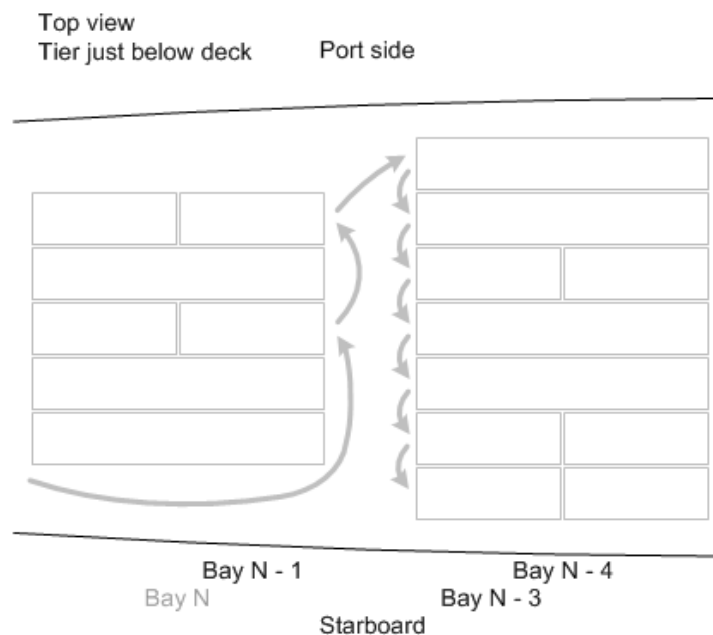
- Having completed bay 01 above deck, you continue with bay 02/03, starting just above deck in port side. When you reach the starboard side, you turn around and inspect possible 20' containers in bay 05, starting in starboard side. Back in port side, you continue to the tier just above. Etc.



- Having completed bay 02/03/05, you repeat this in bay 06/07/09, 10/11/13, etc.
- Having completed the last bay astern on deck, you continue with the last bay astern below deck. If for example the last bay below deck is bay 54, you first inspect bay 54/55, starting in port side just below deck. Having completed the uppermost tier, you step down to the tier just below, starting in port side. Etc.



- Having completed bay 54/55 (in the example), you continue with bay 53, starting just below deck at starboard side. When you reach the port side, you turn around and inspect bay 50/11, starting in port side. Back in starboard side, you continue to the tier just below. Etc.



- Having completed bay 53/50/51 (in the example), you repeat this in bay 49/46/47, 45/42/43 etc.

15.1.3 Creating a custom walk route

Key subjects in this topic:

[Overview](#) ^[254]

[Start: Exporting the default walk route](#) ^[254]

[Customisation](#) ^[254]

[Final: Importing your text file](#) ^[255]

Related topics:

[RDC Preparation Selection](#) ^[247]

[How the Default Walk Route is organised](#) ^[250]

[Operating RDC from REFCON](#) ^[238]

[RDC introduction](#) ^[236]



Overview

On basis of the default walk route, you can create any number of customised walk routes using a text editor like Notepad. Any such custom walk route is kept in an external text file. It is your responsibility to name the file, to decide its location and to back it up. Having created such a file (or more), you use it in the [RDC Preparation Selection](#)^[247] dialog box.

Start: Exporting the default walk route

To create a custom walk route, you need a complete list of all possible locations. The safe and easy way to obtain this list is to export the default walk route. This also gives a few other advantages as described below. Proceed as follows:

1. Open the RDC Preparation Selection dialog (only possible with an RDC connected).
2. Make sure the "As defined by default walk route" radio button is selected. If this is not possible, unfortunately no default walk route exists.
3. Select the "Save as file" button.
4. Choose folder and file name.

Customisation

To do your customisation, proceed as follows:

1. Open Notepad - or a similar plain text editor.
2. Open the file saved above or a previously created walk route file.
3. Follow the instructions at the top of the exported file and/or the hints below.
4. Save the file in a position that can be read from the REFCON Monitor PC, for example on a USB key.

A walk route file opened in Notepad typically looks like this:

```
// This is a copy of the default walkroute, exported by user John on 2011-08-12
// You may create a customised walkroute by editing this file and then importing it to REFCON.
// Please carefully follow these guidelines:
// - Any line like this, starting with //, is a comment, being ignored by the import software.
// - You may add (or delete) comment lines anywhere in the file.
// - There normally are four lines starting with ##. These lines define what to call the
//   concepts "area" and "subarea" in the RDC+ display, in various contexts.
//   You may change the word(s) within the quotes, however, keep them short!
// - A line starting with :: defines an area, like this example (though without comment //):
//   ::Area "East" "Eastern quay"
//   that is, the word Area followed by
//   - a very short name for the area, max 4 characters, used in left column on RDC+
//   - a longer (but still quite short) name for the area, used in RDC+ caption.
// - Most lines define locations. Each such line must contain one complete location string,
//   and nothing else. Every single location defined in your Container Plan must be found.
//   Do not include a location that is not in the Container Plan. Locations are not allowed
//   to be duplicated. Any location belongs to the last area defined above it in the file.
// - You may add a new area by inserting a :: line where appropriate.
// - You may remove an area by deleting the :: line. All locations in that area will then
//   relate to the previous area. Do not delete the very first area!
// - You may MOVE locations using cut and paste. We recommend that you never type in or edit
//   location strings, and you never delete location strings. Always paste immediately after cut!
//
##Area is called singular: "Area"
##Area is called plural: "Areas"
##Subarea is called short: "Location"
##Subarea is called long: "Location"
//
::Area "East" "Eastern quay"
004RA01A
004RA02A
005RA01A
005RA02A
006RA01A
```

The first about 25 lines are comments (defined by the //).

- The first line describes the file. You could choose to edit this line, or add one below, when you make changes to the file, for registration.
- The next about 23 lines is a guiding instruction on how to edit the file. Please read the instructions carefully.

There are in fact four "types" of lines in the file:



Line starting with	is a	Description
//	comment line	Being ignored by the importing software - you may add or delete comment lines to wish.
##	designation line	There should always be exactly 4 designation lines - never change anything but the words within the quotation marks.
::	area definition line	Defines that here starts another area, and the short and long title of this.
location string	location line	There must be exactly one location line for every single location covered by your REFCON system, that is, as defined in the Container Plan.

Normally you would leave all lines down to the first :: as is, maybe except for some registration at the top.

You would probably mainly do the following operations:

- Move a location, using cut and paste.
- Define a new area by a) copying an existing :: line; b) pasting where the added area is to start; c) editing the words within the quotation marks (remember that the first **must** be very short).

Final: Importing your text file

To utilise your custom walk route, proceed as follows:

1. In the RDC Preparation Selection dialog, select the Browse button.
2. Locate the walk route text file you created above.
3. Click Open. The file is validated, and in case of troubles you are informed.
4. Select the Show button next to the Browse button to make sure the file was imported all right.
5. Select the "Prepare (and save as "Quick")" button to have the imported walk route saved in REFCON.

Note: REFCON can only store one walk route at a time. Importing another custom walk route or starting to use the default walk route makes REFCON forget the one you first imported. Therefore make sure you keep the external text file, if you ever would use it again.

REFCON 6

Reefer Container Monitoring System

Four pole monitoring

Transportation Solutions





16 Four pole monitoring

Key subjects in this topic:

[Overview](#) ²⁵⁸
[Using four pole monitoring](#) ²⁵⁸
["Planned" and "unplanned" four pole containers](#) ²⁵⁸
[Settings and alarms related to four pole monitoring](#) ²⁵⁸
[Container List columns relevant to four pole monitoring](#) ²⁵⁹
[Indication in Container Plan etc](#) ²⁶⁰
[The pre-defined Four pole list](#) ²⁶⁰
[Four pole and the RDC](#) ²⁶⁰

Related topics:

[Four pole related maintenance](#) ²⁶¹
[The Container List](#) ⁶⁶

Overview

Four pole monitoring is an optional extension to REFCON 6, allowing supervision of a reefer container through three binary signals, "Compressor Running", "Defrosting" and "In range". The binary signals are read through dedicated hardware modules, called **FMM** units (Fourpole Monitoring Modules) being connected to REFCON by power line communication.

A four pole monitored container may also be power line monitored if it has a power line modem attached. Also it can be additionally monitored by the optional RDC.

Using four pole monitoring

All you have to do in your daily work is to plug in (and later out) the four pole cables to the containers that are not otherwise automatically monitored. Note that it is **important** that you use the correct cable for the specific container, matching the location.

Having plugged in, the container will start being four pole monitored after some time. Note, however, that if all three signals are Off, REFCON is **not** able to detect the container. If/when later a signal gets On, the detection will occur.

"Planned" and "unplanned" four pole containers

When a four pole cable is plugged in, REFCON checks if it has information about a container on that location from the planning system in advance. If so, it is called a "planned" four pole container, and it will occur in the REFCON Monitor with the container ID stated in the planning data.

If the location is not found in planning data, it is called an "unplanned" container. In such case REFCON does not know the real container ID. It is therefore given a "pseudo" container ID starting with the letters "FPMC" (for Four Pole Monitored Container) followed by the location. In the case that the location is shorter than 7 characters, one or more additional figures 9 are added. For example, an "unplanned" container at location 020182 will get the ID "FPMC0201829".

If later planning data arrives defining that specific location, the "pseudo" container ID is replaced by the real ID within a short time.

Settings and alarms related to four pole monitoring

You can control the four pole related alarms from the Four pole tab in the REFCON Setup window:



REFCON Setup

Configuration

Out-of-range alarm cut-off time after plug-in: 01:05 hours

Out-of-range alarm cut-off time after defrost: 01:05 hours

Time from all signals off to 'disconnected': 01:05 hours

☒ Cut-off 'out-of-range alarm' while 'disconnected' timer is running

☒ Enable 'defrosting too long' alarm

Defrosting too long time limit: 01:05 hours

☒ Enable 'FMM offline' alarm

OK Cancel Apply

As you see, all time settings are a little more than an hour as default. This is because in the worst case the updating of the four pole signals can take an hour. You may try smaller values if you think your system runs faster.

- When the "In range" signal goes Off, REFCON releases an alarm unless it is cut off. In the settings window, we refer to the "Out-of-range alarm". It is always cut off for the selected time after four pole cable plug-in, and for the selected time after defrosting has completed according to the "Defrosting" signal.
- Because REFCON can't distinguish between "all three signals gone Off" and cable unplug, a "disconnected timer" is used. It is started when change to all signals Off. While the timer is running, the container is still considered connected and four pole monitored. At timeout, it is considered disconnected and no longer monitored. Depending on other parameters this may mean that it is deleted from REFCON.
- You decide whether the "out-of-range alarm" is to be cut off while the disconnected timer is running by setting the first checkbox in the window.
- You decide if you want an alarm on "defrosting too long" by setting the second checkbox. If selected, you decide the time limit.
- The third checkbox decides whether you will have a system alarm in the case that an FMM stops communicating.

Container List columns relevant to four pole monitoring

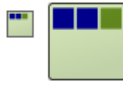
A few columns are available in Container Lists being relevant with the four pole monitoring option enabled. These columns are:

Column header / abbreviation	Explanation
Cmpr.Runn	● means that the compressor is currently running, while ● means that it is not so.
Defrosting	D means that the container is currently defrosting, while empty field means that it is not so.
In range	● means that the temperature is currently in range, while ● means that it is not so.
Plug No	The plug number on the FMM (1..10) reading the binary signals.
4P Station ID	The user decided description of the FMM.



Indication in Container Plan etc

When a container is four pole monitored, the three signals are graphically indicated in the Container Details and the optional Container Plan like this:



From left to right, the signals are indicated as follows:

Cmpr.Runn	 means that the compressor is currently running, while  means that it is not so.
Defrosting	 means that the container is currently defrosting, while  means that it is not so.
In range	 means that the temperature is currently in range, while  means that it is not so.

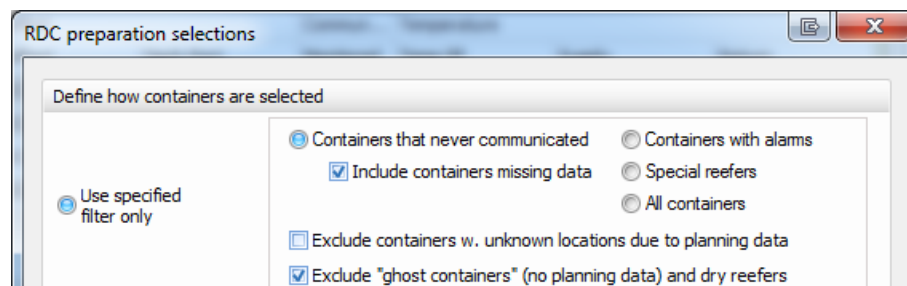
The pre-defined Four pole list

You may [activate](#) ⁸¹ the pre-defined "Four pole list". Afterwards you can modify the column selection etc. to meet your specific needs.

Four pole and the RDC

You may wish to use the optional hand-held RDC to manually retrieve data additional to the four pole signals. For detailed information about using the RDC, please refer to the separate online help module chapter.

The standard RDC selection looks like this:



This will make all "planned" containers that are not Monitored by PCT sent to the RDC, including "planned" four pole containers, but excluding "unplanned" four pole containers.

If you want "unplanned" four pole containers to be included, simply remove the checkmark labeled "Exclude "ghost containers" (no planning data) and dry reefers".



16.1 Four pole related maintenance

Key subjects in this topic:

[The four pole configuration](#)^[261]
[FMMs in the System Devices list](#)^[261]
[Inspecting the four pole configuration](#)^[261]
[Sorting the lists](#)^[262]
[Changing the four pole configuration](#)^[262]
[Editing the User's Description](#)^[263]
[Replacing a station](#)^[263]
[Exchanging plugs](#)^[264]
[Understanding the FMM](#)^[265]

Related topics:

[Four pole monitoring](#)^[258]
[Understanding and Maintaining REFCON](#)^[302]
[System Devices List](#)^[333]

The four pole configuration

Your REFCON system includes a specific number of four pole monitoring stations. In each station you find one FMM.

On each FMM, a specific number (1..10) of plugs are in use. Each plug is attached to one specific container location, meaning that when the cable in that plug is connected to some container, it is presupposed that this container is situated at that exact location.

All this is defined in the four pole configuration, being stored in a file named Location.dat on the REFCON Main PC. Please do never make any changes directly to this file!

The four pole configuration specifies all the stations. First of all, each station has a unique number, from 1 to N, N being the number of stations. Additionally, each station refers to an FMM ID and a list of plug specifications.

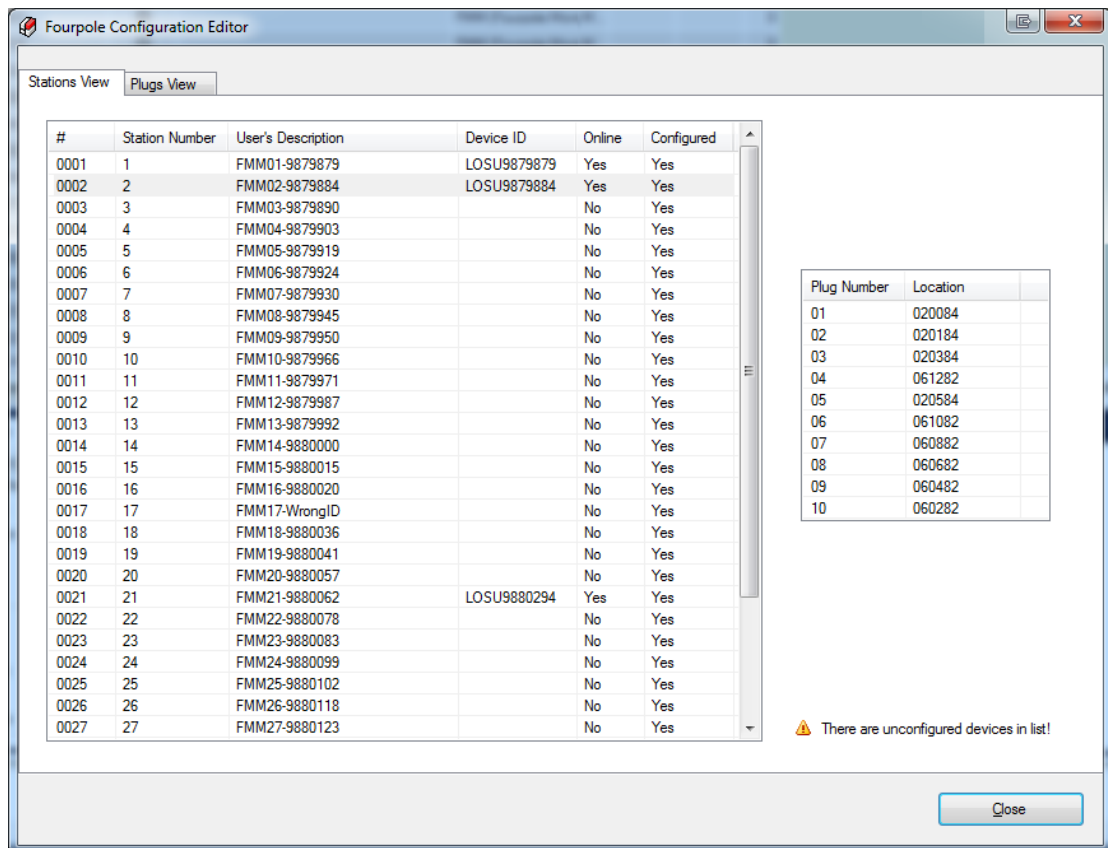
FMMs in the System Devices list

Each FMM occurs as a single line in the [System Devices](#)^[333] list. It appears as follows:

Column header / abbreviation	Explanation
Device	Example "FMM001", the digits indicating the number of the station in the four pole configuration.
User's Description	The free text description of the station, changeable.
Online	Indicates if the FMM is currently online.
Software	Indicates the software version running in the FMM, if known.
Hardware	If the FMM has not started communicating, you will see the static text "FMM (Fourpole Monitoring Module)". Having started communicating, the ID of the FMM is added, for example "FMM LOSU9879879...".
Activity	The date and time when last receiving data from the FMM.
Connected	Number of plugs on the FMM being in the connected state.

Inspecting the four pole configuration

If you right-click an FMM in the [System Devices](#)^[333] list and select "Configuration" from the Shortcut menu, you can view how the plugs are configured:



Note that the unit that you did select in the System Devices list is highlighted (light gray background colour) and is having it's list of plugs displayed to the right.

While having the Editor window open, you can click any other station to have it's plugs displayed in stead.

Note the warning lower right: "There are unconfigured devices in list!". This appears if/when you have installed a new FMM in stead of an existing/a configured. It would tell you that there is a need for changing the configuration as described in the following.

Also note that the Editor has two tabs. The "Stations View" tab shown above is always open when opening the Editor. You may select the "Plugs View" tab to see a list of all plugs as described below.

Sorting the lists

When opening the Editor,

- The Stations View list is always sorted like this: The configured stations come first, sorted by Station Number. After this, the not configured stations are listed, if any.
- The Plugs View list is sorted primarily by Station Number, secondarily by Plug Number.

You may change the sorting of either of the lists by clicking any field in the list's header, except the # header. Click once to sort alphabetically (ascending) by that column. Click once more to sort reverse alphabetically (descending).

Changing the four pole configuration

In a commisioning situation and when replacing equipment there will be a need for editing the four pole configuration. You can do three types of changes: Edit the User's Description, Replace a station, and exchange two plugs' locations.



Editing the User's Description

To change the text shown in the "User's Description" column, select a single station in the "Stations View" list, right-click it and select the first option:

0020	20	FMM20-988005 /	NO
0021	21	FMM21-988005 /	Yes
0022	22	FMM22-988005 /	No
0023	23	FMM23-988005 /	No
0024	24	FMM24-988005 /	No

Change User's Description

Replace Device

Make your changes in the small window and click OK.

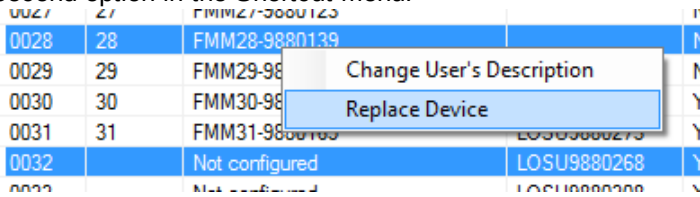
Replacing a station

The wording "replace a station" refers to the situation that you have been replacing an FMM with a spare. The function is also relevant in the commissioning situation when you install an FMM the first time.

To do a replacement, you must **always** proceed as follows:

Step	Action	Comment
1	Pick a spare FMM, and do carefully write down it's ID found on the label at the end of unit: 	Note that the labeled ID forms the FMM Device ID: Remove the first digit and the hyphen, and put the letters "LOSU" up front. For example, ID: 0624-0500 makes it "LOSU6240500".
2	Take the spare to the station needing replacement. Remove the existing FMM. Do carefully write down it's ID, too - maybe make an "Old" note next to the ID.	If you do more replacements, make sure you know what "new" ID matches what "old".
3	Install and power up the spare.	
4	Return to the REFCON PC. In the System Devices list, await that a new FMM reading "Not configured" in the "User's Description" column gets online, with Device ID formed by the ID you noted in step 1.	It may take up to a couple of hours from power up until online.
5	Open the Fourpole Configuration Editor  .	
6	Locate and click your new "not configured" unit at the bottom of the "Stations View" list. If there are more "not configured", carefully select the one with the relevant Device ID.	
7	Hold down the Ctrl button on the keyboard, and click the replaced FMM in the list, reading the Device ID formed by the ID you noted in step 2.	You should now have the two units highlighted. Note that the replaced unit might still be online because it can take a few hours for REFCON to detect offline.

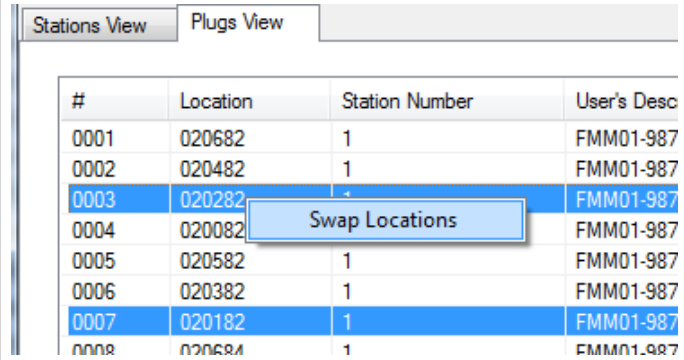


Step	Action	Comment
8	Right-click either of the two highlighted rows, and select the second option in the Shortcut menu: 	
9	You get a warning. To proceed, answer Yes.	You should now see that the "Not configured" unit disappears from the list (and from the System Devices list), and that the replaced unit has got the new Device ID.

Exchanging plugs

It may happen that a location configured for a plug does not match the location stated on the four pole cable. You would then normally have at least two plugs with such mismatch. There can be many reasons, but the point is that easiest way to solve it may be to change the four pole configuration. You do this by exchanging the locations of two plugs. You may have to repeat this if the problem is more complex.

Proceed as follows:

Step	Action	Comment
1	Open the Fourpole Configuration Editor , and click the "Plugs View" tab.	You now see quite a long list of all configured plugs in the system.
2	Locate and click the first of the plugs with wrong location. You would probably locate it by this wrong location!	You should now have one plug highlighted.
3	Locate the other plug. You may need to scroll the list to find it.	
4	Hold down the Ctrl button on the keyboard, and then click the other plug.	You should now have both plugs highlighted.
5	Right-click either of the two highlighted rows, and select the option in the Shortcut menu: 	You will see that the highlights disappear from the list, and that the locations of the two plugs that you did select have swapped. In case you made a mistake, simply select the two again and repeat the operation, to swap back.



Understanding the FMM

Please be aware that REFCON communicates with the FMM through the PCT (Power Cable Transmission). This means that logging in and out as well as data polling work very much as if the FMM were a container with PCT modem.

REFCON 6

Reefer Container Monitoring System

The Email Notification function



17 The Email Notification function

Key subjects in this topic:

[Overview](#)^[268]
["Action categories" and "Notification groups"](#)^[268]
[Deciding email receiver by container location](#)^[269]
[About email templates](#)^[269]
[Customer's contribution](#)^[269]
[Possible delay of received emails](#)^[269]
[Installation](#)^[269]

Related topics:

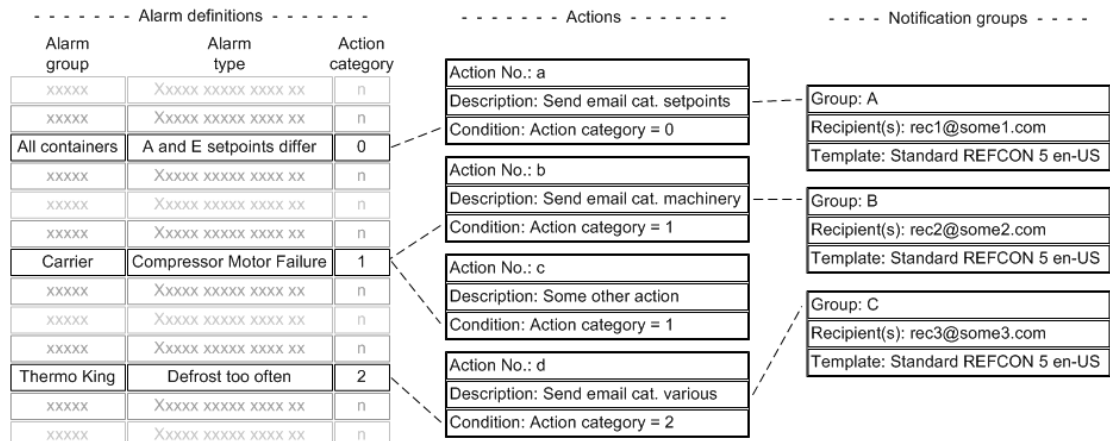
[Email Notification Setup](#)^[269]

Overview

The optional *Email Notification function* allows REFCON to alert about container alarms and possible other decided events by sending emails. The different types of alarms can be categorised so that, for instance, only selected alarms do cause emails to be sent. Also it is possible to have alarms of different categories sent to different email addresses.

"Action categories" and "Notification groups"

The Email Notification function is based on a general "action" function. The connection between these functions can be illustrated as follows:



- Any alarm type (for example Carrier's "Compressor Motor Failure") is related to exactly one *Action category*. The Action category is specified as a non-negative integer and can be edited by the user. In the present REFCON 6 version, this can only be done by editing the "AlarmPrioritiesAndCategories.xml" file located in the "Data Engine\Data" folder on the REFCON Main PC. It is extremely important to be very careful when editing the file!
- Also in the present version, only 4 Action categories are available, numbered 0, 1, 2 and 3. The Action category 3 is reserved for system alarms.
- REFCON generally includes a number of *actions*. Examples of actions would be sending an email, or alerting the user on screen. In the present REFCON 6 version, only 4 email actions are available, corresponding to each of the 4 Action categories.
- Each action that involves sending emails is attached to exactly one *Notification group*. Each Notification group defines the list of email recipients (one or more) and the [email template](#)^[269].



Deciding email receiver by container location

As standard, all recipients defined for a group will receive all emails regarding alarms related to that group.

But if your [Container Plan](#)^[96] has more than one overview (tab) - which is typically relevant for large sites split in different areas - you can decide that the recipients of container alarm emails depend on the containers locations. Please refer to [Email Notification Setup, multiple areas](#)^[272].

About email templates

Any email being sent is based on a decided *template*. The template generally defines

- the "subject"; and
- the "body" of the email.

In the present REFCON 6 version, you do not find a template editor. Please select one of the provided templates, or ask Copeland Transportation Solutions to implement one for you.

Note: If you have upgraded REFCON to a version where you expect the factory templates to having been updated, please ask Copeland Transportation Solutions for guidance on how to update.

Customer's contribution

To have it work, the customer needs a mail account at some Internet company. Depending on this, it is possible to arrange that the emails are received on the user's cell phone. Note that all this is not a part of Copeland Transportation Solutions's delivery!

Possible delay of received emails

Notification emails are sent by REFCON immediately when the alarms or other events occur. Depending on the Internet company and the general load on the Internet, there can be some delay before the email is actually received.

Installation

Having installed REFCON, and having achieved a mail account, you need to [make a few settings](#)^[269]. After this, alarms and other events will automatically be forwarded as decided. You can of course change the settings at any time. Note: In case of a re-installation of REFCON, remember to import your backup of the settings, and to verify that they actually are as desired.

17.1 Email Notification Setup

Key subjects in this topic:

[Overview](#)^[269]
[Common Configuration](#)^[270]
[Notification Group Configuration](#)^[270]
[Adding or editing an email recipient](#)^[271]
[Editing the Recipient List, single Container Plan area](#)^[271]
[Editing the Recipient List, multiple Container Plan areas](#)^[272]

Related topics:

[The Email Notification function](#)^[268]

Overview

The Email Notification function is controlled through the Alarm Options tab of the [REFCON Setup](#)^[330] dialog:



To have email notifications transmitted, make sure the "Enable" checkbox is checked. Also decide the configuration, see [Common Configuration](#)^[270] and [Notification Group Configuration](#)^[270] below.

If you further want an email in the System alarm notification group in case REFCON keeps refusing to start for a selected time, check the second checkbox as well and select a time value.

Remember to apply your changes.

Common Configuration

Note that when enabling Email Notification (at the moment you click the checkbox), the "Email Server" tab shows up in the [REFCON Setup](#)^[330] dialog box, if not already visible. In addition to the settings described below, you need to set up your email server, please refer to ["Email Server" settings](#)^[177] for information.

Notification Group Configuration

To decide the email recipients and more, click the "Configure notification groups" button. This makes the following dialog box appears:

In the left side, the available Notification groups are listed. In the present REFCON 6 version, there is a fixed number of 4 Notification groups available. Out of the box these are named as shown above.

At each notification group in the list a status indication is found:



- --no recipient-- means that no emails will be sent in this group because no email address has been selected.
- --disabled-- means that no emails will be sent in this group because the "Group enabled" check mark has been removed.
- Neither of the above means that the group is active.

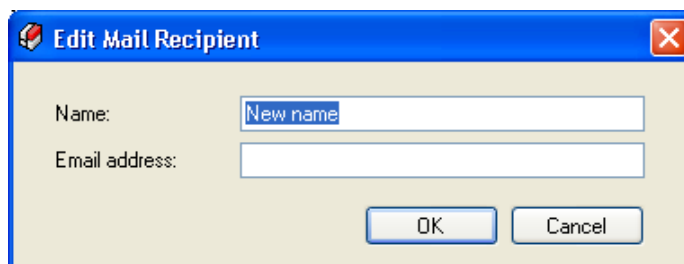
To make any change to a notification group, select it in the list. This makes the details of the group appear in the right side of the dialog box. Here you can:

- See the Group ID, being the link to the specified [action category](#) ^[268] of each alarm.
- Change the Group name. When you select this field, you get a change warning, because the out-of-the-box text actually matches the specified out-of-the-box action categories, so change with care! Apart from this, you can enter any string to wish.
- Decide if the group is to be enabled. If you remove the check mark, no emails will be sent in this group.
- Choose the template by selecting one in the list.
- Decide the email recipients by setting or removing check marks to the already defined recipients. If a recipient is missing in the list, use the ["Add recipient"](#) ^[271] ^[271] button.

Your possible changes are **not** applied when selecting OK. Having returned to the REFCON Setup, select OK or Apply here to make your changes take effect. You may alternatively select Cancel to dismiss your changes.

Adding or editing an email recipient

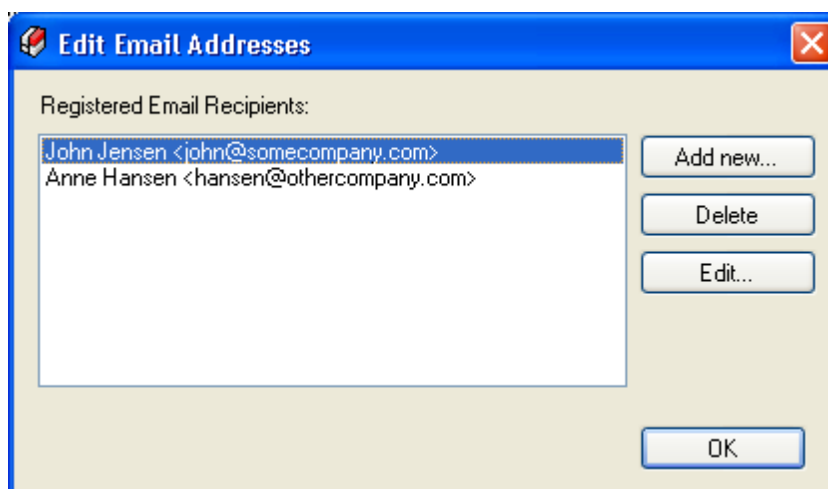
When you decide to Add or Edit a recipient, the following dialog box opens:



- It is optional but recommended that you enter a Name for the recipient. It could be a persons name or a job title, maybe. If you do not want to define a name, clear the field.
- You must enter a valid email address as two strings separated by @.
- Click OK to finish your change.

Editing the Recipient List, single Container Plan area

Selecting the "Edit receiver email addresses" button in the REFCON Setup makes the following dialog box appear when your Container Plan has one primary page only covering all locations:





You will recognize the Recipient List as a simplified list of Contacts, as known from most email programs.

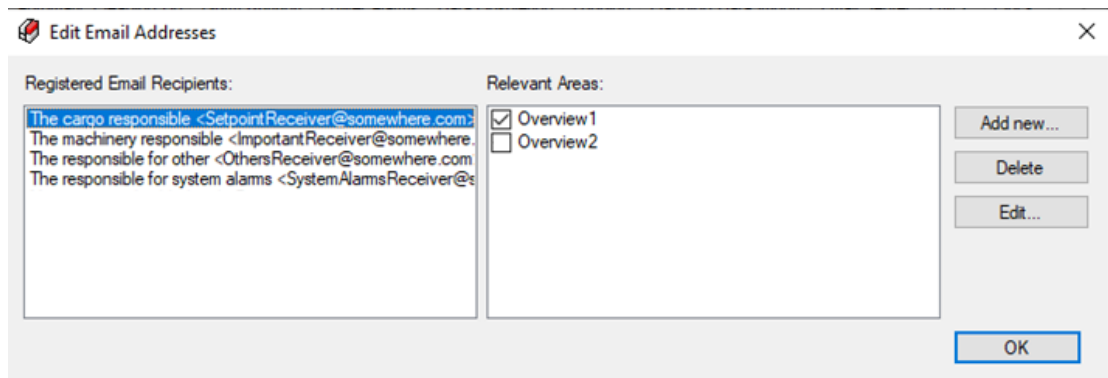
In the dialog you can:

- Add a new recipient by clicking the "Add new" button. This opens the [Edit Mail Recipient](#)^[271] dialog box described above.
- Remove the selected recipient by clicking the "Delete" button. Note that if the recipient is in use in one or more Notification groups, you are warned about this.
- Change the name or the email address of an existing recipient by clicking the "Edit" button. This opens the [Edit Mail Recipient](#)^[271] dialog box described above.

Your possible changes are **not** applied when selecting OK. Having returned to the REFCON Setup, select OK or Apply here to make your changes take effect. You may alternatively select Cancel to dismiss your changes.

Editing the Recipient List, multiple Container Plan areas

Selecting the "Edit receiver email addresses" button in the REFCON Setup makes the following dialog box appear when your Container Plan has more than one overview:



In this example, the Container Plan has two areas called "Overview1" and "Overview2". When a recipient is created it is assigned to all areas. In the example, "Overview2" has been deselected for first recipient, meaning that if a container in alarm is located in the "Overview2" area, first recipient will have no email.

To change the area selections, select recipients one by one and check/uncheck the areas appropriately.

Apart of this, recipients can be added, deleted and edited as described above.

REFCON 6

Reefer Container Monitoring System

Digital input/output signals



18 Digital input/output signals

Key subjects in this topic:

[Overview](#) ²⁷⁴

[Digital outputs](#) ²⁷⁴

[Digital inputs](#) ²⁷⁴

Related topics:

[The Digital IO viewer](#) ²⁷⁴

Overview

Your REFCON Main PC is optionally equipped with hardware allowing writing/reading digital outputs/inputs. As standard a relay card type LPCI-7250 is used, but for specific details, please refer to "Instruction Manual - Hardware Descriptions.pdf" being as standard found on the "Customer disk" supplied with REFCON.

Digital outputs

The following table shows how the digital outputs are used as standard:

Channel	Type	Name / usage
DO 0	NO and NC	System Alarm
DO 1	NO and NC	Reefer Container Alarm, Normal
DO 2	NO and NC	Reefer Container Alarm, Special
DO 3	NO and NC	Common Alarm (/ Buzzer On)
DO 4	NO	System Alarm
DO 5	NO	Reefer Container Alarm, Normal
DO 6	NO	Reefer Container Alarm, Special

Digital inputs

The following table shows how the digital inputs are used as standard:

Channel	Type	Name / usage
DI 0	24VDC input	Reset (/ Buzzer Silence)

18.1 The Digital IO viewer

Key subjects in this topic:

[Overview](#) ²⁷⁵

[Opening the Digital IO viewer](#) ²⁷⁵

[Description of the Digital IO viewer](#) ²⁷⁵

Related topics:

[Digital input/output signals](#) ²⁷⁴

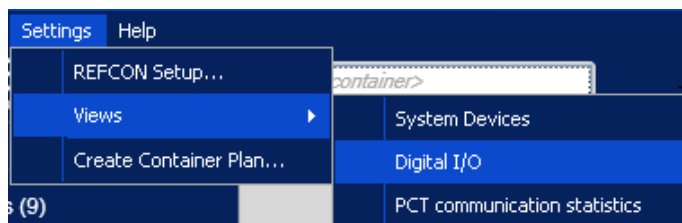


Overview

The *Digital IO viewer* shows the status of each individual digital output and input on the optional relay card. Note, however, that it is in fact the internal software status that is shown; should a relay or external cabling fail, the physical signal might differ.

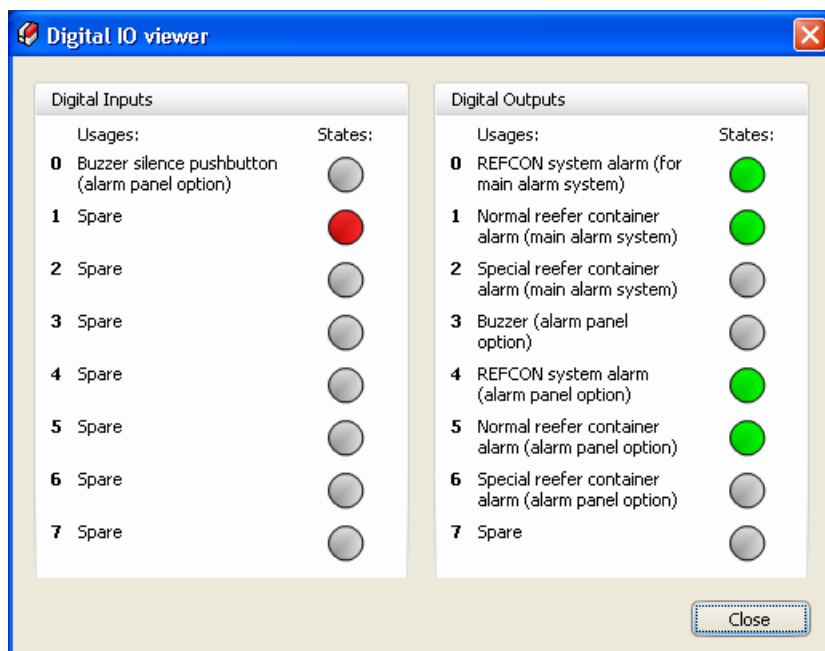
Opening the Digital IO viewer

The Digital IO viewer is available from the Settings menu under the Views entry (if enabled in the [license](#)^[12]):



Description of the Digital IO viewer

In the Digital IO viewer you see one circular indicator for each digital output and input:



The digital input indicators appear as follows:

Symbol	Colour	Explanation
	Grey	The input is not active (= 0VDC).
	Red	The input is active (= 24VDC).

The digital output indicators appear as follows:

Symbol	Colour	Explanation
	Grey	The output is not active.
	Green	The output is active.



REFCON 6

Reefer Container Monitoring System

PCT Modem Troubleshooting



19 PCT Modem Troubleshooting

Key subjects in this topic:

[Overview](#) ^[278]
[Communication path](#) ^[278]
[The REFCON system must work](#) ^[278]
[Updating Planning data](#) ^[278]

Related topics:

[Checking the Containers](#) ^[279]
[Checking modem-related alarms](#) ^[283]
[Checking the Modem Remedy List](#) ^[292]
[Checking the Power Cabling](#) ^[295]
[Typical Errors](#) ^[295]
[Locating the Modem](#) ^[296]
[RMM Modem Facts](#) ^[298]

Overview

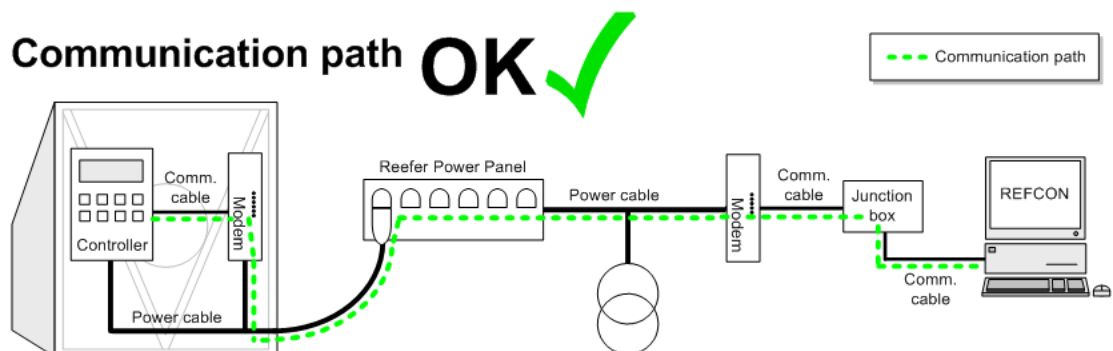
To have the full benefits of your REFCON system, it is very important that the highest possible number of reefer containers are communicating on the PCT (Power Cable Transmission) or through a [REFCON 6 extension](#) ^[218].

This section focuses on optimizing the PCT by having the electronics – modem, controller etc – on every single container maintained. This Guide intends to show you how to check the containers, and how to act in case of troubles.

Please note that this guide assumes that the conditions are stable for several hours for the reefer container being investigated. Stable conditions means that it is constantly plugged in to power, and it's Planning data are up-to-date.

Communication path

The communication only works if the entire communication path is OK:



The REFCON system must work

It is essential to modem troubleshooting that the REFCON system as such works well. If you get an error here, it might appear as a high number of containers have communication problems, maybe located in same physical area. Please consult [Understanding and Maintaining REFCON](#) ^[302] to learn how to check and fix system errors.

Updating Planning data

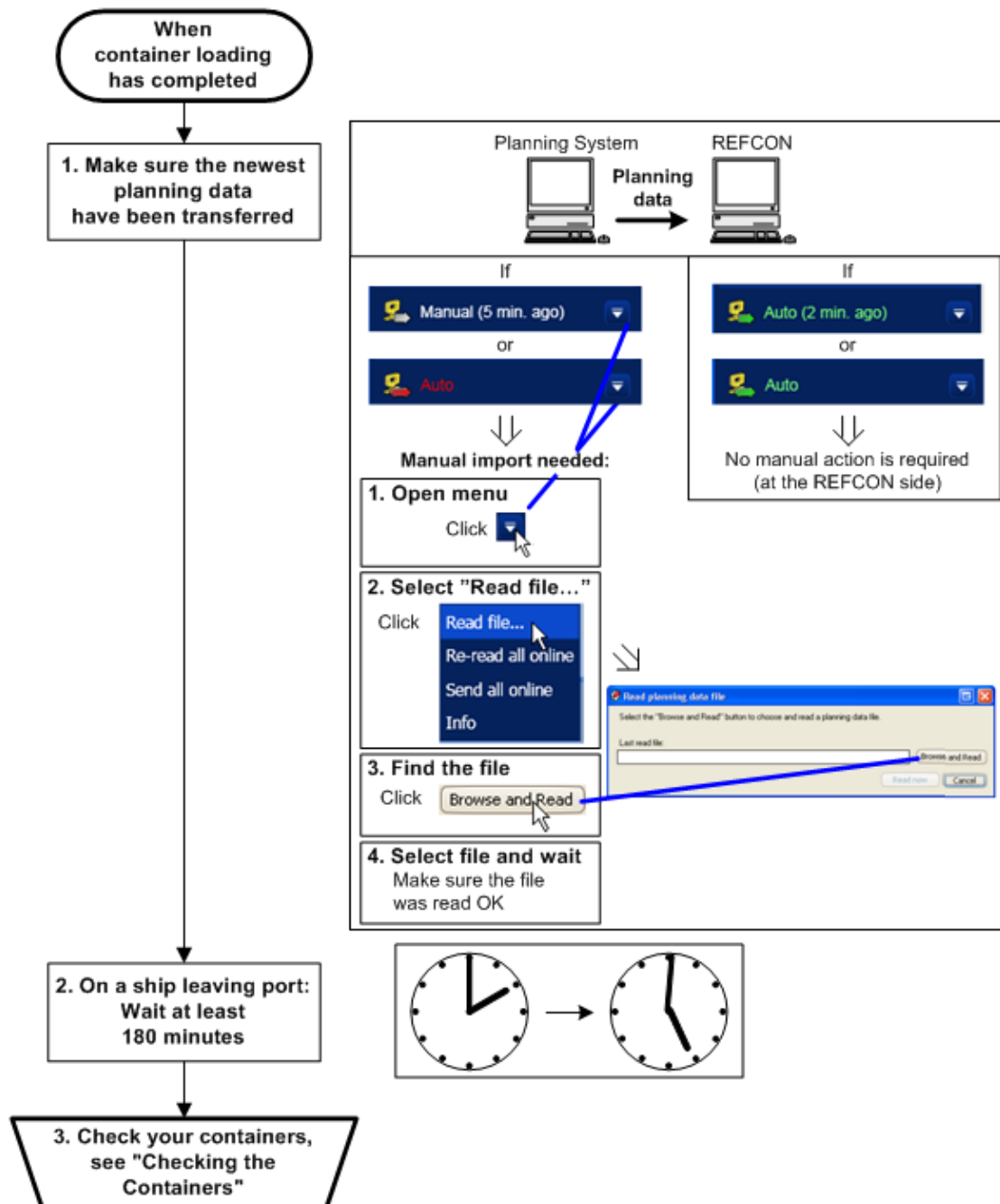
Modem troubleshooting highly depends on updated and reliable Planning data.

First of all this is a requirement to your Planning System. It **must** always have the complete list of reefer containers and their locations.

If manual action is required in the Planning System to produce the data for REFCON, make sure you do this whenever significant changes have occurred.



The following figure sketches how to import Planning data into REFCON if this does not run automatically:



19.1 Checking the Containers

Key subjects in this topic:

[Overview](#) ^[280]
[General checking tools](#) ^[280]
[Primary check procedure](#) ^[280]
[Understanding the flow charts](#) ^[281]
[Activating the "Modem remedy" list](#) ^[281]

Related topics:

[Checking modem-related alarms](#) ^[283]
[Checking the Modem Available List](#) ^[292]
[Modem Troubleshooting](#) ^[278]



Overview

How do you find out that something is wrong, as regards the communication equipment on the containers? Depending on the problem, it will be visible in one of the following ways:

- Certain types of errors give alarms.
- The other types can be located in various lists as described below.

General checking tools

In addition to [handling the modem-related alarms](#)^[283], we recommend that you regularly check your containers and attempt to fix the problems.

The job to fix a problem can be more or less easy. As reflected by the [Performance Bar](#)^[27], we think of two categories of containers:

- "White" containers - that appear to have never had a modem installed - are assumed not practicable to remedy; but of course, you are very welcome to install modems. The "white" containers are not described any further in this guide.
- "Light blue" containers - that have modems according to history - should be repaired.

Note that having connected power to a container, you should always wait at least an hour before deciding that it needs trouble-shooting. It will often happen that a "white" or "light blue" container gets "green" automatically after some time.

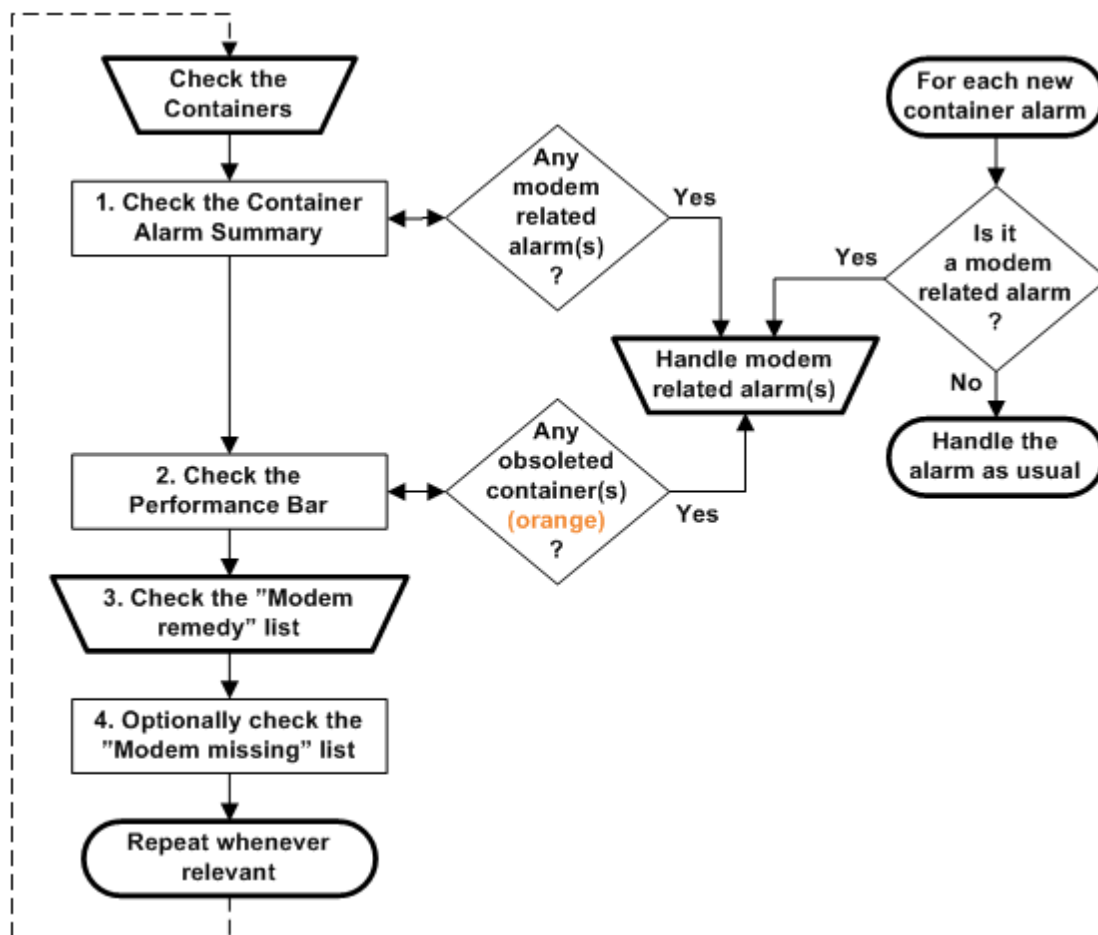
"Light blue" containers appear in the "Modem available" list as the starting point. This list can be opened from the [Performance Bar](#)^[27].

Unfortunately, the "Modem available" list does not in all cases give a complete overview of the containers to be handled.

In this guide we assume that you do instead use the provided "Modem remedy" list.

Primary check procedure

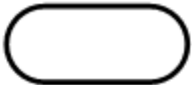
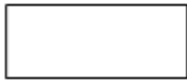
Pre-supposing that Planning data and the PCT system are all right, you should check your containers regularly and in case of alarm as follows:



Note: If "Possible loss of power" alarms are enabled, the check of the Performance Bar (step 2 above) should not be necessary, as you would then handle the alarms as they occur.

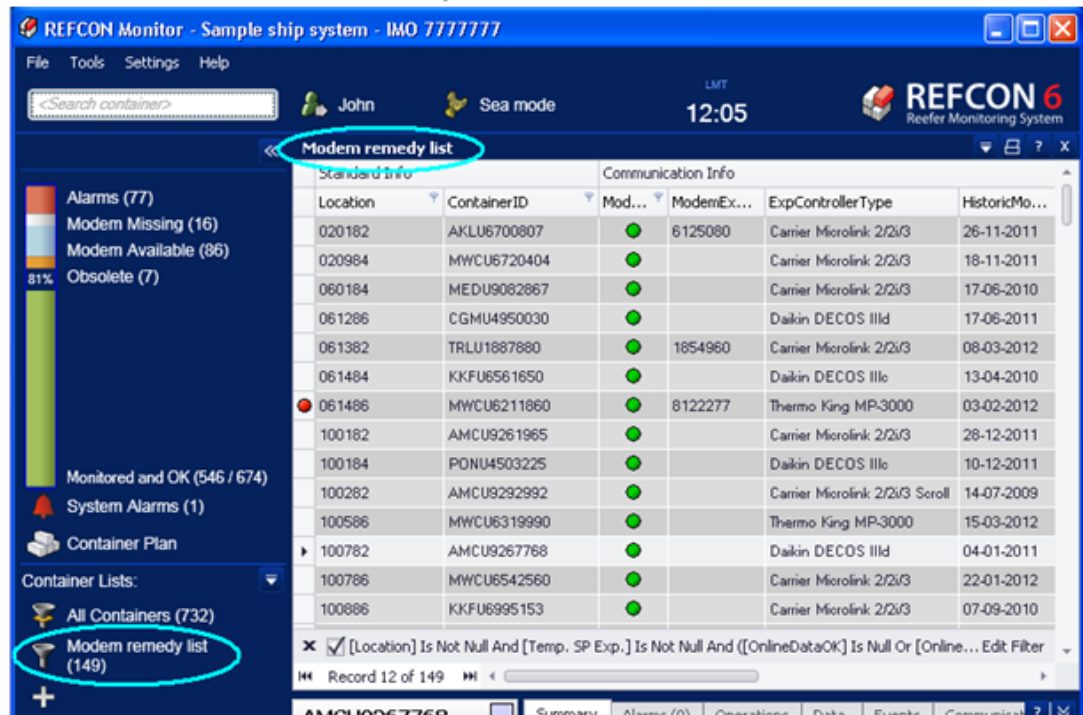
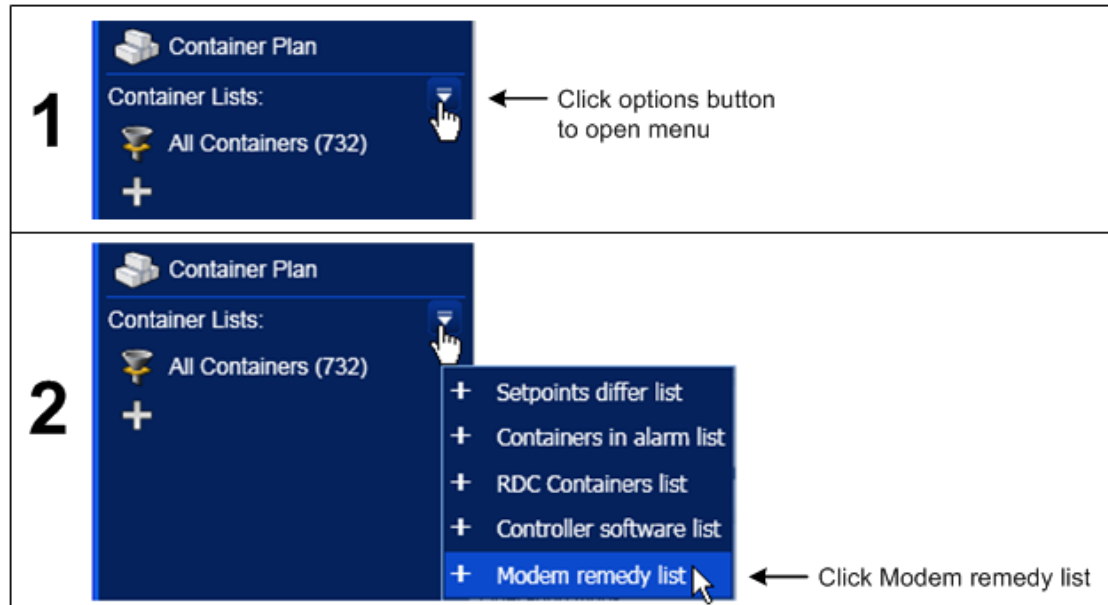
Understanding the flow charts

In the flow charts shown in this guide, like the one above, the following elements are used:

Element	Description
	Action that • either is referred to from another flow chart • or is described in detail in another flow chart / topic.
	Flow chart termination, either start or stop. Shows event or action that is not further described in this guide.
	Elementary action that needs to further explanation.
	Decision, typically evaluating to either Yes or No.

Activating the "Modem remedy" list

The "Modem remedy" list is a pre-defined list that needs to be activated once for all. Please proceed as follows:



Having activated list, you may [edit the filter](#)^[87] or [add or remove columns](#)^[82]. If you do so, and later want to return to the original list, you need to delete the list and then activate it anew.

Note that this guide presupposes that the Modem remedy list is **not** modified.



19.2 Checking modem-related alarms

Key subjects in this topic:

[Overview](#)^[283]
[Modem not connected to controller](#)^[283]
[Container ID is wrong or Planning Data missing](#)^[283]
[Container ID is missing](#)^[283]
[WARNING – Possible loss of power](#)^[283]

Related topics:

[Symptom: No Location – Wrong or No ID in Controller \("Ghost"\)](#)^[284]
[Symptom: No Data from Container](#)^[289]
[Symptom: Stopped Communicating](#)^[290]
[Checking the Modem Remedy List](#)^[292]
[Checking the Containers](#)^[279]
[Modem Troubleshooting](#)^[278]

Overview

A few types of communication equipment related problems give alarms, appearing in the [Container Alarm Summary](#)^[112]. Examples are shown in this figure:

Container Alarm Summary						
	Location	Origin	Text	Alarms	User	Time UTC
▼	Containers having alarms (Count=4)					
+		MWCU6508870	Modem not connected to controller	1		2012-06-27 13:30:07
+		XXXU0006921	Container ID is wrong or Planning Data missing	1		2012-06-27 13:30:07
+		LOSU0002111	Container ID is missing	1		2012-06-27 13:28:55
+	391082	MSCU7304207	WARNING - Possible loss of power	1		2012-06-27 13:27:01

When such an alarm occurs, we highly recommend that you take action as soon as possible because

- your cargo might be at risk;
- the remedy will surely improve your monitoring performance; and
- it should be practicable.

See also [Possible loss of power alarm](#)^[119] and [Alarms for missing or wrong ID](#)^[122].

Modem not connected to controller

Should this alarm occur, please follow the remedy procedure described in [Symptom: No Data from Container](#)^[289].

Container ID is wrong or Planning Data missing

Should this alarm occur, please follow the remedy procedure described in [Symptom: No Location - Wrong or No ID in Controller \("Ghost"\)](#)^[284].

Note: If the [wrong ID alarm](#)^[122] is **not** enabled, we recommend that you regularly check the [Modem remedy list](#)^[292].

Container ID is missing

Should this alarm occur, please follow the remedy procedure described in [Symptom: No Location - Wrong or No ID in Controller \("Ghost"\)](#)^[284].

WARNING - Possible loss of power

Should this alarm occur, please follow the remedy procedure described in [Symptom: Stopped Communicating](#)^[290].

Note: If the [Possible loss of power alarm](#)^[119] is **not** enabled, we recommend that you regularly check the [Performance Bar](#)^[27]. If there are any obsolete containers, indicated by orange colour, you should carefully check the Obsolete list:



- If you are using the optional RDC, and obsolescence is due to a delayed RDC inspection round, please make sure this is completed as soon as practicable.
- Otherwise, for each container, please follow the remedy procedure described in [Symptom: Stopped Communicating](#)^[290].

19.2.1 Symptom: No Location - Wrong or No ID in Controller ("Ghost")

Key subjects in this topic:

[Overview](#)^[284]

[Ghost explanation](#)^[285]

Related topics:

[Finding a Ghost Container](#)^[286]

[Change of Container ID](#)^[287]

[Checking modem-related alarms](#)^[283]

[Checking the Modem Remedy](#)

[List](#)^[292]

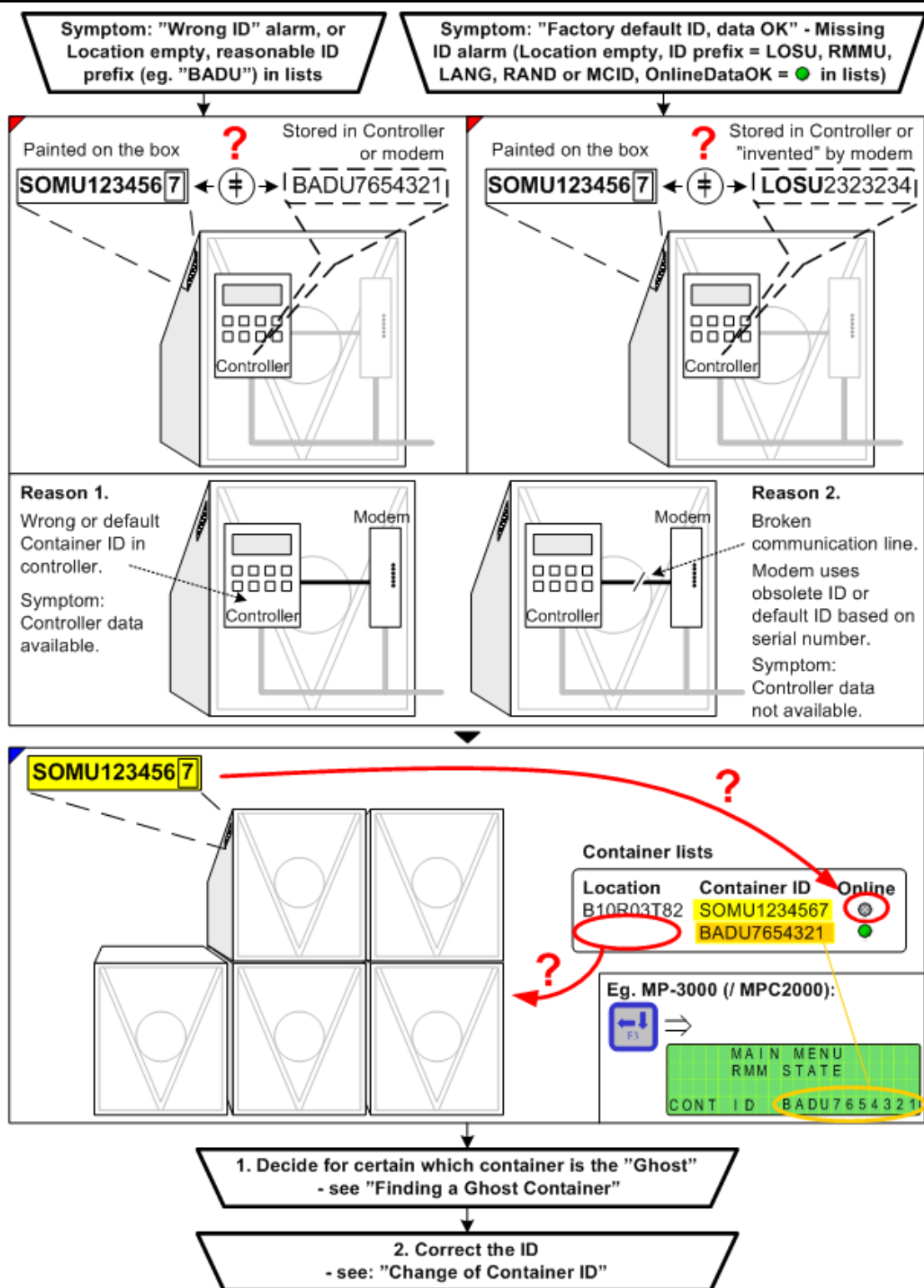
[Modem Troubleshooting](#)^[278]

Overview

A so-called "ghost container" is most likely a container with wrong container ID, or no ID at all. It is communicating but has no valid location. Correspondingly, a container reported by the Planning System has got a location, but it never started communicating. The two are one and the same - but how do you match them?



Ghost explanation



Please refer to [Finding a Ghost Container](#) ^[286] below to solve the problem.



19.2.1.1 Finding a Ghost Container

Key subjects in this topic:

[Overview](#) ²⁸⁶
[Looking for container ID similarity](#) ²⁸⁶
[Comparing Actual Setpoint and Expected Setpoints](#) ²⁸⁶
[Using the RMM Master number](#) ²⁸⁶
[Manual inspection](#) ²⁸⁷
[Final check for 100 % certainty](#) ²⁸⁷

Related topics:

[Change of Container ID](#) ²⁸⁷
[Symptom: No Location – Wrong or No ID in Controller \("Ghost"\)](#) ²⁸⁴
[Container Plan](#) ⁹⁶
[Container Details, Communication tab](#) ¹⁴⁸
[Modem Troubleshooting](#) ²⁷⁸

Overview

If you have (at least) one ghost container, how do you physically locate it?

In the **ideal** situation, all your reefer containers are equipped with power cable modems, and the number of communicating containers is as expected. Then you would look in the "Modem missing" list, the "Modem available" list or the "Modem remedy" list for non-communicating containers with valid location. Or you would open the optional Container Plan and look for **white** and **light blue** container positions. In both cases we will refer to **white or light blue positions** in the following. The number of such positions would match the number of ghosts. This number is usually very small, so the further investigation would be practicable.

In many cases you will, however, face a non-ideal situation. Just a few reefer containers without modems will make your job much more difficult because these will appear as white or light blue positions, exactly like the ghosts. So rather than deciding which white or light blue position is which ghost, you must find out which **of** the white or light blue positions are actually ghosts!

Looking for container ID similarity

Some wrong IDs are caused by misprint when the ID was originally entered into the controller – maybe just a single character out of 11 is wrong (note: the check digit also must be correct). Look for white or light blue position IDs that are similar to the Ghost ID. If there is only one, you probably found it.

Comparing Actual Setpoint and Expected Setpoints

In some cases, controller data like actual setpoint, temperatures and controller type are available. You would discover this in the Modem remedy list. In this case, maybe try going through the white or light blue positions, looking for expected setpoint values that are close to the actual setpoint. Do not expect an exact match. In some cases this would reduce the number of "suspects" dramatically.

Using the RMM Master number

Note: This method is only possible for certain REFCON systems.

The communication on the power lines is similar to radio communication. On some REFCON systems, signals from any point on the power lines are reaching literally any other point – and then the RMM Master number is useless. On certain other REFCON systems, there are barriers so that a certain RMM Master modem only covers a limited area. In the latter case, you may take advantage of this to find your ghost.

To know if your REFCON system provides this advantage, proceed as follows:

1. Inspect the System Layout drawing. Do you have only one or two RMM Masters in the system? If so, the RMM Master number is useless – please jump to [Manual inspection](#) ²⁸⁷ below.
2. Look in the Container Plan. Walk through the Bays/Stacks one by one.



3. In each Bay/Stack, look up the Container Details for at least 10 communicating containers (green positions) – click the "Communication" tab and write down the "CurMasterNet" and the "CurMasterNode" numbers.
4. Are the master numbers different in one Bay/Stack? If so, the RMM Master number is useless – please jump to [Manual inspection](#)^[287] below.
5. Having walked through all Bays/Stacks, do you see a clear correlation between Bay/Stack numbers and RMM Master numbers (one master may cover more than one Bay/Stack)? If so, the RMM Master number can be useful – please keep your notes for future use.

If the above test is successful in your REFCON system, you might open the Container Details for the Ghost, and click the "Communication" tab. Notice the "CurMasterNet" and the "CurMasterNode" numbers. From your notes written above, you should now be able to decide in which Bay(s)/Stack(s) to look for the Ghost.

Manual inspection

Eventually you have a number of white positions that can be your ghost. Now you have to manually inspect the containers. It is a great advantage to be two with walkie-talkies, one at the REFCON PC and one at the container.

For each suspected container, try using the exclusion method:

If you know the controller type of the ghost (controller data available), does it match the actual container?

If you know the actual setpoint of the ghost (controller data available), does it match the Actual Setpoint read on the controller of the actual container. Remember: If the one is in Celsius, and the other in Fahrenheit, there might be a small conversion error.

Check the Communication sub-page of the Container Details of the ghost. Does the last part of the "Modem" string match the serial number of the modem at the actual container?

If the first 4 letters of the container ID read "LOSU", you are probably dealing with a Thermo King MP-3000 or MP-4000. The 7 digits of the container ID will then normally reflect the last 7 digits of the serial number of the controller.

If the first 4 letters of the container ID read "RMMU", you are probably dealing with an RMM Mk IV modem. You will then normally find this default container ID printed at the rear of the modem.

Final check for 100 % certainty

How many containers do you have that matches those criteria? Typically there will be only one.

To be absolutely sure you suspect the right one, do a final test: Try sending a setpoint change of 0.1 degree from REFCON, and then go to the container and check that it has responded. Or try making the change at the container, and check that it updates in REFCON (it might take a few minutes). If so, you got it! Remember to change back the setpoint!

Please refer to [Change of Container ID](#)^[287] below to clear the ghost.

19.2.1.2 Change of Container ID

Key subjects in this topic:

[Change procedure](#)^[288]

Related topics:

[Finding a Ghost Container](#)^[286]

[Symptom: No Location – Wrong or No ID in Controller \("Ghost"\)](#)^[284]

[Container Details. Sub-page](#)

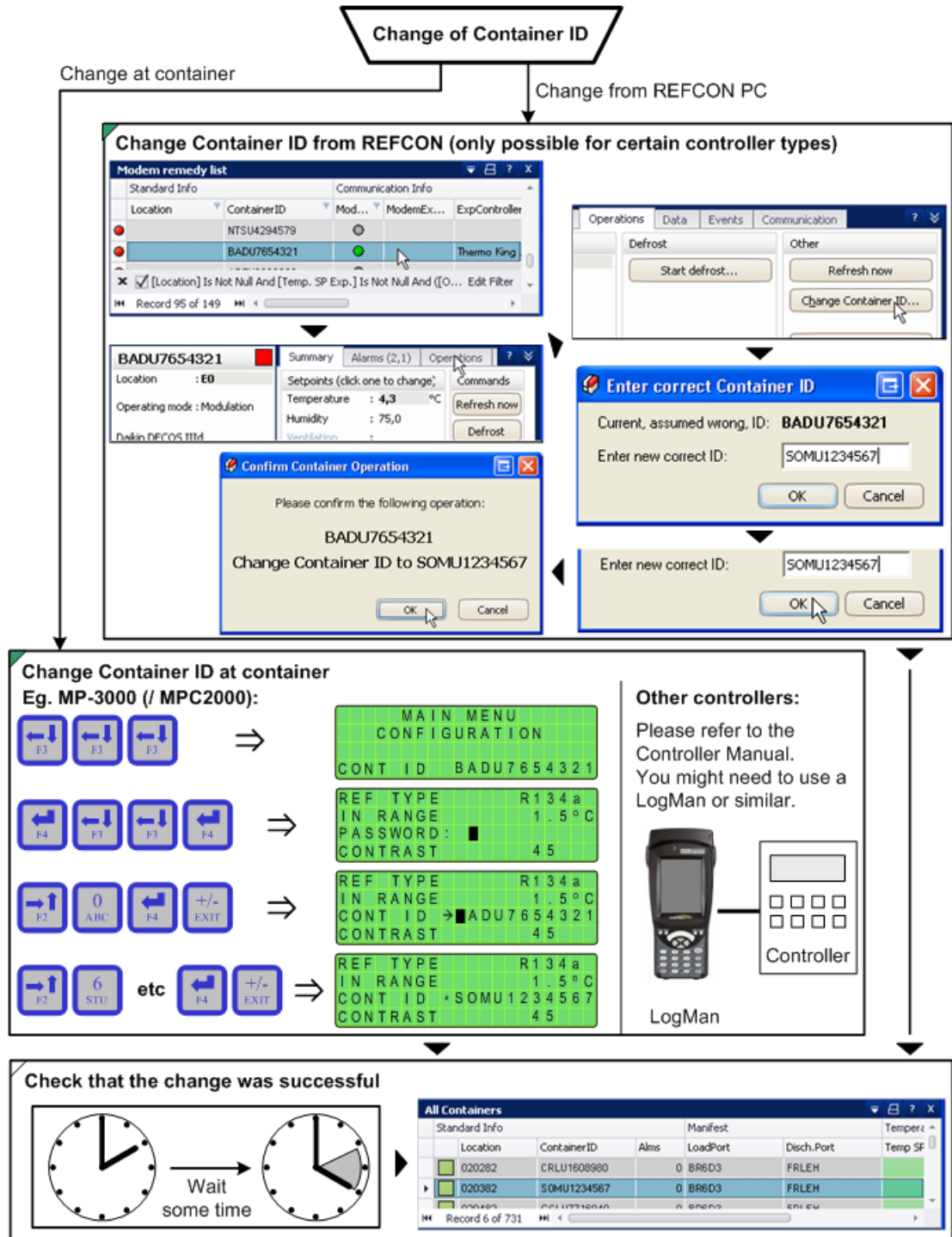
[Operations \(optional\)](#)^[146]

[Modem Troubleshooting](#)^[278]



Change procedure

When you are absolutely sure which container to rename to what ID, proceed as follows:





19.2.2 Symptom: No Data from Container

Key subjects in this topic:

[Overview](#) ^[289][Check procedure](#) ^[289]

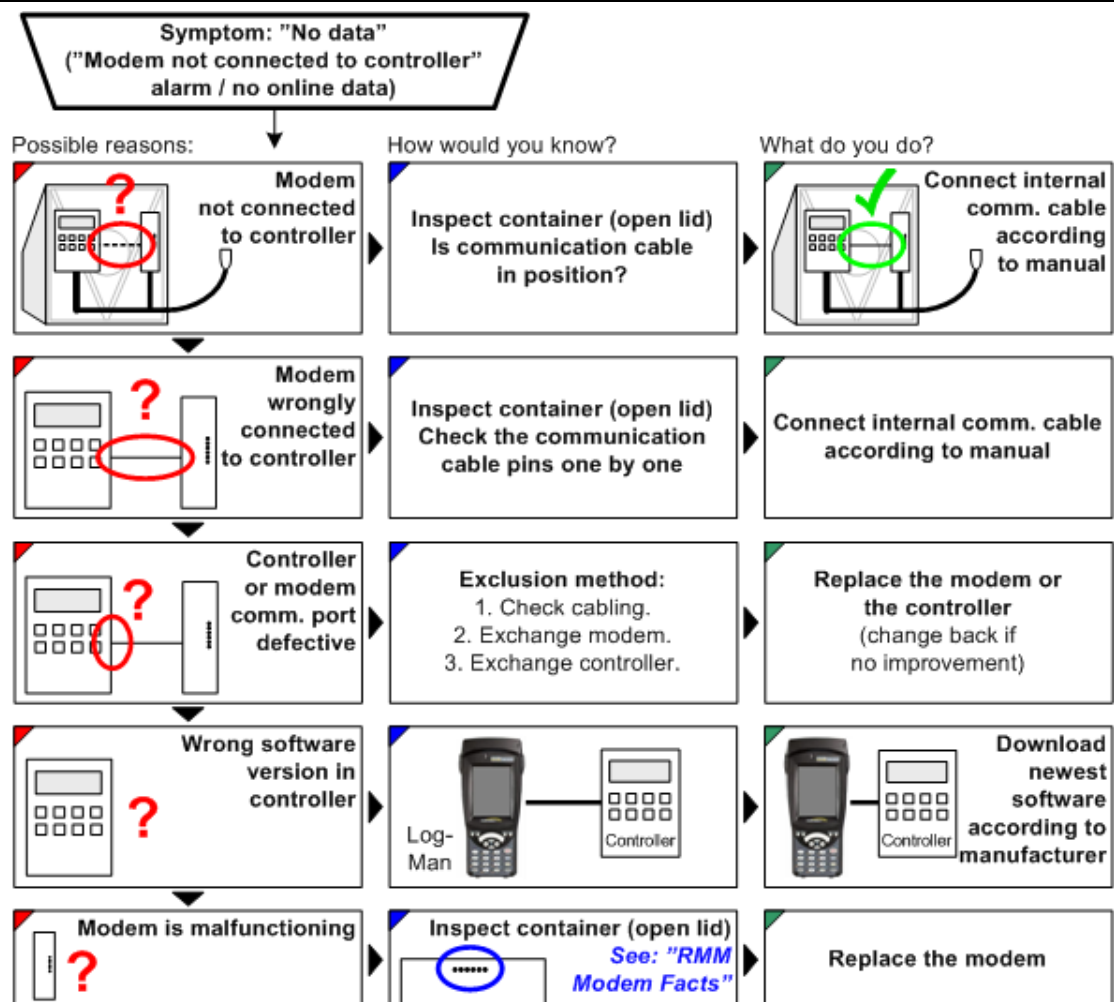
Related topics:

[RMM Modem Facts](#) ^[290][Checking modem-related alarms](#) ^[283][Checking the Modem Remedy](#)[List](#) ^[292][Modem Troubleshooting](#) ^[278]

Overview

The figure below shows the most common reasons when no data are read from a container, even though it is communicating. Note that the container ID might be correct (and thereby the location probably is known) because certain modems remember the ID even if the connection to the controller fails.

Check procedure





19.2.3 Symptom: Stopped Communicating

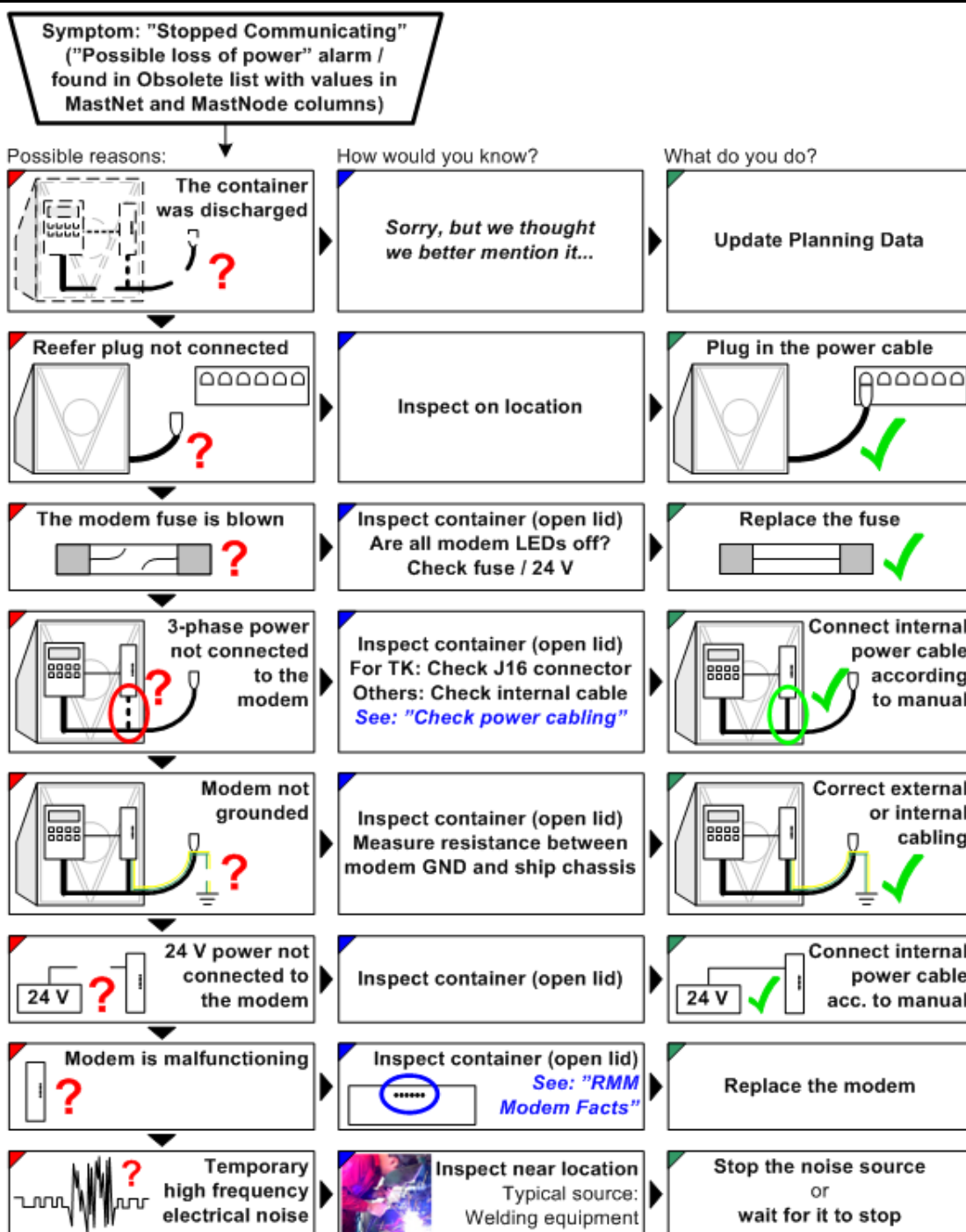
Key subjects in this topic:[Overview](#) ²⁹⁰[Check and repair procedure](#) ²⁹¹**Related topics:**[Checking the Power Cabling](#) ²⁹⁵[RMM Modem Facts](#) ²⁹⁸[Checking modem-related
alarms](#) ²⁸³[Checking the Modem Remedy
List](#) ²⁹²[Modem Troubleshooting](#) ²⁷⁸

Overview

The following figure shows the most common reasons when a container used to communicate but does not do so anymore, and tips for diagnoses and repairs.



Check and repair procedure





19.3 Checking the Modem Remedy List

Key subjects in this topic: Overview^[292] Check procedure^[293]	Related topics: Symptom: Never Communicated^[293] Symptom: No Location – Wrong or No ID in Controller ("Ghost")^[284] Symptom: No Data from Container^[289] Modem Troubleshooting^[278]
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Overview

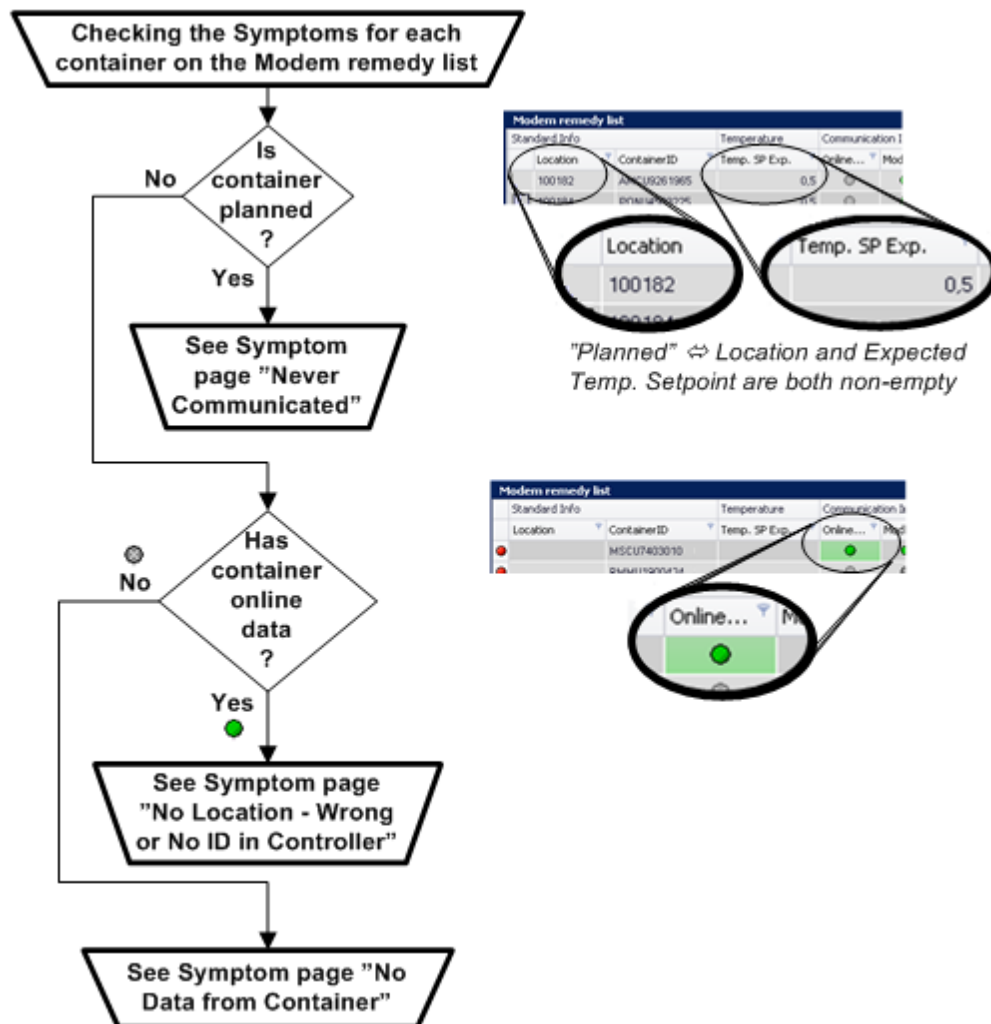
The pre-defined *Modem remedy list* attempts to list all containers with fixable communication equipment problems. It includes problems that will or can give [alarms](#)^[283] as well as all containers that are basically "light blue", belonging to the "Modem available" group in the [Performance Bar](#)^[27].

Out-of-the-box it does **not** include "white" containers, belonging to the "Modem missing" group - because these do probably require a modem installation rather than a modem repair.

As regards "orange" containers, belonging to the "Obsolete" group, PCT-obsolete containers are **not** included, because the reason for these is most likely not related to the equipment on the containers.

Hence there are overlaps between on the one side checking the Container Alarm Summary, the Modem available list and the Obsolete list, and on the other side the Modem remedy list. The idea about the Modem remedy list is to provide a common working list. To utilise the Modem remedy list, you will once for all have to [activate](#)^[281] it.

In general, when a container appears on the Modem remedy list and it stays there for some time (1 - 3 hours), you should check it and fix the problem, if possible:

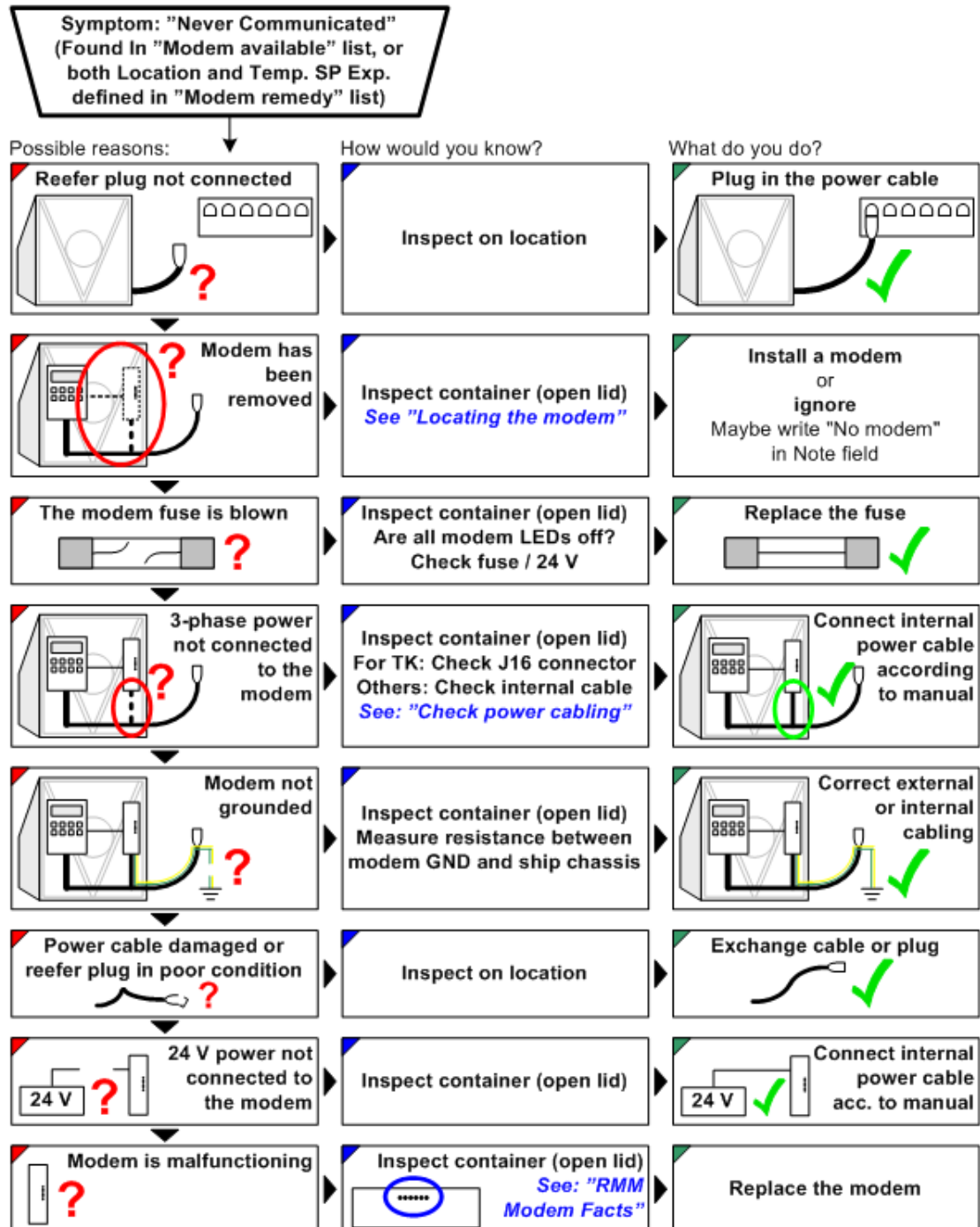
**Check procedure****19.3.1 Symptom: Never Communicated****Key subjects in this topic:**[Overview](#) ^[294][Check and repair procedure](#) ^[294]**Related topics:**[Locating the Modem](#) ^[296][Checking the Power Cabling](#) ^[295][RMM Modem Facts](#) ^[298][Checking the Modem Remedy](#)[List](#) ^[292][Modem Troubleshooting](#) ^[278]



Overview

The following figure shows the most common reasons when a container does not communicate, and tips for diagnoses and repairs.

Check and repair procedure



Another possible reason is described in [Symptom: No Location - Wrong or No ID in Controller \("Ghost"\)](#) ²⁸⁴.



19.4 Checking the Power Cabling

Key subjects in this topic:

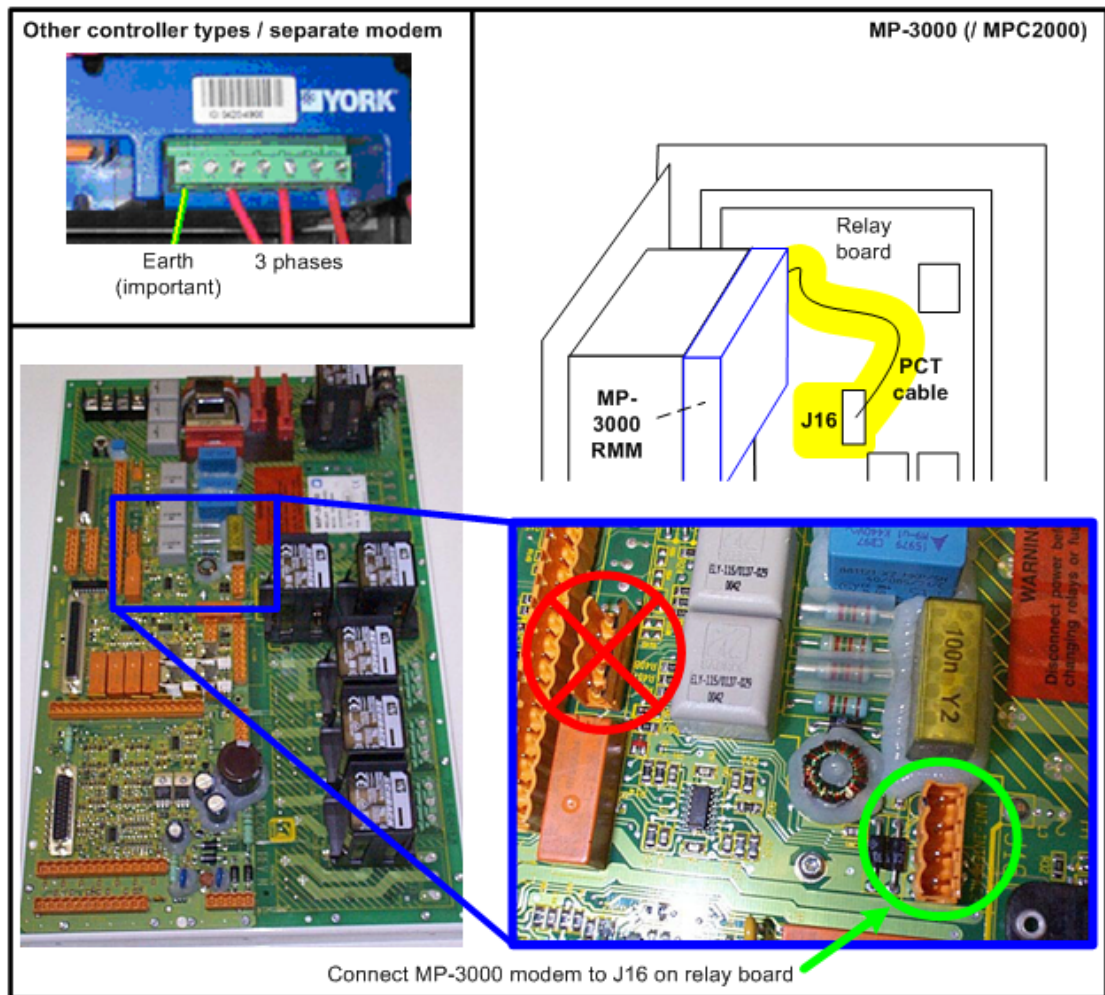
[Overview](#) ²⁹⁵

Related topics:

[Modem Troubleshooting](#) ²⁷⁸

Overview

The power cable communication only works if the internal cabling at the container is correct:



19.5 Typical Errors

Key subjects in this topic:

[Overview](#) ²⁹⁵

[Inexpedient cabling example](#) ²⁹⁶

[Forced plug example](#) ²⁹⁶

[Actual setpoint above 51 C](#) ²⁹⁶

Related topics:

[Modem Troubleshooting](#) ²⁷⁸

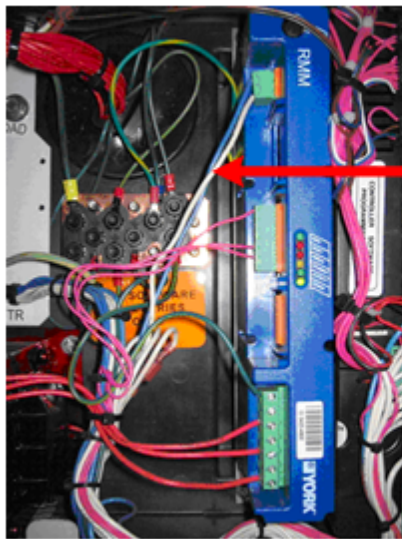
Overview

This section describes some typical errors that have been observed on containers.



Inexpedient cabling example

Example of inexpedient cabling: A small plug carrying a heavy bundle of wires => vibrations make the plug fall out, sooner or later.



Cables before (after re-inserting the 24 V power plug)



Cables after re-arrangement

Forced plug example

Example of plug turned 180 degrees - which requires a little force



Turned wrong
Easy to see because the plug doesn't fit well

Actual setpoint above 51 C

In rare cases you may get the alarm "Actual and expected setpoints differ" caused by the actual setpoint reported by the container controller reading 51.1 or 51.2°C, being obviously wrong.

This is a known potential problem when the container is equipped with a Daikin Decos controller type IIIc or III d in combination with an older RMM modem. In this case the problem shows up if the connection between the controller and the RMM modem fails (plug fallen out, cable cut, or electronic problem).

The recommended way to solve the problem is to replace the modem with an RMM MK IVb, HW-revision H, or newer.

19.6 Locating the Modem

Key subjects in this topic:

[Overview](#) ²⁹⁷

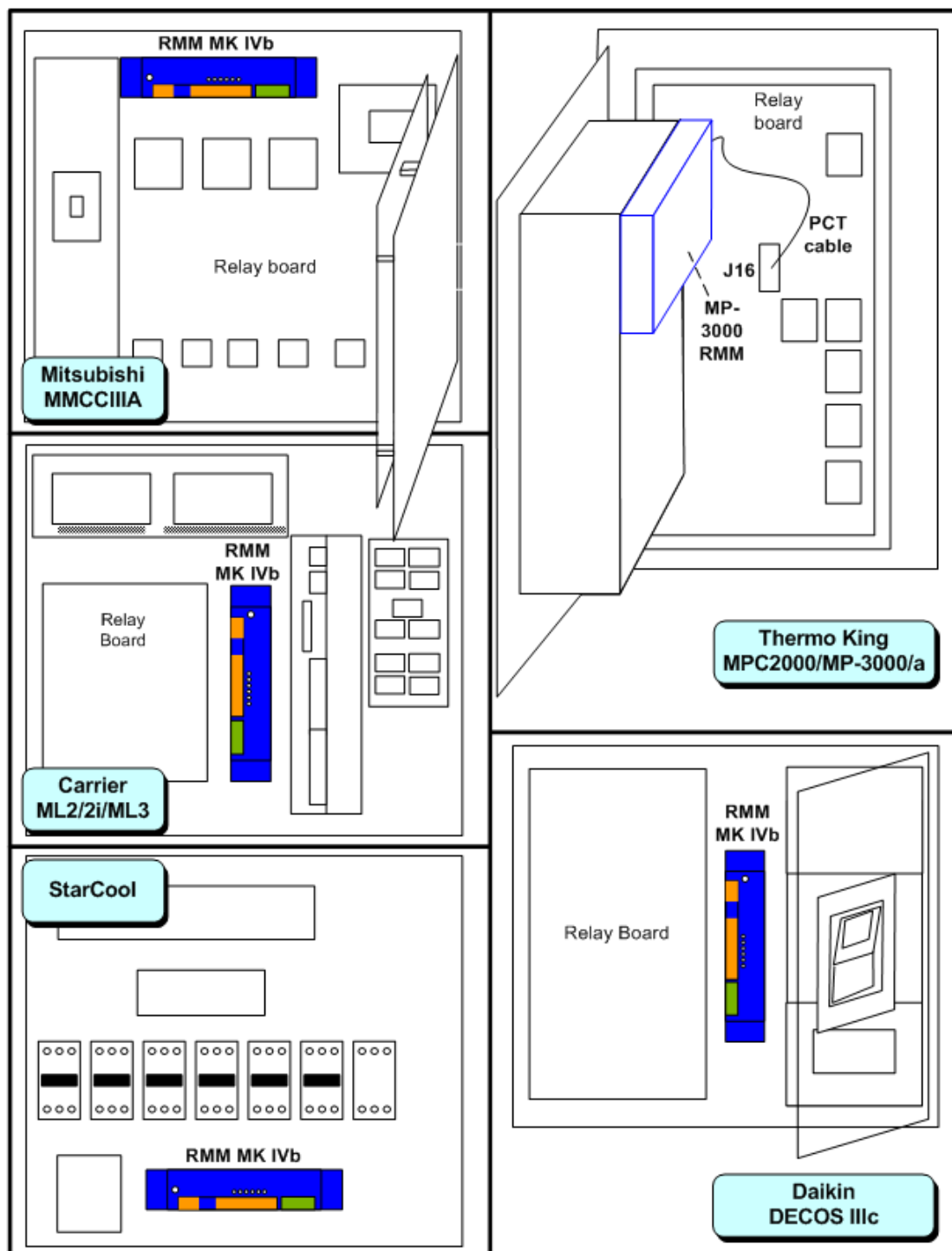
Related topics:

[Modem Troubleshooting](#) ²⁷⁸



Overview

The modem can be an integrated unit (MPC2000 / MP-3000) or an independent unit attached to various controller types. The following figure shows the most common examples:





19.7 RMM Modem Facts

Key subjects in this topic:

[Overview](#) ^[298]

[LED interpretation](#) ^[299]

[LED appearance examples](#) ^[299]

[Connector Specifications RMM](#)

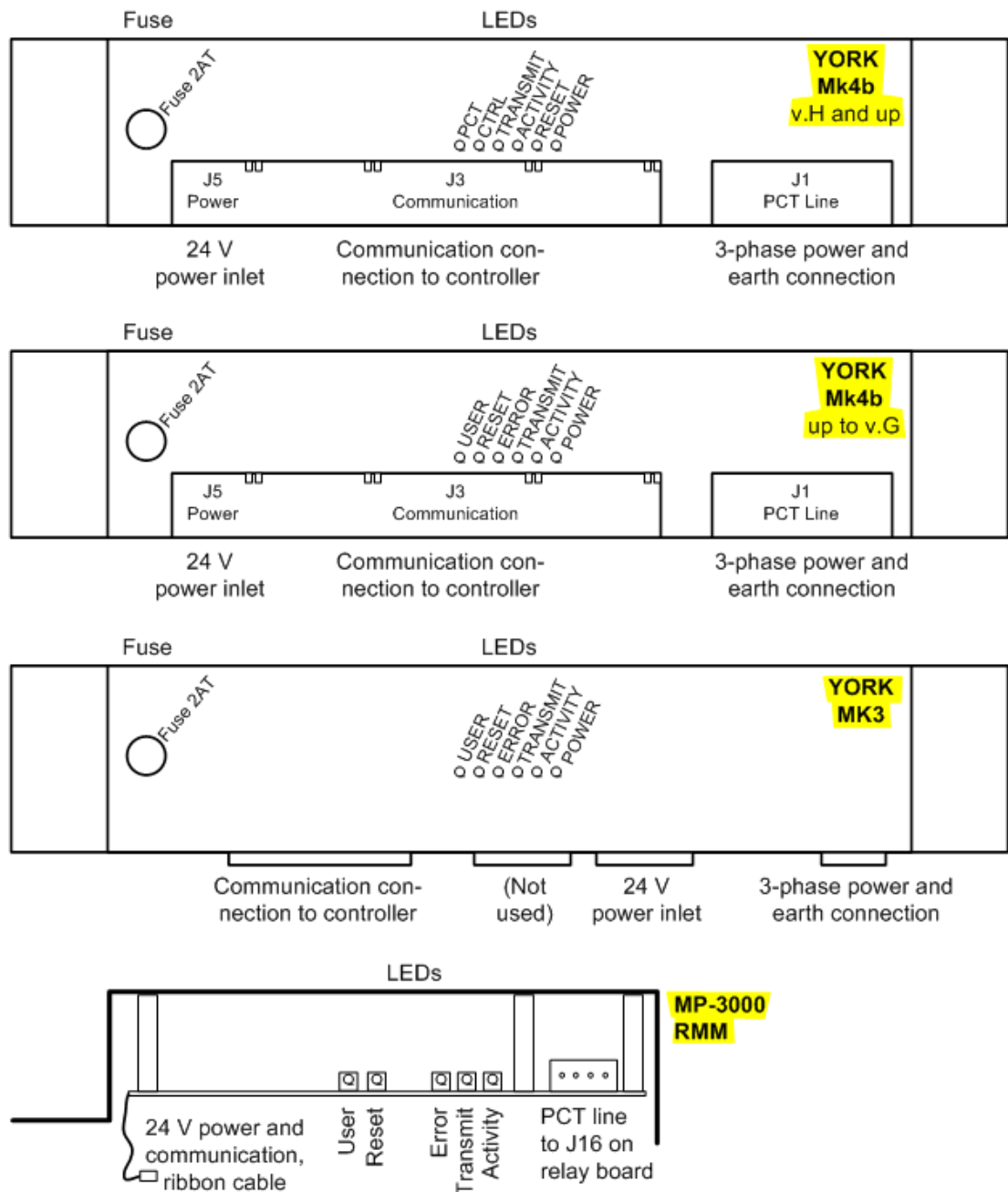
[Mk4b](#) ^[300]

Related topics:

[Modem Troubleshooting](#) ^[278]

Overview

Below you find facts about the three most common modem types.





LED interpretation

The LEDs can be interpreted as follows:

LED	Color	Status	Description
POWER	Green	On	24V AC or DC power on. Power cable is correct wired.
		Off	No power.
ACTIVITY	Green	Flashing	High Data Rate activity is detected at the power line.
		On	High Data Rate activity is detected at the power line.
		Off	No power cable communication is detected.
TRANSMIT	Yellow	Flashing	Communication in progress with a Master on the PCT network.
		Off	No communication in progress.
ERROR	Red	On	Internal malfunction has occurred.
		Off	Normal operation.
RESET	Red	Off	Normal operation.
		On	The RMM is reset. Turns off after a while.
CTRL	Green	Off	Searching for a connected controller.
		Flashing	Controller connected – controller does not have a valid ID.
		On	Controller connected – valid container ID found.
USER or PCT	Green	On	The container is logged on to a master, with normal parameters.
		Flashing	Ready to log on.
		Off	Between 15 and 60 minutes has passed since last device poll. As 60 minutes is reached, the unit resets and gets ready to log on.

LED appearance examples

You often can obtain useful information simply by watching the Light Emitting Diodes (LEDs) of the RMM modem.

The following figures show the most common examples. Note, however, that in case of malfunctions, the LED behaviour can not always be foreseen.



Typical modem LED appearances, Mk4b v.H and up

 = LED flashes



Typical modem LED appearances, other types

 = LED flashes



Connector Specifications RMM Mk4b

The connectors on the modem RMM Mk4b are specified as follows. For other modem makes, please refer to the respective manuals.

J1: 3-phase power and earth connection

Terminal (pin)	Description	Terminal (pin)	Description
1	Earth/Ground	5	AC Phase 2
2	No connection	6	No connection
3	AC Phase 1	7	AC Phase 3
4	No connection		

J3: Communication connection to controller

Terminal (pin)	Description	Terminal (pin)	Description
1	GND	5	RxD
2	De-select	6	TxD
3	RTS	7	No connection
4	CTS		

J5: 24 V power inlet

Terminal (pin)	Description	Terminal (pin)	Description
1	AC/DC input	3	AC input/DC ground
2	No connection		

REFCON 6

Reefer Container Monitoring System

Understanding and Maintaining REFCON



20 Understanding and Maintaining REFCON

Key subjects in this topic: Overview ^[302] The Reefer Container ^[302] The hardware ^[303] The software ^[303] The external "partners" ^[303]	Related topics: The REFCON system hardware ^[303] The REFCON 6 software package ^[306] The REFCON Data Service and plugins ^[309] Third-party Software on the REFCON PC ^[312] The External interfaces ^[313] Understanding installation and update ^[314] Standard installation and update ^[316] Remote installation and update ^[324] Automatic backup and restore ^[328] REFCON Setup ^[330] System Devices List ^[333] The Settings Editor ^[338] Container Modems list ^[340] PCT Communication Statistics List ^[345] Master groups ^[347]
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Overview

This topic and the sub-topics describe the REFCON system, its possible components and external connections, and its [settings](#) ^[330]. They also describe the screen pictures and functions in REFCON Monitor dealing with these subjects. The descriptions are mainly meant for users taking the [REFCON maintainer role](#) ^[11] or the [IT maintainer role](#) ^[11] and should be useful in situations where you have to install or troubleshoot REFCON.

Some of the items described here are optional and controlled by your [license file](#) ^[11].

The descriptions found here - in the online help this manual - are general, with a medium level of details. When you need more details and/or information specific to *your* REFCON, please refer to "Instruction Manual - Hardware Descriptions.pdf" being as standard found on the "Customer disk" supplied with REFCON.

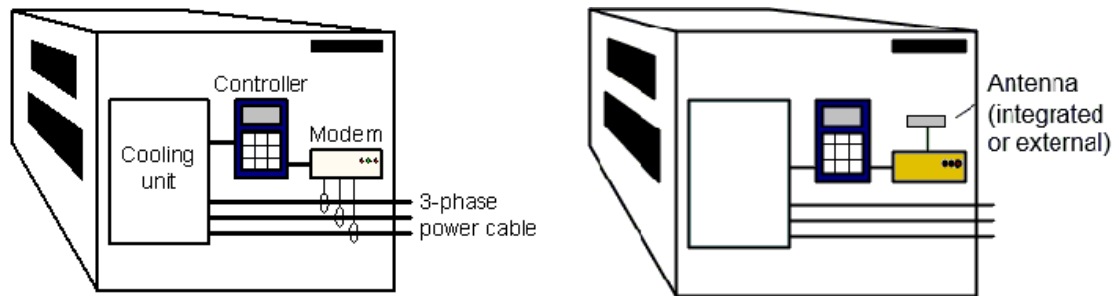
Every single component that has been included in your specific REFCON system, as well as every single network connection and every single external system interfacing with REFCON, are and must be considered *important*. It is therefore vital for the performance of REFCON that you, in case of errors, remedy these as soon as possible.

The Reefer Container

Though it is not a part of the REFCON system as such, it is important to know about the equipment on every single reefer container.

A reefer container is an independent unit that is completely self-sustained, being supplied with 3-phase power. The onboard controller makes sure that the temperature and other parameters are kept at their respective setpoint(s), and it continuously logs the conditions.

Many reefer containers are equipped with a *power cable modem*, allowing the container to communicate most parameters via *Power Cable Transmission* (PCT).



The power cable modem can be an independent unit or integrated with the controller. If the modem is manufactured by Copeland Transportation Solutions, it is named *RMM* (Remote Monitoring Module) or *RMM+*.

Lately, quite many reefer containers are equipped with modems that are not able to communicate via PCT. You can still make them updated using a [REFCON 6 extension](#)^[218].

If the container has the newest version of the Copeland Transportation Solutions' RMM-W modem, this can communicate with REFCON via Bluetooth.

You can see all the modems in play in the [Container Modems list](#)^[340].

The hardware

In it's most common basic configuration, the REFCON system hardware consists of a PC and two PCT master modems; but there is a high number of possible additions and variants as described in [The REFCON system hardware](#)^[303].

The software

The REFCON 6 PC software is a package with a number of [configuration options](#)^[11]. The software is described in [The REFCON 6 software package](#)^[306].

The external "partners"

In the very most cases, REFCON has one or more interfaces to external systems. An important example is the Planning System from which REFCON receives important information about the containers. The Planning System can therefore be seen as a "partner" that significantly increases your advantages of using REFCON.

20.1 The REFCON system hardware

Key subjects in this topic:

[Overview](#)^[303]

[The most simple power line solution](#)^[304]

[Simple solution with RS2LAN](#)^[304]

[RS2LAN 2](#)^[304]

[More than one REFCON Monitor](#)^[305]

[REFCON Data Collector](#)^[305]

[RMM Master extensions](#)^[305]

Related topics:

[Understanding and Maintaining REFCON](#)^[302]

Overview

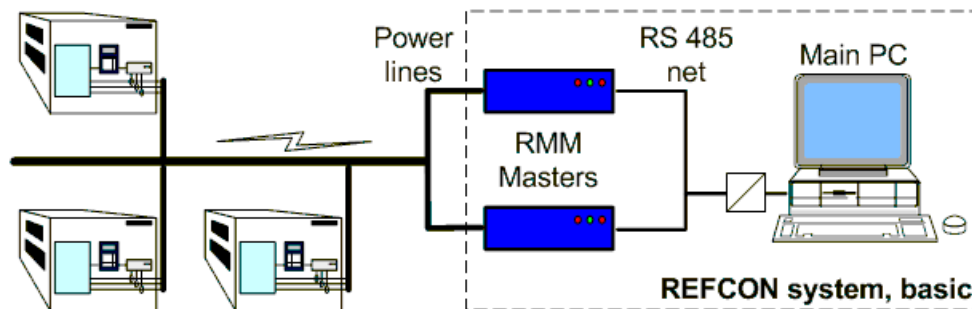
The description below aims at giving you a useful understanding of the standard REFCON system hardware; what are the possible components, how are they connected, what are their roles, and how can they be trouble-shooted. You may also have one or more [REFCON 6 extensions](#)^[218] available, please refer to the documentations for these.



REFCON can be hardware configured in a very high number of ways. If you need to know in details exactly how your specific REFCON system is configured, you would consult the System Overview drawings in the "Instruction Manual - Hardware Descriptions.pdf" being as standard found on the "Customer Disk" supplied with REFCON.

The most simple power line solution

As sketched in the following figure, the most common basic configuration includes three components:



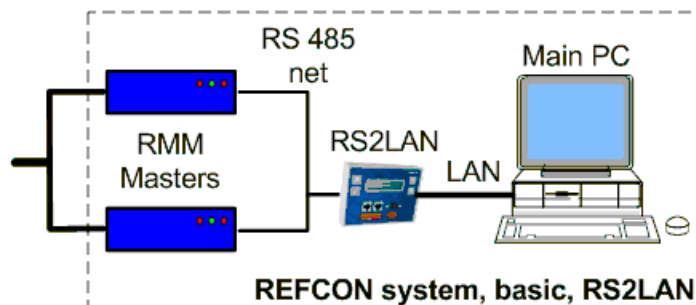
Any REFCON system includes one and only one Main PC. You would then probably know it as the REFCON PC.

The RMM Masters communicate with the Main PC on an RS 485 network. However, standard PCs do not support RS 485. If the PC is equipped with an RS 232 port, the problem often is solved by inserting a converter as indicated in above sketch and applying the so-called Danbuss-driver to the RS 232 port.

The standard solution of today is, however, to use the RS2LAN:

Simple solution with RS2LAN

In the following sketch, the obsolete RS 232 solution was replaced by a single RS2LAN unit:



This solution is far more extendable than the RS 232 solution:

- The LAN can (for example by using fibre optic cable) be extended to literally any length.
- You can apply a high number of RS2LAN units and thereby separate the RMM Masters - though still there normally must be at least two on each RS2LAN.

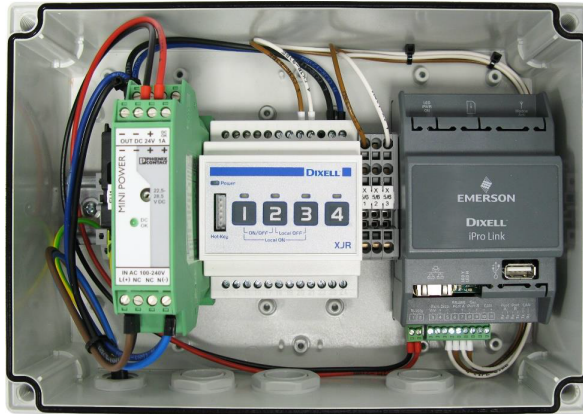
To make this solution work, it is required that

- the Main PC has a fixed IP address
- that IP address must be typed into the RS2LAN using it's display/keyboard
- each RS2LAN must have a unique fixed IP address.

A separate user's manual is supplied with the RS2LAN as standard.

RS2LAN 2

The above picture and description relates to the original version of the RS2LAN. Copeland LP has now released a new version, RS2LAN 2:

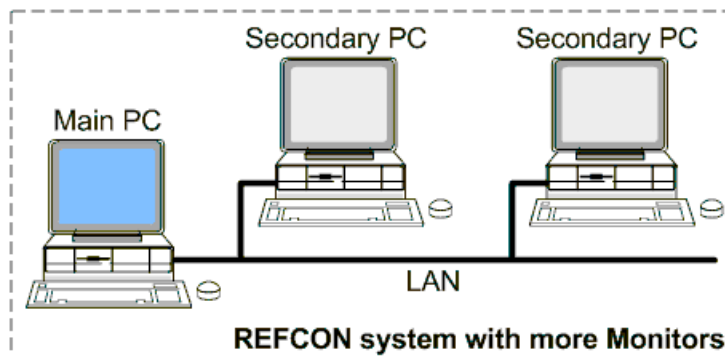


Please refer to the "RS2LAN 2 User Manual" for descriptions, installation instructions and more.

More than one REFCON Monitor

As standard you operate REFCON through the REFCON Monitor, being a standard application that can run on any PC with a supported, updated version of Microsoft Windows (Windows 10 or similar Windows Server version). It is always possible to run the REFCON Monitor on the Main PC.

You can run the REFCON Monitor from any PC meeting the base requirements, provided that it is connected to the Main PC via LAN:



The max number of permitted concurrent running REFCON Monitors is decided by the [license file](#)¹¹. If, for example, the maximum is 3, and you have already three users working with the REFCON Monitor on three different PCs, should a fourth user try to start a REFCON Monitor, a warning will appear on screen saying that this is not possible. The solution is to have one of the other users close down a REFCON Monitor.

You find a list of the running and previously running REFCON Monitors in the [System Devices List](#)³³³.

REFCON Data Collector

The optional [RDC](#) must also be seen as a part of the REFCON system, when included.

RMM Master extensions

The RMM Master modem can be connected directly to the 3-phase power lines; however, in many cases you will find electronic devices in between. The purposes of these devices are

- Cheaper equipment - you save RMM Masters even when the power system is highly sectioned (as regards high frequency signals). The RMM Bridge is an example of such a device.
- Optimized communication - in some cases the power system properties make direct high frequency communication difficult. The ISC (inductive signal coupler) in many cases is used as a very effective solution to this problem.



Please find detailed information about the possible devices in your specific case in the "Instruction Manual - Hardware Descriptions.pdf" mentioned above.

20.2 The REFCON 6 software package

Key subjects in this topic:

[Overview](#) 306

[Version indication](#) 306

[Software parts on REFCON Main](#)

[PC](#) 307

[Special notes about the Main PC](#) 308

Related topics:

[The REFCON Data Service and plugins](#) 309

[Understanding and Maintaining REFCON](#) 302

Overview

The REFCON 6 software package installed on the REFCON Main PC consists of a number of services and applications as described in the following.

The essential parts of the software are all **services** (in the Microsoft Windows terminology). Being a service means that

- it is started automatically and almost immediately when the PC is powered up, even if no one logs on to the PC;
- it has no user interface in itself;
- it fully complies with Microsoft's rules for services regarding safety etc.

The other parts of the software are **applications** with their own user interfaces. Being an application means that

- a user must be logged on to have it running;
- you can start and stop it;
- you can interact directly with it.

Version indication

You should always run the newest REFCON 6 software. The version looks like this:

➤ **6.xx.y.zzzz** or **6.xx.y-zzzz** (example: 6.13.0.d191853 or 6.13.0-d191853).

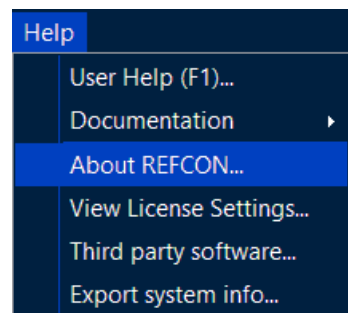
- **xx** is "Major", being increased when a version is released with new feature(s) and/or significant changes to existing features.

- **y** is "Minor", being increased when a version is released with minor changes and/or bug fixes only.

- **zzzz** is the unique reference ID of the version, being essential should you need Copeland LP's help to trouble-shoot.

Along with REFCON 6, the REFCON System Bundle is installed. This has a similar version indication, for example 1.5.2-1.

You can check the version of both in the About Box:

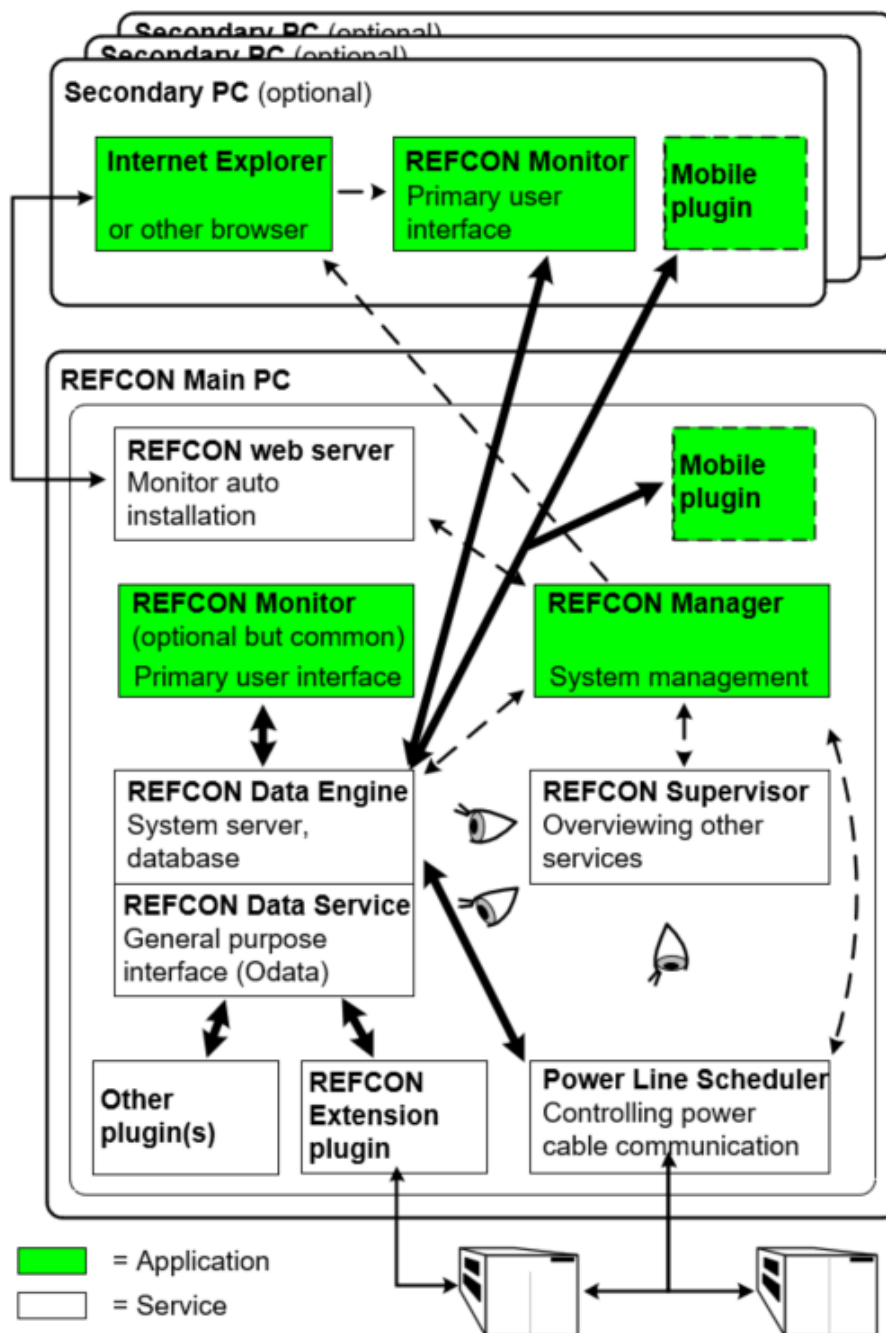


REFCON 6 version: 6.13.0.d191853
Release date: 2021-03-26 11:16
REFCON System Bundle version: 1.5.2-1
Release date: 2020-10-23 13:44



Software parts on REFCON Main PC

The following drawing illustrates the different parts of REFCON 6 that are or can be running on the REFCON Main PC. Descriptions are found in the table below.



The software parts are:

Part name	Type	Function	Description
Data Engine	Service	System server, database	The heart of the REFCON PC software system, where all container data are stored and all background activities are running. Provides data for the REFCON Monitors, and receives data via the PLS.



Part name	Type	Function	Description
REFCON Data Service	Integrated in Data Engine	OData interface	Part of Data Engine, providing a general purpose data interface, see The REFCON Data Service and plugins ^[309] .
PLS	Service	Power Line Scheduler	Makes sure the Data Engine can effectively communicate with the containers.
Web server	Service	Monitor auto installation	On a Secondary PC, all you need to install the REFCON Monitor is entering the link to the Web server into a browser. The Web server then provides all that is needed to install the REFCON Monitor (in Microsoft terminology: ClickOnce).
Supervisor	Service	Keeping eye on above services	Does nothing but watch and take care of that all required services are continuously running.
REFCON Manager	Application	Controlling above services	Described in The REFCON Manager ^[350] .
REFCON Monitor	Application	REFCON's primary user interface	See REFCON Monitor ^[22] and many other topics in the online help manual.
Various plugins	Service	Interfacing	REFCON runs one or more plugins responsible for specific interfaces or REFCON 6 extensions ^[218] depending on license.
Mobile plugin	Application	Optional plugin for the REFCON Mobile solution ^[226]	Interface application, available only if enabled in license.

Special notes about the Main PC

Please be aware of the following points regarding the REFCON Main PC:

- We recommend that the Main PC is exclusively used for running REFCON, because this is a demanding and time critical application that could be disturbed by running other programs on the PC. In contrary you are free to use possible Secondary PCs for other purposes.
- The Main PC must be running 24 hours every day (unless you for a period has no reefer containers at all). It must therefore never be powered off, shut down, hibernated or turned stand-by.
- Even if you set the power options on the Main PC to turn stand-by after some time of inactivity, you will see that it does not. This is because the Data Engine service works to Windows just as if a user were active.



20.3 The REFCON Data Service and plug-ins

Key subjects in this topic: Overview ^[309] How to make software use the REFCON Data Service ^[309] Plugins ^[309] How a plugin is installed and updated ^[310] How to supervise a plugin ^[310] Plugin details and update info ^[311] Plugin settings ^[311] Plugin list in REFCON Manager ^[311]	Related topics: The REFCON 6 software package ^[306] Understanding and Maintaining REFCON ^[302]
---	---

Overview

As illustrated in [Software parts on REFCON Main PC](#)^[307], the REFCON Data Service is a part of the Data Engine.

The REFCON Data Service allows any external software or a [REFCON 6 extension](#)^[218] to read data from and/or write data to REFCON. Both reading and writing does require specific permissions being agreed with Copeland Transportation Solutions and specified in the [license file](#)^[11].

This means that, given these permissions, any software developer can create an application that uses data from or changes data in REFCON. This can be done without any further accept from Copeland Transportation Solutions. Of course Copeland Transportation Solutions takes no responsibility for and does not service any such applications.

Copeland Transportation Solutions does, however, deliver a variety of **plugins** that are or can be parts of your REFCON system. A plugin works together with REFCON 6 through the REFCON Data Service.

How to make software use the REFCON Data Service

To develop third-party interfacing software, please request the "*REFCON Data Service - Programmer's guide*" from Copeland Transportation Solutions. Also request the specific *roles* to be taken by the interfacing software. These roles must be registered in the license for the specific REFCON system.

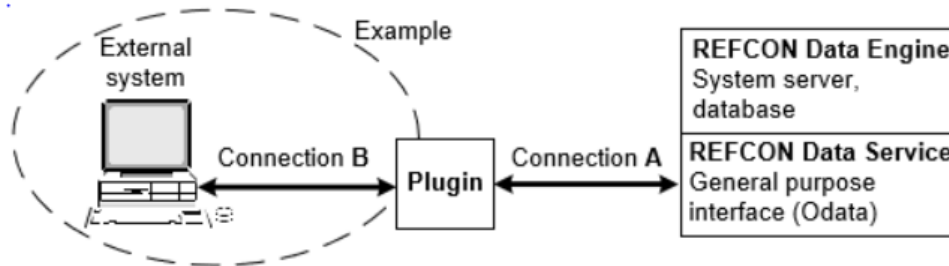
Third party software can run on any computer connected to the REFCON Main PC by LAN. Note that the same is true for plugins delivered by Copeland Transportation Solutions, however, these are as standard installed on the REFCON Main PC.

Plugins

Normally a plugin is a service running on the REFCON Main PC. You can check that it is running by inspecting the [System Devices list](#)^[333] or "Services" in Windows' Task Manager.

There can be zero, one or more plugins installed and running.

Most plugins work as *middleware*. A middleware plugin provides an interface between two systems, of which REFCON is the one:



How a plugin is installed and updated

As standard, plugins delivered by Copeland Transportation Solutions are included in the REFCON System Bundle, please refer to [Understanding installation and update](#)³¹⁴.

It is possible to install a plugin on another PC, provided that this is connected by LAN to the REFCON Main PC. This requires extra manual action. Should this be desired, it should be specifically agreed between you (the customer) and Copeland Transportation Solutions ApS.

How to supervise a plugin

As any plugin must be considered vital, it will as standard be indicated in the REFCON Monitor as a line in the [System Devices List](#)³³³. An example would be:

System Devices					
Device	Address	User's description	Online	Software	Hardware
RS2LAN (Net 21)	Net: 21		●	127.0.0.1 port 12001	RS2LAN
RMM Master (21,10)	Net: 21, Node: 10		●	Mk.IV: E:01070300, F:0305...	RmmMasterMKIV
RMM Master (21,11)	Net: 21, Node: 11		●	Mk.IV: E:01070300, F:0305...	RmmMasterMKIV
DKNST01-100852	10.68.54.64, ::1	REFCON Main PC (HPLund)	●	REFCON Monitor	PC
REFCON Wireless Hub plugin		Wireless Hub synchronization...	●	1.0.0.33	DKNST01-100852

Here, the plugin is the last device, with User's description "Wireless Hub sync...". "Online" is green, indicating that the plugin is running. There is no alarm indication, which as standard means that the plugin works as intended. The current software version of the plugin is 1.0.0.33, and it is running on computer "DKNST01...".

Should the plugin fail to run, or should the Connection A in the above figure break, it causes an alarm:

●	REFCON Wireless Hub plugin	Wireless Hub synchronization...	●	1.
---	----------------------------	---------------------------------	---	----

As seen, "Online" is grey, indicating no connection to the plugin. If the plugin runs on the REFCON Main PC, you might try to restart this. If the plugin runs on another PC, check the LAN connection, or try restarting the other PC.

If, in contrary, the plugin works fine, but it has no connection to the external system (Connection B in the above example), or it otherwise fails to do its duty, it creates an alarm in REFCON:

●	REFCON Wireless Hub plugin	Wireless Hub synchronizati...	●	1.
---	----------------------------	-------------------------------	---	----

Note that it looks the same except that the "Online" indicator is green.

To see the alarm description, you need to check the System Alarm Summary. If both the above alarms are active at the same time (which is unlikely in practice), it looks like this:

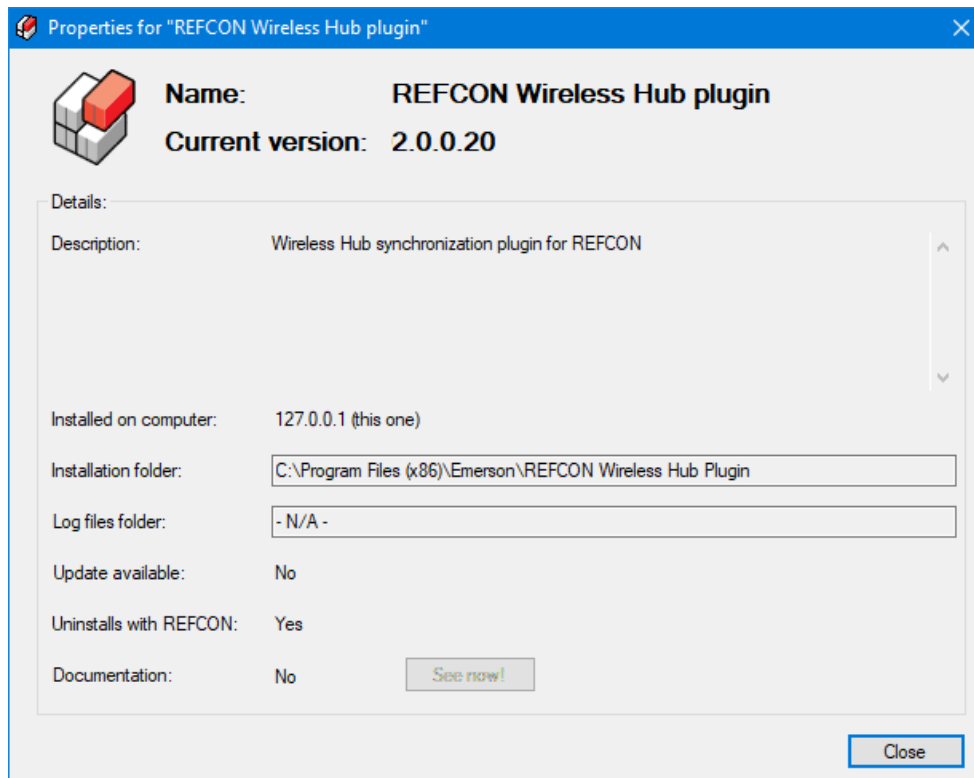
System Alarm Summary								
Location	Origin	Text	Alarms	U...	T...	Time LMT	User's note	Generated By
System alarms (Count=3)								
●	REFCON Wireless Hub (HubDevice2)	System device alarm	1	202...	2020-10-06 11:54:13			REFCON Wireless Hub plugin
●	REFCON Wireless Hub plugin	Device went offline	1	202...	2020-10-06 11:53:24			REFCON

Notice the "Generated By" column. The "Connection A" alarm is generated by REFCON, while the "Connection B" alarm (or any other function alarm) is generated by the plugin. In the latter case, please refer to the plugin manual, see below.



Plugin details and update info

You can get more information about the plugin by right-clicking it in the [System Devices List](#)³³³ and selecting "Properties" from the drop-down menu:



Among others, you can here

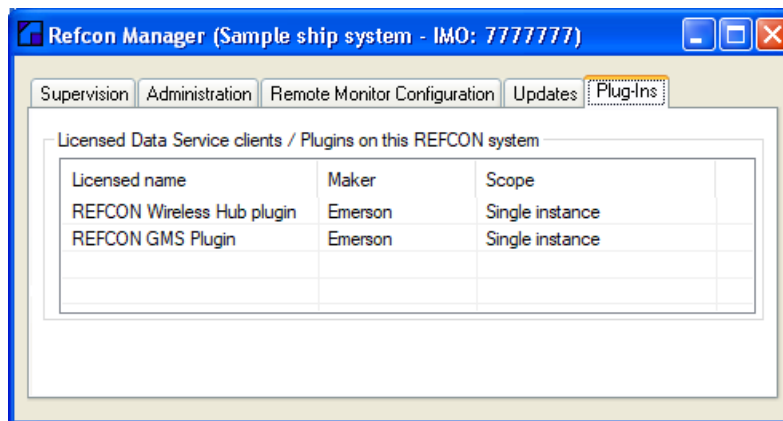
- See what version of the plugin is running.
- Read a description of the plugin.
- In case of trouble, see where to find the possible log files that could help you solve the problem.
- See if a plugin update is available (which it is not in above example).
- See that if you uninstall REFCON, is also the plugin uninstalled?
- See if the plugin provides some documentation - and click "See now" to open it immediately. In the above example there is no such, as for this plugin, the documentation is included [here](#)²³³.

Plugin settings

For some plugins you can use the [Settings Editor](#)³³⁸ to change parameters used by the plugin.

Plugin list in REFCON Manager

In the REFCON Manager you can see a list of all licensed plugins and similar:



In this example, two plugins are found. Note that this picture does **not** show if the plugins are in fact installed and running well - it is nothing but an extract of the license.

20.4 Third-party Software on the REFCON PC

Key subjects in this topic:

[Overview](#) ³¹²
[REFCON on public networks](#) ³¹²
[Protection against computer viruses and similar](#) ³¹²
[Windows updates](#) ³¹³

Related topics:

[Understanding and Maintaining REFCON](#) ³⁰²

Overview

Your system includes a REFCON Main PC and maybe one or more REFCON Monitor PC(s).

As standard the REFCON Main PC is reserved for the task of running REFCON. Being a demanding task, **you should not install additional software** on the REFCON Main PC without acceptance from Copeland Transportation Solutions, with the exceptions described below. Be aware that REFCON may be using COM ports and other PC resources in special ways, which might conflict with third-party software. In worst case this could cause REFCON to malfunction.

In contrary, the REFCON Monitor often runs on a PC that also runs other applications. As the REFCON Monitor is less critical and less demanding, this rarely causes troubles; however, if your REFCON Monitor does not work well, please try closing the other applications.

REFCON on public networks

Some REFCON systems are stand-alone systems that are not connected to the Internet or similar. If your REFCON is such a system, please stop here, as virus software etc. is not an issue in this context.

If, however, your REFCON PCs are somehow connected to a public network, please be aware of the following:

Protection against computer viruses and similar

We strongly recommend that you – the REFCON customer – make sure the PC (or PCs) running REFCON software is protected against all kinds of damaging threats. If necessary you should install and run anti-virus programs, firewalls and similar on each REFCON PC.

If such software is installed, you should update it periodically as recommended by the vendor.

Note, however, that full harddisk scans and similar functions should be disabled, as such activities will significantly influence REFCON's performance.



Windows updates

When Microsoft does release Windows upgrades due to security reasons, we recommend that you perform the update. Do **always** reset the PC after an update.

Should this cause troubles to REFCON – and Windows upgrades sometimes do cause troubles on PCs – please uninstall the upgrade and inform Copeland Transportation Solutions about the experience.

20.5 The External interfaces

Key subjects in this topic:

[Overview](#)^[313]
[Planning System interface](#)^[313]
[Global Monitoring Server interface](#)^[313]
[Alarm extension interfaces](#)^[313]
[The REFCON Data Service](#)^[313]

Related topics:

[The REFCON Data Service and plugins](#)^[309]
[Understanding and Maintaining REFCON](#)^[302]

Overview

This topic describes the possible external interfaces that can be applied to a REFCON system. Any of the specific interfaces has a certain purpose and should always be considered vital. It is therefore important that **all** interfaces are kept running well.

Planning System interface

Essential REFCON functions depend on data received from your Planning System (loading calculator, terminal operating system or what you name it). Also, for some REFCON systems, data are transmitted the other way, from REFCON to the Planning System.

There are a number of ways that this interface can work:

- Manual file transfer from Planning System to REFCON - see [Transferring the Planning data File](#)^[192].
- Automatic file transfer from Planning System to REFCON - see [Automatic Planning data File Transfer](#)^[187]. This requires a LAN connection.
- Automatic two-way online communication - see [Planning System Interface](#)^[202].
- Through a dedicated plugin when so agreed with customer.

Global Monitoring Server interface

To most REFCON owners it is essential that data are continuously transmitted from REFCON to the Global Monitoring Server. Learn more about this in [GMS Log Interface](#)^[176].

Alarm extension interfaces

In addition to displaying alarms on screen, REFCON includes a number of alarm extension options:

- [Email Notification](#)^[268].
- [Digital Outputs](#)^[274] for main alarm system, REFCON Alarm Panel or similar.

The REFCON Data Service

The REFCON Data Service allows other interfaces than the ones described above, please see [The REFCON Data Service and plugins](#)^[309]. It is likely that in future REFCON 6 versions, plugins will even replace some of the above.



20.6 Understanding installation and update

Key subjects in this topic:

[Overview](#)^[314]
[The Installation Package](#)^[314]
[Installation versus Update](#)^[314]
[The Installation Set](#)^[315]
[Updating an existing REFCON 6 installation](#)^[315]
[Detecting that an update is available](#)^[315]

Related topics:

[Standard installation and update](#)^[316]
[Remote installation and update](#)^[324]
[Automatic backup and restore](#)^[328]
[Understanding and Maintaining REFCON](#)^[302]

Overview

This topic attempts to give you an understanding of the installation and update processes. The linked topics merely explain how to execute installation or update.

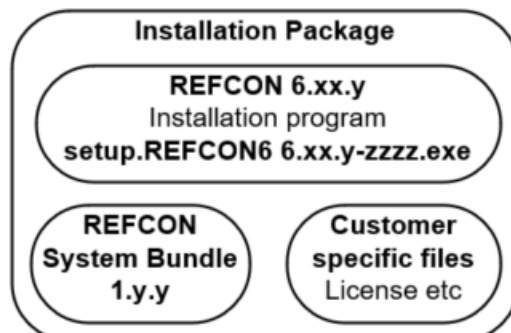
Note that you need to be administrator on the computer to run installation and update.

If you have been using REFCON 6 for long time, you may already have a CD called the Customer Disk. If you have purchased **no** new features, this can still be used - you will be asked for it during installation if required.

With REFCON 6.11 and newer, the Installation Package is as standard no longer based on CDs, see [The Installation Set](#)^[315].

The Installation Package

It may be helpful to be aware that the REFCON installation package consists of 3 major elements:



The Installation Package is uniquely prepared for the specific REFCON 6 system and must **never** be copied or used in any other system, as at least the included unique [License file](#)^[12] was prepared for this specific system.

The REFCON 6.xx.y (see [Version indication](#)^[306]) and the License are mandatory. Also other customer specific files (Container Plan, maybe more) are normally required.

The REFCON System Bundle includes software and data adding functionality to REFCON 6 itself. If you are using the REFCON Portable Modem Solution, but also in other cases, the REFCON System Bundle is mandatory. If you are updating to REFCON 6.11.4 or newer from an old version without having purchased any added functionality, the REFCON System Bundle may not be required.

Installation versus Update

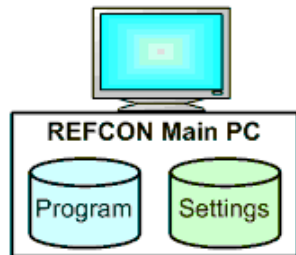
You will typically run an **installation** if REFCON is going to run on a new PC, or if you change to REFCON 6 from an older version, typically REFCON 5. In such case, be sure to uninstall the old version before installing REFCON 6.



You start an installation by executing the REFCON 6 setup.exe from the installation set. The installation will, as standard, automatically load and read the license and other customer specific files. Then it will install the licensed items from the REFCON System Bundle, if any. Normally you only have to deal with one question during installation, [Automatic backup and restore](#)³²⁸.

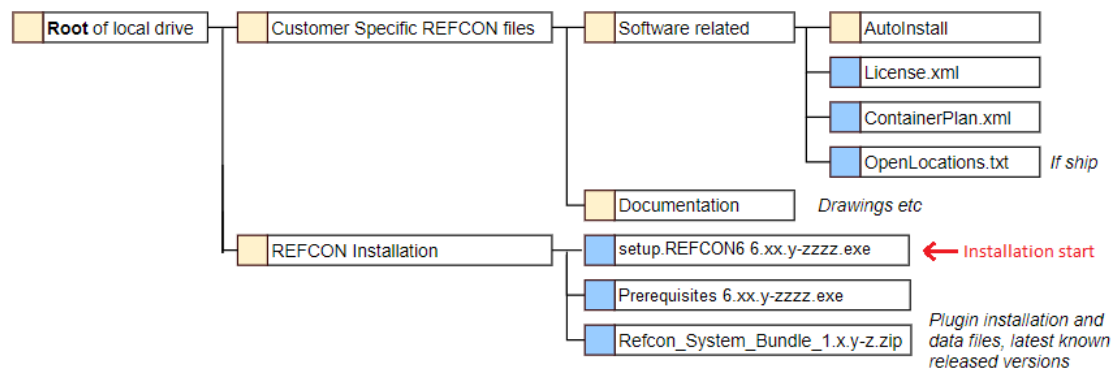
You would run an **update** to upgrade to a newer release of REFCON 6, or to a new release of the REFCON System Bundle, or to new customer specific files. Updating one of these will normally not affect the others.

Whether you run an installation or an update, the goal is a fully running REFCON with program and settings in place on the Main PC harddisk:



The Installation Set

An **installation set** is delivered with any REFCON 6 system. You will as standard receive it as a customer specific USB memory stick organised as follows:



Note: There may be more folders than shown above.

You may instead receive the entire contents as a .zip file or similar. If so, copy the contents to the

- **root** of a USB memory stick and attach this to the REFCON Main PC, or
- **root** of a hard disk drive in the REFCON Main PC, eg. c:\.

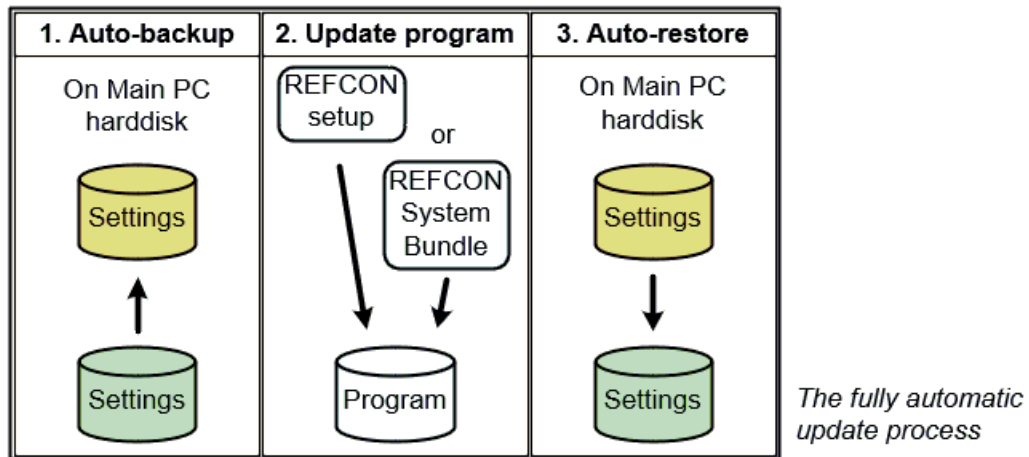
Updating an existing REFCON 6 installation

Having once installed REFCON 6 as described above, it is likely that you at some point of time decide to upgrade to a newer release of either REFCON 6 or REFCON System Bundle or both.

If you need to update both, **always update REFCON first**.

You may know the update concept from common PC products. For example, Microsoft's Windows Update can be used to update your Windows - and after an update everything seems to work as before, that is, all your data are intact, and your programs work as you are used to.

The REFCON update is designed to work exactly the same way - details are changed, in the background or visible, but your data are safe, and essentially everything works as before. All this is taken care of automatically and can be illustrated like this:



Detecting that an update is available

You can be informed that a new REFCON 6 release is available in the following ways:

- You receive a new Installation set or an Update set from Copeland Transportation Solutions.

20.7 Standard installation and update

Key subjects in this topic:

[Overview](#) ³¹⁶
[Installing REFCON 6 on new PC](#) ³¹⁶
[Running a REFCON 6 update](#) ³¹⁸
[Running a REFCON System Bundle update](#) ³¹⁹
[Uninstalling REFCON](#) ³²⁰
[Cleaning up after uninstallation](#) ³²²
[Getting access to c:\ProgramData](#) ³²²

Related topics:

[Understanding installation and update](#) ³¹⁴
[Automatic backup and restore](#) ³²⁸
[Understanding and Maintaining REFCON](#) ³⁰²

Overview

This topic describes how to conduct manual installation and update.

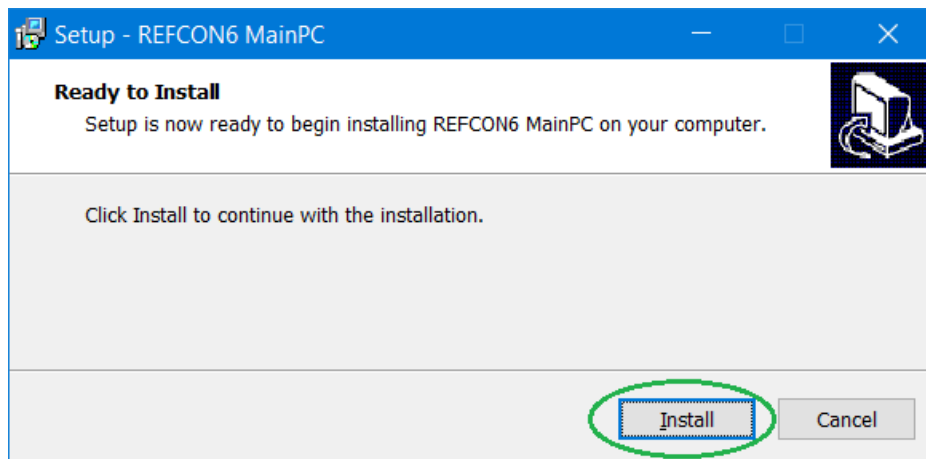
Installing REFCON 6 on new PC

To install REFCON on a computer where it is not installed in advance, proceed as follows:

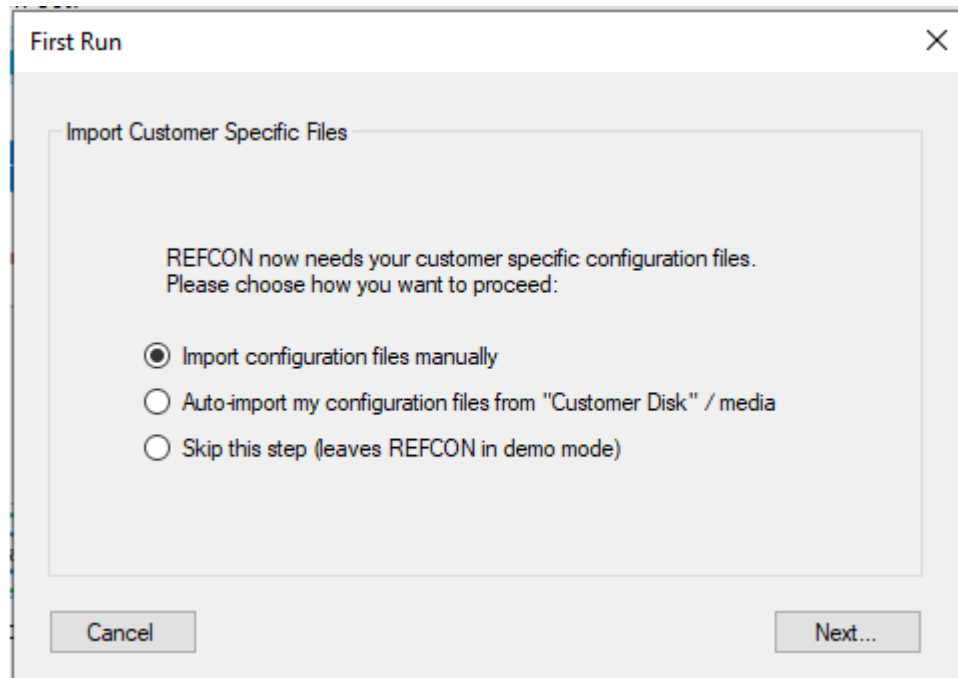
1. Run setup.REFCON6 6.xx.y-zzzz.exe from the installation set.



2. Confirm by selecting Install.

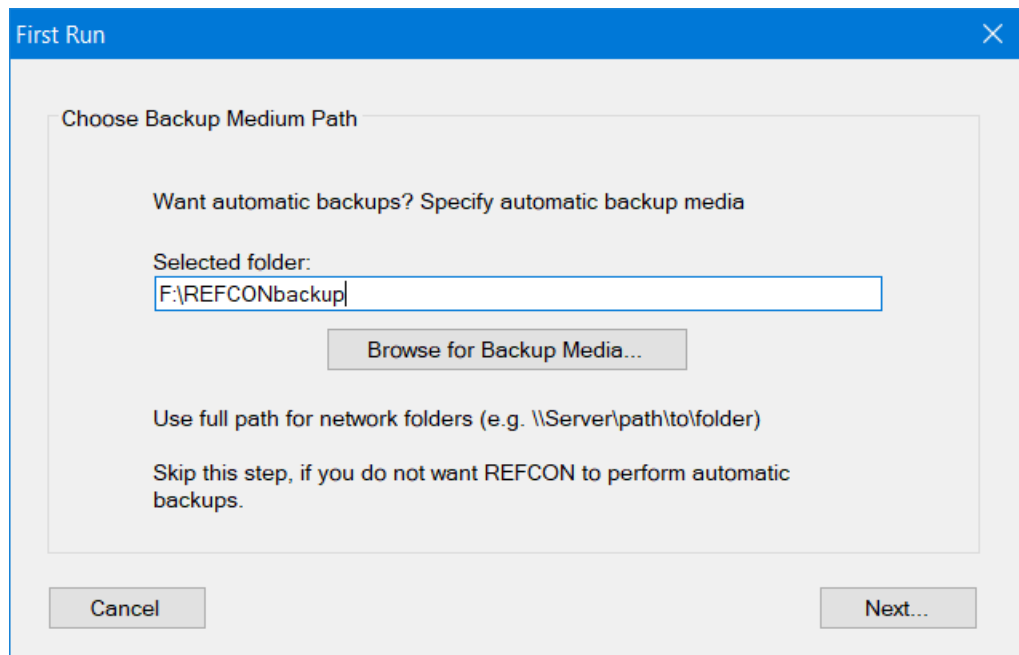


3. Please wait for a minute or so while various windows indicate progress.
4. **REFCON** license and container files are required for a successful installation. There are 3 options to choose from:

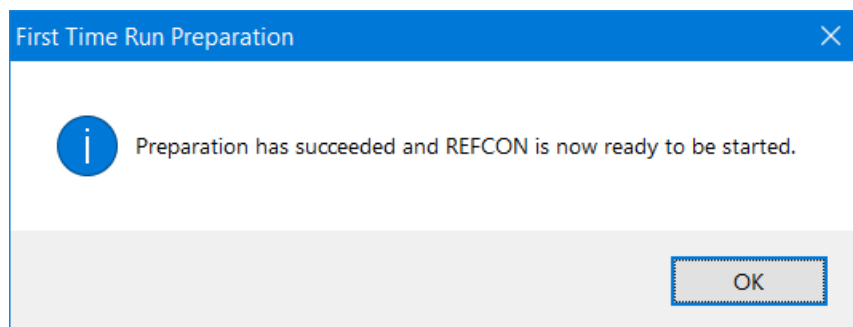


Choose the folder where the container plan and license file is present and select Next.

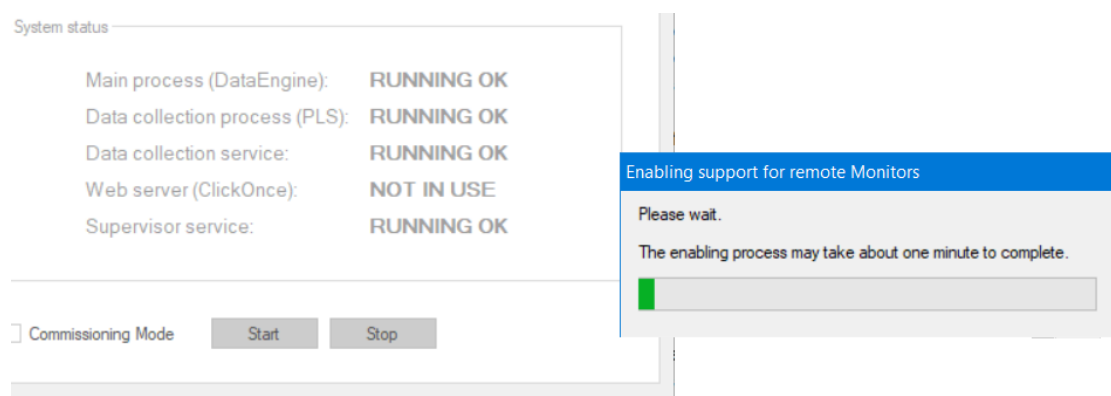
5. When requested, decide about [Automatic backup](#)³²⁸. Be careful not to use drive mappings! When done select Next...



6. The installation reports Finished. Select OK.



7. Please wait for a while remote monitors are enabled, if defined by license.



8. Start REFCON Monitor and make sure everything is running as expected. Check the settings, and change if necessary.

9. Set up initial users and container lists, etc.

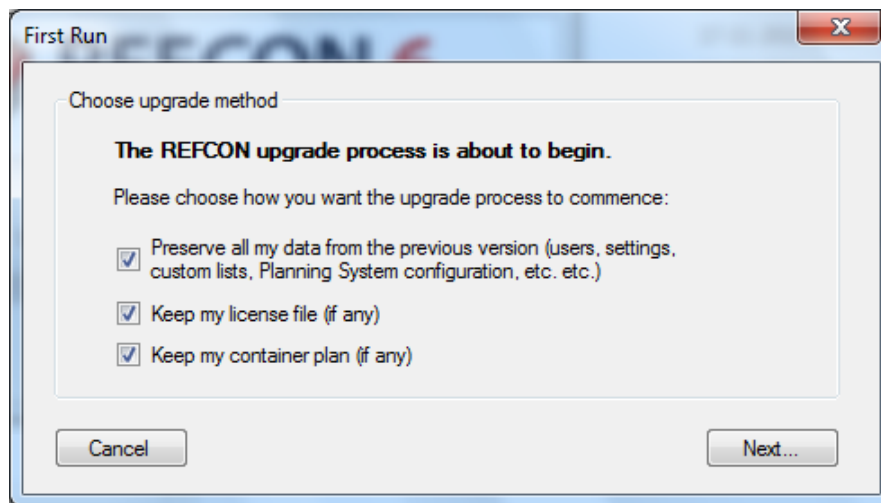
10. Having completed the above we strongly recommend that you make a [backup](#)^[45].

Running a REFCON 6 update

It is *your* decision whether and when to run the update.



If you use a new Installation set, you start the installation just as if you were [Installing REFCON 6 on new PC](#)³¹⁶ (steps 1 and 2). When doing so, the installation program actually checks if REFCON 6 is already installed on the PC. If so, you will (after some time) see the following dialog box:

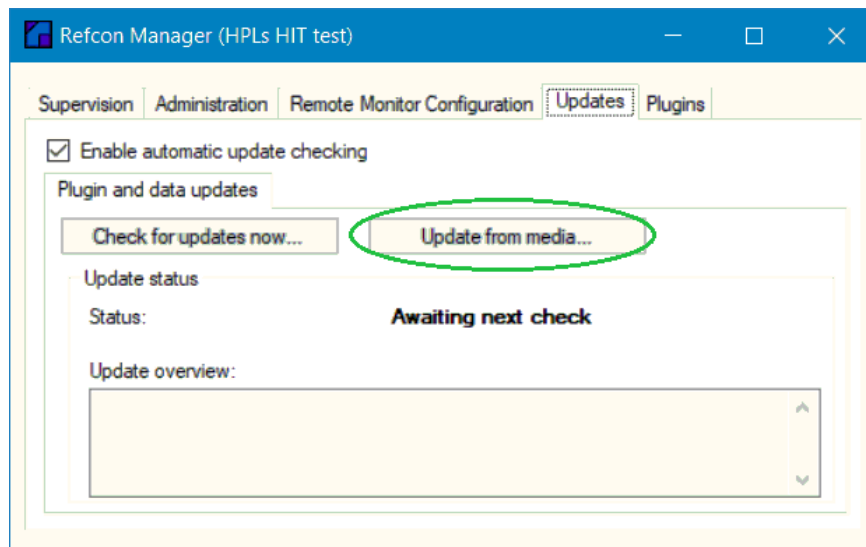


Here you in fact decide whether to run an automatic update (by keeping all check marks) or a new installation (by clearing all check marks).

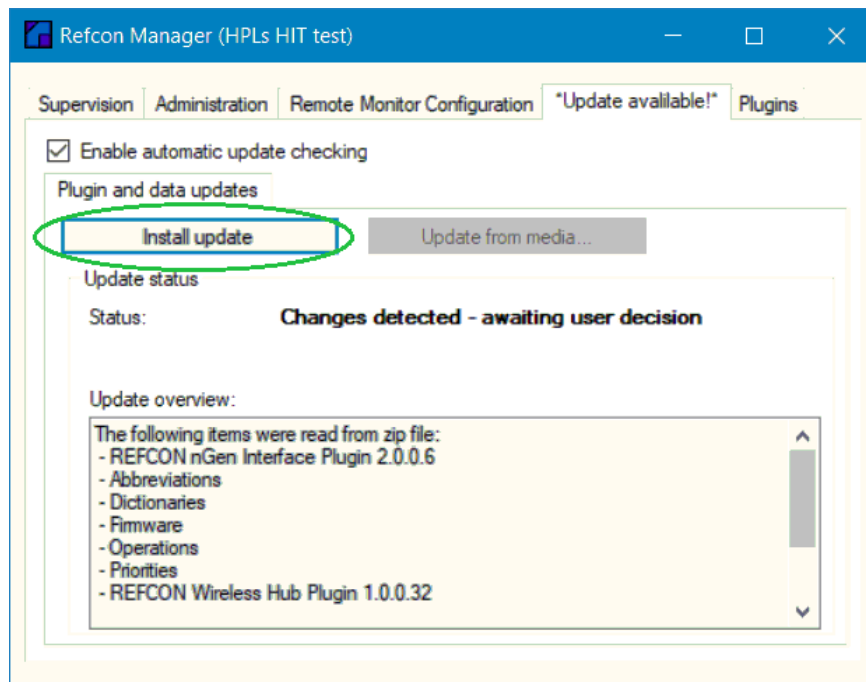
Running a REFCON System Bundle update

When you have received a new REFCON System Bundle, proceed as follows:

1. In REFCON Manager, open the Updates tab and select "Updates from media...".

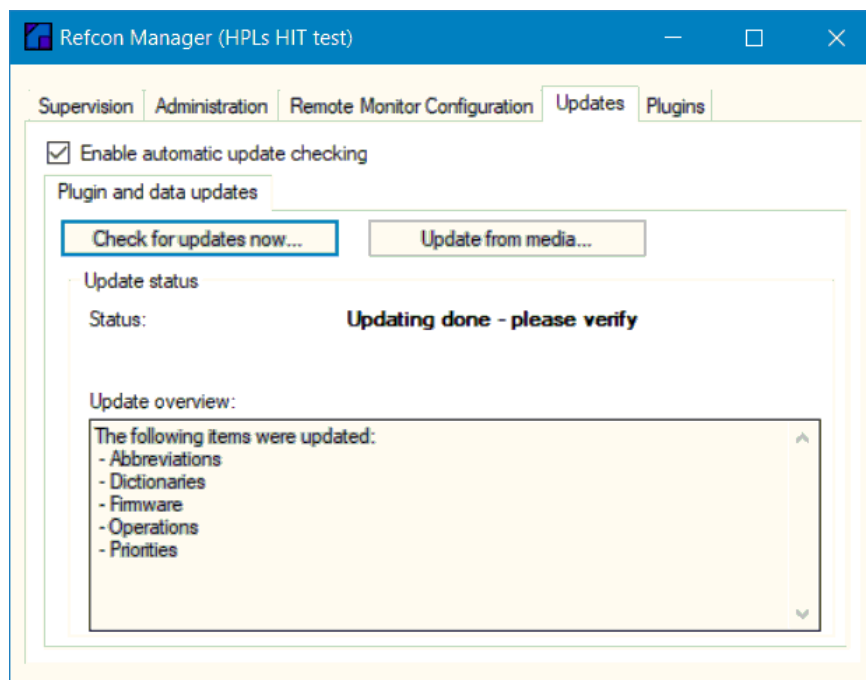


2. In the file dialog, point to the received REFCON System Bundle .zip file and open it.
3. After a while, the result of the check is shown.
 - 3a. If no relevant changes found, the dialog keeps showing the above picture - "Awaiting next check".
 - 3b. If changes are found to software installed on the REFCON Main PC, you will see the list of changed items:



Here, select "Install update" when you find it appropriate. No, there is no Cancel button - you have to do the installation as soon as possible.

3c. If changes are found, but not to software installed on the REFCON Main PC, the changed files are copied immediately, and you see the result:



Uninstalling REFCON

Note: If your running REFCON is

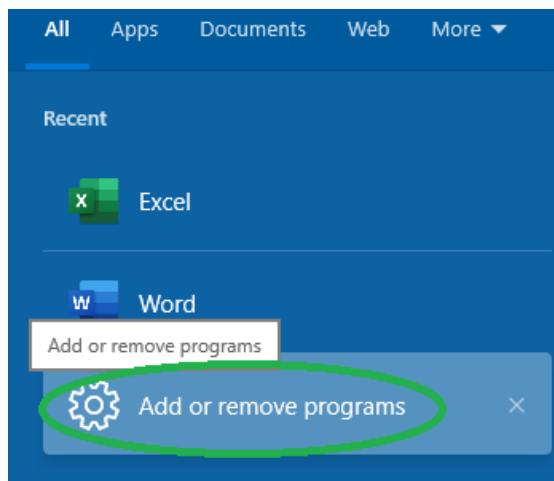
- a newer version, 6.9 or later, it is normally not necessary to uninstall it - you should be able to go ahead [upgrading](#)^[318].
- very old, such as REFCON 5, you need to uninstall this before switching to REFCON 6.

To uninstall REFCON 6, you need to be administrator in Windows.

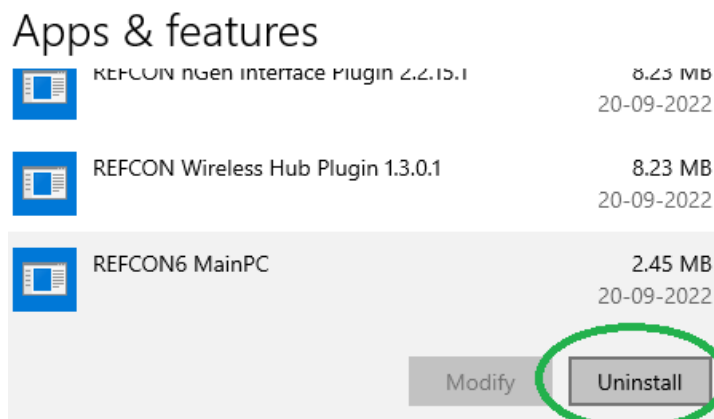
Please proceed as follows:



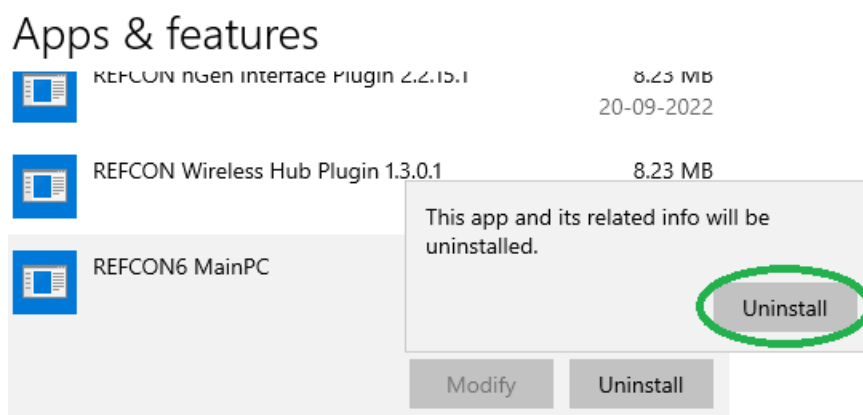
1. Open the uninstallation tool:



2. Select "REFCON6 MainPC" and then select [Uninstall]:

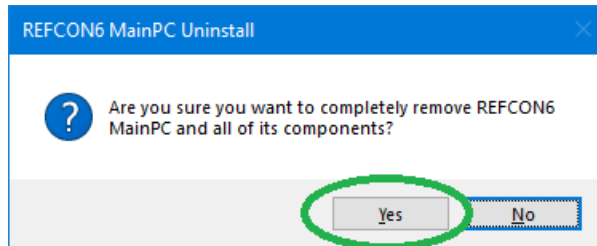


3. Select [Uninstall]:



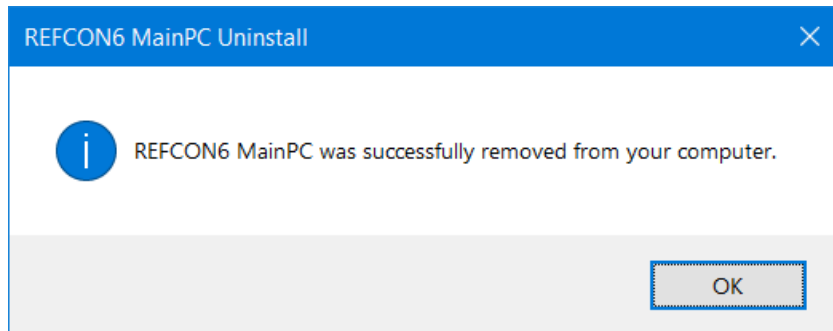
4. Accept to run "unins000.exe" from unknown publisher.

5. Accept to uninstall:



Now please wait a minute or two while you see uninstallation progress and several windows opening and closing.

Finally the result is indicated:



Cleaning up after uninstallation

If you are going to install newest REFCON 6 after having uninstalled an old version of same, we recommend that you clean-up as follows before installing the new version:

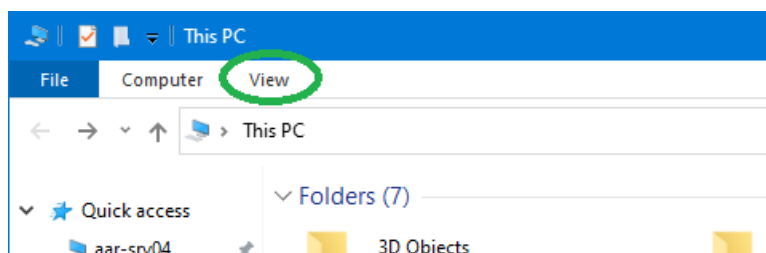
1. Make sure the folder "Refcon" is **not** found in folder "c:\Program Files (x86)" (64-bit Windows) nor "c:\Program Files" (32-bit Windows). If found, delete the folder and all contents.
2. Delete folder "c:\ProgramData\Emerson" including all files and subfolders. Note that c:\ProgramData is hidden as default; below it is described how to unhide.

In some cases you will have to restart the computer before being able to delete above folders.

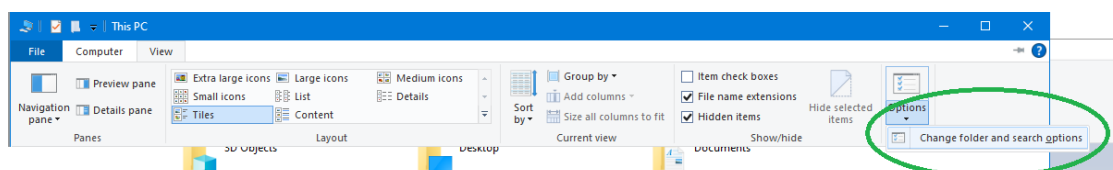
In any case, after cleaning up, make sure to restart the computer before starting a fresh installation.

Getting access to c:\ProgramData

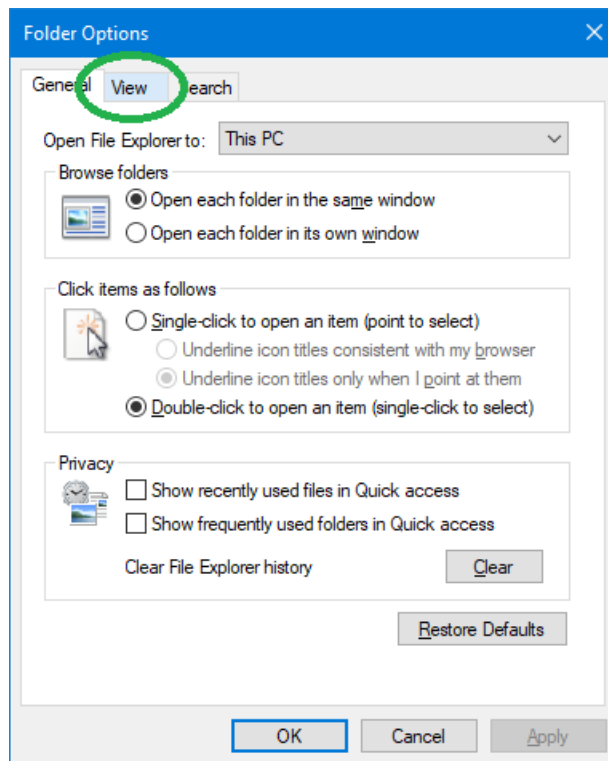
1. Open Windows File Explorer and select "View":



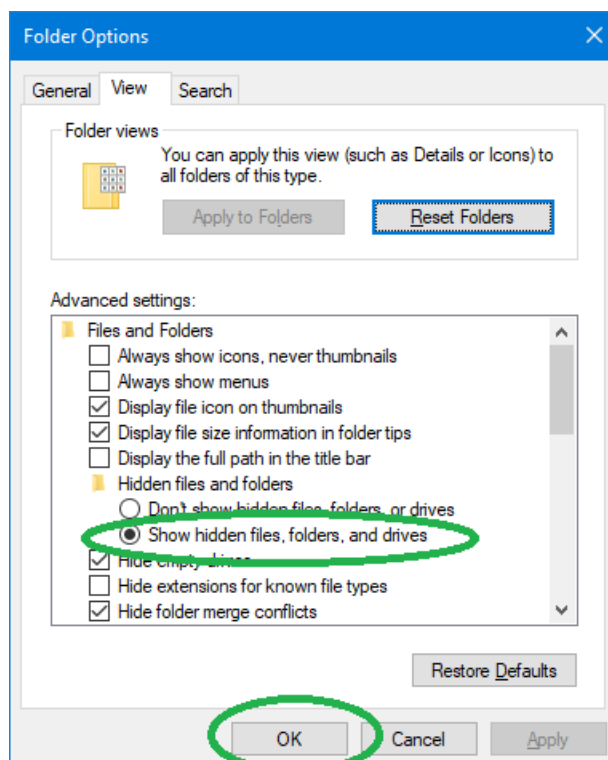
2. Select "Options" and then [Change folder and search options]:



3. In "Folder Options" dialog, select "View":



4. Make sure "Show hidden files, folders and drives" is selected, and then select [OK]:





20.8 Remote installation and update

Key subjects in this topic:

[Overview](#) ³²⁴
[Customer's basic preparations](#) ³²⁵
[Remote users' preparations](#) ³²⁵
[Preparing the Powershell script files](#) ³²⁵
[Enabling Automatic backup and restore](#) ³²⁶
[Executing a new installation](#) ³²⁶
[Executing a software upgrade](#) ³²⁷
[Checking the RefconManagerStartLog.txt](#) ³²⁷
[Possible other script parameters](#) ³²⁷

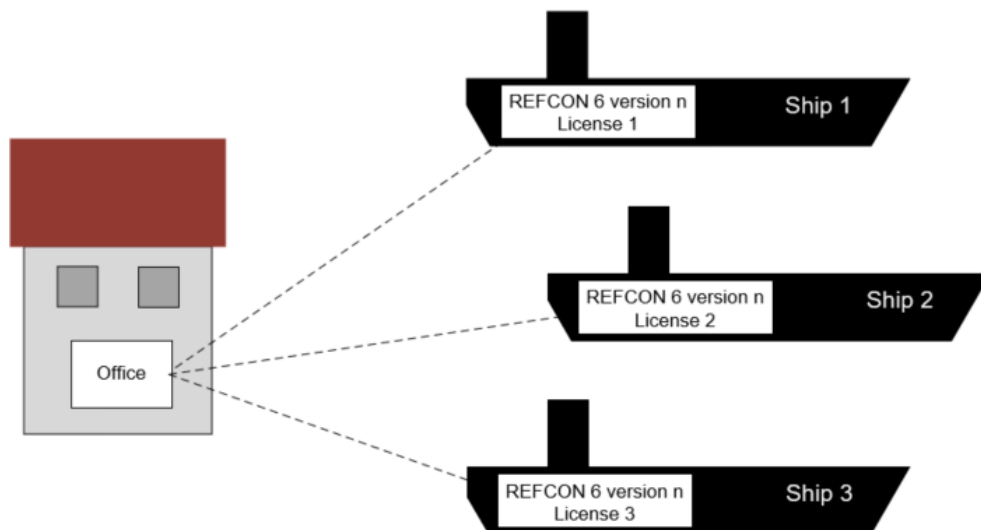
Related topics:

[Understanding installation and update](#) ³¹⁴
[Standard installation and update](#) ³¹⁶
[Automatic backup and restore](#) ³²⁸
[Understanding and Maintaining REFCON](#) ³⁰²

Overview

This topic explains how to prepare and conduct a remote installation or update of REFCON.

Remote execution of installations and updates means that user does not have to be physically situated at the REFCON computer. A typically use case is that a shipping line can install or update REFCON on a number of ships from the office.



We assume that you will use Windows Powershell and execute a script as suggested below; however, you may decide for a similar solution, suiting you better.

The user going to execute remotely can look into any of the following cases:

Ref	Use case	Important notice	Remote user's schedule
1	Installation from scratch on "new" computer	Please make sure to use the unique License.xml received from Copeland LP for each individual installation (ship or terminal). Also ContainerPlan.xml and other files <i>may</i> differ from one installation to another.	1. Copy REFCON installation files and customer specific files to the harddisk of the remote REFCON computer. 2. Run the Windows Powershell script described below.



Ref	Use case	Important notice	Remote user's schedule
2	Upgrade of REFCON software on running installation	Here it is presupposed that you have a running REFCON and do not need to have license changed.	1. Copy REFCON installation files to the harddisk of the remote REFCON computer. 2. Run the Windows Powershell script described below.
3	Upgrading licensed functionality	Here it is presupposed that you have already the newest REFCON running and need to have license changed.	1. If necessary, first upgrade the software as described in above case 2. 2. Copy the updated License.xml and possible other updated customer specific files to the harddisk of the remote REFCON computer. 3. Run the ???

Customer's basic preparations

To facilitate the remote installation/update, customer needs to

- Make sure there is a network connection between the office and the REFCON computer. The connection can exist always or time limited.
- Make sure the user at the office can operate remotely with administrator privileges on the REFCON computer, and thereby copy files and execute scripts or similar.

Remote users' preparations

As a remote user it may be useful to

- Understand the required file placement on the local drive on the REFCON computer, as explained in [Understanding installation and update](#)³¹⁴.
- Understand the manual installation or update described in [Standard installation and update](#)³¹⁶, as the remote installation/update does the same, however, without requiring user's interaction.

You should, probably once for all, prepare one or more script files as described below.

Note that you need to have administrator privileges on the REFCON computer to execute the script.

Preparing the Powershell script files

To remotely execute an installation or a software upgrade (first two use cases in above table), you need script file A.

We suggest that you produce an initial version of script file A by copying the following text to a file - you could name it Remote_REFCON_software_installation_and_upgrade.ps1.

```
#Script for installing and upgrading REFCON software remotely

#$Path variable accepts the REFCON setup path with setup file name.
# Example $path='C:\REFCON Installation\setup.REFCON6 6.14.5-e39e064.exe'
$path='c:\REFCON Installation\setup.REFCON6 6.14.5-4e3f7b9.exe'

#$args variable accepts the arguments which are mandatory for remote installation/update.
# Example $args='/VERYSILENT /args="-pp C:\REFCON Installation\Refcon_System_Bundle_1.7.0-10.zip"'
$args='/VERYSILENT /args="-q true -pp "c:\REFCON Installation\Refcon_System_Bundle_1.8.0-3.zip"'
```



```

echo $path
echo $args

$process=Start-Process -FilePath $path -ArgumentList $args -PassThru

if($args.Contains("-q")) {
'REFCON installation/upgrade process is started'
}
if($args.Contains("-pp")) {
'Plugins installation/upgrade process is started'
}

$process.WaitForExit()
if ($process.HasExited) {
    echo 'Installation/upgrade process is in progress, please check the log file after two minutes in
path="C:\Users\{YourUserName}\AppData\Roaming\RefconManagerStartLog.txt"'
    echo 'for installation/upgrade status'
}

```

As you see,

- you need to edit the file each time to get a new REFCON 6 version or REFCON System Bundle version;
- we recommend and assume you are using the default folder "c:\REFCON Installation", but you may choose otherwise.

Enabling Automatic backup and restore

Please consider enabling [automatic backup](#)³²⁸ on each REFCON system. Backups are very valuable in case of re-installation or computer problems - and users may forget to execute manual backups.

To enable automatic backup, you define a drive and folder where to store the backup.

Preferably, chose a USB drive "permanently" attached to the REFCON computer, or a network drive if possible. If neither of these are available, we recommend choosing a folder on a local harddisk (c: or other), as automatic backups here are still valuable.

To enable automatic backup, add the -b optional parameter to the \$args - like this:

```
$args='/VERYSILENT /args="-q true -b "\\OurServer17\REFCON\Backup" -pp "c:\REFCON Installation\Refcon System Bundle 1.8.0-3.zip"'
```

or

```
$args='/VERYSILENT /args="-q true -b "d:\REFCONbackup" -pp "c:\REFCON Installation\Refcon System Bundle 1.8.0-3.zip"'
```

where d: is a local drive. Please do not use drive mappings.

When upgrading a REFCON installation where automatic backup is already enabled, leave out the -b parameter to keep the existing.

You may disable automatic backup by including -b "" in your script (empty string).

Executing a new installation

Note: "New" installation means that no REFCON version is currently installed on the REFCON computer. You may have to uninstall before proceeding.

Please proceed as follows:

1. Receive / retrieve the REFCON 6 and the REFCON System Bundle installation files from Copeland Transportation Solutions.



2. For **each** ship or terminal where you need to install REFCON, receive / retrieve the set of customer specific files from Copeland Transportation Solutions, including the unique license file.

For **each** ship or terminal,

3. Copy the files "setup.REFCON6 6.nn.n-nnnnnnn.exe" and "Prerequisites 6.nn.n-nnnnnnn.exe" to same folder on a local drive of the REFCON computer, "c:\REFCON Installation" as default. *If received as zip file, make sure you unzip before copying.*
4. Delete any possible "Refcon_System_Bundle_v.v.v-v.v.zip" file from the folder chosen in next step.
5. Copy the file "Refcon_System_Bundle_x.x.x-x.x.zip" to a folder on a local drive of the REFCON computer, "c:\REFCON Installation" as default. Do **not** unzip before copying.
6. Copy the the customer specific files and folders prepared for the specific ship or terminal to folder "Customer Specific REFCON files" in the root of a local drive of the REFCON computer, "c:\Customer Specific REFCON files" as default.
7. Copy the prepared script file to the REFCON computer.
8. Run Windows Powershell as administrator.
9. Execute the script.
10. Wait for a couple of minutes and then check "C:\Users\{YourUserName}\AppData\Roaming\RefconManagerStartLog.txt" on the REFCON computer for the result.

Executing a software upgrade

Note: "software upgrade" means that a newer REFCON version is currently installed and running on the REFCON computer.

Please proceed as follows:

1. Receive / retrieve the REFCON 6 and the REFCON System Bundle installation files from Copeland Transportation Solutions.

For **each** ship or terminal,

2. Copy the files "setup.REFCON6 6.nn.n-nnnnnnn.exe" and "Prerequisites 6.nn.n-nnnnnnn.exe" to same folder on a local drive of the REFCON computer, "c:\REFCON Installation" as default. *If received as zip file, make sure you unzip before copying.*
3. Delete any possible "Refcon_System_Bundle_v.v.v-v.v.zip" file from the folder chosen in next step.
4. Copy the file "Refcon_System_Bundle_x.x.x-x.x.zip" to a folder on a local drive of the REFCON computer, "c:\REFCON Installation" as default. Do **not** unzip before copying.
5. Copy the prepared script file to the REFCON computer.
6. Run Windows Powershell as administrator.
7. Execute the script.
8. Wait for a couple of minutes and then check "C:\Users\{YourUserName}\AppData\Roaming\RefconManagerStartLog.txt" on the REFCON computer for the result.

Checking the RefconManagerStartLog.txt

<To be added>

Possible other script parameters

You can optionally decide for a combination of the above use cases 2 and 3, that is, you upgrade REFCON and optionally the REFCON System Bundle, and at the same time upgrade license and/or container plan.

To **update the license** along with a software upgrade,



1. For each ship or terminal that you want to update, copy the new unique license received from Copeland Transportation Solutions to a folder on a local drive of the REFCON computer.
2. Edit the Powershell script to include the -lp parameter:

```
$args='/VERYSILENT /args="-q true -lp d:\NewLicenseFolder'
```

where you include the path to the folder where the license was copied - as the license file must be named License.xml. If the path has any spaces, include it in quotation marks.

When you run the script, the normally existing License.xml will be overwritten by the new.

To **update the container plan** along with a software upgrade,

1. For each ship or terminal that you want to update, copy the new container plan received from Copeland Transportation Solutions to a folder on a local drive of the REFCON computer. Note that often the container plans are different from one installation to another, but if ships are identical as this regards, they can use same container plan. Note that for ships you also need to copy the related OpenLocations.txt.
2. Edit the Powershell script to include the -cp parameter:

```
$args='/VERYSILENT /args="-q true -cp d:\NewContainerPlanFolder'
```

where you include the path to the folder where the container plan was copied - as the container plan file must be named ContainerPlan.xml. If the path has any spaces, include it in quotation marks.

When you run the script, the normally existing ContainerPlan.xml - as well as the OpenLocations.txt on ships - will be overwritten by the new.

20.9 Automatic backup and restore

Key subjects in this topic:

[Overview](#)^[328]
[What is backup and restore?](#)^[328]
[Understanding automatic backup and restore](#)^[328]
[Understanding automatic backup of archived log files](#)^[329]
[Decisions to make by customer](#)^[329]
[Setting up automatic backup](#)^[330]
[Missing backup folder](#)^[330]

Related topics:

[Understanding and Maintaining REFCON](#)^[302]

Overview

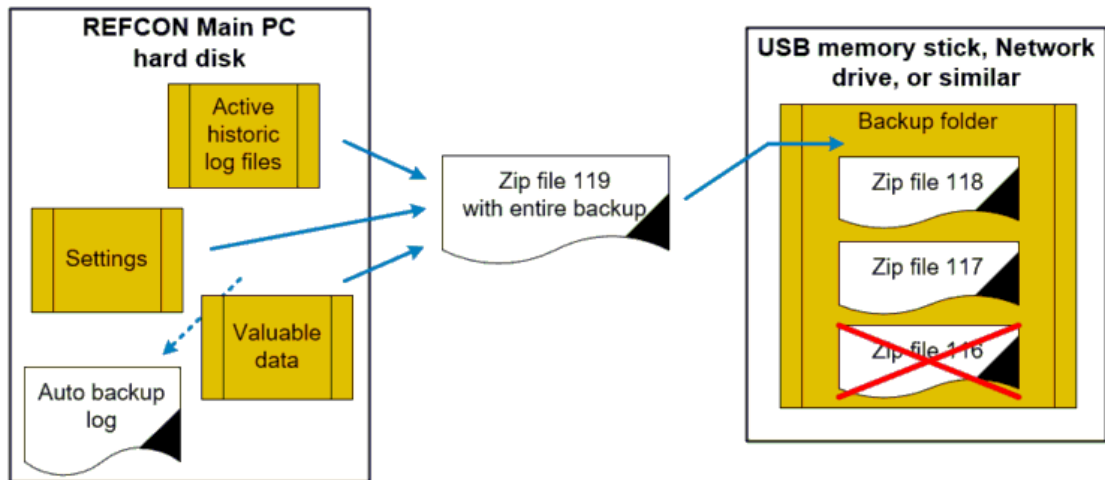
The automatic backup and restore functionality is using the manual backup and restore functions having been part of REFCON 6 for many years. Basically it helps you so that you will not have to remember to carry out backups manually.

What is backup and restore?

Please refer to [Backup function](#)^[43] and [Restore function](#)^[355]. The backup includes the first two chunks, marked S (Basic System Data) and L (Active Container log files). Also, archived log files are copied, see [Understanding automatic backup of archived log files](#)^[329].

Understanding automatic backup and restore

When decided during installation, automatic backup is performed once an hour **and** when certain settings or data are changed:



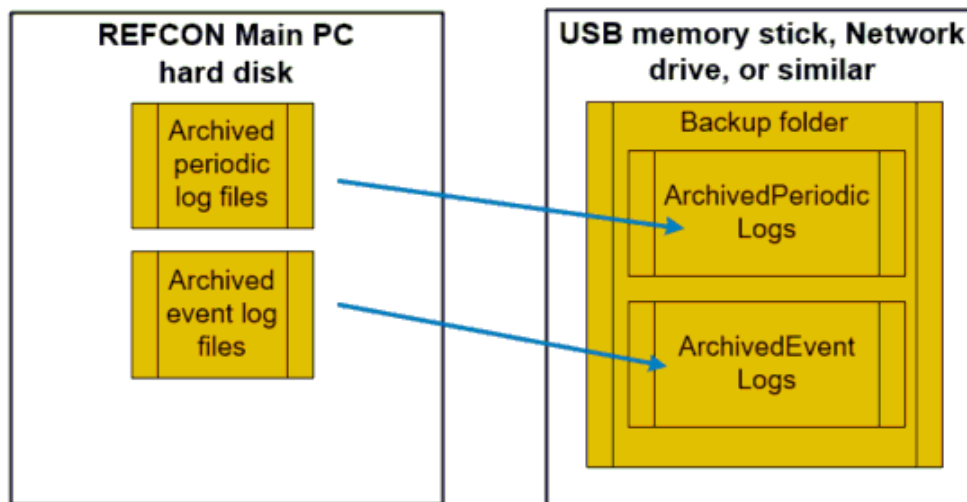
Each time an automatic backup is completed, the oldest backup file is deleted for housekeeping, so that maximum 3 files are kept. Normally only the newest will have relevance for restore.

During a restore, the files from the newest zip are copied back to their original folders.

Automatic restore is considered each time REFCON is started. The automatic backup log shows if latest backup was created by this REFCON. If it is, **no** restore is done as the files are already present on the hard disk. So, restore is done if a backup file is found newer than the last one logged, or if no automatic backup log file is found.

Understanding automatic backup of archived log files

In addition to the backup described above, REFCON creates copies of archived container log files as these are being archived:



Please refer to [Historic data logging](#)¹⁶⁰ for a description of log file archiving.

Each time a log file is archived, it is also copied to the relevant subfolder of the automatic backup folder within a few seconds.

Decisions to make by customer

You need to decide how to utilize this function.

Solution	Description	Path example
USB memory stick	The most simple solution is to use a USB memory stick or similar. The idea is that you	F:\AutoBackup



Solution	Description	Path example
	keep the USB stick "permanently" mounted in the REFCON Main PC. If one day you need to swap to another computer, you move the USB stick and thereby can be sure the new computer has access to latest backup.	
Network drive folder	Another solution is possible if the REFCON Main PC is connected to a network with some server attached. Here you could create a backup folder and share it, so that it can be accessed by REFCON.	\\YourNetworkServer\AutoBackupShareName
Local harddisk folder	If neither of the above are possible, we recommend to have automatic backups created on the local harddisk or similar, as backups here still are valuable.	C:\AutoBackup
No automatic backup	Though not recommended, you <i>can</i> decide to run without automatic backup. To choose this, leave the decided path empty.	

Having decided your solution you need to make sure the backup folder is created and, if on network, shared, and that you know the path to the folder.

Setting up automatic backup

Having prepared as described above, you choose the backup destination folder during REFCON installation or update, please refer to [Installing REFCON 6 on new PC](#)^[316] step 4.

If you have picked the network folder solution, **be aware that you cannot use drive mapping**. Even if you as Windows user maybe can refer to \\YourNetworkServer as, say, H:, do **not** use this, as the automatic backup and restore functions are run by the REFCON Data Engine service which most likely cannot see H:.

Missing backup folder

Copeland Transportation Solutions does consider the automatic backup function important.

Therefore, should the specified backup folder become unavailable (for example, USB stick removed or network connection broken), a system alarm is created.

When the folder recovers, a backup is immediately done if such was unable to run while folder was unavailable.

20.10 REFCON Setup

Key subjects in this topic:

[Overview](#)^[331]

[Automatic Cleaning Up parameters](#)^[331]

[Unplanned container – why?](#)^[331]

[Cleaning up – why?](#)^[331]

[Data Obsolescence](#)^[332]

[Various settings](#)^[333]

Related topics:

[Periodic logging parameters](#)^[161]

[Setting up automatic load data file transfer](#)^[188]

[Setpoint alarm parameters](#)^[119]

[Understanding and Maintaining REFCON](#)^[302]



Overview

The REFCON Setup dialog box is used for setting up system parameters.

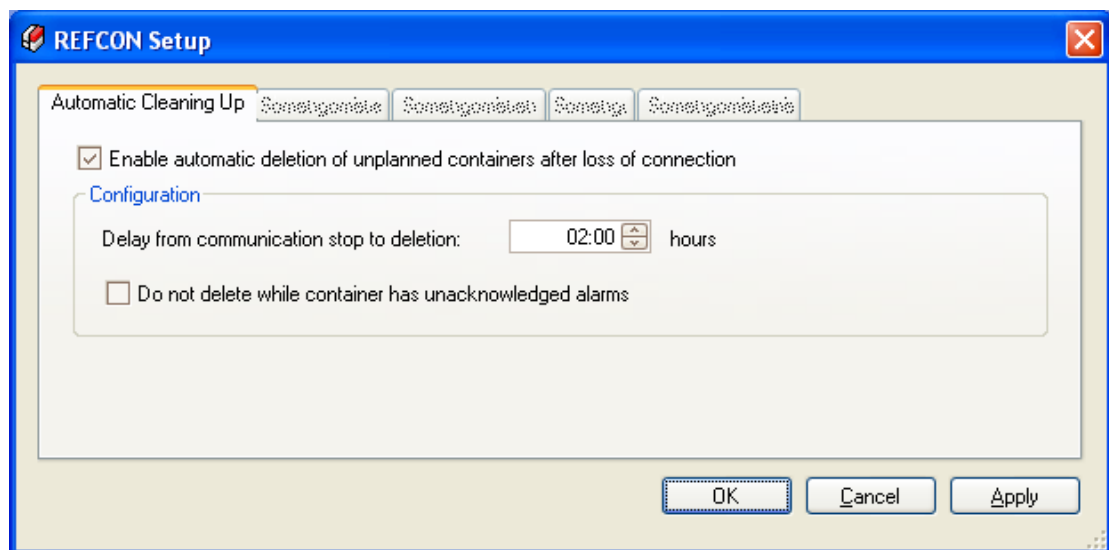
As you see (eg. in the figure below) the dialog box consists of several tabs, each including a number of parameters and options. Some of the tabs are described below, others are described in relevant parts of the online help manual.

Certain tabs are only available if the corresponding options are enabled in the [License file](#)^[12].

The number of tabs might be so high that not all are visible. In this case, use the two small arrow buttons upper right to scroll horizontally.

Automatic Cleaning Up parameters

Under the *Automatic Cleaning Up* tab of the REFCON Setup dialog box, you can enable or disable automatic deletion of [unplanned](#)^[183] containers:



When enabled - as in the example in the figure - you can further choose:

- The delay field: The time (hours:minutes) to wait before doing the deletion; in the example in the figure, 2 hours.
- The second checkbox: Whether the presence of any unacknowledged alarm(s) shall postpone the deletion.

Please find information about [cleaning up](#)^[331] below.

Unplanned container - why?

An [unplanned](#)^[183] container is a container that starts communicating even though it has not been reported by the Planning System.

Four explanations of containers being unplanned can be mentioned:

- The container is plugged in and starts communicating before REFCON has received information from the Planning System. When later this information arrives, it changes "status" from unplanned to planned.
- There is no Planning System attached, or data are not being updated.
- The site has (smaller) areas not covered by the Planning System where containers can be located.
- It is a *ghost container* that communicates using a wrong ID.

Cleaning up - why?

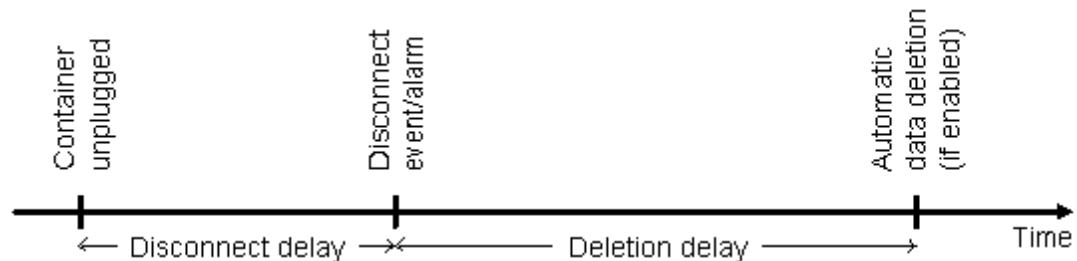
The REFCON database includes an amount of information for each container. Even when a (communicating) container is disconnected, this information stays in the database – for two reasons: It could be just a temporary disconnection, or you might need the data for your record.



But when you do certainly no longer keep the container, and you have certainly saved data as needed, the container must be "deleted", that is, the corresponding data are removed from the database. While not "deleted", the container will still appear in the Container Plan and in the Container List – so if you don't make sure containers are "deleted", you will very soon have a confusing mix of relevant and irrelevant containers.

As standard, [planned](#)^[183] containers are deleted automatically on information from the Planning System. The cleaning up parameters described above thus only relate to unplanned containers.

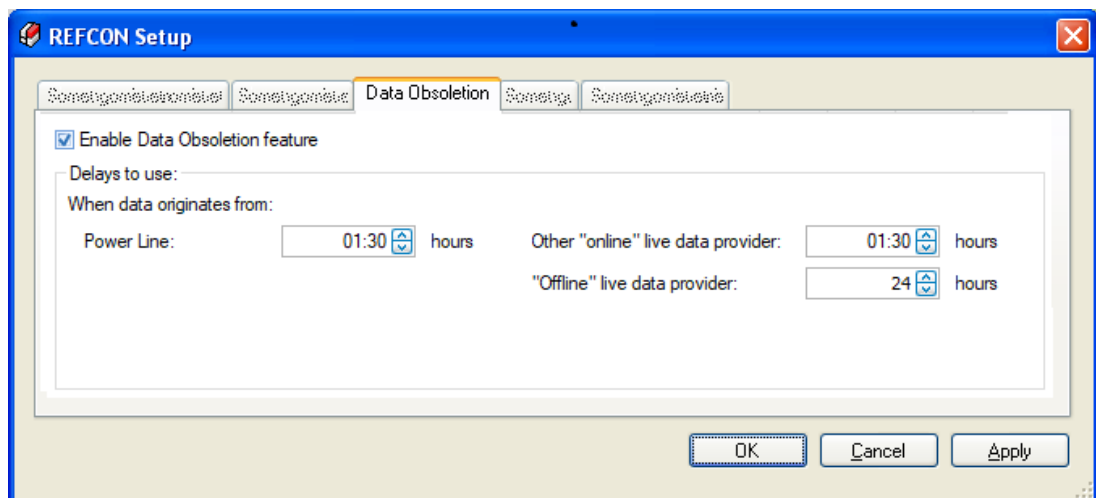
Note that there are two delays involved:



At leftmost time mark, the container is actually unplugged. Then, after the time "Disconnect delay", it is noted "disconnected" by REFCON. Next, after the "Deletion delay", if still disconnected, the container's data are removed. Note that possible log files are **not** removed by this function.

Data Obsolescence

Under the *Data Obsolescence* tab of the REFCON Setup dialog box, you can enable or disable the Data Obsolescence function:



When enabled - as in the example in the figure - you can further choose three delays, the *Power Line*, the *Other "online" live data provider*, and the *"Offline live data provider" delays*. These can be explained as follows:

The Data Obsolescence function decides if values received from containers are to be shown and used as being up to date or not.

With the introduction of [REFCON 6 extensions](#)^[218], data can have passed different routes to reach REFCON 6. These routes may have different nature and therefore different expected refresh intervals. The idea is that you select a Data Obsolescence delay a little higher than the refresh interval of the route:

- Data read via Power Cable Transmission are normally updated at least once an hour. Consequently, the Power Line is a little higher, 90 minutes as default.



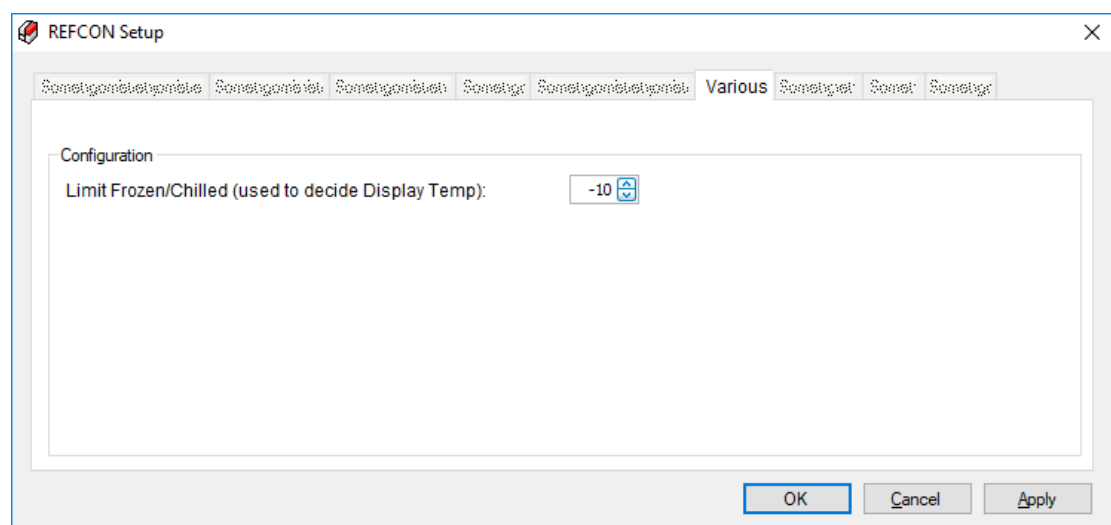
- Data read via other routes considered online, typically the [REFCON Wireless Hub solution](#)^[219], may have a different expected refresh interval, and you may fine-tune the Data Obsolescence delay correspondingly.
- Data read via devices being considered offline, such as the RDC and the [REFCON Mobile and REFCON Mobile Plugin solution](#)^[219], are likely to have a longer refresh interval due to the manual rounding effort required. You may select a delay lower than the default 24 hours to use it as a warning to the user that it is time to round.

The delay being used is the one related to latest Source, that is, the latest route for the data.

When a data point is marked "obsolete", the data field on the screen will indicate "obsolete", for example by changing colour. The periodic logging function and the forwarding of data to the Global Monitoring Server will omit the data point in this case. Please refer to other parts of the online help manual for detailed descriptions.

Various settings

Under the *Various* tab of the REFCON Setup dialog box, you find following setting:



- The Display temperature equals the Return temperature when the temperature setpoint is below the decided frozen/chilled limit, being -10 deg.C as default.
- The Display temperature equals the Supply temperature when the temperature setpoint is above the limit.

20.11 System Devices list

Key subjects in this topic:

[Overview](#)^[334]
[Opening the System Devices list](#)^[334]
[The System Devices list window](#)^[334]
[Column specifications](#)^[334]
[Shortcut menu in System Devices list](#)^[336]
[Remove offline devices](#)^[336]
[Reset statistics](#)^[337]
[How to get log files from a supporting device](#)^[337]

Related topics:

[Master groups](#)^[347]
[Understanding and Maintaining REFCON](#)^[302]

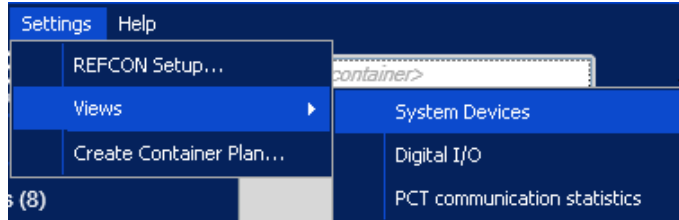


Overview

The *System Devices list* shows all units connected to your REFCON system, including software (e.g. REFCON Monitor) and hardware (e.g. REFCON Wireless Hub). This allows you to check if everything is all right. In case of troubles, the list should help you deciding what is wrong. Here the included statistics might be helpful.

Opening the System Devices list

The System Devices list is available from the Settings menu under the Views entry:



The System Devices list window

In the System Devices list you find one line for each unit:

System Devices							
Device	User's description	Online	Software	Hardware	Activity	Group	Connected
COM1	Main PC's RS 232 port	●	Danbuss	SerialPort		0	0
RS2LAN (Net 22)	Placed in junction room	●	127.0.0.1 port 12002	RS2LAN		0	3
RMM Master (22.18)	Placed in junction room	●	Mk.IV: E:01070300, F:03050212, Flash	RmmMasterMKIV	><[[f">	1	88
RMM Master (22.19)	Placed at swbd 14	●	Mk.IV: E:01070300, F:03050212, Flash	RmmMasterMKIV	[o"<	1	88
RMM Master (22.20)	Placed at swbd 19	●	Mk.IV: E:01070300, F:03050212, Flash	RmmMasterMKIV	><[[f">	1	102
STAGINGSERVER (10.6...	REFCON Main PC (StagingU...	●	REFCON Monitor	PC		-	
C5175W478 (10.63.139...	Container Office PC (lundh)	●	REFCON Monitor	PC		-	

Remove offline devices Reset statistics ☒ Standard view ☐ Statistics view ☐ Both

As shown in the lower right corner this is the "Standard view". If you instead select the "Statistics view", the included columns change:

System Devices											
Device	User's description	Online	Activity	Con...	Total	Retrans	Rejects	Errors	Deads	Lates	No ans
COM1	Main PC's RS 232 port	●			0						
RS2LAN (Net 22)	Placed in junction room	●			3						
RMM Master (22.18)	Placed in junction room	●	><[[f">		155	679	0 (0%)	No data	No data	No data	0 (0%)
RMM Master (22.19)	Placed at swbd 14	●	<">]]>		185	1419	0 (0%)	No data	No data	No data	0 (0%)
RMM Master (22.20)	Placed at swbd 19	●	[o"<		102	967	0 (0%)	No data	No data	No data	0 (0%)
STAGINGSERVER (10.6...	REFCON Main PC (St...	●									
C5175W478 (10.63.13...	Container Office PC [...	●									

Remove offline devices Reset statistics ☐ Standard view ☒ Statistics view ☐ Both

Selecting "Both" includes all columns from the two views.

Column specifications

The following table specifies the columns included in the System Devices list:

Column header / abbreviation	Explanation	Origin
Device	Textual, automatically created short identification/description of the device. In possible parentheses: Device ID: - For RMM Master: Network, Node number. - For others, when relevant: Serial number or MAC address.	REFCON / Device



Column header / abbreviation	Explanation	Origin
Address	Where relevant, the address used for communication with the device - eg. IP address.	REFCON / Device
User's description	Description of the device. Decided by user, but default texts are created by the system. As soon as user makes any change, this will not be overwritten by the system.	REFCON / Device / User. For a REFCON Monitor device it is the PC description followed by the Windows user shown in parentheses.
Online	● means that the device is online, while ● means that it is offline.	REFCON / plugin
Software	Name of software and/or version in device as relevant.	Device
Hardware	The type and version of device hardware in question.	REFCON / Device
Activity	For RMM Masters: A "fish" - when moving it indicates that the master is actively communicating at the moment. Note that in each Group, only one master can be active at a time. For other devices: Some activity indication, when relevant. Always expect inactivity for shorter periods.	REFCON / Device
Group	The master group ³⁴⁷ , if defined.	REFCON
Connected	- For RS2LAN device: Number of connected RMM Masters. - For RMM Master and REFCON Wireless Hub device: Number of "connected" containers. - For plugins: Number of external system connections or devices.	REFCON / Device
Total	The total number of messages sent to or received from this device (since Data Engine start or statistics reset).	REFCON / plugin / upstream node
Retrans	Number of messages re-transmitted to this device (since Data Engine start or statistics reset).	REFCON / plugin / upstream node
Lates	Number of messages received from that RMM Master after timeout (since Data Engine start or statistics reset).	REFCON
No answ	Number of times (since Data Engine start or statistics reset) that a response message never arrived from that RMM Master.	REFCON
Errors	Number of messages with errors received from this device (since Data Engine start or statistics reset).	REFCON / plugin / upstream node
Up Time	Duration of time that the device has been working and available in format days:hours:minutes	REFCON



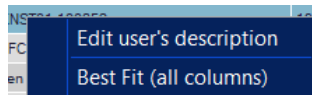
Column header / abbreviation	Explanation	Origin
Downstream	Data regarding the device to device communication, according to the device, in direction towards REFCON.	Device
Upstream	Data regarding the device to device communication, according to the device, in direction away from REFCON.	Device

The following columns are irrelevant when the RMM Master is connected to an RS2LAN, while they are used when it is connected through the COM port:

Column header / abbreviation	Explanation	Origin
Rejects	Number of messages rejected by that RMM Master (since Data Engine start or statistics reset).	REFCON
Deaths	Number of dead nodes.	REFCON

Shortcut menu in System Devices list

At any device you can right-click to have the Shortcut menu opened. This is an example:



The available options depend on the device, and for some you need to check the separate device documentation for user guidance.

The commonly seen options are:

Menu option	Explanation	
Best Fit (all columns)	Adjusts the widths of all columns so that longest text is fully visible.	
Edit user's description	Opens a form where you can change the text shown in the User's description column.	
Properties	Opens a window with detailed information about the device.	
Settings	Where available, opens the Settings Editor ^[338] .	
Get Log Files	See How to get log files from a supporting device ^[337] .	

Remove offline devices

REFCON automatically detects the devices. Once a device has been detected, it does not automatically disappear, even if it goes offline. For the case that you have been experimenting with extra devices or have given a device a wrong network ID, the "Remove offline devices" button is available.

Selecting the button makes REFCON check each device group - the RMM Masters, the RS2LANs etc.

- If all devices in a device group are online there is nothing to remove in that group.
- If the number of devices of a specific type is less than or equal to the optional count stated in the [license file](#)^[12], there is nothing to remove in that group.

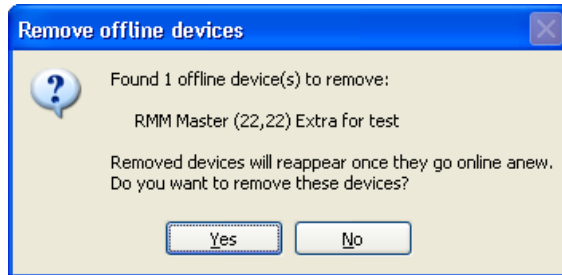


- When neither of the two conditions above are met, all offline devices of the type exceeding the optional count stated in the [license file](#)^[12] are listed on the remove candidate list.

If the check reveals no remove-candidates at all, you are informed like this:



Otherwise, if there are one or more candidate, these are listed:



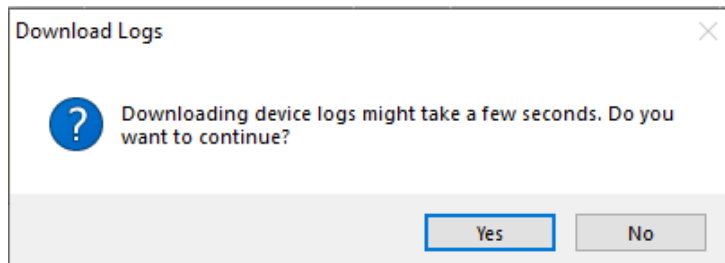
Select Yes to confirm the removal.

Reset statistics

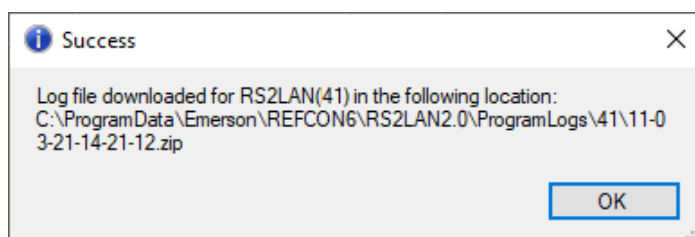
You can at any time select "Reset statistics" to have all the statistics counters reset to 0.

How to get log files from a supporting device

If the device supports log file retrieval, you find the "Get Log Files" option in the [Shortcut menu](#)^[336]. Selecting this displays a confirmation dialog:



After successful download of the log file you will be notified about the path of the log file:





20.12 The Settings Editor

Key subjects in this topic:

[Overview](#) ^[338]
[Opening the Settings Editor](#) ^[338]
[The Settings Editor window](#) ^[338]
[Plugin settings permissions](#) ^[338]
[Using the Settings Editor](#) ^[339]

Related topics:

[System Devices list](#) ^[333]
[Understanding and Maintaining REFCON](#) ^[302]

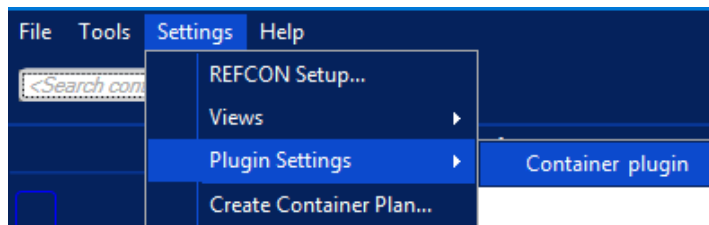
Overview

The *Settings Editor* is available for any plugin defining *configurable settings*, that is, settings that can be viewed and changed by user.

The plugin fully controls the contents of the editor, so what is described here is the general guide lines and an example.

Opening the Settings Editor

If any plugin(s) defines any settings, the Settings Editor can be opened from the Settings menu:

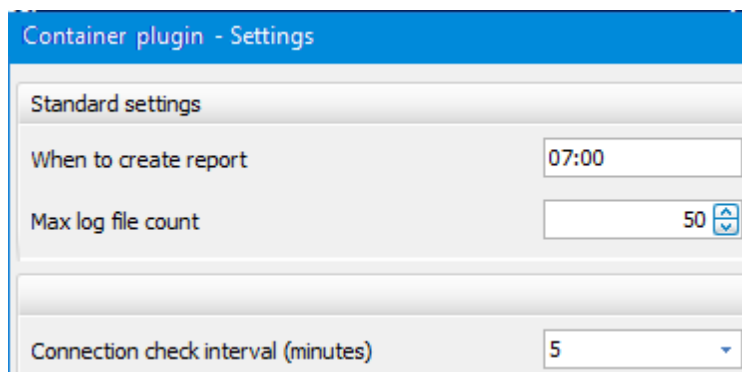


Select the plugin name to open the editor.

Alternatively you can right-click the plugin in the [System Devices list](#) ^[333], and select the Settings entry from the Shortcut menu.

The Settings Editor window

In the Settings Editor you find one line for each setting defined by the plugin, in groups if so decided by the plugin:



In this example you see two settings in the "Standard settings" group and one in the no-name group.

Plugin settings permissions

To be allowed to make changes, the REFCON user must be entitled to a role that includes either or both of the following user actions (DS being short for REFCONs **D**ata **S**ervice):

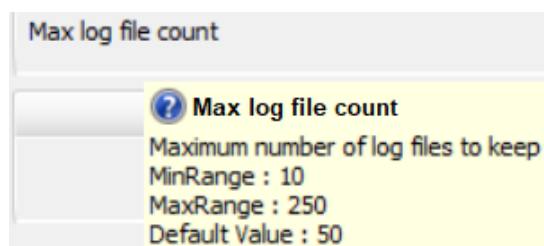


User action	Explanation
Update settings for DS client (REFCON setup group)	This allows changing "normal" settings, meaning parameters that adjust the behavior of the plugin, normally without risk of critical changes.
Update privileged settings for DS client (REFCON setup group)	This allows changing "special" settings that should be changed with extra care, maybe requiring deeper knowledge.

Using the Settings Editor

In the editor you see short label texts and the current values being actively used by the plugin now.

If you want details, move your mouse pointer over the label of the setting. A tool tip with additional information shows up:

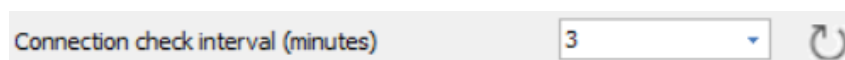


Here,

- Next to the question mark, the label is repeated.
- Below this you will normally find an explanatory description.
- The Default Value is the factory setting, as suggested by the plugin manufacturer, shown for your reference. When the plugin has just been installed the first time, all current values will be equal to their default values.

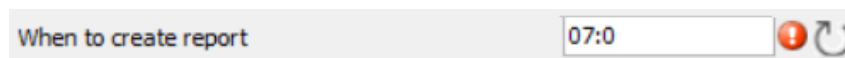
Your changes are forwarded to the plugin when you select the [OK] button. You may discard all changes by selecting [Cancel].

When you have made a change (and not yet selected [OK] or [Cancel]), a revert button shows up:



Use this button to discard your changes to this individual setting.

If you make a non valid selection, a red ! symbol shows up:



Move the mouse cursor over the red symbol to get an explanation.



20.13 Container Modems list

Key subjects in this topic:

[Overview](#)^[340]
[Mounted versus Portable modem](#)^[340]
[Opening the Container Modems list](#)^[340]
[The Container Modems list window](#)^[340]
[A modem's location](#)^[340]
[When does a modem disappear from the list?](#)^[342]
[Column specifications](#)^[342]
[Modems' Communication State](#)^[343]

Related topics:

[Understanding and Maintaining REFCON](#)^[302]

Overview

The *Container Modems list* shows all modems registered in REFCON. This allows you to check if your monitoring infrastructure is all right. In case of troubles, the list will help you troubleshoot what is wrong.

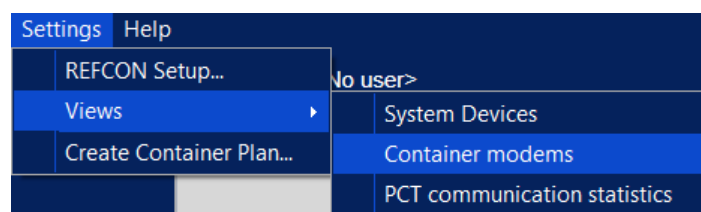
Mounted versus Portable modem

A *Mounted modem* is any modem being a part of the container's electronics. It can be integrated in the controller, or it can be fastened by screws and permanently connected to the controller. A mounted modem arrives at and leaves your site with the container. There are many types of mounted modems on the market, including the RMM PCT modem and the RMM-W modem manufactured by Copeland Transportation Solutions.

A *Portable modem* is owned by the site. It is connected to the container when this arrives, and disconnected from the container before this is shipped. It allows the site to remotely monitor **all** reefer containers. If you need this, please contact Copeland Transportation Solutions about the REFCON Portable Modem Solution.

Opening the Container Modems list

The Container Modems list is available from the Settings menu under the Views entry:



The Container Modems list window

In the Container Modems list you find one line for each container modem known to REFCON:

Container Modems														☰ ? x
Modem name	Serial	Comm State	Software	Hardware	First ever UTC	First after change UTC	Last comm UTC	Current location	Modems location	Loc_match	Container	Category	Source	Comm Start Time
REFCON Portable Modem	7270003	Not communicating	1.0.7	1.1.4	2022-08-02 08:33:51	2022-08-02 08:33:51	2022-08-02 09:05:47	A026-70-A.2	A026-70-A.2	●	HUBU0010015	Portable	Wireless ...	2022-08-02 09:03:03
REFCON Portable Modem	7270004	Communicating	1.0.7	1.1.4	2022-08-02 08:33:51	2022-08-02 08:33:51	2022-08-02 09:08:27	A026-70-A.3		○	HUBU0010020	Portable	Wireless ...	2022-08-02 09:03:03
REFCON Portable Modem	8020003	Communicating	1.0.6	1.1.4	2022-08-02 08:58:44	2022-08-02 09:06:59	2022-08-02 09:07:21	?		○	TSTU0010033	Portable	Wireless ...	2022-08-02 09:03:03
REFCON Portable Modem	8020004	Poor communication	1.0.7	1.1.4	2022-08-02 08:58:44	2022-08-02 09:08:13	2022-08-02 09:08:13	A026-70-A.5		○	TSTU0010049	Portable	Wireless ...	2022-08-02 09:03:03

When opening the window, only the registered [REFCON Portable Modems](#)^[222] are shown. At the bottom of the window you can in stead choose to see only Mounted modems or All.

A modem's location

In the Container Modems list, you find 3 location-related columns.



The "Current location" shows the location of the communicating container as reported by the Planning System. The location remains shown after the container stops communication, and until it is replaced by another, or the modem record is removed.

For [mounted modems](#)^[340], only the "Current location" is in use.

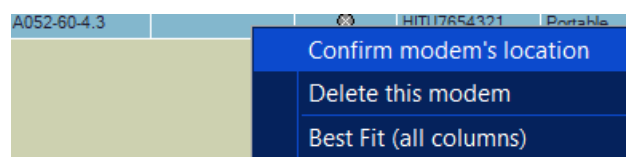
Customers handle REFCON Portable Modems in different ways. If you move them around, maybe keeping them in a common box when not used, also only the "Current location" is relevant.

If your REFCON Portable Modems are fixed to the locations by wires, so that a certain modem can only be used on the various containers in a specific location, the "Modems location" and the "Loc.match" columns are important, as they can help you investigate "ghost containers", that is, containers having been positioned wrongly or have wrong container IDs programmed into the controllers.

Please have a look at the following cases (relevant to fixed REFCON Portable Modems only):

No.	Case	Display	Explanation
1	Modem starts communication the very first time.	If container's location is known, this is displayed as "Current location", otherwise "?". "Modems location" is empty.	"Modems location" has not yet been confirmed.
2	Container is powered off and is shipped.	No change to the location columns.	Even when the container record is deleted, the modem remains in the list, and its location data are unchanged.
3	Modem starts communication on another container.	If container's location is known, and it is equal to old "Current location" (not "?"), "Modems location" is set equal to "Current location". "Loc.match" gets green.	This is the automatic confirmation : Two containers in succession have had same location, so we assume this is correct and therefore confirm the modem's correct location.
4	Container is powered off and is shipped.	No change to the location columns.	
5	Modem starts communication on yet another container, claimed by Planning System to have a different but valid location.	"Modems location" keeps unchanged as set above. "Current location" is set to container's location. "Loc.match" gets gray.	Here you have a "ghost". You spot the gray "Loc.match" and note the "Modems location", which is then where you walk to resolve the problem.

If you wish, you can confirm modem's location manually. This requires that the "Current location" is valid, and that the location is not already confirmed. In such case, right-click the modem:



Select the Confirm command and accept.

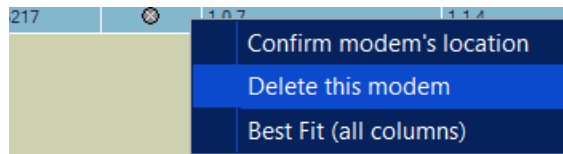


When does a modem disappear from the list?

The Container Modems list is first of all a tool for you to manage your REFCON Portable Modems. Your site has invested in a number of these modems, supposed to be used for a long time on various containers. Therefore, a fundamental idea is that REFCON Portable Modems stay in the list "forever".

We have chosen also to list [mounted modems](#)^[340]. A mounted modem per definition leaves your site with its container; therefore, any mounted modem is automatically removed from the list when its container is deleted from REFCON.

It will happen that a REFCON Portable Modem is no longer available - maybe because you forgot to dismount it before the container left, or because of a defect. In such case you should manually delete it from the list. This requires that the modem is not communicating. In such case, right-click the modem:



Select the Delete command and accept.

Column specifications

The following table describes the columns included in the Container Modems list:

Column header	Explanation	Origin
Modem name	The type of modem.	Modem
Serial	Serial number of the modem. Can normally be recognized at the exterior of the modem. Together with the Modem name, this uniquely identifies the modem.	Modem.
Comm State	Refer to Modems' Communication State ^[343] below.	REFCON
Software	Version of the software running in the modem. When a new software version for Portable modems is released and gets known to your REFCON system, the software is as standard automatically updated in the modem; however, this will take some time (days, likely) and requires modems to be within reach of a REFCON Wireless Hub or a phone running the REFCON Mobile - so you may observe different versions in the list.	Modem
Hardware	The hardware version of the modem.	Modem
First ever UTC	The very first time REFCON received information from the modem identifying it.	REFCON
First after change UTC	For REFCON Portable Modem: The time when the modem started communicating after connection to a different container, or after a period of being <i>Not communicating</i> . For PCT modems: The time when the modem data set completed - normally within a minute after First ever UTC. In fact this is the time when the record is created in the list. Other mounted modems: Normally same as First ever UTC.	REFCON
Last comm UTC	The latest time REFCON communicated with this modem.	REFCON
Current location	The location of the modem's container according to the Planning System. See also A modem's location ^[340] above.	REFCON / Planning



Column header	Explanation	Origin
	If the container has started communication before having been reported by the Planning System, you will read a question mark.	System
Modems location	See A modem's location ^[340] above.	REFCON / User
Loc.match	See A modem's location ^[340] above.	REFCON
Container	The ID of the container to which the modem is connected. If the container has been deleted from REFCON, the ID is shown in parentheses.	Modem / REFCON
Category	Portable or Mounted	REFCON
Source	Similar to Source for container.	REFCON
Comm Start Time	When not empty, the Communication Start Time would normally be as close as possible to the modem power-on time. Using the REFCON Wireless Hub solution ^[232] , the Communication Start Time may be earlier than the "First after change UTC" time because REFCON did register communication from the modem before having received sufficient information to fully identify the modem. In such case you will see that at the moment the modem shows up or is updated in the list, the Comm Start Time is earlier.	REFCON

Modems' Communication State

The Comm State column exists both in the Container Modems list and in the [Container List](#)^[66]. In the Container Modems list, Comm State indicates the status of the communication between the modem and REFCON.

The following time settings are involved when determining the Comm State:

Time setting	Explanation
t_1	Maximum time allowed for no "life signs" or no live data updates from modem. The value of t_1 depends on the solution; for PCT it is 1 hour; for REFCON Portable Modem Solution, please refer to the description of this ^[219] .

The following table shows all the Comm States possible for modems.

Priority	Comm State option	Explanation	Recommended action
1	Communicating	If your REFCON includes an online solution (PCT, REFCON Wireless Hub ^[232] solution or similar), <i>Communicating</i> is the desired, "normal" state	None



Priority	Comm State option	Explanation	Recommended action
		for any modem being on power. <i>Communicating</i> means that REFCON has received modem data from this modem within last t_1 minutes through an online source.	
2	No controller	The modem is on power but is unable to communicate with any controller, and therefore did report "no connection to controller" less than t_1 minutes ago.	Check the plugs and cable between the modem and the controller.
3	Poor communication	REFCON has not received any complete live data update nor any indication of no connection to controller for this modem within last t_1 minutes; however, REFCON has in fact registered some communication related to this modem within last t_1 minutes.	If this state persists, check if there are any objects or signal noise that can affect the communication. If so, attempt to move objects / reduce signal noise.
4	Communication started	REFCON has registered some communication related to this modem but has not yet received any complete live data update nor any indication of no connection to controller. Normally this state remains only for a short time, after which REFCON receives complete live data and then changes to <i>Communicating</i> (or <i>No controller</i>). If REFCON keeps registering some communication but still no complete live data	None



Priority	Comm State option	Explanation	Recommended action
		for t_1 minutes, the state changes to <i>Poor communication</i> . If, after change to <i>Communication started</i> , REFCON registers no communication at all for t_1 minutes, the state changes to <i>Not communicating</i> .	
5	Not communicating	REFCON has registered no communication related to this modem for the past t_1 minutes after having seen communication before.	If power off was expected: None. Otherwise it <i>could</i> indicate unintended loss of power.

The Comm State options are listed in prioritized order. For example, if Comm State *Communicating* is determined, this generally overrules the lower prioritized options; similar, if *Poor communication* is determined, but *Communicating* was determined less than t_1 minutes ago, *Communicating* is kept.

20.14 PCT Communication Statistics list

Key subjects in this topic:

[Overview](#)³⁴⁵

[Opening the PCT Communication Statistics list](#)³⁴⁵

[The PCT Communication Statistics list window](#)³⁴⁶

[Column specifications](#)³⁴⁶

Related topics:

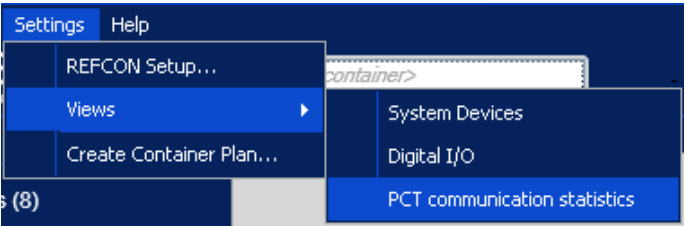
[Understanding and Maintaining REFCON](#)³⁰²

Overview

The *PCT Communication Statistics list* is a locked [Container List](#)⁶⁶ having been configured out-of-the-box to show communication statistics for each individual container.

Opening the PCT Communication Statistics list

The PCT Communication Statistics list is available from the Settings menu under the Views entry:





The PCT Communication Statistics list window

In the PCT Communication Statistics list you find one line for each container:

PCT communication statistics														
Standard Info			Communication Info			Controller		Diagnostics						
Loc...	ContainerID		Online	HasData	Modem	CtrType		MastNet	MastNode	MsgCount	Invalids	NoResp	Logins	%Online
	MWCU6745130		●	●	Mk.IVb, 0539-3720	StarCool RCCU5		22	18	9	0	0	1	100 0 (0%)
	MWMU6417276		●	●	MPC2000/MP3000, 0233-9002	Thermo King MP3000		22	18	7	0	0	1	100 0 (0%)
	MWCU6520999		●	●	Mk.III/SEA, 0348-4310	Carrier Microlink 2/2i/3		22	18	17	0	0	1	100 0 (0%)
	SEAU4678140		●	●	Mk.III/SEA, 0538-5890	Carrier Microlink 2/2i/3 Scroll		22	18	17	0	0	1	100 0 (0%)
	MWCU6830269		●	●	Mk.III/SEA, 0148-4310	Carrier Microlink 2/2i/3		22	18	17	0	0	1	100 0 (0%)
	WFHU5019482		●	●	Mk.III/SEA, 0148-4310	Carrier Microlink 2/2i/3		22	18	17	0	0	1	100 0 (0%)
	MCHU8972961		●	●	Mk.III/SEA, 0148-4310	Carrier Microlink 2 Everfresh		22	18	24	0	0	1	100 0 (0%)

Column specifications

Some of the columns in the PCT Communication Statistics list are recognised from other Container Lists, see the [general column description](#)^[69]. The following columns can also be used in other Container Lists, but most likely they are only used in the statistics list:

Column header / abbreviation	Explanation
MastNet	The network number of the RMM master that communicates with this container.
MastNode	The node number of the RMM master that communicates with this container.
MsgCount	Total number of messages sent to this container.
Invalids	Number of messages with errors received from this container.
NoResp	Number of times that a response message never arrived from this container.
Logins	Number of times this container has logged in to the RMM Master.
%Online	Since first login, the fraction of the time that this container has been online.
MsgErrors	Sum of Invalids and NoResp. In parentheses: Same in fraction of MsgCount.
FirstLoginUTC	Time when this container logged in (the first time).
LastLoginUTC	Time when this container logged in (the last time).
LastPollTryUTC	Time when this container was last attempted polled (= requested for data).
LastPollAnswerUTC	Time when valid data were last received from this container.

20.15 Other trouble shooting tools

Key subjects in this topic:

[Overview](#)^[346]

Related topics:

[Understanding and Maintaining REFCON](#)^[302]



20.16 Master groups

Key subjects in this topic:

[Overview](#) ³⁴⁷

[Viewing the master group](#) ³⁴⁷

[Editing the master groups](#) ³⁴⁷

Related topics:

[System Devices List](#) ³³³

[Understanding and Maintaining REFCON](#) ³⁰²

Overview

A *master group* is a set of RMM Masters that are treated as if they are connected to a common power cable network in the sense that the individual RMM Master can "hear" the signals from the other RMM Masters. To avoid that the RMM Masters disturb each other, REFCON makes sure that only one RMM Master in a master group is active at a time.

As standard there is one and only one master group in a REFCON system. In this case there is always only one RMM Master active at a time.

The maximum communication performance is obtained by defining as many master groups as possible; however, it is vital that all RMM Masters and containers in one master group are completely separated from all RMM Masters and containers in all other master groups as regards Power Cable Transmission, that is, a unit in one master group must in no way be able to "hear" signals sent by any unit in any other master group. Please contact us for advice about your specific power cable configuration.

Viewing the master group

Each RMM Master always belongs to one and only one master group. You can see the number of the group in the Group column in the [System Devices List](#) ³³³.

Editing the master groups

It is possible to change the used master groups by

1. Editing the file intended for this purpose.
2. Stopping and Starting REFCON from the REFCON Manager.

Warning: Because of the separation requirement mentioned above, do never change the master group file unless you are highly qualified in Power Cable Transmission.

REFCON 6

Reefer Container Monitoring System

The REFCON Manager

Transportation Solutions





21 The REFCON Manager

Key subjects in this topic:

[Overview](#) ^[350]
[When do you use the REFCON Manager?](#) ^[350]
[Starting the REFCON Manager](#) ^[351]
[The REFCON Manager taskbar icon](#) ^[351]
[Opening and closing the REFCON Manager](#) ^[351]
[Understanding the REFCON Manager](#) ^[352]

Related topics:

[Supervision](#) ^[352]
[Administration](#) ^[354]
[Remote Monitor Configuration](#) ^[356]

Overview

The *REFCON Manager* is an extra software program running on the REFCON Main PC only. It provides a number of tools for controlling and maintaining the entire REFCON software package installed and running on the REFCON Main PC and the optional REFCON Monitor PCs.

When do you use the REFCON Manager?

First of all: When REFCON runs perfectly well, you do normally not worry about the REFCON Manager at all. It does not even have to run.

You must or can use the REFCON Manager in the following cases:

Case	Typical situation(s)	What to do
New License file	You purchased extended functionality.	Import the License file ^[355] .
New Container Plan	Maybe you had Copeland Transportation Solutions create a new design.	Import the Container Plan ^[356] .
Import Configuration	- You need your REFCON to work exactly like a specific other REFCON system from which you received an exported configuration. - Copeland Transportation Solutions did organise your configuration in advance.	Import configuration ^[355] .
Install software updates	You received an update package.	See how to check for updates and how to install in Understanding Installation and Update ^[314] .
Restore backup data	You performed a re-installation of REFCON 6.	Use the Restore function ^[355] .
Back up data	- You changed your settings. - You regard log files valuable and therefore make back up regularly.	Use Backup command from REFCON Monitor ^[43] or from REFCON Manager ^[355] .
Prepare REFCON Monitor installation	You want the REFCON Monitor to run on a Secondary PC.	Save the link on a USB key or similar as described in



Case	Typical situation(s)	What to do
		Remote Monitor Configuration ^[356]
Trouble-shooting the REFCON system	System hardware is failing.	Watch the REFCON Manager taskbar icon (see below) or the Supervision ^[352] tab.

Starting the REFCON Manager

To start the REFCON Manager, you need administrator rights in Windows.

If the taskbar icon described below is **not** available when you need it, you must start the REFCON Manager using the shortcut being normally found on the desktop:



You do as standard have to log on to run the REFCON Manager, similar to [Logging on to REFCON Monitor](#)^[57].

The REFCON Manager taskbar icon

The REFCON Manager is identified by the following taskbar icon:



The icon can display the following colours:

Icon colour	in words	Explanation
	Green	Everything is fine - all parts of REFCON on the REFCON Main PC work well.
	White	REFCON was stopped by a user and does not run.
	Orange	Temporary state: REFCON is being started or being stopped. Normally the colour changes within a short time.
	Red	Something is wrong - one or more parts of REFCON on the REFCON Main PC is malfunctioning. Note: The icon might be red temporarily for a few seconds, this should be considered OK.

Opening and closing the REFCON Manager

You open the REFCON Manager's user interface by

- Right-clicking the taskbar icon and selecting Show from the menu:



- Double-clicking the taskbar icon.

You can hide the REFCON Manager by clicking either the Minimize or the Close button upper right:



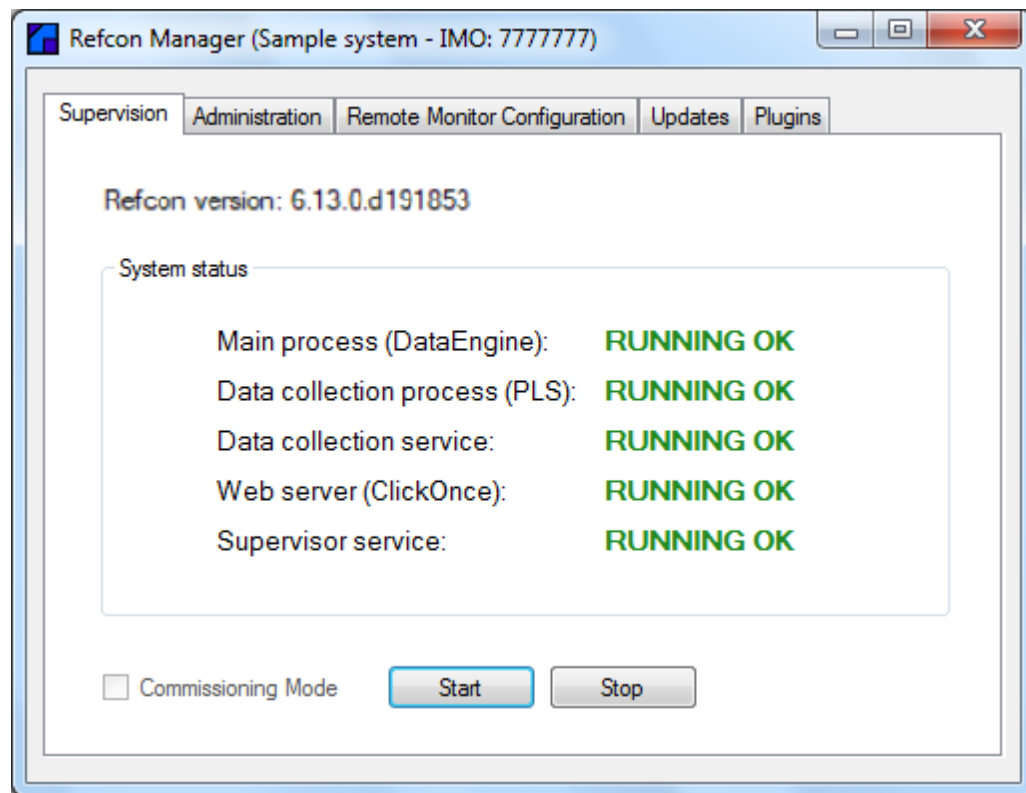
In both cases, the program keeps running, and the taskbar icon keeps visible.



If you for some reason want the REFCON Manager to close completely, you need to select "Exit Refcon Manager" from the taskbar icon menu (see above).

Understanding the REFCON Manager

When just opened, the REFCON Manager appears like this:



The window is organised with five tabs: [Supervision](#)³⁵², [Administration](#)³⁵⁴, [Remote Monitor Configuration](#)³⁵⁶, Updates and Plugins.

21.1 Supervision

Key subjects in this topic:

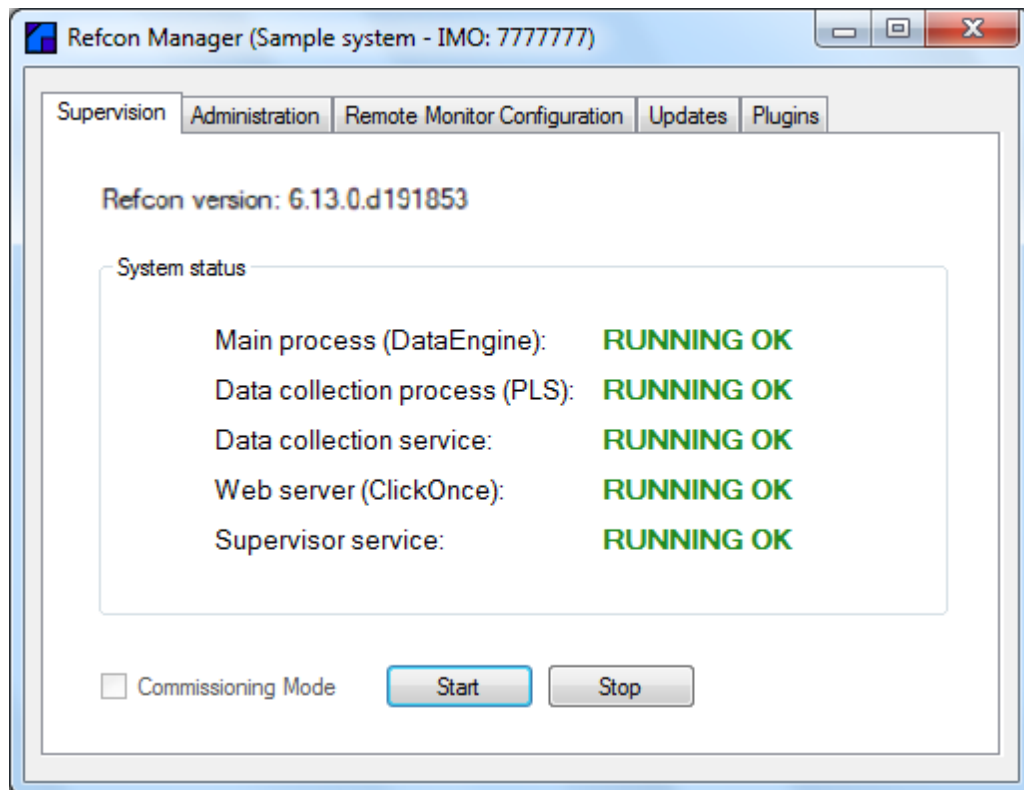
- [Overview](#)³⁵²
- [The states of REFCON's services](#)³⁵³
- [Why Stop REFCON](#)³⁵³
- [Commisioning mode](#)³⁵⁴

Related topics:

- [The REFCON Manager](#)³⁵⁰

Overview

The REFCON Manager's Supervision tab shows the [REFCON 6 version](#)³⁰⁶ and the states of REFCON's services.



Also it allows you to stop REFCON and to start it in normal mode or commissioning mode.

The states of REFCON's services

Each service can be in any of the following states:

State	Colour	Explanation
RUNNING OK	Green	The service is running as expected.
NOT IN USE	Grey	The service is not running, and it is not expected to be running according to License options.
STARTING	Orange	The service is starting right now, should last for a few seconds only.
STOPPING	Orange	The service is stopping right now, should last for a few seconds only.
STOPPED	Dark red	The service is not running though it should be. If not stopped by user, this indicates a fault.
RUNNING AS APP	Red	Only PLS can be in this state. It indicates commissioning mode. Before logging off from the PC you must make sure you return to normal mode.

Why Stop REFCON

REFCON is supposed to run "permanently", supervising your containers 24 hours - so normally you would not stop it.

You do, however, need to select Stop to be able to perform the functions under the Administration tab: Backup/restore and import of new License, Container Plan or configuration. Note: You do not need to Stop REFCON to make the backup from the REFCON Monitor.

Remember that you have no container supervision while REFCON is stopped.



Commissioning mode

Having stopped REFCON, you have the option to check the "Commissioning Mode" checkbox. You will in such case get a warning saying that Commissioning mode should only be selected by service technicians from Copeland Transportation Solutions or our co-operating partners.

Please **be aware** that in Commissioning mode, PLS is running as an application rather than a service. This is potentially a sensible situation because if you (or another user) logs off from Windows without having cancelled the Commissioning mode, PLS will simply stop, and REFCON will therefore not be able to communicate with the containers.

The advantage about Commissioning mode is that PLS, being an application, has a separate user interface useful for trouble-shooting and similar purposes.

To **cancel Commissioning mode**, proceed as follows:

1. Click the Stop button.
2. Uncheck the Commissioning mode checkbox.
3. Click the Start button. You should now see the green words RUNNING OK at the PLS.

21.2 Administration

Key subjects in this topic:

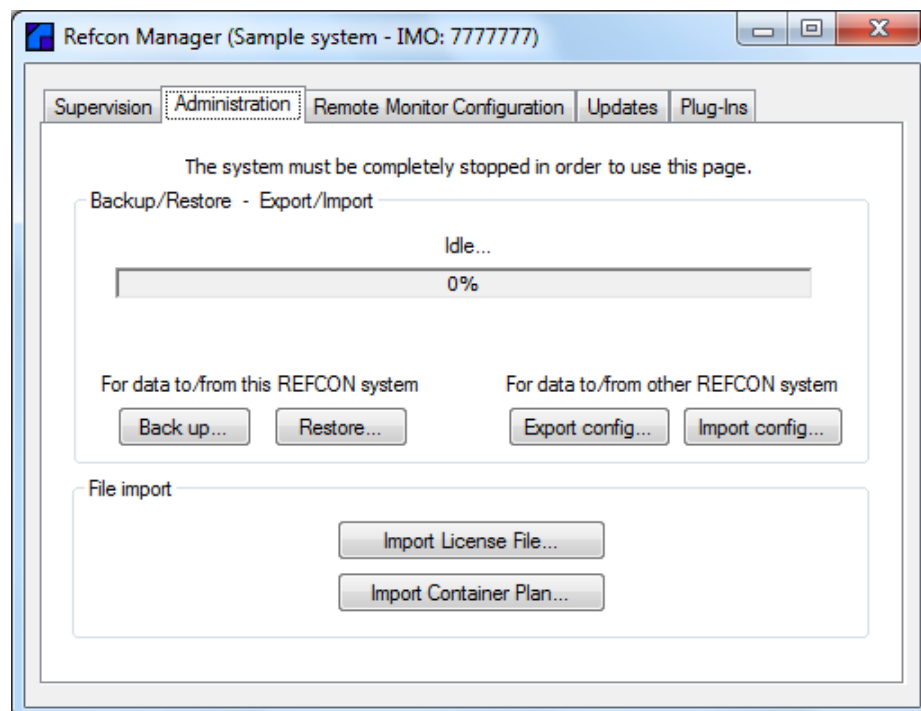
[Overview](#) ³⁵⁴
[Backup function](#) ³⁵⁵
[Restore function](#) ³⁵⁵
[Export configuration](#) ³⁵⁵
[Import configuration](#) ³⁵⁵
[Import License file](#) ³⁵⁵
[Import Container Plan](#) ³⁵⁶

Related topics:

[The REFCON Manager](#) ³⁵⁰

Overview

The REFCON Manager's Administration tab provides file input and output functions:





Note that REFCON must be stopped for these functions to be enabled. This is done from [The Supervision tab](#)^[352].

Backup function

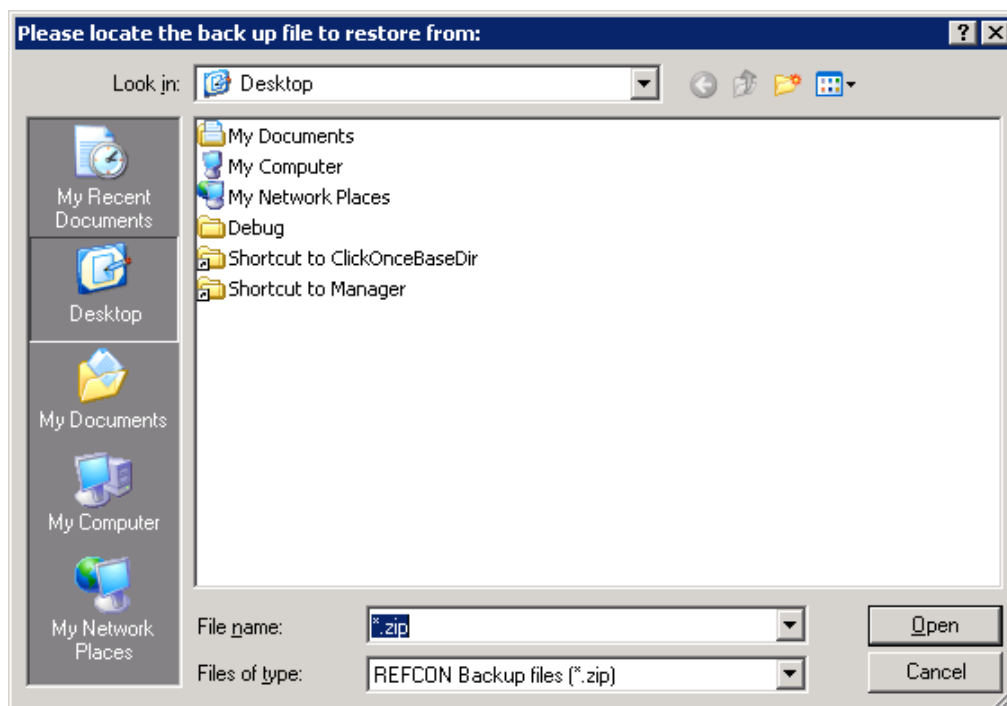
The Backup function available from the REFCON Manager works exactly like the one available from the REFCON Monitor, see [The Backup Function](#)^[43].

Restore function

The Restore function is your tool to have data saved with the Backup function reloaded into REFCON. You would normally need this after a REFCON re-installation only.

Proceed as follows:

1. Stop REFCON.
2. Click "Restore". The following dialog appears:



3. Locate the backup zip file and click Open.
4. Wait for the data to load.
5. Start REFCON.

Export configuration

The Export configuration function available from the REFCON Manager works exactly like the one available from the REFCON Monitor, see [Configuration Export](#)^[46].

Import configuration

The Import configuration function is your tool to have data saved with the Configuration Export function imported into REFCON. You would typically import a configuration received from another REFCON system or from Copeland Transportation Solutions.

The procedure is quite similar to the [Restore function](#)^[355] above.

Import License file

Any REFCON system needs a unique [license file](#)^[11], defining exactly what standard features are enabled for the specific application. If you decide to purchase a new license, or if you for



other reason receive a new license file, you must import it using the REFCON Manager as follows:

1. Stop REFCON.
2. Select "Import License File".
3. In the dialog, locate the license XML file and select Open. If you choose an XML file of wrong type you are warned.
4. Wait for import to complete.
5. Start REFCON.

Import Container Plan

You can make a new or changed Container Plan available at the REFCON Monitor in the following ways:

- By selecting "Create Container Plan" from the REFCON Monitor menu. On ships, you will need a [baplie file](#)^[102], while in terminals, a special [XML file](#)^[100] provided by Copeland Transportation Solutions is required. In both cases, the result is a Container Plan, stored as a file in the format used by REFCON 6.
- By importing a Container Plan file in the format used by REFCON 6.

In the latter case, proceed as follows:

1. Stop REFCON.
2. Select "Import Container Plan".
3. In the dialog, locate the Container Plan XML file and select Open. If you choose an XML file of wrong type you are warned.
4. Start REFCON.

21.3 Remote Monitor Configuration

Key subjects in this topic:

[Overview](#)^[356]
[Common case: REFCON Main PC has 1 network connector](#)^[357]
[If the automatic assumptions are not OK](#)^[357]
[How to install certificate on remote computer](#)^[358]
[How to use a custom certificate?](#)^[360]
[Disabling HTTPS for Remote Monitor installation](#)^[360]

Related topics:

[The REFCON Manager](#)^[350]

Overview

To have the REFCON Monitor running on a secondary PC, first of all this must be allowed in the License file. In the following we assume that it is allowed.

For the REFCON Monitor to work on a secondary PC, this must be connected to the REFCON Main PC by a network of ethernet type (LAN). Most commonly, the REFCON Main PC is equipped with one network connector only, and then things are very straightforward.

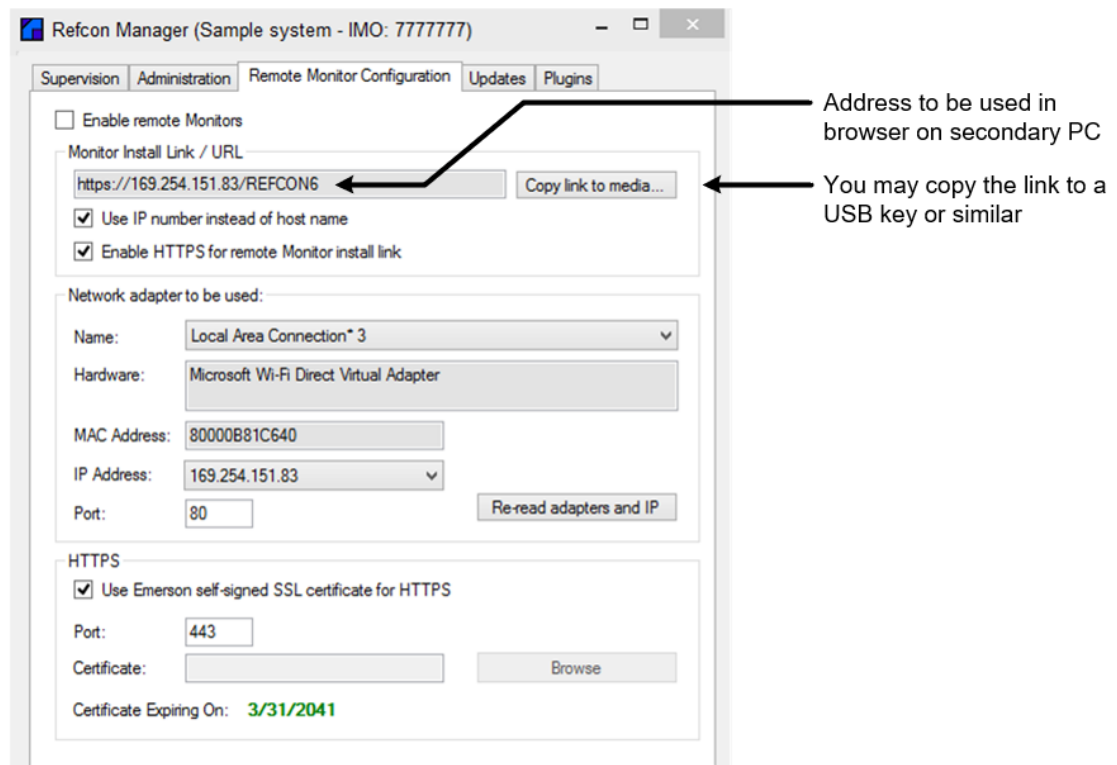
Should the REFCON Main PC be equipped with more than one network connector, you need to decide which one to use for REFCON Monitor connection in the Remote Monitor Configuration tab of the REFCON Manager.



Common case: REFCON Main PC has 1 network connector

During installation, REFCON always picks "the first the best" network connector on the REFCON Main PC and assumes that this is the one to be used for REFCON Monitor connection - and of course, if there is only one, this assumption is always correct. It also assumes that Port number 80 is used, which is the standard web service Port being generally accepted by firewalls.

If both assumptions are OK, everything is set up automatically. All you need to do to have one or more REFCON Monitors installed on secondary PCs is to either write down or save the Monitor Install Link/URL:



Then take your note or your USB key to the secondary PC.

Here, if the specified SSL certificate is not already installed, follow [the instructions below](#)³⁵⁸.

When the SSL certificate has been installed, either

- start Internet Explorer or another suitable browser and type in the URL noted above, or
- invoke the link directly from the USB key - which will start the browser and automatically insert the correct address.

If the automatic assumptions are not OK

If you need to select another network adapter or another Port number, proceed as follows:

1. Uncheck "Enable remote Monitors" (to get editing enabled).
2. Make the required changes in the lower part of dialog box (see below).
3. Check "Enable remote Monitors" (to get your changes applied).

During installation, REFCON automatically detects if the REFCON Main PC has more than one network connector (= adapter). Therefore you will normally be able to simply pick the right adapter from the drop-down list (labelled "Name"). Watch that selecting another makes the other data change, and you should then be able to recognize the correct adapter. Note that also the Monitor Install Link/URL changes, so you may have to make changes on the secondary PCs.

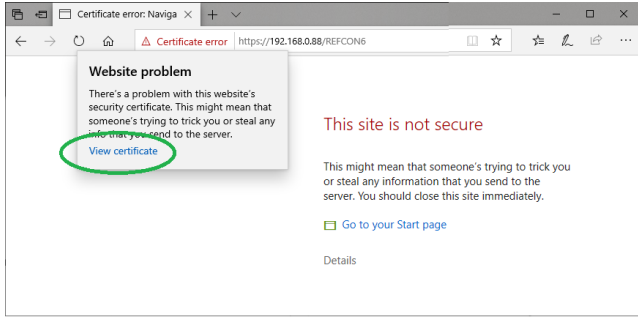
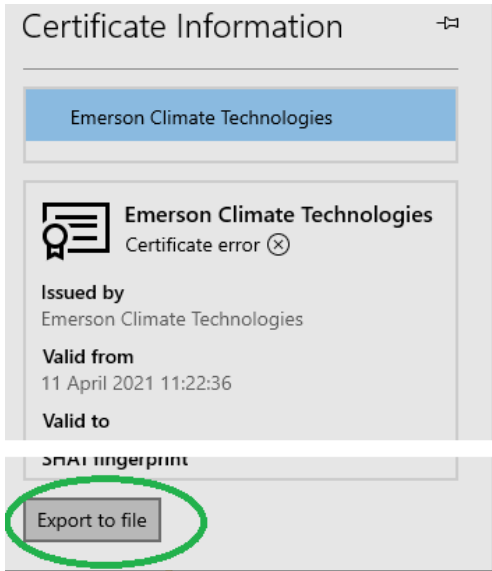


If necessary select another Port number. You might have to make changes to firewalls if doing so.

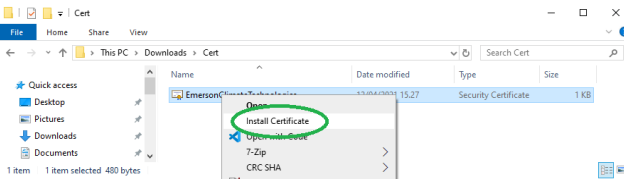
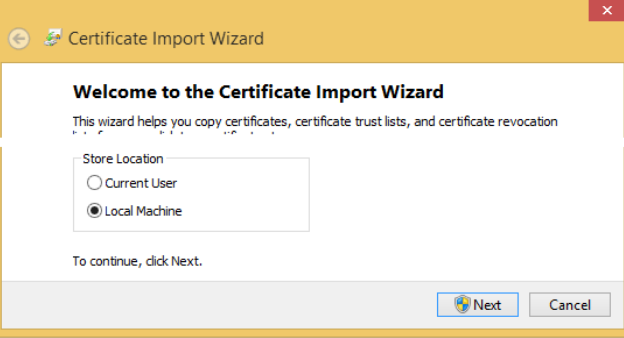
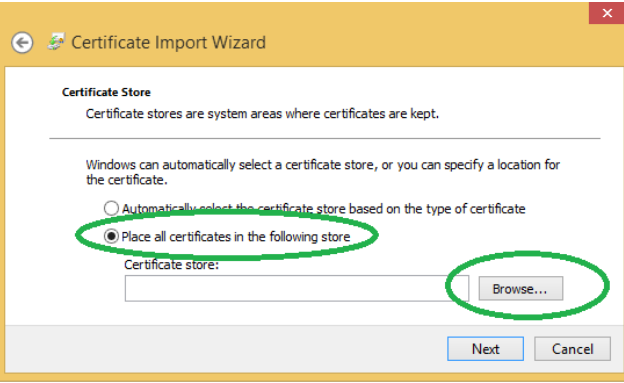
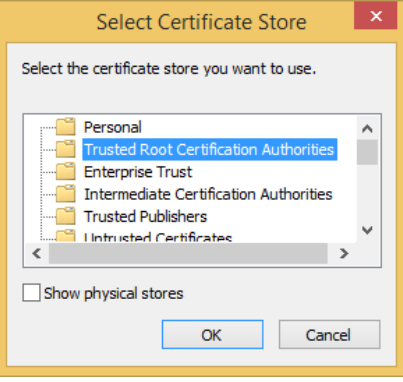
You may even want to select a different IP Address from the drop-down. Normally there is only one available, but if using virtual IP addressing, there can be more.

How to install certificate on remote computer

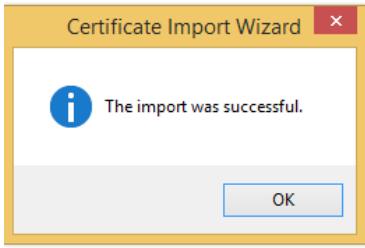
The used certificate needs to be installed on each remote computer that may be used to run the REFCON Monitor. This can be done as follows:

Step	Action	Result
1	Log on to Windows on the remote computer. If you need to prepare the certificate for more users, log on as administrator.	
2	Navigate to the Remote Monitor URL in your browser (Microsoft Edge in this example).	The Certificate error appears.
3	Select "View certificate": 	Certificate information is displayed.
4	Select [Export to file] and choose a folder where to store the file: 	The file is saved.
5	Navigate to where you stored the certificate. Right-click the file and select "Install Certificate".	The Certificate Import Wizard opens.



Step	Action	Result
		
6	In the Wizard, select "Local Machine" if you are administrator (re. step 1), otherwise "Current User": 	
7	Select "Place all certificates in the following store" and click [Browse...]: 	The list of certificate stores opens.
8	Select "Trusted Root Certification Authority" and click [OK]: 	The Store is shown in the Wizard.
9	In the Wizard, select [Next] and then [Finish]. You should see the result:	



Step	Action	Result
		
10	Return to your browser and try the Remote Monitor URL again.	You do not get Certificate Error. The REFCON Monitor installation page opens, and you can continue.

How to use a custom certificate?

If you want to use your own certificate, proceed as follows after the above:

1. Uncheck "Enable remote Monitors" (to get editing enabled).
2. Uncheck "Use Emerson self-signed SSL certificate for HTTPS", or select [Browse].
3. In the file selection dialog, browse to your valid customer certificate (.pfx file), and select [Open].
3. Check "Enable remote Monitors".
4. Make sure your certificate is trusted on all remote computers.

Disabling HTTPS for Remote Monitor installation

If you want Remote Monitor installations to not use an encrypted connection, you can disable HTTPS by unchecking "Enable HTTPS for remote Monitor install link". This will make it an http URL.



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